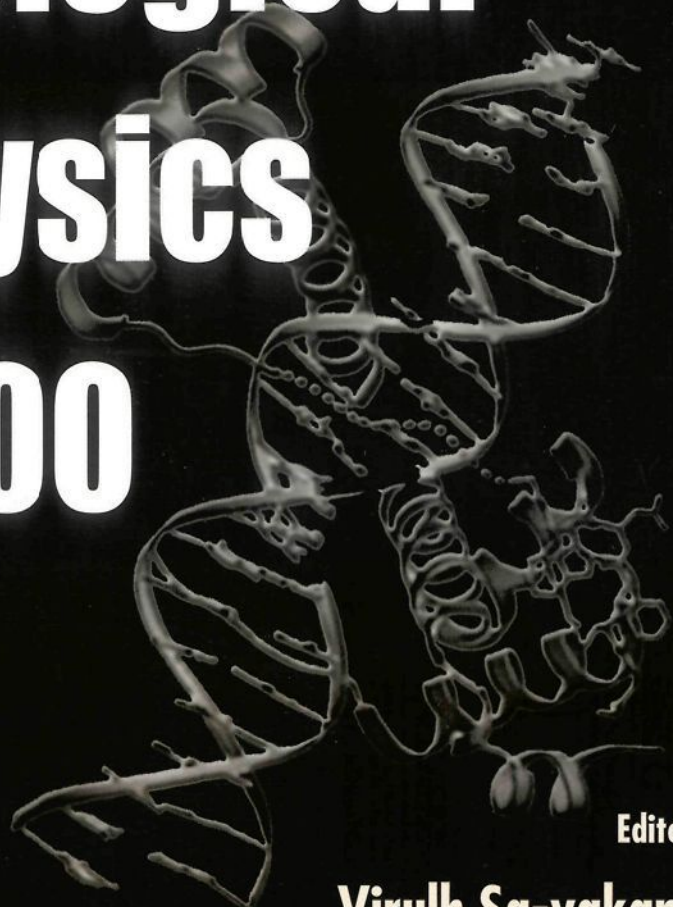


Proceedings of the First Workshop on

Biological Physics 2000



Editors

Virulh Sa-yakanit
Leif Matsson
Hans Frauenfelder

World Scientific

Biological Physics 2000

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Proceedings of the First Workshop on

Biological Physics 2000

Chulalongkorn University, Bangkok, Thailand

September 18 – 22, 2000

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PREFACE

A workshop on Biological Physics was held in Bangkok, Thailand, on 18-22 September 2000. Biological Physics, once a small part of physics in which a few devoted and possibly far-sighted physicists tried to build a new direction, has become a rapidly growing field. The workshop in Bangkok was planned to survey the field.

Biological Physics covers an enormous number of interesting subfields. A living cell, for instance, is a chemically open system in which many essential biological functions are nonequilibrium nonlinear collective phenomena, driven by chemical reactants such as ATP, GTP, different ligands and receptors. The living cell and many of its subsystems such as proteins, membranes, cytoskeleton with the microtubules, and the nucleus with the DNA double helix and the centrosome, are hence lyotropic systems depending on various reactant concentrations.

One central problem is what molecules can do collectively in living matter that they cannot do in inanimate matter. In other words, what is the phenomenon of life expressed in terms of the collective behavior of the constituent molecules. Without correlations between biomolecules, for instance receptors engaged by their ligands, there would be no life. Both nonlocal and local correlations are prerequisites for instance for signal transduction and DNA replication. A description of biological phenomena, experimentally studied at a molecular level, thus requires an extensive physical modeling with the full arsenal of tools and knowledge of many-body physics derived from studies of inanimate matter. However, biological systems are far more complex than the systems encountered in ordinary condensed matter physics. A central problem of biological physics thus is to start from what is known and then to study experimentally, theoretically, and computationally the new phenomena encountered in biosystems in order to create novel nonequilibrium models. This approach may lead to an understanding of the nature of biological functions. Thus one of the aims of the conference was to bridge the gap between experiment and theory and to bring together researchers from different fields.

The order of contents of this book follows that in which the talks were given at the meeting. One of the contributions, the one by Wokyung Sung, was published previously and is therefore included here in the form of an extended abstract. The meeting was economically supported by National Research Council of Thailand (NRCT), Thailand Research Fund (TRF), Asia Pacific Center for Theoretical Physics (APCTP), Third World Academy of Sciences (TWAS), and Chulalongkorn University, which are here gratefully acknowledged. We also want to thank Forum for Theoretical Science (FTS), Chulalongkorn University, for all the arrangements

before and during the conference. Many cordial thanks are also directed to Associate Professor Chaiyute Thunpithayakul, Associate Professor Wichit Sritrakool, Mr. Chanin Churrahmun, Mr. Piyapol Anubuddhangkura, and students of the FTS whose help made the Workshop a fruitful event.

On behalf of all visiting participators, two of us, Hans Frauenfelder and Leif Matsson, particularly want to thank the hosts, Virulh Sa-yakanit and all others in the FTS, for all the care extended to us. It created a warm atmosphere, which contributed to the quality of the meeting, and which we carried home with us as an unforgettable memory.

Finally, we sincerely thank Mr. Piyapol Anubuddhangkura for his valuable assistance in preparing and editing the manuscripts for the proceedings.

Virulh Sa-yakanit

Leif Matsson

Hans Frauenfelder

TRIBUTE TO PROF. STIG LUNDQVIST

One of the initiators of this meeting, Stig Lundquist, died on 6 April 2000. He was born in Gudmundraa near Kramfors in Sweden on 9 August 1925. After studies in Haerndesand and Uppsala he started research in physics as a pupil of Ivar Waller and became a doctor in 1955 with a thesis on properties of ion crystals. During his studies he made a living as a jazz-musician, playing trumpet and piano. He became a professor in theoretical physics in Gothenburg in 1961. He retired from this position in 1990.

Stig played a significant role in the rapid development of modern physics in Sweden, and in particular for the condensed matter physics in Gothenburg. His ability to inspire pupils and collaborators in this field is broadly witnessed. His contributions to international activities in physics, for instance at NORDITA in Copenhagen and ICTP in Trieste, are well known. With over a hundred published works he was an outstanding theoretical physicist in his field. As a leader of the Adriatic Research Conferences in condensed matter physics, parallel to the activities at the University of Gothenburg and Chalmers, the Royal Swedish Academy of Sciences, the Nobel Committee for Physics as a member 1971-1985 and as chairman 1980-85, he also stimulated interdisciplinary research activities such as in Biological Physics. He enriched our scientific community. His contributions to science were recognized by his election to the Royal Swedish Academy of Sciences, to the American Philosophical Society, and election to fellowship in the American Physical Society. The passing of Stig leaves us with a deep sense of loss.

Virulh Sa-yakanit

Leif Matsson

Hans Frauenfelder

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Opening Report

Prof. Virulh Sa-yakanit

Chairman of the Organizing Committee and FTS Director

Your Excellency, Distinguished Speakers, Ladies and Gentlemen:

It is a great pleasure for me as the chairman of the organizing committee to welcome you all to our first workshop on Biological Physics. The idea of organizing this workshop dates back to 1995 when Prof. Leif Matsson visited us at Forum for Theoretical Sciences as a visiting scientist. At that time Prof. Leif Matsson had informed us about the new developments in the field of Biological Physics in the international physics communities.

With the support of the Thailand Research Fund, I and Associate Professor Chaiyute Thunpithayakul visited Prof. Leif Matsson in Gothenburg. We discussed details about the possibility of organizing the first workshop on Biological Physics in Thailand. After consultation with Prof. Stig Lundqvist we had the confidence to go ahead in organizing the workshop and we invited Prof. Lundqvist to serve as the chairman of the International Advisory Committee. During our one week visit in Sweden we had worked out the tentative program of the first workshop. We are very fortunate that all distinguished speakers that we had intended to invite during our discussion in Gothenburg have accepted our invitation to attend the workshop.

The concept of Biological Physics workshop was also raised in the annual meeting of the National Research Council of Thailand with the prime minister as the chairman early this year. This workshop is also considered as an external activity of the Asia Pacific Center for Theoretical Physics. I would like to express our appreciation to the Third World Academy of Sciences for sending the TWAS prize recipient in Biological Physics from Mexico as one of the invited speakers.

In this workshop there are 30 participants from abroad such as the United States, Mexico, European countries, and Asian countries and 70 local participants from different disciplines ranging from mathematics, physics, biology, biotechnology and medicine.

Finally, I would like to thank all distinguished speakers for their participation in this workshop which enable us to organize this workshop smoothly and efficiently. We expect this to be a grand opening of ensuing more workshops and conferences on Biological Physics in Thailand where we all will have opportunities to attend in the future.

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Opening Remark

Prof. Vichai Boonsaeng

Director of Academic Research Division, TRF

Your Excellency, Distinguished Speakers, Ladies and Gentlemen:

It is a great honor for me to give some opening remarks on behalf of the Thailand Research Fund. I would like to express my appreciation to Prof. Virulh Sayakanit for initiating this Biological Physics workshop in Thailand and I am very happy to say that the Thailand Research Fund has played an important role in the early stages of the development of this workshop.

On behalf of Prof. Vicharn Panich, the director of the Thailand Research Fund, who is unable to attend due to other commitment abroad, I would like to inform you that our granting agency has established a number of research funds and research promotion agencies in Thailand such as the Senior Scholar Fund of which Prof. Virulh is one of such Senior Scholars. I am very pleased to say that there are several biochemists from different universities participating in this workshop. I myself was trained as a biochemist and therefore, I am sure that this workshop will open up a new dimension for biologists and medical doctors to look into the microscopic world where physicists and mathematicians will tell us their sides of scientific knowledge during the workshop.

Finally, I extend my best wishes for the success of the workshop. I am certain that with such eminent speakers and participants gathered here now, the workshop will succeed with flying colors.

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Opening Address

H. E. Ampol Senanarong

Privy Councilor and Chairman of National Research Council of Thailand

Distinguished Guests, Participants, Ladies and Gentlemen:

It is a pleasure to be here and to join you in opening the First Workshop on Biological Physics 2000. Biological Physics is considered as one of the most important fields in this century because it integrates all disciplines, both basic and applied sciences.

I would like to commend and congratulate the organizing committee for the choice of this topic of the workshop. Research in Biological Physics has contributed a revolutionary step in the application of science and technology in making life better. This is a fast growing field so a workshop of this kind is necessary in order that new ideas can be shared which I believe will certainly lead to more new ideas. Multidisciplinary areas of science are the most powerful areas of science because they integrate various technologies from different areas. Biological Physics, among all other disciplines of science, is unquestionably the most all-inclusive area of science that we have nowadays so we all are fortunate to be here today.

It is also a delight to see that this workshop is a gathering of scientists from many countries in America, Europe and Asia. This opportunity for scientists to come together and share their views is certainly a meaningful step for more future international cooperations. Now, I declare the opening of the First Workshop on Biological Physics 2000 and offer all of you my best wishes for a successful meeting.

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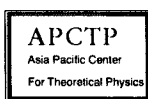


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BIOLOGICAL PHYSICS: AN OVERVIEW

HANS FRAUENFELDER

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1 Introduction

The best definition of biological physics was given by the famous mathematician Stanislaw Ulam during a dinner with HF. HF sketched the ideas of his work and Ulam said: "I understand. Ask not what physics can do for biology, ask what biology can do for physics." The first is biophysics, the second biological physics. What then, is the approach of the physicist who uses biological systems? It is of course not different from other branches of physics and can be characterized by a remark that Pauli made to Heisenberg: "The history of physics is a history of concepts. For an understanding of the phenomena, the first condition is the introduction of adequate concepts." Our problem thus is clear we must look at biological systems and, out of the incredible richness of phenomena, distill concepts. The field of biological physics is very large and no complete overview exists at present. One review covers part of the field. There is also a Virtual Journal of Biological Physics which can be found on the net at www.vjbio.org. Here I will only present a brief overview to sketch some topics.

2 Complex Systems

Physicists used to be proud to work on simple systems and leave the complexity to chemists and biologists. No longer. The study of complex systems has become hot even in physics. First we have to distinguish between what looks complex and what is complex. Simple nonlinear equations can produce beautiful pictures that look complex but are not. On the other hand, complex systems can appear to function simply. Biological physics can help here, because biomolecules, in particular proteins, are prototypes of complex systems. In contrast to entire living systems, they are simple enough so that progress in quantitative understanding can be made, yet they are still so complex that we can hope to discover laws of complexity. Moreover, they connect biology, chemistry, and physics and for a true understanding, we need experiment, theory, and computation.

Every biology, biochemistry, or biophysics textbook describes how proteins are constructed: Proteins are built from 20 different building blocks, the amino acids, also called residues. These have all the same backbone, but differ in their side chains. Of the order of 150 amino acids are linked covalently and then fold into a three-dimensional structure, the working protein. The arrangement of the amino acids in the chain determines structure and function. The information about the arrangement is coded in the genome. The code of the human genome has now been determined in approximately. This unexpectedly fast progress is opening new vistas for biological physics.

The structure of a large number of proteins has been determined by X-ray diffraction and by NMR. These structures at first give the impression that all copies of a particular protein are all exactly alike, with each atom in exactly the same place in each copy. Proteins seemed to look like miniature crystals and color plates in textbooks reinforce this impression. Reality, however, is different, and shows clearly how complex proteins are. A given protein can assume a very large number of different conformations. This property leads to two important concepts, conformational substates and energy landscapes. A protein in a given conformation is said to be in a particular conformational substate. Substate, because a protein can usually assume more than one state, it can for instance be reduced or oxidized. The ensemble of substates is described by the energy (or conformation) landscape. Substates are valleys in this landscape, separated by energy barriers.

The energy landscape is actually a construct in a very-high dimensional space. Assume that a protein contains about 1000 atoms. To describe the position of each atom, including the positions of water molecules that are inside the protein or cling to the outside, at least 3000 coordinates are needed. It is difficult, or even impossible, to picture such a landscape. Only the computer can deal efficiently with it. A one- or two-dimensional cross section only gives a poor approximation. Motions of the protein correspond to transitions among substates in the energy landscape. The energy landscape and the multitude of possible motions are the expression of the complexity of proteins.

Another concept that emerges from this picture is that of a protein glass transition, characterized by a glass temperature T_g . Above T_g , motions among the substates are faster than a characteristic time, say a thousand seconds. Below T_g the motions are essentially "frozen," they are slow compared to typical measurement times and the protein appears to be rigid.

The existence of an energy landscape, and of corresponding motions, is important for understanding the workings of proteins. Proteins can, for a short time, say 10^{-6} sec, assume a structure that is very different from the X-ray structure. This excited state structure may, however, just be the one that permits a reaction to occur or a drug molecule to dock. The exploration of the energy landscape, and its properties, is consequently important and relevant. A number of questions come immediately to mind: What are the experiments that prove and explore the energy

landscape? How is the energy landscape organized? How is the energy landscape involved in protein function? I will briefly treat these questions.

3 The Energy Landscape

One of the first clear observations that implied the existence of substates and an energy landscape was the time dependence of the binding of CO to myoglobin. Myoglobin (Mb) is a protein built from 153 amino acids. It was the first protein whose structure was determined. It contains a heme group with an iron atom at its center. Textbooks state that its function is only storage of dioxygen, which can bind to the heme iron. Mb also binds carbon monoxide, CO, and the study of this binding process has led to a deeper understanding of protein dynamics. The technique is straightforward. MbCO, Mb with CO bound to the heme iron, is hit with a laser flash. The light breaks the Mb-CO bond, the CO moves away and returns later to bind again. The binding process is followed spectroscopically. A detailed investigation of the rebinding process at temperatures below about 150 K shows that the process is highly nonexponential in time and can extend over more than 10 orders of magnitude in time. Such nonexponential processes have a long history, going back to Gauss. Multi-flash experiments prove that the non-exponentiality is caused by the fact that each Mb molecule rebinds with a different rate coefficient and that the protein ensemble is inhomogeneous. Thus Mb must be described by an energy landscape.

The existence of conformational substates was confirmed by many observations, for instance the large Debye-Waller factors in X-ray diffraction and the Mössbauer effect, spectral hole burning, specific heat data, neutron scattering, and computer simulations. Thus the existence of substates and of a conformational energy landscape is assured. Questions, however, remain as to the organization, the connection to structure, and to function.

Glasses also possess an energy landscape, but it is simpler than that of proteins. In glasses, essentially only one glass temperature occurs. In proteins, however, some transitions between substates cease at about 150K, whereas others proceed even at temperature well below 1 K. This wide range of "glass temperatures" suggests that the energy landscape of proteins is organized in a hierarchy, with valleys within valleys within valleys. Three classes of substates can be recognized:

- *Taxonomic substates.* Mb can assume at least three different conformations that are so different that they can be unambiguously identified. A clear signal is the stretch frequency of the bound CO, which differs for the three taxonomic substates. They have different properties and, for instance, rebind CO with different rates. A look at stretch frequencies in proteins other than Mb suggests that taxonomic substates may not be rare. It is conceivable that prions are taxonomic substates.

- *Statistical substate.* Each taxonomic substate is furcated into a very large number of substates that cannot be characterized individually. The number of such substates is larger than 10^4 , as deduced from hole burning experiments. Their properties must therefore be characterized by distributions.
- *Few-level substates.* Statistical substates are further furcated into a small number of few-level substates. These may be similar to the famous two-level states in glasses.

Our knowledge of the energy landscape of proteins is still in its infancy, comparable to the knowledge of nuclear energy levels at the beginning of nuclear physics. But we also know that in nearly every field of physics--atomic, nuclear, solid, elementary particle--progress in understanding started with, and was guided by, the exploration of the relevant "energy landscape." We hope that similar progress will also happen in biological physics.

4 Reactions

Very often biological phenomena involve "chemical" reactions in some form. Typical examples are transfer and transformation of energy, charge, matter, and information. A quantitative understanding of such reactions involves reaction theory. Studies of such processes in proteins and protein assemblies may lead to an insight into the concepts that govern reactions in complex systems. Even in apparently simple reactions, complex effects can occur. A few examples exhibit the richness of the field.

4.1 Friction and the Kramers Equation

Many studies of biochemical reactions are performed at a single temperature and without changing pressure. If studies are performed as a function of temperature, it is usually the Arrhenius relation that is used to evaluate the data. Such an approach may, however, lead to wrong conclusions. Kramers showed more than 60 years ago that friction is a crucial factor in controlling a chemical reaction. At low friction, the reaction rate increases, at high friction it decreases with increasing friction. This effect may be crucial for the control of biological reactions, because temperature in a living system is usually well controlled, but friction can be changed. To obtain insight into the control mechanism of biological reactions, reaction rates should therefore be studied as function of both temperature and friction or, in other words, viscosity: $k(T)$. Cuts in this surface at constant viscosity then provide information about the barrier height. Unfortunately, this technique has not found widespread application and much work remains to be done, theoretically, experimentally, and

computationally, before protein reactions and their dependence on friction are well understood.

4.2 *Molecular Tunneling*

The quantum-mechanical tunneling of electrons and protons in biological systems is well known. The tunnel effect of larger molecules such as CO or O₂, however, has been much less investigated. Heme proteins form an excellent laboratory where such tunneling processes, including the isotope effect, can be studied. Recently, tunneling of a water molecule to aquometmyoglobin has also been observed.

4.3 *Pressure Effects*

Pressure affects both the equilibrium and the rate of chemical reactions. The existence of an energy landscape in proteins implies that the effect of pressure is more involved than in ordinary liquids or solids because it affects the distribution of substates. This effect is seen in the strong influence that pressure has on, for instance, the binding of CO to Mb.

4.4 *Complexity*

In glasses, the rate coefficients for reaction or relaxation processes are not given by the Arrhenius or the Kramers equations, but can often be approximated by the Vogel-Fulcher relation,

$$k(T) = A \exp[E/(T - T_0)],$$

where A, E, and T₀ are adjustable parameters. Because of the similarity of glasses and proteins, it is to be expected that reaction and relaxation phenomena should also not strictly follow an Arrhenius relation. Indeed, measurements over wide ranges of temperature show that a Vogel-Fulcher relation fits the data better²⁰. Complexity and the energy landscape are important not just in protein function, but also in the protein folding problem. Here again is a field where more experiments, theory, and computation will lead to deeper insight both into the physics of proteins and the physics of reactions and relaxations.

5 New Physics from Proteins

I have only covered a very small part of biological physics, but even this sketch shows that the field is rich and offers many possibilities for new insights, obtained by experimentalists, theorists, and computer scientists. A brief summary will show where new physics is likely to emerge, and where this new physics will help to understand the workings of biomolecules:

- *Structure.* Schrödinger called biomolecules aperiodic crystals, but they are not crystals. They are prototypes of complex systems, flexible, moving, with functional states. They assume structures that may be different from the average “X-ray” structure, but that are important for the function and that may be crucial for the design of efficient drugs. They may be able to assume, with the same primary sequence, structures with different functions. Structural genomics implies learning about these “excited states.”
- *Energy levels.* Energy levels for biomolecules must be generalized to the energy landscape. The knowledge of the energy landscape of even the simplest protein is still rudimentary, but it suggests that the landscape contains an extremely large number of conformational substates, energy valleys in the landscape. The landscape is organized into at least three different tiers, a small number of taxonomic substates, each furcated into a very large number of statistical substates which in turn are furcated into a few tiers of few-level substates.
- *Dynamics.* Proteins share similarities with glasses, but are much more highly organized. Their equilibrium fluctuations, non-equilibrium relaxations, and reactions are in general nonexponential in time and non-Arrhenius in temperature. While the general features of these processes are emerging, a great deal of work will be needed before a complete characterization will be available. Question like the existence of stochastic resonances, chaos, and breathers, will also have to be studied.

6 Acknowledgments

I thank Professor Virulh Sa-yakanit and his collaborators for the invitation to attend this interesting meeting and for their hospitality. The research was supported under the auspices of the U.S. Department of Energy through the Center for Nonlinear Studies at the Los Alamos National Laboratory, contract W-7405-ENG-36.

FLUCTUATIONS AND RELAXATIONS IN PROTEINS

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At suitable experimental conditions, a metastable intermediate state of a protein between two equilibrium conformations can be determined by X-ray structure analysis. Two different temperature regimes can be distinguished. Below a characteristic temperature, T_c , the intermediate is stable for a long time while above T_c the relaxation occurs into the equilibrium conformation. The low temperature regime and the physiological temperature regime above T_c are also investigated by Mössbauer absorption spectroscopy and phonon-assisted Mössbauer effect. While essential features of the low temperature dynamics can be understood by normal modes, quasi-diffusive motions have to be taken into account in the physiological regime.

1 Introduction

The structure determination of a protein by X-ray or NMR is one of the most important steps for the understanding of the function on atomic level. However, it is not sufficient, even in a relatively simple molecule like myoglobin. The X-ray structure shows no channel which is wide enough to allow the oxygen molecule to reach its binding position at the heme iron deep in the molecule. Structural fluctuations are necessary to make possible the reversible binding of oxygen [1]. In several proteins with enzymatic activity two conformations exist, one with a substrate bound to the catalytic center and one where this center is unoccupied. Such different conformations can be well characterized by X-ray structure analysis. However, not much is known how the molecule changes from one conformation to the other. But it is obvious that, for a conformational change, the molecule needs some flexibility.

For many years, the study of the dynamical properties of proteins has been the center of interest of biological physics. An important impact came from the flash photolysis experiments on CO ligated myoglobin, performed by Frauenfelder and coworkers [2]. In these experiments the rebinding kinetics of CO to the heme iron in myoglobin has been investigated as a function of temperature. At temperatures below 160 K the rebinding of CO after photodissociation was found to be nonexponential following a power law over 8 orders of magnitude in time. These experiments have shown that in an ensemble of myoglobin molecules the individual molecules are not identical. Small structural differences may cause different barrier heights for the rebinding of CO. For a description of the rebinding kinetics one

needs a barrier height distribution [2,3]. While the overall structure is a characteristic for a conformation, the small structural differences within a conformation are called conformational substates. This picture was supported by a X-ray structure analysis, where individual mean square displacements of the atoms in myoglobin have been determined [4]. It is reasonable that at physiological temperatures the individual molecules are not trapped in specific conformational substates but fluctuate between these states. This is also indicated by the rebinding kinetics above 180 K [5].

Early hints for different dynamics of myoglobin below and above 180 K came from a comparison of mean square displacements obtained from the Debye Waller factor in X-ray structure determination and the Lamb Mössbauer factor in Mössbauer spectroscopy at the heme iron [6]. Mössbauer experiments indicated a dynamical transition at about 180 K with a strong increase of the $\langle x^2 \rangle$ values at the iron with higher temperatures.

Meanwhile, nearly all experimental and theoretical methods have been used to study protein dynamics. A consistent physical understanding emerges gradually. Nevertheless, many details are still unclear. Moreover, the physics of protein dynamics is mainly developed from experiments on myoglobin. It is already clear that many rules obtained from the study of myoglobin are quite general. But strong efforts are still necessary to compare different types of molecules in order to find the specific influence of dynamics on function in each case.

This paper reviews only some specific aspects of protein dynamics. It concentrates on the physical nature of the dynamical transition at about 180 K in myoglobin and its consequences for the functional property. Supplementary and additional information is given by Nienhaus *et al.* in this issue.

2 X-ray Structure Determination of Metastable States in Myoglobin

The study of intermediate structures which are on the path between two well defined conformations are of crucial importance for an understanding of conformational relaxations. In the case of myoglobin there exist two conformations: the ligated (r) and the unligated (t). An intermediate structure can be created by photolysis of CO in CO-ligated myoglobin (MbCO). This photolysis triggers the $r \rightarrow t$ transition. At conditions where the CO cannot go out of the protein moiety one may stabilize an intermediate between r and t, often called Mb*CO. Schlichting *et al.* [7] succeeded in determining a Mb*CO structure at 20 K. The CO molecule was still in the heme pocket 3.6 Å away from the iron. The iron had moved some tenth of an Angstrom out of the heme plane towards the position which it occupies in deoxymyoglobin (Mbdeoxy). It was claimed that the distal histidine had slightly moved. No other conformational changes were seen. Teng *et al.* [8] found only a small displacement of the CO molecule with no conformational changes. In the experiments of Hartmann *et al.* [9] at 40 K two possible positions of the CO molecule were

determined, one in accordance with Schlichting *et al.* and one close to the position found by Teng *et al.* Besides a movement of the iron out of the heme plane, no conformational changes could be detected.

Significantly new information was obtained by using a point-mutated myoglobin for the experiment [10]. Here, the mutant L29W was used where the lysine 29 is replaced by the bulky tryptophan. In this way the rebinding kinetics of CO was drastically slowed down. Two types of experiments have been performed. In the first experiment, the CO was photolysed by continued illumination of the sample with a light of 532 nm of a frequency-doubled Nd-YAC laser. The X-ray structure was determined at the same temperature. In the other experiment, the irradiation was performed slightly above 180 K. With continuous illumination, the temperature and the light intensity were lowered ending at 105 K where the structure was determined. The CO migration and binding is discussed by Nienhaus *et al.* in this issue. Here we want to concentrate on conformational changes. While a photolysis of CO at 105 K causes no significant conformational change, a photolysis above 180 K gives rise to a strong conformational relaxation at the distal side. Below 180 K the structure is frozen in. After the photolysis of CO it cannot relax in the direction of the new equilibrium structure. At about 180K a dynamical transition occurs. Now the structure is able to relax. These results suggest the division of the temperature dependence of the dynamics into two regimes. In the low temperature regime functional processes like conformational relaxations are blocked. In the “physiological regime” the molecule fulfils its basic functional properties. Low temperatures only slow down the rate. In the following we want to discuss the physical nature of this dynamical transition.

3 Mössbauer Absorption Spectroscopy

Iron containing proteins are excellent candidates for Mössbauer spectroscopy on ^{57}Fe . In this method one uses the 14.4 keV radiation of an excited ^{57}Fe nucleus (wavelength $\lambda = 0.86 \text{ \AA}$) populated by the decay of ^{57}Co . Its average lifetime, τ_N , is 140 ns. It determines the time sensitivity of the method. Figure 1 shows the Mössbauer spectrum of MbCO crystals at two different temperatures. As it is common, the energy scale (x axis) is given in unit of velocity. It is just the Doppler velocity v which is responsible for the energy variation ΔE . For a conversion of ordinary energy units one has to use the relation $\Delta E = E_\gamma v/c$ with $E_\gamma = 14.4 \text{ keV}$ and $c =$ velocity of light. $\Delta E = 0$ is fixed by definition which is not important in this context [11]. At 80 K one sees two transmission lines which can be fitted by Lorentzians. These lines build a so-called quadrupole doublet. The size of its splitting proves that the iron is CO-ligated. The second Mössbauer spectrum taken at 245 K can no longer be fitted by two Lorentzians. Additional broad lines have to be taken into account. Of great importance is the area A of the Lorentzians. It is given by

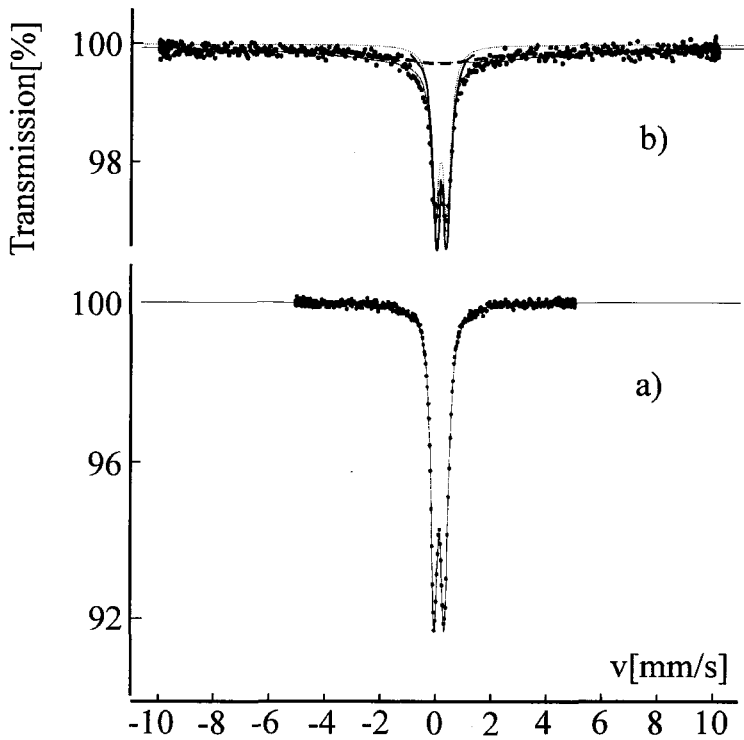


Figure 1. Mössbauer spectra of CO-ligated myoglobin crystals. a) $T = 80\text{K}$, solid line: least squares fit with two Lorentzians; b) $T = 245\text{K}$, the Lorentzians broad lines have to be taken into account in order to give a satisfying description of the experimental data.

$$A = \text{const} \cdot f \quad \text{with } f = e^{-k^2 \langle x^2 \rangle^\gamma} \quad (1)$$

The f is the Lamb Mössbauer factor while $k = 2\pi/\lambda$ and $\lambda = 0.86 \text{ \AA}$. The mean square displacement, $\langle x^2 \rangle^\gamma$, of the iron has an internal time dependence. Only displacements taking place on a time scale faster than τ_N contribute. Figure 2 shows the $\langle x^2 \rangle^\gamma$ values obtained from Eq. (1) from Mössbauer spectra between 80 K and 245 K (filled circles). Below 180 K the mean square displacement increases linearly with temperature indicating harmonic motions. At about 180 K there is a strong additional increase indicating a new channel of motions. It is only the temperature that causes the additional broad lines in the Mössbauer spectrum. The analogy of the

findings of the structure investigation of the L29W mutant is striking. Again we obtain different dynamics below and above 180 K. The low temperature regime is characterized by harmonic motions while in the “physiological regime” the shape of the Mössbauer spectrum changes in combination with a strong increase of the amplitude of motions.

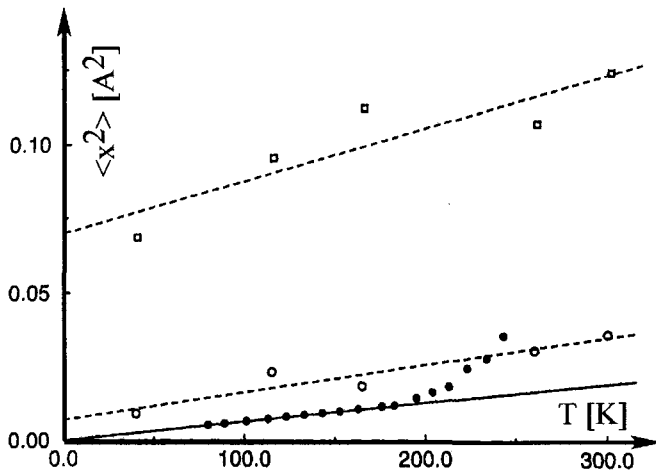


Figure 2. Mean square displacements, $\langle x^2 \rangle$, at the position of the iron as a function of temperature. The full circles are obtained from Mössbauer spectroscopy on CO-ligated myoglobin according to Eq. (1). The solid line was obtained by a normal mode analysis. Open Symbols: X-ray structure analysis of metmyoglobin; squares: total value including TLS modes; circles: normal mode refinement yielding only displacements, which do not change the center of gravity. dashed lines: linear regressions [12].

Let us first discuss the low temperature regime. Melchers *et al.* have performed a normal mode analysis of myoglobin [13]. It provides us with the amplitude and the polarization of each normal mode at each atom of the molecule. This allows the calculation of the mean square displacements at the iron, coming from normal modes

$$\langle x_{\text{Fe}}^2 \rangle_{\text{NM}} = \sum_{k=7}^{3N} \langle x_{k\text{Fe}}^2 \rangle . \quad (2)$$

N gives the number of atoms in the molecule yielding $3N$ degrees of freedom. The index k counts the modes. The first 6 modes represent the translation and the rotation of the whole molecule. As shown in Fig. 2, the $\langle x_{\text{Fe}}^2 \rangle_{\text{NM}}$ values are in

good agreement with the experimentally obtained $\langle x^2 \rangle^Y$ values below 180 K. At this state we postpone the discussion of the physiological regime to a later chapter.

4 Phonon-Assisted Mössbauer Effect

Pulsed X-rays from a synchrotron in combination with the Mössbauer effect on ^{57}Fe proved to be an excellent tool for the investigation of phonons in condensed matter. It was also applied for the investigation of myoglobin. Figure 3 shows the setup of the experiment (confer [14]). Pulsed synchrotron radiation with a pulse length of 100 ps and a pulse sequence of 176 ns is monochromatized by 4 silicon crystals. This allows the selection of a radiation with an energy E_{syn} of $14.4 \text{ keV} \pm 70 \text{ meV}$ and a width of 4.4 meV. With this radiation a scattering experiment is performed on a myoglobin sample, where the heme iron is replaced by the Mössbauer isotope ^{57}Fe . Let us assume that E_{syn} overlaps with E_γ , the energy of the ^{57}Fe Mössbauer transition. In this case the following scattering processes occur: i) Rayleigh scattering by the electrons in the sample, ii) inelastic scattering by the electrons of the sample, iii) nuclear resonant scattering by the ^{57}Fe nuclei (compare with the insertion in Fig. 3). Time-delayed nuclear resonance scattering is observed in forward direction and in 4π . If the monochromator system is adjusted in a way such that E_{syn} no longer overlaps with E_γ ($E_\gamma - E_{\text{syn}} = \varepsilon$, i.e., 35 meV) one would believe that no time delayed nuclear scattering occurs. However, the contrary is true. Time-delayed scattering is counted in the detector APD1. The missing energy ε of the synchrotron radiation is supplemented by phonons. In this way the intensity of the time delayed radiation becomes a measure for the number of phonons with the energy ε . Figure 4 shows the phonon spectrum of a hydrated metmyoglobin sample. Note the “Stokes” and the “anti-Stokes” lines. It should be mentioned in passing that this method has a different selectivity than Raman scattering [15]. It allows the determination of the density of phonon states. It has to be emphasized that the quasi-elastic scattering, as seen in the Mössbauer absorption spectrum as broad line, cannot be resolved by this method and is hidden below the elastic scattering ($\varepsilon = 0$). It occurs within an energy band of about 0.1 μeV which has to be compared with the energy resolution of 4.4 meV of the experiment discussed here.

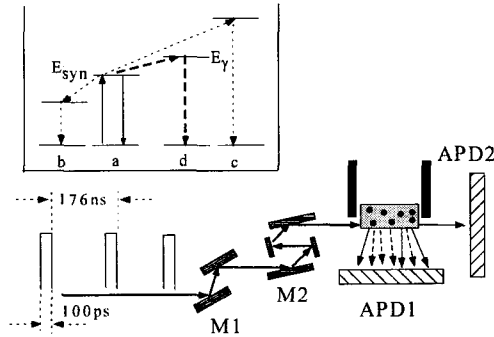


Figure 3. Phonon-assisted Mössbauer effect. Synchrotron radiation with the energy E_{syn} (flash length 100ps, repetition time 176 ns) is monochromatized to $\Delta E = 4.4\text{meV}$ by the sets of silicon crystals M1 and M2. The radiation scattered by a myoglobin sample having ^{57}Fe in the heme group is counted in the avalanche photo diode APD1. Three types of scattering processes occur in the sample: elastic Rayleigh scattering (insertion a), inelastic scattering by the electrons of the sample (insertion, b: phonon emission, c: phonon absorption) and Mössbauer scattering at the ^{57}Fe nuclei if phonons coupling to the iron just balance the energy $E_\gamma - E_{\text{syn}}$ (insertion d). Processes a to c are prompt while d occurs with the time delay τ_N . The photo avalanche diode APD2 measures the nuclear forward scattering and is used for calibrations. For details confer [14] and [15].

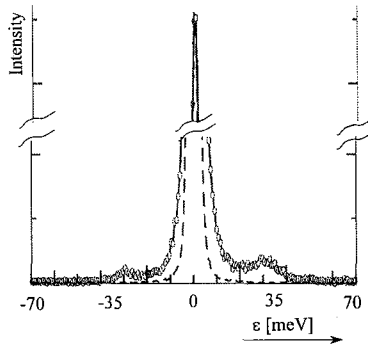


Figure 4. Number of phonons coupling to the iron in myoglobin as a function of their energy ϵ . Negative energies: Phonon absorption (anti-Stokes), positive energy: Phonon emission (Stokes); $T = 300\text{ K}$; the dashed line gives the resolution function.

5 Mean Square Displacements Obtained from X-ray Structure Analysis

Frauenfelder *et al.* showed that individual mean square displacements, $\langle x^2 \rangle^x$, of all non-hydrogen atoms of myoglobin can be obtained from the Debye Waller factor of the X-ray structure analysis [4]. There was, however, the difficulty to separate internal molecule displacements from rigid body displacements of the whole molecule by translations and rotations. Kidera and Go have solved this problem by

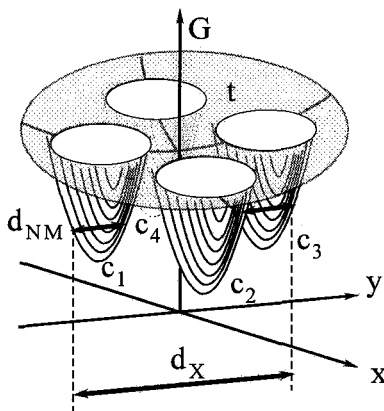


Figure 5. Schematic drawing of the energy landscape in two dimensions (Gibbs free energy as function of the coordinates x and y). Four conformational substates (c_1 to c_4) are shown together with the transition state t . The displacement measured by normal mode analysis is given by $d_{NM} = \langle x^2 \rangle_{NM}^{1/2}$. The normal mode refinement yields $d_X = (\langle x^2 \rangle_{intra}^X)^{1/2} = (\langle x^2 \rangle_{NM} + \langle x^2 \rangle_c)^{1/2}$. Mössbauer spectroscopy measures below T_c : $(\langle x^2 \rangle^y)^{1/2} = (\langle x^2 \rangle_v)^{1/2} = d_{NM}$. Above T_c , $(\langle x^2 \rangle^y)^{1/2} = (\langle x^2 \rangle_v + \langle x^2 \rangle_c)^{1/2}$ approaches d_X and can even overshoot this value due to diffusive TLS motions.

the normal mode refinement [16,17]. This method separates TLS modes (Translation, Rotation, Screwing) from internal displacements which do not change the center of gravity of the molecule. Figure 2 shows the results at the position of the iron [12]. Total mean square displacements, $\langle x^2 \rangle_{total}^x$, intra molecular displacements, $\langle x^2 \rangle_{intra}^x$, as obtained from X-ray structure analysis and $\langle x^2 \rangle_{NM}$ calculated by normal mode analysis are compared. It is obvious that most of the displacements at the iron position stem from TLS modes. Nevertheless, the intra molecular displacements of the normal mode refinement are still about two times larger, than those from normal mode analysis. This difference is due to the conformational substates. Figure 5 gives the principal picture. For simplicity, let us assume that there exist only 4 conformational substates of myoglobin. The normal modes in all substates are very similar so that their $(\langle x^2 \rangle_{NM})^{1/2}$ values are practically

the same. X-ray structure analysis cannot differentiate between dynamic displacements caused by normal mode vibrations and static displacements due to the separation of the conformational substates in space. As a consequence one obtains

$$\langle x^2 \rangle_{\text{int ra}}^{\text{X}} = \langle x^2 \rangle_{\text{NM}} + \langle x_c^2 \rangle. \quad (3)$$

The X-ray structure analysis of metmyoglobin was performed at temperatures between 40 K and 300 K. Within the limit of error, the $\langle x^2 \rangle$ values increased linearly with temperature. No indication of different behavior at low temperatures or at physiological temperatures is seen as in the Mössbauer experiments. This apparent discrepancy is discussed in the following section.

6 Protein Specific Motions in the Physiological Regime

We now come back to the discussion of the Mössbauer spectra above 180 K in the physiological regime. Broadened lines are well known in Mössbauer spectroscopy. In general they characterize diffusive processes. As an example, we refer to Fe^{2+} ions in glycerol [18]. In pure free diffusion the area of the spectrum remains constant with decreasing height of the maximum and increasing width. The Mössbauer spectra of myoglobin are clearly different and indicate no free diffusion. Broad and narrow lines are present at the same time. This is characteristic for a diffusion in limited space. The dimension of the space depends on the wavelength of the radiation used for the investigation. In our special case the limit is in the order of 1 Å. There are several ways for a theoretical description of a bound diffusion. They all have in common that the reduction of the absorption area which exceeds the lost due to $\exp[-k^2 \langle x^2 \rangle]$ is shifted into the broad lines. In the following we use a modified Brownian oscillator as the model [19,12].

Let us assume that a molecule is trapped in a conformational substate. If the temperature is high enough, it may reach the transition state. We now assume that the molecule is not trapped immediately again, but performs a Brownian oscillation in the transition state. The space for the Brownian oscillation is characterized by $\langle x_c^2 \rangle$, the distribution of conformational substates.

Free diffusion is described by the Langevin equation. In order to limit the diffusion in space one has to add a back driving term

$$\ddot{x}_{t0} + 2\beta_{t0}\dot{x}_{t0} + \langle \omega_{t0}^2 \rangle x_{t0} = \tilde{F}[t]. \quad (4)$$

$\tilde{F}[t]$ is a stochastic force representing the thermal bath. The diffusive motions of water molecules around the protein push the protein. Since protein crystals are used

in the experiments, a free diffusion of the molecule is not possible. Instead, segments of the molecule like α chains can make a diffusive motion around their average position as known from X-ray structure analysis. This motion is damped by β_{10} . The back driving force is approached by only one average frequency $\langle \omega_{10}^2 \rangle$. From this equation it is possible to calculate the Mössbauer spectra shown in Fig. 1:

$$\text{Trans} \propto \{ 1 - \text{const} \cdot e^{-k^2 \langle x_v^2 \rangle} e^{-k^2 \langle x_{10}^2 \rangle} \sum_{N=0}^{\infty} \sum_{l=1}^2 \frac{(k^2 \langle x_{10}^2 \rangle)^N}{N!} \frac{\frac{\Gamma_{al} + \Gamma_s}{2} + N\alpha_{10}}{(E_{al} + E_s(1 + \frac{v}{c}))^2 + (\frac{\Gamma_{al} + \Gamma_s}{2} + N\alpha_{10})^2} \} \quad (5)$$

The trivial parameters of a Mössbauer spectrum are the Lorentzians of the source and the absorber described by the centers of the emission and absorption energy, E_s and E_{al} , and their line widths, Γ_s and Γ_{al} respectively where v is the Doppler velocity of the source. With $N = 0$ and $\langle x_v^2 \rangle = 0$ the equation describes the low temperature case, where the molecule is trapped in a conformational substate and cannot jump into the transition state. In the physiological temperature regime $\langle x_{10}^2 \rangle$ is not zero and the summation over N has to be carried out. Note, that this summation from $N = 0$ to ∞ yields $\exp[+k^2 \langle x_{10}^2 \rangle]$. This proves that the reduction of the intensity of the narrow lines due to quasi-diffusive motions is compensated by the intensity of the broad lines. The Mössbauer spectrum does not depend directly on the parameters β_{10} and $\langle \omega_{10}^2 \rangle$ of the Brownian oscillator but on their quotient $\alpha_{10} = \omega_{10}^2 / 2\beta_{10}$.

Now one has to take into account the probability, $p_t[T]$, that the molecule is in the transition state:

$$p_t[T] = (1 + e^{\frac{\Delta G_{ct}}{k_B T}})^{-1} . \quad (6)$$

Here, ΔG_{ct} is the Gibbs free energy difference between the conformational substates and the transition state and k_B denotes the Boltzmann constant. We now obtain

$$\langle x_t^2 \rangle = \langle x_{10}^2 \rangle p_t[T] \quad \text{with} \quad \langle x_{10}^2 \rangle = \frac{k_B T}{m \langle \omega_{10}^2 \rangle} . \quad (7)$$

The mean square displacement, determined by the Mössbauer experiment becomes

$$\langle x^2 \rangle = \langle x_v^2 \rangle + \langle x_t^2 \rangle . \quad (8)$$

At low temperatures, only $\langle x_v^2 \rangle$ contributes. This can be reproduced by a normal mode analysis. At physiological temperatures quasi-diffusive motions are activated, yielding $\langle x_t^2 \rangle$. These motions are undeniable for the function. In the data analysis the $\langle x_v^2 \rangle$ values at physiological temperatures are obtained by a linear extrapolation from the low temperature values. However, it has to be emphasized that the degrees of freedom used in the quasi-diffusive motions have to be taken from degrees of freedom of the normal mode analysis.

7 Acknowledgment

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References

1. Elber, R. and Karplus, M., Enhanced sampling in molecular dynamics: use of the time-dependent Hartree approximation for a simulation of carbonmonoxide diffusion through myoglobin. *J. Am. Chem. Soc.* 112 (1990) pp. 9161-9175.
2. Austin, R. H., Beeson, K. W., Eisenstein, L., Frauenfelder, H. and Gunsalus, I. C., Dynamics of ligand binding to myoglobin. *Biochemistry* 14 (1975) p. 5355-5373.
3. Frauenfelder, H., Parak, F. and Young, R. D., Conformational substates in proteins. *Annu. Rev. Biophys. Biophys. Chem.* 17 (1988) pp. 451-479.
4. Frauenfelder, H., Petsko, G. A. and Demetrius, T., Temperature-dependent X-ray diffraction as a probe of protein structural dynamics. *Nature* 280 (1979) pp. 558-563.
5. Nienhaus, G. U., Mourant, J. R. and Frauenfelder, H., Spectroscopic evidence for conformational relaxation in myoglobin. *Proc. Natl Acad. Sci. USA* 89 (1992) pp. 2902-2906.
6. Parak, F. and Formanek, H., Untersuchung des Schwingungsanteils und des Kristallgitterfehleranteils des Temperaturfaktors in Myoglobin durch Vergleich von Mössbauerabsorptionsmessungen mit Röntgenstrukturdaten. *Acta Cryst. A* 27 (1971) pp. 573-578.
7. Schlichting, I., Berendzen, J., Phillips, G. N. J. and Sweet, R. M., Crystal structure of photolysed carbonmonoxy-myoglobin. *Nature* 371 (1994) pp. 808-812.
8. Teng, T. Y., Srajer, V. and Moffat, K., Photolysis-induced structural changes in single crystals of carbonmonoxy myoglobin at 40 K. *Nature Struct. Biol.* 1 (1994) pp. 701-705.

9. Hartmann, H., Zinser, S., Komninos, P., Schneider, R. T., Nienhaus, G. U. and Parak, F., X-ray structure determination of a metastable state of carbonmonoxy myoglobin after photodissociation. *Proc. Natl. Acad. Sci. USA* 93 (1996) pp. 7013-7016.
10. Ostermann, A., Nienhaus, G., U. and Parak, F. G., Ligand binding and conformational transitions in myoglobin. *Nature* 404 (2000) pp. 205-208.
11. Parak, F. and Reinisch, L., Mössbauer effect in the study of structure dynamics. *Methods in Enzymology* 131 (1986) pp. 568-607.
12. Parak, F., Ostermann, A., Gassmann, A., Scherk, C., Chong, S.-H., Kidera, A. and Go, N., Biomolecules: Fluctuations and relaxations. *Biological Physics: Third International Symposium, ed.: H. Frauenfelder, G. Hummer, R. Gracia;* (1999) pp. 117-127.
13. Melchers, B., Knapp, E. W., Parak, F., Cordone, L., Cupane, A. and Leone, M., Structural fluctuations of myoglobin from normal-modes, Moessbauer, Raman, and absorption spectroscopy. *Biophys. J.* 70 (1996) pp. 2092-2099.
14. Keppler, C., Achterhold, K., Ostermann, A., van Bürck, U., Potzel, W., Chumakov, A. I., Baron, A. Q. R., Rüffer, R. and Parak, F., Determination of the phonon spectrum of iron in myoglobin using inelastic X-ray scattering of synchrotron radiation. *Eur Biophys J* 25 (1997) pp. 221-224.
15. Parak, F. and Achterhold, K., Protein dynamics studied on myoglobin. *Hyperfine Interactions* 123/124 (1999) pp. 825-840.
16. Kidera, A. and Go, N., Normal mode refinement: Crystallographic refinement of protein dynamic structure I. Theory and test by simulated diffraction data. *Journal of Molecular Biology* 225 (1992) pp. 457-475.
17. Kidera, A., Inaka, K., Matsuhima, M. and Go, N., Crystallographic refinement of protein dynamic structure II. Application to human lysozyme. *Journal of Molecular Biology* 225 (1992) pp. 477-486.
18. Nienhaus, G. U., Frauenfelder, H. and Parak, F., Structural fluctuations in glass forming liquids: Mössbauer spectroscopy on iron in glycerol. *Phys. Rev. B* 43 (1991) pp. 3345-3350.
19. Knapp, E. W., Fischer, S. F. and Parak, F., Protein dynamics from Mössbauer spectra. The temperature dependence. *J. Phys. Chem.* 86 (1982) pp. 5042-5047.

QUALITATIVE METHODS IN IMMUNOLOGICAL MODELLING

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The immune system is a complex system that learns, remembers what it has learned and acts to protect the organism against various diseases (antigens). Here we discuss simple mathematical concepts that can be used to model these features of the immune response in a quantitative way. We identify two design principles for immune system recognition. Using the concept of shape space we analyze the reaction of the immune system to invasion by a foreign antigen. This analysis leads to a distinction between specialist and generalist antibodies. We then consider the case of a mutating antigen which is "hunted" by the immune system. The relevant biology is explained in a highly simplified way, and the mathematical analysis is outlined in a global way.

1 Introduction

In these pages we shall review some recent concepts in the field of mathematical modelling of the immune system. The common feature of these concepts is their qualitative character: the intricate biological structure of the immune system is replaced by highly simplified mathematical entities, which are then used in a qualitative way.

Our references to the relevant part of the literature shall be simplified in a similar way: instead of listing the hundreds of papers which could be quoted in connection with this topic, we shall only mention a few of them. Namely, those which we ourselves found useful for a theoretical physicist who ventures into the terra incognita of the immune system. Here is the main reference of this type: the review which was published a few years ago by Perelson and Weisbuch [1]. This paper gives and discusses about 250 references which are relevant to immunological modelling.

We shall proceed along the following path. First, the immune system will be introduced. Its function is to protect the body from disease-causing organisms (antigens). In order to do this job, the immune system manufactures a special set of biological polymers which are called antibodies. We shall then briefly discuss the steps in the antibody-antigen recognition process and we shall argue that the immune system is an evolutionary system: it is similar to an ecology which is involved in a continuing process of evolution. Next we shall introduce the concept of shape space, which can help us to describe this evolutionary process in a mathematical way. The rest of this paper is devoted to demonstrating how one can do simple calculations, using the concept of shape space, to describe the biological phenomena of infection, somatic hypermutation, and the "immune hunt" which goes on in patients who suffer from diseases which are

caused by mutating antigens (of which AIDS, malaria and sleeping sickness are well-known examples).

2 Cells Perform Chemoreception

Before we focus on the immune system it should be pointed out that just about any living cell continuously monitors the presence of certain molecules in its environment. These chemical compounds (usually called ligands) are either naturally present in the cell's vicinity, or they have been emitted previously by other cells of the same or another organism. They are present in the extracellular medium in small concentrations, and they are transported from cell to cell by means of Brownian motion, hydrodynamic convection, electromagnetic fields or other forces. A cell can usually detect those ligands which are important to its proper functioning, by means of receptors. A receptor is a protein, or a complex structure consisting of proteins and other biopolymers, which somehow binds specifically with a certain ligand. Usually the receptors are embedded in the outer membrane of the cell. Each receptor has a binding site which has the property that a ligand which is specific for this receptor is captured almost immediately, i.e., a ligand-receptor complex is formed almost immediately. This process of a highly selective interaction of the cell with specific ligands in its environment is called chemoreception.

The biological and biochemical aspects of chemoreception are fairly well understood, and the physical principles involved have been explored by a small number of physicists and biophysicists. These physical principles, as well as their mathematical modelling, form the subject of a small monograph [2] to which the reader is referred for more details.

Before we turn to the immune system (which performs a very sophisticated form of chemoreception) we want to introduce the reader to some order of magnitude estimates concerning the "average" cell. This fictitious cell has a spherical shape, a radius of $5 \times 10^{-6} \text{ m} = 5 \times 10^4 \text{ \AA} = 5 \mu$. It has a volume of $5 \cdot 24 \times 10^{-16} \text{ m}^3$ and a mass of $5 \cdot 40 \times 10^{-13} \text{ kg}$. These estimates show that the "average" human, who might weigh about 75 kg, would consist of a number of cells which is of the order of magnitude $\approx 10^{+14}$.

3 The Immune System

The immune system is that part of the body which is uniquely devoted to protecting it against disease-causing organisms. In the human body about 5% of all cells belong to it; this leads to the estimate of about 5×10^{12} cells in our immune system.

These cells are not all similar, but they can be divided in various categories, of which the B-cells and the T-cells are the two most important ones. In this paper we shall only consider the B-cells; their number is of the order of magnitude $\approx 10^{12}$ for humans. These B-cells circulate throughout the whole body. They are not all identical, but they consist of a certain number of different clones; each clone consists of a number of genetically identical cells. It is believed that the human immune system consists of $\approx 10^7$ different clones, each clone has $\approx 10^5$ cells.

Before we say a bit more about the human immune system it should be noted that in the mouse the immune system is estimated to contain $\approx 10^8$ B-cells, which are divided in $\approx 10^7$ clones, each of ≈ 10 cells. These numbers, which come from the experimental immunological literature, seem to suggest that 10^7 is a "complete repertoire" of B-cell clones for any mammal.

The immune system protects the body against disease-causing organisms. The latter are usually called antigens; an antigen can, for example, be a virus, a bacterium or a parasitic cell. The immune system performs this work in three stages: (i) Learning; (ii) Action; (iii) Memory. In the learning stage the system learns to recognize the antigen; this is called immune recognition. In the action stage it proliferates the appropriate clone of B-cells, which cells will then shed a specific type of biopolymer (the antibody) which binds selectively to the antigen in such a way that the antigen can be eliminated by the body. In the memory stage the immune system remembers the event in such a way that the next time the same antigen tries to invade the body, the immune response will be almost immediate.

In the following pages we shall focus on the first stage, immune recognition. Before we enter upon a more detailed discussion, a few remarks about antibodies and B-cells are pertinent. Antibodies are medium-sized proteins which usually have two identical binding pockets of a specific form. An antibody will only bind to an antigen if the shape of its binding site, and the shape of the antigenic site where binding occurs, are complementary (this is often called the lock-key relation between an antigen and its corresponding antibody).

Antibody molecules are always made inside a B-cell; they are then located at the outer membrane of that cell, in such a way that the two binding sites are exposed on the outside of the cell, whereas the rest of the antibody is anchored to the inside of the cell.

Now, one of the marvels of the immune system is the empirical fact that a B-cell seems to have about four stages in its mature life:

1. If no antigen is bound to the antibodies which it carries on its surface, the B-cell remains stable, that is, not much happens to it.
2. If some antigen gets bound to the antibodies, it “triggers” the B-cell into activity and it will start to divide rapidly. A dividing B-cell is also prone to mutations, which show up in small changes in the shape of the binding pocket of the antibodies which that B-cell produces. Hence, the clone of activated B-cells generates a large variety of new clones of B-cells. This corresponds with the “learning” stage of the immune response.
3. Once sufficient antigen is bound to a B-cell, this cell will stop mutating, but it starts to produce huge numbers of antibodies which it “sheds” into the extracellular space where they help to eliminate the antigens. This corresponds to the “action” stage of the immune response.
4. Once the infection caused by antigens is over, most of these activated B-cells will disappear from the body, but some of them will be stored for an indefinite time, to serve as a memory in case the same antigen invades the body in the future. This, of course, takes care of the “memory” stage of the immune response.

4 Two Design Principles for Immune System Recognition

The paradigm which was originally used to understand the immune system was a rigid interpretation of the “lock-key” relation between antigen and antibody: it was assumed that one specific type of antigen was recognized by one specific type of antibody. This simple idea was gradually shown to be wrong. The main argument used to show why it must be wrong consist of the following two orders of magnitude estimations. The number of different types of antibodies in the (human) immune system was shown to be of the order 10^7 . Compare this with the number of possible antigens, which follows from simple biochemical considerations, and which is at least of the order 10^{130} . Hence one is led to the obvious notion:

First design principle. The immune system has to use “sloppy” antibodies, in such a way that a single clone of B-cells can eventually recognize a huge number of different types of antigens.

How does it do this? The new paradigm for immune system recognition compares the human immune system with an evolutionary system which consists of $\approx 10^7$ different species (the various clones of B-cells); each species is represented by $\approx 10^5$ individuals (the individual B-cells in a clone). Another way to phrase this is the following:

Second design principle. Upon infection by an antigen the immune system becomes an evolutionary system.

Recall stage (2) in the life of a B-cell, as discussed in section 3: the weak binding of some antigen to a B-cell triggers it to divide and mutate rapidly. Hence the generation time in this evolutionary system is the time between two successive B-cell divisions, say six to seven hours. The rate of mutation is also very high. For the sake of convenience we shall assume that one mutation occurs during each cell division.

Now the question arises: how could one possibly model this evolution mathematically?

5 Shape Space

The mathematical modelling of evolution in the immune system became feasible only after the introduction of the concept of shape space by Perelson and Oster [3] in 1979. Shape space S is an n -dimensional space which consists of those sequences of n real numbers $s_1, s_2, \dots, s_n$ which one needs to characterize the properties of the binding site of an antibody which are relevant from an immunological point of view. For example, the first seven such numbers might be:

s_1, s_2, s_3 : the length, width and height of the binding site;

s_4 : the electrical charge of the region of the antibody which surrounds the site;

s_5, s_6, s_7 : the three components of the electrical dipole moment of this region.

Note that one clone of B-cells in the immune system corresponds with one type of antibody and hence with one representative point in shape space. The number N of representative points in S equals the number of B-cell clones; in section 3 we have argued that $N \approx 10^7$ for men as well as for mice and presumably for all mammals.

In the absence of infection, i.e., in the absence of antigen, these N representative points will be at fixed locations in S ; this is the shape space aspect of stage (1) in the life of a B-cell, as discussed in section 3.

Before we ask what happens to these N representative points in S when antigen enters the organism, we would like to mention the remarkable fact that empirically the best fit of the shape space model to immunological reality seems to occur for values of n in the range 5 to 10. The smallness of n reflects the fact that immune recognition works with "sloppy" antibodies; that is, the small value of n reflects the first design principle of section 4.

6 What Happens When New Antigen Enters the Organism?

When the body is infected by a new antigen there usually is no antibody in the existing immune repertoire which has a binding site complementary in shape to the binding region of the antigen. The shape of the latter's binding region can also be plotted as a point in shape space; let it be denoted by $\vec{s}_A$. The existing repertoire is represented in S by a "swarm" of N points; let $\vec{s}_B$ be the point in this swarm which is nearest to $\vec{s}_A$.

Now two possibilities can occur. Either $\vec{s}_B$ is so near to $\vec{s}_A$ that an effective lock-key relation between antigen (A) and antibody (B) is realized. In this case an immediate, powerful immune response will occur [compare stage (3) in section 3]. We shall call B a *specialist* antibody. We shall model this situation by introducing a (small) distance ε_1 , such that $|\vec{s}_A - \vec{s}_B| < \varepsilon_1$ implies rapid elimination of the antigen. In other words; around each point in the swarm we imagine a small ε_1 -sphere; if the new antigen's representative point $\vec{s}_A$ falls within one of these ε_1 -spheres, the immune response is practically immediate.

The other possibility is that $\vec{s}_A$ is located outside all the N different ε_1 -spheres. In this case we find the point in the swarm which is nearest to $\vec{s}_A$. Let this point be denoted by $\vec{s}_B$. We shall assume that there is a (large) distance ε_2 , such that $|\vec{s}_A - \vec{s}_B| < \varepsilon_2$ implies stimulation of the clone of cells which exhibit antibody B on their surface [stage (2) in section 3]. We shall call this antibody a *generalist* antibody. The cells in this clone are now activated, which means that they will divide and mutate very rapidly. This activity is called somatic hypermutation.

Before we model somatic hypermutation in a mathematical way, we ask the reader's attention to the following immunological experiment. Make a "random" antigen. Inject it into a mouse. Then measure the fraction of all B-cells which are stimulated by this new antigen. Such experiments can actually be performed in the immunological laboratory, and the resulting fraction is found to be of the order of magnitude 10^{-5} to 10^{-4} . If we assume that all clones have roughly the same number of cells, and that the representative points in the mouse's immune repertoire are scattered more-or-less at random throughout the relevant part of shape space, then one has found the order of magnitude estimate,

$$\frac{V_{\varepsilon_2}}{V_S} \approx 10^{-5} \text{ to } 10^{-4}, \quad (6.1)$$

where V_{ε_2} denotes the volume of an ε_2 -sphere, and V_S denoted the volume of the relevant part of shape space. As the total number of clones is $\approx 10^7$ we have found that the number of different clones which are stimulated by this new antigen is $\approx 10^2$ to 10^3 . In other words, although a new antigen (will have a shape of its binding region which) falls *outside* of all 10^7 different ε_1 -spheres, it will usually fall simultaneously *inside* $\approx 10^2$ to 10^3 different ε_2 -spheres. It will weakly activate ≈ 100 to 1000 different clones. The clone which is nearest in shape space will be activated most strongly; this is the clone whose history we shall now describe mathematically.

7 A Random Flight Model for Somatic Hypermutation

The cells of the clone corresponding to the point $\vec{s}_B$ which is nearest to the point $\vec{s}_A$ will divide and mutate very rapidly. Each mutation will slightly change the shape of the antibody's binding site. In terms of shape space this means that the representative point of the next generation of B-cells will be shifted slightly away from $\vec{s}_B$. This shows that somatic hypermutation will lead to a large number of random flights in shape space, which all start from $\vec{s}_B$ at generation 0, and which take small steps in random directions, successively visiting the positions $\vec{s}_B = \vec{s}_0, \vec{s}_1, \vec{s}_2, \dots$. Let these random flights be characterized by an effective diffusion coefficient D . We shall now sketch the mathematics of this random flight model of somatic hypermutation and refer the reader to the paper by Perelson and Wiegel [4] for details.

First of all we note that the order of magnitude of D can be related to the other parameters in this model. A random walk in n -dimensional space will cover a (distance)² during time T_R which is on the average given by

$$\langle (\text{distance})^2 \rangle = 2nDT_R. \quad (7.1)$$

If we take for T_R the time for the immune response to mature, then the distance covered in shape space must be of the order $(\varepsilon_2 - \varepsilon_1)$. Taking $T_R = 2$ weeks, $n = 5$ and using $\varepsilon_1 \ll \varepsilon_2$, one finds the order of magnitude estimate

$$D \approx \frac{\varepsilon_2^2}{20} (\text{week})^{-1}. \quad (7.2)$$

A more accurate calculation will proceed as follows. We imagine an ε_1 -sphere around $\bar{s}_A$ and we put the nearest antibody at $t=0$ at a distance $\frac{1}{2}d$, where d equals the average distance between the N representative points for the antibody clones in shape space. Then we ask for the average time (T_o) which lapses till this antibody's representative point hits the ε_1 -sphere for the first time. A rough calculation gives

$$D \frac{T_o}{\varepsilon_1^2} = \varphi(x), \quad (7.3)$$

where

$$x = \frac{d}{2\varepsilon_1}, \quad (7.4)$$

and φ is a simple polynomial:

$$\varphi(x) = \frac{x^n}{n(n-2)} - \frac{x^2}{2(n-2)} + \frac{1}{2n}. \quad (7.5)$$

The average distance d between the N representative points in shape space is given by

$$\frac{d}{R} \equiv \left\{ \frac{2\pi^{n/2}}{nN\Gamma(n/2)} \right\}^{1/n}. \quad (7.6)$$

if the biologically relevant part of shape space is modelled by an n -dimensional sphere of radius R . The gamma-function is denoted by Γ . If instead the relevant part of shape space were modelled by a cube with side R , then the last equation should be replaced by

$$\frac{d}{R} \equiv N^{-1/n}. \quad (7.7)$$

It is instructive to illustrate these results with a numerical example, which holds for humans and mice equally. The first-passage time T_o will, of course, again be identified with T_R , the time typical for immune maturations; say $T_o \cong 2$ weeks. Using $n=5$ and $N=10^7$ in eq. (7.6) one finds $d/R \cong 0.055$. Equation (6.1) tells us that $(\varepsilon_2/R)^5 \approx 10^{-5}$ to 10^{-4} , which gives $\varepsilon_2/R \cong 10^{-1}$. With these numbers eq. (7.3) reads

$$\frac{1}{10} \left(\frac{\varepsilon_2}{\varepsilon_1} \frac{R}{R} \right)^2 = \varphi \left(\frac{0.055}{2\varepsilon_1 / R} \right). \quad (7.8)$$

Solving this equation numerically one finds $\varepsilon_1 / R \cong 10^{-2}$. From these numbers we draw the following quantitative conclusions:

(α) The volume of an ε_1 -sphere as a fraction of the total relevant volume of shape space equals $(\varepsilon_1 / R)^5 \cong 10^{-10}$.

(β) As there are $N=10^7$ such ε_1 -spheres, their combined relative volume equals $\cong 10^{-3}$.

(γ) The probability that a "new" antigen will land inside one of these existing ε_1 -spheres is therefore also $\cong 10^{-3}$.

(δ) In the course of a human life there are of the order of magnitude 1000 infections with a "new" antigen. The probability quoted under (γ) shows that $\cong 1$ of these will immediately be dealt with by a specialist antibody; the remaining $\cong 999$ will be eliminated as a result of an evolutionary process in which a generalist antibody, through hypermutation, is transformed into a specialist antibody.

8 Mutating Antigens

The model for the maturing of the immune response, which formed the subject of sections 5-7, will now be applied to the case of infection by mutating antigen. This situation is very common in nature: by mutating rapidly the antigen tries to avoid recognition by the immune system. Some of the most important examples of rapidly mutating antigens are:

- (i) the HIV virus (which causes AIDS);
- (ii) the malaria parasite (which causes malaria);
- (iii) trypanosomes (which cause sleeping sickness).

We shall not incorporate into our model the intricate biological details of the mechanisms used by these antigens to change their immunological "appearance." For example, parasites like the trypanosomes usually change their surface proteins, and other antigens use other mechanisms. We shall assume that the point $\vec{s}_A$ in shape space, which characterizes the shape of the antigenic binding region, is now a time dependent point:

$$\vec{s}_A(t) = \vec{s}_A(0) + \vec{v}_A t \quad (8.1)$$

In other words, we shall model this mutating antigen by a shape which transforms itself in such a way that it moves through S with a constant velocity $\vec{v}_A$. Let $t=0$ indicate the moment at which the infection started, and let $\vec{s}_B$ be the representative point in the immune repertoire which is nearest to $\vec{s}_A(0)$.

Now consider all the B-cells which belong to this clone, as well as all their descendants. This whole group of cells is being stimulated by the antigen and is subject to rapid cell division as well as to somatic hypermutation. In this group of cells we now represent each cell by a corresponding point in shape space S . The resulting "swarm" of representative points can be characterized by a continuous density distribution $b(\vec{s}, t)$ in such a way that $b(\vec{s}, t)d^n\vec{s}$ equals the number of cells- the representative points of which are located, at time t , in a small volume element of shape space which has a volume $d^n\vec{s}$ and which is centered on position $\vec{s}$.

Next one asks: how does $b(\vec{s}, t)$ change in time, for a fixed positions $\vec{s}$? The reader might want to consult ref. [4] for all details of the answer. In principle the temporal change of b (which equals $\frac{\partial b}{\partial t}$) can have three causes:

- (1) Diffusion of shape due to somatic mutation, as discussed in section 7. This leads to an expression of the form $D\Delta b$, where Δ denotes the n -dimensional Laplace operator and where D is the diffusion coefficient, cf. (7.1,2).
- (2) Proliferation of the cells in this swarm. This gives an expression of the form $f(|\vec{s} - \vec{s}_A(t)|)b$, where the function f is a measure for the proliferation rate. We assume that this rate depends on the distance, $|\vec{s} - \vec{s}_A(t)|$, between $\vec{s}$ and the representative point $\vec{s}_A(t)$ which characterizes the shape of the antigen's binding region at time t . As this rate increases when the distance gets shorter the function f should be >0 and its value should increase for decreasing values of its argument.
- (3) Death of the cells in this swarm. For simplicity we shall represent this effect by a term $-d_o b$, which states that each cell in the swarm has a constant probability to die which equals d_o per unit of time.

The mathematical analysis now proceeds in three steps [for details consult ref. (4)]:

(α) The density distribution $b(\vec{s}, t)$ is a solution of the partial differential equation

$$\frac{\partial b}{\partial t} = D\Delta b + f(|\vec{s} - \vec{s}_A(t)|)b - d_o b. \quad (8.2)$$

This equation must be solved under the initial condition

$$b(\vec{s}, o) = \delta(\vec{s} - \vec{s}_B), \quad (8.3)$$

where δ denotes Dirac's delta function. This condition expresses the fact that the swarm consists at $t=0$ of a single B-cell which carries in its outer membrane antibodies which all have a binding pocket of shape $\vec{s}_B$. If the original clone consists at $t=0$ not of a single B-cell, but of 10^5 B-cells, one should multiply the right-hand-side of (8.3) with a factor 10^5 .

(β) Solve the problem posed in (α) in an approximate way, for example with the use of path integrals.

(γ) Calculate the number of cells in this swarm, the representative points of which are located inside an $\mathcal{E}_1$ -sphere around the point $\vec{s}_A(t)$. Call this number $N_B(t)$. Obviously

$$N_B(t) = V_{\mathcal{E}_1} b(\vec{s}_A(t), t), \quad (8.4)$$

where $V_{\mathcal{E}_1}$ denotes the volume of an $\mathcal{E}_1$ -sphere. Using the approximate result of step (β), one finds for long times the asymptotic expression:

$$N_B(t) \equiv C \frac{2\pi^{n/2}}{n\Gamma(n/2)} \mathcal{E}_1^n \exp[\{f(0) - d_o - \frac{\vec{v}_A^2}{4D}\}t], \quad (8.5)$$

where C is a constant.

In order to make contact between the mathematics of our model and the sometimes grim reality of the diseases mentioned at the beginning of this section, we assume that the antigen will be eliminated by the immune system if and only if at some time t the number $N_B(t)$ exceeds some critical value N_0 . The actual value of N_0 could only be determined from intricate biological considerations. It should be clear from the exponential factor in (8.5) that for parameters such that

$$f(0) - d_o > \frac{\vec{v}_A^2}{4D}, \quad (8.6)$$

the antigen will always be eliminated after a while. If, however,

$$f(0) - d_0 < \frac{\vec{v}_A^2}{4D}, \quad (8.7)$$

then the situation is potentially very dangerous indeed: in the long run the antigen will escape detection.

9 Concluding Remarks

The model which formed the subject of this paper has the advantage that it catches some of the most essential features of the immune system and that it is simple enough to be accessible to mathematical treatment.

The price one has to pay for it is an oversimplification of the reality of the immune system. For example, the phenomenon of auto-immunity is ignored altogether. The actual immune system has to keep a delicate balance between being sufficiently aggressive towards foreign antigens, and being sufficiently tolerant towards the biomolecules of the organism itself. This balance is reflected in the number $\approx 10^7$ different clone of B-cells in the immune system: had this number been substantially larger, auto-immune disease would be too likely; had it been substantially smaller, foreign antigen could not be detected rapidly enough.

Another important neglected aspect of the immune response is the way in which its cells and antibodies are distributed throughout the organism, and the effect this has on the size of clones and the dynamics of the numerous reactions that go on continuously. This point touches upon the scaling aspect of the architecture of the immune system: a topic which up till now has received less attention than might be expected.

This list of immunological shortcomings could be continued. The more mathematically inclined reader, who might have consulted the calculations in ref. [4], would also have realized that we treated the mathematical problems only in a rough and almost qualitative fashion. We feel that many refinements of the theory are very well possible indeed, but that the global features of these more accurate solutions would not differ significantly from those which were found here.

10 Acknowledgments

I am indebted to the organizers of this conference for giving me an opportunity to talk about the mathematical modelling of biological reality, a subject that has been dear to me since 1974.

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References

1. Perelson, A.S. and G. Weisbuch. "Immunology for physicists." *Review of Modern Physics* 69 (1997): 1219-1267.
2. Wiegand, F.W. *Physical Principles in Chemoreception*. Berlin and New York, NY: Springer-Verlag, 1991.
3. Perelson, A.S. and G.F. Oster. "Theoretical studies of clonal selection: minimal antibody repertoire size and reliability of self-nonself discrimination." *Journal of Theoretical Biology* 81 1979: 645-670.
4. Perelson, A.S. and F.W. Wiegand. "Some design principles for immune system recognition." *Complexity* 4 1999: 29-37.

PROTEIN FOLDING: PHYSICS ON PRODUCTS OF EVOLUTION

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Proteins are self-assembling molecular systems. A polypeptide chain of a protein molecule folds into a globular three-dimensional structure, which is specific to the amino acid sequence of the chain. A protein molecule is in the "native state" when folded into its specific three-dimensional structure. Only in the native state, a protein molecule carries out its biological function. This extraordinary self-assembly ability of proteins can be explained based on the three generally accepted empirical observations in proteins: (1) Two-state character; Folding and unfolding transitions in small globular proteins are generally of the two-state character. (2) Consistency principle; Various components of intra-molecular interactions responsible for stabilizing the native state of globular proteins are consistent to each other in their native state. (3) Principle of marginal stability; The native folded states of globular proteins are generally only marginally stable against their unfolded states. Deduction of the self-assembly ability from the three observations is a problem of physical nature. Very sophisticated theories have been developed recently as to this point. I shall give a very simple and intuitive discussion on this point. Asking why protein molecules show the three observations is another problem. Observation (1) can be derived from the globularity of native states. Observations (2) and (3) can be understood only by considering the evolutionary history of protein molecules, i.e., only polypeptide chains with very specific amino acid sequences selected during the history of evolution show properties of observations (2) and (3). Here we see a case where the mechanism of an extraordinary ability of biopolymers is elucidated in terms of physics, and physics expects that only a very small fraction of amino acid sequences have such an ability. Nature has left the job of finding able sequences to the history of evolution.

1 The Consistency Principle Revisited

About twenty years ago I was concerned about the statistical mechanical characteristics of the process of protein folding, and was using computer simulation of folding and unfolding transitions in lattice model protein as a main tool of the study. At that time no other people in the world were using such a simplified lattice model of protein for their study. Fortunately my work was recognized in the field of protein physico-chemistry to the extent that I was invited to talk on this work in a Gordon Conference. At that time, no one else used the lattice model of protein. A very famous American professor commented personally after my talk in the Gordon conference. "I was amazed about how much one can say about protein from such a simple model." This comment of course meant that you, young guy, do not know how complex proteins are and therefore blindly bold enough to use such a simplified model. Even though I was sure about what I was saying in my work, still I was discouraged by the fact that the results of my work almost a decade ago received such a cynical comment. Partly due to this situation I decided to leave the field in the year 1983 after writing a review article [1]. There were two other fields

that made me busy then. At that time I was embarking into the new field of theoretical studies of conformational dynamics, especially normal mode analysis of native state proteins. Also during that time I showed for the first time that NMR experimental data had enough amount of information to determine protein conformations in atomic details and worked out a theoretical algorithm for actual conformation determination from experimental data. This work in a sense forced me to be involved in the field of determination of protein conformations in solution from NMR experimental data.

Anyway I left the field of protein folding in 1983. Quite ironically the field became to attract much attention just after that. Many people began to use lattice models of protein in their theoretical studies of protein folding. Among the results of such “new” theoretical studies of protein folding, there were many ingredients that were re-discovery of what had been already described in my old papers expressed now in a slightly different language. An example is “The consistency principle.” An idea, which appears essentially identical to the consistency principle, was proposed by Wolynes in year 1987 under the name of “The principle of minimum frustration” [2]. Of course he arrived at the idea from a different point of view, and therefore used a different name. However, it appears that they express essentially the same idea. Wolynes and coworkers later developed the funnel idea for the shape of potential energy surface responsible for the phenomenon of protein folding based on the principle of minimum frustration. Even though I do not understand the funnel idea very clearly, some of my views of protein folding process are different from what the funnel idea appears to tell us. Therefore I would like to re-visit the idea of “The consistency principle” and examine it in the modern view. At the same time in the later part of this essay I would advocate another new principle, the principle of mutational robustness, as a principle which all protein molecules appear to satisfy.

2 The Two-State Character of Folding and Unfolding Transition

I would start my discussion of protein folding by pointing out that the two-state character is one of the most important basic characteristics of the phenomenon. The equilibrium folding- and unfolding process in all small globular proteins is known to undergo the two-state transition. This means that during all the process of folding- and unfolding transition observed under the equilibrium condition, i.e., under the experimental condition in which the external parameter to cause the transition is changed very slowly, only two conformational species, totally folded and unfolded, are observed with significant populations. Population of molecules with half of a molecule folded and the other half unfolded within a single molecule is negligibly small. At the midpoint of the equilibrium transition half of the molecules in the ensemble are in the totally folded conformational state and the other half in the unfolded state. It is a mixture of molecules, each of which is in

either of the two states. This situation of the two-state character of the folding- and unfolding transition in globular proteins can be compared with the alpha helix-random coil transition in some of homopolypeptide molecules. There the transition occurs as a process in which each molecule in the ensemble undergoes a gradual conformational change from the totally helical to the totally random coil state within a molecule. At the midpoint of the transition all of the molecules in the ensemble are in the state in which half of the molecule is in the helical state and the remaining other half of the molecule in the random coil state. It has been shown that this non-two-state characteristics of the helix-coil transition comes basically from the one-dimensional character of the phenomenon. In contrast, the two state character of the folding and unfolding transition is a result of three-dimensional character of the phenomenon.

When I say that the two-state character is one of the most important basic characteristics of the folding and unfolding transition in globular proteins, I may have to comment on cases of some larger protein molecules, which clearly show non-two-state folding and unfolding transitions. They are mostly cases of large molecules consisting of two or more domains loosely bound together. In such cases each domain undergoes the folding and unfolding transitions as in a small proteins. Therefore the most basic characteristics of the transition manifests itself in small globular proteins.

For the purpose of later discussions, I would translate the above characteristics of the folding- and unfolding transition in globular proteins into the language of free energy profile against the order parameter. The free energy profile for the folding and unfolding transition has two minima corresponding to the folded and unfolded states, respectively. The one for the unfolded state is wide and shallower, reflecting a wide range of unfolded conformations, and the one for the folded state is narrow and deeper, reflecting a more-or-less unique character of the native state. There is a barrier in between. As the external parameter for causing the transition changes, the relative depths of the two minima change, resulting in the change of relative populations of the two states. This is how the folding and unfolding transition takes place under the equilibrium condition in the free energy space. In the process of folding each individual molecule fluctuates for some time within the unfolded state. It may sometime climb up the free energy barrier purely as a random fluctuation. If it climbs up high enough to go over the barrier, the molecule may eventually go over into the folded state.

3 The Consistency Principle

Now I proceed to describe the consistency principle. I first recognized the consistency between the short- and long-range interactions responsible for the stability of the native state. Here interactions between atoms that are separated shortly along the polypeptide main chain of a protein are defined to be of the short-

range. Those between atoms that are separated long along the main chain but usually short through space are defined to be of the long-range. I was forced to recognize the consistency by the strange observation that both of the short- and long-range interactions appear individually to have dominant role in determining the three-dimensional structures of proteins.

I first paid attention to the fair success of various methods of prediction of secondary structures in globular proteins. Because all of the methods of secondary structure prediction are based on information about short-range interactions, their fair success appears to mean that the short-range interactions have dominant role in determining protein structures.

However, if really only the short-range interactions were operative in proteins, then they should essentially be one-dimensional system, in which the transition should be of the type of helix-coil transition. Therefore, the two-state character of transition in globular protein indicates that long-range interactions do in fact have very important role in maintaining the three-dimensional structures of proteins. The importance of long-range interactions can also be observed in the fact that atomic packing within protein interior is very good. Considering the great varieties of shapes of twenty amino acid side chains, the degree of good atomic packing should be amazing. It gives us an impression that the requirement of the good packing should determine the three-dimensional structures.

Now we have observed that both of the short- and long-range interactions appear individually to have dominant role in determining the three-dimensional structures of proteins. This strange observation can be reconciled only by recognizing that the two types of interactions are designed to be consistent to each other. We are here assuming a long history of molecular evolution through which protein molecules with such amino acid sequences are selected that satisfy the requirement of consistency between the two types of interactions. This historical aspect of the consistency principle will be discussed later again. Also consequences of this principle on the mechanism of folding will be discussed in detail below. Before that, I would also comment that the consistency is observed not only between short- and long-range interactions but also between more general various energy terms.

An example is found between intra-residue interactions and all other interactions. Occurrence of side-chain conformations in various rotameric states within protein three-dimensional structures has been found to agree very well with those expected from conformational energies of such rotameric states of isolated amino acid residues. In protein three-dimensional structures side-chain atoms are in close contact with atoms from other amino acids. Even with these obvious interactions side-chain rotameric states are realized as if they are determined only due to intra-residue interactions. This strange finding can be reconciled only by assuming that these two types of interactions are designed to be consistent to each other. In the 1983 review I proposed to assume that the consistency should be observed for more general divisions of interactions responsible for the three-

dimensional structures of proteins. The word “principle” is used partly to mean this generality.

So far the consistency principle is concerned with static roles of various energy terms in maintaining the three-dimensional structures of proteins. One of the questions I was addressing by carrying out simulations of lattice model of protein is specific roles of various energy terms in the kinetic process of protein folding. One specific study [3] was carried out by intentionally changing relative weights of the short- and long-range interactions in the lattice protein model in order to know specific kinetic roles of respective energy terms. It was found that the long-range interactions were important for endowing the two-state character for the transition. The short-range interactions were found to accelerate the process of folding, when they are consistent with the long-range interactions. Thus, various energy terms which are consistent to each other for stabilizing cooperatively the native three-dimensional structures may have different specific roles in the kinetic process of folding.

Wolynes and coworkers studied the problem of protein folding theoretically by taking advantage of an analogy with spin glass problem. They found that, even though protein is a system with highly non-uniform elements, its ground state (or its global minimum), unlike spin glass systems, has no frustration. They appreciated the importance of this finding, and gave a name of “the principle of minimum frustration.” It appears that their “principle of minimum frustration” means essentially the same as my “consistency principle.”

4 Search for Successful Paths of Folding

By combining the two important characteristics of protein folding, the two-state character and the consistency principle, we can arrive at a significant picture of paths of protein folding. As outlined above folding is a random process of climbing up the hill of free energy barrier between the unfolded and folded states. Climbing up the hill occurs when inter-atomic interactions between various parts of the molecule come in to form a small “structure.” This process of formation of a small structure is associated with energy and entropy decreases. While the structure is small, an amount of energy decrease would be smaller than an amount of free energy increase due to entropy decrease, resulting in the increase of total free energy (climbing up the hill). To further climb up the hill, the structure must grow in size by further assembling other parts of the molecule. As long as they are climbing up the hill, they are intrinsically unstable against the unfolded state. The partially structured states are possible only as fluctuation. Various energy terms operative in such a partially structured state may be either consistent to each other or inconsistent. When inconsistent, energy stabilization of such a structure is weak and it would not be able to grow in size significantly further by random fluctuation, and its intrinsic instability would cause it to be disintegrated into the unfolded state.

When consistent, further growth by fluctuation may be possible, and in fortunate cases it may grow large enough to go over the free energy barrier finally to fold into the native structure. Thus, the intrinsic instability of partially folded structures is a built-in proofreading mechanism to exclude inconsistent and therefore non-native structures. Process of folding is characterized by a numerous number of searches for correct consistent paths. It is not like simply rolling down a funnel.

5 Marginal Stability

It is well known that the native states of proteins are only very marginally stable against the unfolded state. Typically the amount of free energy stabilization of the native state over the unfolded state is around 10kcal/mol. Considering the large sizes of molecules, this is a very small amount. There should be a good reason for it. I think in the following way.

The fact that native states of proteins are always only very marginally stable against unfolded states is a result of design so that the native state conformation is unique. If the native state conformation is substantially stable against the unfolded state, conformations other than the native one may also happen to have free energy lower than that of the unfolded state. If such is the case, the native state conformation may cease to be unique. If the amount of stabilization is only very marginal, chances would be very small for other conformations to have free energy lower than that of the unfolded state. Thus, my contention is that the marginal stability is a device developed during the molecular evolution to insure the uniqueness of the native state conformation.

Because I am assuming here that the marginal stability is a result of design, there is an implicit assumption behind this contention that for most proteins designing amino acid sequences with larger stability would be easy. Experience from numerous studies of designing stable proteins appears to indicate that this is in fact the case. However, if one designs a sequence with too much stability, then it may allow other non-native conformations to have free energy lower than that of the unfolded state. Such non-native conformations are likely to have some undesirable hydrophobic patches exposed on the surface making the molecule to aggregate and precipitate.

6 Why Consistent?

The native state conformation and other non-native conformations are respectively characterized by whether or not various energy terms operative in them are consistent to each other. Therefore the native state conformation with consistent interactions is expected to have distinctively stable against other possible folded structures. Free energy of this distinctively stable state is further designed to be only

marginally stable against the unfolded state.

The role of short-range interactions to accelerate the process of folding has been already mentioned above. The mechanism for this observation is easy to understand. The short-range interactions would enhance the probabilities of correct substructures which should be integrated into various structures tried during the search for correct paths of folding. This is the mechanism explaining why folding process is fast when interactions are consistent to each other.

7 Selection during Evolution

The two most essential features used to construct the picture of the mechanism of protein folding are a) the consistency principle and b) the marginal stability of the native state. Due to these two features proteins can have a unique native state conformation and can fold into it quickly.

I described both of these features as results of design developed during molecular evolution. Implicit in this statement is that only very special amino acid sequences can satisfy these two features and such sequences are selected during the evolution.

Protein folding problem is not just a physico-chemical study of poly-amino-acid molecules. To understand it in depth we have to go into their history.

8 Consistency is not Perfect

A few words of comments should be necessary at this point about the consistency principle.

Even though I am claiming that various energy terms responsible for maintaining the native state conformation are consistent to each other, I do not mean that they are perfectly consistent. It is known that in some cases consistency is in fact not perfect.

The first type is cases where non-native secondary structures are formed temporarily in the initial phase of the process of folding. Especially in proteins rich in β -sheets sometimes α -helices are observed to be formed first in the initial phase, but eventually taken over by β -sheets. When one analyzes these proteins by methods of prediction of secondary structure, these secondary structural segments are often found to have very similar propensity to assume either of α -helix and β -strand. When α -helices are observed experimentally to be formed first, it means that α -helical structures are slightly more favored than beta-sheets before inter-segment interactions for stabilization for sheet structures come in. If the consistency principle were perfect, it should mean that intra-segment interactions alone should already favor the β -strand structures over the α -helical structures even before inter-segment interactions come in. In this sense these cases of transient formation of

non-native secondary structures are examples of imperfect consistency. However, the degree of imperfection is small, because such segments assume either structure with very similar propensity.

The second type of imperfection is related to conformational resolution. So far I have discussed the phenomenon of folding, in which we can treat the native state as a single unique state. However, in a finer conformational resolution, the native state conformation is known to be fluctuating dynamically in a very complex manner within a certain conformational range. Such conformational dynamics is known to be essential for carrying out certain biological functions of proteins. Complexity of the dynamics in the native state is a reflection of complexity of the conformational energy surface. In fact the energy surface is known to be very rugged within the range of native state fluctuation. Such rugged energy landscape should be a result of frustration of component energy terms. However, it should very clearly be understood that such frustration of component energy terms is observed only at the high conformational resolution to see the native state as a dynamical ensemble of fluctuating conformations. In the folding problem, we see the native state at a resolution of seeing it as a unique one structure. It is in such a resolution that we observe the consistency principle.

Why do I propose to call the consistency the principle, when there are clear exceptions? The spirit is that I am proposing to understand real proteins in reference to the idealized protein with hypothetical perfect consistency. This attitude is similar to use the concept of ideal gas to understand real more-or-less non-ideal gases. When the consistency is partially unsatisfied, it may be a constructive question to ask why it is not perfect.

9 Principle of Mutational Robustness

Before closing this essay I would like to discuss one new idea. I pointed out that amino acid sequences have been selected for to satisfy the two conditions of the consistency principle and the marginal stability. From a naïve physico-chemical intuition meeting both of these two conditions appears to be very difficult. Let us consider for example a case of introducing a single amino acid mutation into a certain protein. Change of a side-chain chemical structure would cause many changes in conformations and intra-molecular interactions. Each of such changes should usually be expected to be associated with free energy changes of at least a few kcal/mol. Adding them all for all possible changes caused by a single amino acid mutation would be much larger than the order of 10kcal/mol. I think that this reasoning is reasonable from a naïve physico-chemical sense. If so, a single mutation would very often disrupt the condition of marginal stability. Yet, the reality is that most proteins are amazingly robust against amino acid mutations. Because we are so much accustomed to see that proteins are generally so generous for amino acid mutations, it may be difficult to amaze at this fact. To amaze at this

fact, which we are so much accustomed to, we have to have a fresh physico-chemical look at the fact. At first I am proposing to amaze at this fact.

Then, I would like to propose to understand this amazing fact by considering that all protein molecules are designed to satisfy another principle, which I propose to call "the principle of mutational robustness." Amino acid sequences should have been selected also to satisfy this principle. This principle tells us that amino acid sequences have been selected not only to satisfy the two conditions of the consistency principle and the marginal stability, but also to ensure that even after introduction of many different amino acid mutations these two conditions remain to be satisfied. These three conditions put together to satisfy at the same time appears to be very difficult. But nature appears to have done it and the existing proteins are examples of solutions of such a difficult problem.

The reason why nature has found a solution to such a difficult problem should be to endow proteins an ability to mutate easily in order to ensure the ability to tune its function for changing environment. Existence as a globular protein is a prerequisite for carrying out function. Therefore, such sequences should have been selected for that even when amino acid mutations are introduced it is likely that the new sequence generates a globular protein with a unique native conformation.

References

1. Go, N. "Theoretical Studies of Protein Folding" *Ann. Rev. Biophys. Bioeng.* **12**, 183, (1983).
2. J. D. Bryngelson and P. G. Wolynes, *Proc. Natl. Acad. Sci. USA* **84**, 7524, (1987).
3. N. Go and H. Taketomi, *Proc. Natl. Acad. Sci. USA* **75**, 559-563, (1978).

MOVEMENTS OF MOLECULAR MOTORS

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The movements of cytoskeletal motors such as kinesin or myosin V cover many length and time scales. When such a motor is bound to a filament, the consumption of a single fuel molecule leads to a certain motor displacement or step which is of the order of several nanometers. The motor typically makes about a hundred such steps in its bound state and, in this way, covers a walking distance which is of the order of micrometers. On even larger length scales, the motor undergoes random walks which consist of alternating sequences of bound and unbound motor states, i.e., of directed walks along the filaments and nondirected diffusion in the aqueous solution.

1 Introduction

The living cell is a rather complex structure built up from macromolecules and supramolecular aggregates (or colloids) in aqueous solution. From the physical point of view, the most amazing property of this structure is its high degree of spatial organization or 'order' which is maintained in a stationary state far from equilibrium.

It has been realized during the last decade that this order is sustained, to a large extent, by molecular motors which consume 'fuel' molecules such as ATP and perform mechanical work within the cell. It is now believed that all transport processes or movements which occur within the cell in a coherent fashion are governed by such motors. Examples are the transmembrane transport of ions and macromolecules, the regulated adhesion and fusion of membranes, the intracellular transport of vesicles and organelles, cell division, and cell locomotion. [1] Thus, these motors act like little mechanical robots and resemble the microscopic demons envisaged by Maxwell.

There are several classes of motors which fulfill different functions within the cell: (i) DNA and RNA polymerases which move along the strands of these biopolymers; (ii) membrane pumps which transport ions and small molecules across membranes; the resulting concentration gradients may be used in order to drive (iii) rotary motors such as the bacterial flagellar motor and the F_1 -ATPase which are used for cell locomotion and ATP synthesis, respectively; (iv) myosins in muscles which work in groups and collectively displace actin filaments; and (v) cytoskeletal motors which bind to cytoskeletal filaments and then walk along these filaments in a directed fashion. The latter class of motors is responsible for the directed transport of vesicles and other types of cargo across the cell. This nonequilibrium transport on

the micrometer scale is powered by the free energy released from ATP hydrolysis on the subnanometer scale.

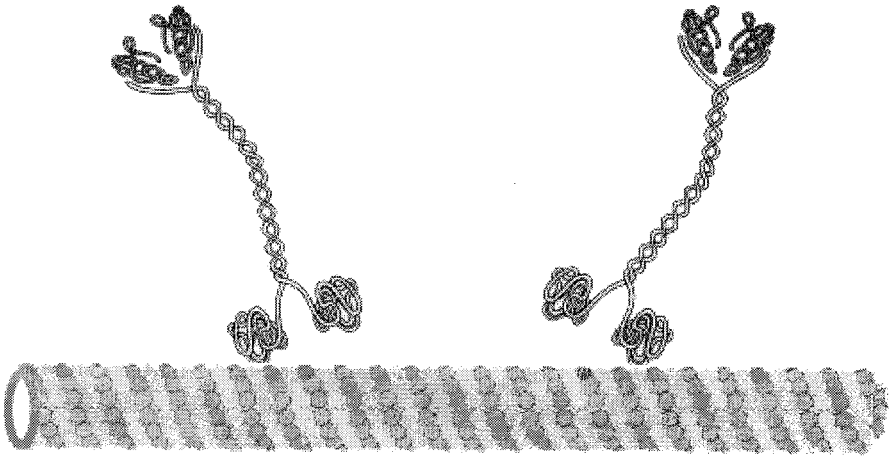


Figure 1. Cartoon of two kinesin molecules bound to a microtubule. The latter tubule has a thickness of about 25 nm. It consists of tubulin dimers which have a linear size of about 8 nm. The kinesin molecule has two heads or motor domains which can bind to the microtubule; its stalk is about 50 nm.

In the following, I will focus on cytoskeletal motors which are processive in the sense that they make many steps before they detach from the filaments. It has been estimated that a typical eucaryotic cell might contain between fifty and a hundred different types of proteins which act as such motors [2]. One example is provided by dimeric kinesin which moves along microtubules. A cartoon of the kinesin-microtubule system is shown in Fig. 1.

Two types of cytoskeletal motors, namely dimeric kinesin on microtubules [3] and myosin V on actin filaments, [4] were observed to walk via discrete steps, the size of which is close to the repeat distance of the filament which is 8 nm and 36 nm, respectively. The average velocity of these walks depends on the ATP concentration and on the applied load force; its maximal value is of the order of 1 μ m per second and is obtained for high ATP concentrations and zero load force.

At each step, the cytoskeletal motor has a small but nonzero probability to unbind from the filament. Thus, its directed movement can be characterized by a

walking time Δt_b , which is the average time period between binding and unbinding, and a corresponding walking distance Δx_b along the filament. For all cytoskeletal motors studied so far, Δt_b was found to be of the order of seconds and Δx_b to be of the order of micrometers [5, 6, 7, 4, 8]. On larger time scales, the motor undergoes random walks which consist of alternating sequences of bound and unbound motor states, i.e., of directed walks along the filaments and nondirected diffusion in the aqueous solution.

This article is organized as follows. First, it is emphasized in Section 2 that the movements of molecular motors involve a wide range of different length (and time) scales. The directed walks of bound motors are discussed in Section 3. It is argued that these walks exhibit some motor properties which are universal in the sense that they are insensitive to many features of the molecular dynamics [9, 10, 11]. The random walks arising from the sequence of bound and unbound motor states are considered in Section 4.

2 Different Types of Movements

The movements of molecular motors cover many length and time scales. For cytoskeletal motors as considered here, one can distinguish three different regimes:

- (i) the molecular dynamics related to the chemomechanical energy transduction;
- (ii) the directed walks along the filaments; and
- (iii) the random walks arising from the unbinding and rebinding of the motors to the filaments.

The molecular dynamics regime (i) covers all length scales up to the motor displacement arising from the ATP hydrolysis. Experiments on dimeric kinesin strongly indicate that the hydrolysis of one ATP molecule leads to a center-of-mass displacement or step size $l_b = 8$ nm which corresponds to the repeat distance of the microtubule (here and below, the subscript b means 'bound'). The corresponding step time τ_b depends on the ATP concentration and on the external load force. For high ATP concentration and low load force, one finds the step time $\tau_b \cong 6$ ms.

The directed walk regime (ii) contains the intermediate length scales which lie between the step size l_b and the walking distance Δx_b . As mentioned, the walking distance is of the order of micrometers for all cytoskeletal motors studied so far; the corresponding walking times Δt_b are of the order of seconds. For dimeric kinesin derivatives without tails, the experimentally determined values are $\Delta x_b \cong 2 \mu\text{m}$ and $\Delta t_b \cong 2.6$ s [6].

Finally, those length and time scales, which exceed the walking distance Δx_b and the walking time Δt_b , define the random walk regime (iii). This latter regime applies to the intracellular transport of vesicles and organelles which typically

involve transport over tens of micrometers. An extreme case is provided by neurons and the transport along their axons as schematically shown in Fig. 2. Axons are tube-like structures enclosed by the plasma membrane of the neuron. The axon diameter varies between hundreds of nanometers and millimeters. Their length is usually much larger than their diameter and can be up to many centimeters or even decimeters.

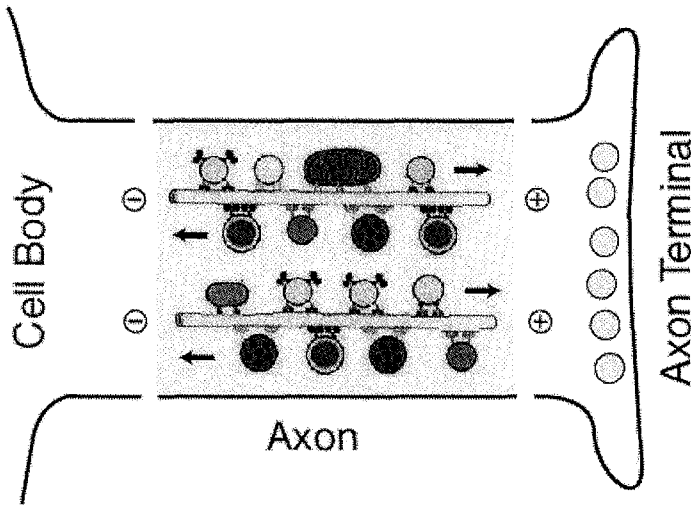


Figure 2. Schematic view through an axon which contains many microtubules, all oriented with their plus ends towards the axon terminal and with their minus ends towards the cell body. Cytoskeletal motors (indicated as small 'feet') are responsible for the transport of vesicles and organelles along these filaments.

3 Directed Walks of Bound Motors

In their bound states, the cytoskeletal motors undergo directed walks along the filaments. For dimeric kinesin, this directed movement has been observed in various types of assays [12, 3, 13, 14, 15, 16, 17, 18]. This experimental work has provided several clues (i) to the biochemical and geometric features, which characterize this motor on molecular scales, and (ii) to the transport properties which characterize the motor performance on supramolecular scales.

3.1 *Experiments on dimeric kinesin*

First, let us look at the kinesin-microtubule system on the molecular scale as schematically shown in Fig. 1. Each dimeric kinesin consists of two identical amino-acid chains. The amino-terminal ends of these chains correspond to the motor domains or heads which have a size of about $7 \times 4.5 \times 4.5 \text{ nm}^3$ [19]. The two heads are connected via a neck region to a long stalk. The neck region consists of ordered coiled-coil domains and a disordered hinge domain, the stalk is a coiled coil with an extension of about 50 nm.

The kinesin molecule can bind to a microtubule. The latter tube-like filament has a thickness of about 25 nm and is built up from tubulin dimers. These tubulin molecules form 13 parallel protofilaments which provide the different 'lanes' on the surface of the microtubule. Each protofilament or 'lane' represents a 1-dimensional lattice of binding sites with a lattice constant of 8 nm.

The kinesin molecule has two binding domains located in its two heads which can adhere to one of the binding sites of the microtubule. In addition, each head can act as an ATPase which adsorbs and hydrolyses ATP. Thus, each head has an ATP-adsorption domain and a microtubule-binding domain.

The free energy released from the ATP hydrolysis is of the order of 20 T (here and below, temperature is measured in energy units, i.e., the Boltzmann constant k_B is adsorbed into T). This free energy is then transduced into a conformational transition of the motor molecule which leads to its directed movement. Dimeric kinesin requires both heads in order to make many successive steps, see, e.g., [15]. Each step corresponds to a center-of-mass movement of 8 nm [3, 13, 14]. In its rigor state, the two heads are bound to two successive lattice sites separated by 8 nm [16].

All of these experimental observations are consistent with the view that dimeric kinesin moves in a 'head-over-head' (or 'hand-over-hand') fashion, i.e., by alternating steps in which one head moves forward while the other one remains bound to the tubule. If the motor does indeed advance by this type of stepping motion, the unbound head and the center-of-mass of the motor would move by 16 nm and by 8 nm, respectively, during each step.

The relative displacement of the kinesin motor against the filament was determined by optical trap experiments. The most direct evidence comes from experiments in which the filament is firmly attached to a solid substrate and the motor molecule is anchored to a bead. This bead is grapped by optical tweezers and then brought into contact with the filament. In these experiments, one can directly measure the time evolution of the displacement of a single motor molecule (plus the attached bead). From a large number of such displacement-versus-time curves, one obtains average motor properties such as the motor velocity which characterize the motor performance on length scales which are large compared to the step size.

For dimeric kinesin, the motor velocity has been measured as a function of two control parameters. The first such parameter is provided by the ATP concentration

Γ , i.e., by the concentration of the fuel molecules. The second control parameter is given by the external load force F usually applied by the optical trap.

Several experiments have shown that the motor velocity v increases monotonically with Γ and exhibits a saturation behavior. In addition, the data for zero or small F could be fitted by the hyperbolic form $v(\Gamma) \equiv v_{\max} \Gamma/[\Gamma_* + \Gamma]$ [12, 20, 14]. More recently, it was found that such a fit is even possible over the whole range of accessible forces as given by $0 \leq |F| \leq 5.6$ pN provided one uses F -dependent fit parameters $v_{\max}$ and Γ_* which leads to [18]

$$v(\Gamma, F) \equiv v_{\max}(F) \frac{\Gamma}{\Gamma_*(F) + \Gamma}. \quad (3.1)$$

As far as the functional dependence on the load force F is concerned, the motor velocity is observed to decrease monotonically with increasing F as one would expect naively. However, the precise functional dependence of velocity versus force has been a matter of some controversy. One problem is related to the vectorial character of the applied force which may, in general, not act parallel to the filament. In fact, if the load force is applied to the motor via an attached bead, one always generates two force components, one which is tangential to and one which is normal to the filament [20]. The normal component of the force acts to increase the probability that the motor unbinds from the filament and may also affect the ATP hydrolysis rate. Furthermore, since the tether between the bead and the kinesin molecule is not expected to behave as a linear spring, the force applied by the optical trap may not be simply proportional to the tangential force acting on the motor molecule. Here and below, the force F is taken to be the tangential component of the force which is conjugated to the motor displacement parallel to the filament. I also use the sign convention that $F < 0$ for a load force which acts against the directed walk of the motor.

The relation as given by (3.1) has two nontrivial features. First, it represents a specific functional dependence of the velocity on the *two* parameters Γ and F . Secondly, the dependence on the fuel concentration Γ is found to be rather simple. These two features raise two general questions: (i) How can one *understand* the simple $v(\Gamma)$ relationship in view of the complex molecular dynamics?; and (ii) The relation as given by (3.1) describes the data for dimeric kinesin, i.e., for one particular cytoskeletal motor. *Should such a relation hold for other motors as well?* In other words, is this relation *specific* to kinesin or is it more *universally* valid? In order to address these questions, we have studied a rather large class of ratchet models [9, 10, 11] as discussed in the next subsection.

3.2 Stochastic (M, K)-ratchets

Within the ratchet models considered here, the directed movement of the motor is described by one spatial coordinate x . For linear motors, this coordinate describes the displacement of the center-of-mass of the motor parallel to the filament (for rotary motors, x would represent an appropriate angular coordinate). For a given value of x , the motor molecule must be bound to the filament but can still attain different conformations or internal states. These different states will be labeled by the discrete index m which can have M different values. In addition, the motor can undergo transitions between these states at the discrete set of K spatial positions per motor cycle. These (M, K)-models [9, 11, 10] represent generalizations of those studied in [21, 22, 23], which were restricted to $(M, K) = (2, 2)$.

Depending on the molecular architecture of the motor, one may identify several discrete subgroups of internal states. If the motor has only one enzymatic domain or head, this head can attain a discrete number of states corresponding to (i) no substrate, (ii) adsorbed ATP, (iii) adsorbed ADP/P, and (iv) adsorbed ADP. In each of these states, the motor may adopt a different conformation which will experience different interactions with the filament. If the motor has two heads, a and b , one has three groups of levels corresponding to (I) two bound heads, (II) bound head a , and (III) bound head b .

In general, the motor conformation also involves internal degrees of freedom which vary in a continuous fashion. For example, one may tilt a two-headed motor molecule, which is bound by one head, and simultaneously move its unbound head without changing the position of its center of mass. In the theoretical framework considered here, these continuous degrees of freedom are also discretized. This is primarily done for computational convenience. However, this discretization can involve a large number M of internal states and a large number K of transition locations corresponding to many intermediate states, and, thus, does not represent a real limitation compared to a continuous description.

Within the (M, K)-ratchets, the time evolution of the motor position x and its internal state m is described by the probability densities $P_m(x, t)$ to find the motor particle at center-of-mass coordinate x and in internal state (or level) m . This probability density may change (i) because of thermally-excited diffusion within the internal state m or (ii) because of transitions between the different internal states. The probability currents arising from these different processes form a 2-dimensional network of transitions as shown in Fig. 3.

* A general theoretical framework in terms of continuous variables or coordinates was recently discussed in [24].

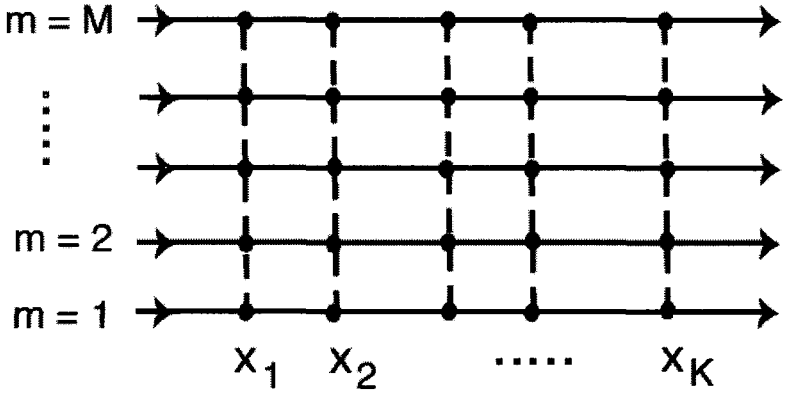


Figure 3. Network of transitions at locations $x = x_k$ with $1 \leq k \leq K$ between different internal states or levels m with $1 \leq m \leq M$. Pairs of vertices $[x_i, m]$ and $[x_j, n]$ are connected by local lateral currents for $n = m$ and $i \neq j$ (corresponding to the full lines), and by local transition currents for $i = j$ and $n \neq m$ (corresponding to the broken lines). These currents obey a vertex rule which corresponds to Kirchhoff's first law for electric circuits. The arrows represent the periodic boundary conditions in the lateral direction.

3.3 Universal or generic aspects

A detailed investigation of the (M, K) ratchets shows that they exhibit some generic or universal transport properties which hold for *arbitrary* values of M and K . [10] In particular, it is possible to determine the explicit dependence of the motor velocity on the transition rates of the unbalanced transitions, i.e., of those transitions which arise from the enzymatic activity and which do not satisfy detailed balance.

In general, there will be Q such transition rates per motor cycle. These unbalanced transition rates will be denoted by Δ_q with $q = 1, 2, \dots, Q$ (see Fig. 4). The simplest situation is provided by motors, such as monomeric kinesin, which have a single enzymatic domain. Such motors should have only one location x_k with only one unbalanced transition rate which implies $Q = 1$ (see Fig. 4). There are several cytoskeletal motors such as dimeric kinesin, dynein, or myosin V which have two identical enzymatic domains or heads. The corresponding ratchet models are characterized by two (usually different) locations with enzymatic activity. If each head can make (i) only forward steps or (ii) both forward and backward steps, it can be activated (i) at only one of these locations or (ii) at both locations, which corresponds to $Q = 2$ and $Q = 4$, respectively (see Fig. 4).

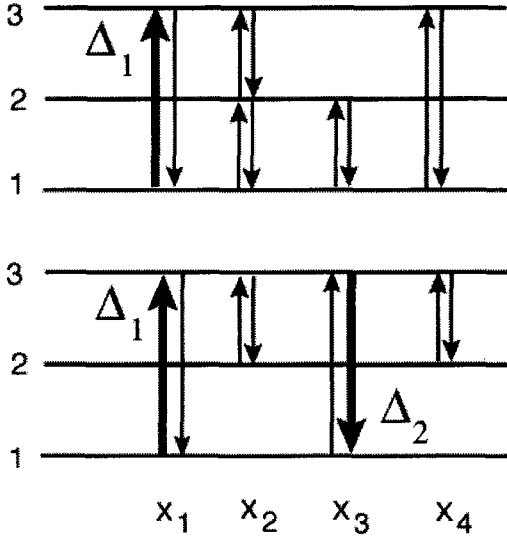


Figure 4. Transition networks with $M = 3$ internal states or levels and (top) $Q = 1$ or (bottom) $Q = 2$ unbalanced transitions. In both examples, the balanced and unbalanced transitions are indicated by thin and thick arrows, respectively. The unbalanced rate constants are denoted by Δ_q with $1 \leq q \leq Q$.

Using a transfer matrix formalism, the motor velocity v is found to have the general form [10]

$$v = \frac{A(\Delta_1, \Delta_2, \dots, \Delta_Q)}{B(\Delta_1, \Delta_2, \dots, \Delta_Q)} \quad (3.2)$$

where both A and B are two polynomials which are *multilinear* in the Q arguments Δ_q . Thus, the highest order terms which can (but need not) appear in these polynomials are proportional to the product $\Delta_1 \Delta_2 \dots \Delta_Q$. The functional relationships as given by (3.2) are *universal* in the sense that they hold for ratchets with arbitrary values of M and K and, thus, for any number of balanced transition rates provided one has the same number Q of unbalanced transition rates. Likewise, these relationships do not depend on the specific shapes of the molecular interaction potentials which the filament exerts onto the motor in its various internal states.

It is also important to note that the velocity-rate relationships as given by (3.2) do not involve *any* assumptions about the enzyme kinetics underlying the

unbalanced transition rates. However, in order to relate the behavior of the (M, K) -ratchets to experiments, one must implement the underlying chemical kinetics which implies a certain dependence of the unbalanced rate constants Δ_q on the concentration Γ of the fuel molecules and on the applied load force F . The simplest scheme for an enzymatic reaction between the fuel molecules and the motor domain (s) is provided by Michaelis-Menten kinetics [25] which implies that $\Delta_q^{-1} = (c_q \Gamma)^{-1} + d_q^{-1}$ with $1 \leq q \leq Q$ where the reaction rates $c_q = c_q(F)$ and $d_q = d_q(F)$ will, in general, depend on the load force F .

If one uses these Michaelis-Menten relations, one obtains the general velocity-concentration relationship as given by [10]

$$v(\Gamma, F) = \left[\sum_{n=0}^Q g_n(F) \Gamma^n \right] / \left[\sum_{n=0}^Q h_n(F) \Gamma^n \right] \quad (3.3)$$

which represents the ratio of two Γ -polynomials of degree Q with F -dependent coefficients. The first polynomial coefficient g_0 in the numerator satisfies $g_0(F=0) = 0$. If the motor cycle exhibits certain symmetries or constraints, some polynomial coefficients may vanish for all values of F .

The relationships as given by (3.3) are again *universal* in the sense that they are valid (i) for any number of balanced transition rates, (ii) for any choice of the molecular interaction potentials, (iii) for arbitrary load force F , and (iv) for any force dependence of the Michaelis-Menten reaction rates c_q and d_q .

As mentioned, the simplest motor cycles are characterized by $Q=1$. In this case, the relation (3.3) simplifies and reduces to

$$v(\Gamma, F) = [g_0(F) + g_1(F)\Gamma] / [h_0(F) + h_1(F)\Gamma]. \quad (3.4)$$

For $F=0$, one has $g_0(F=0) = 0$ which implies

$$v(\Gamma, 0) = v_{\max} \Gamma / [\Gamma_* + \Gamma] \quad (3.5)$$

with the saturation velocity $v_{\max} \equiv g_1(0)/h_1(0)$ and the characteristic concentration $\Gamma_* \equiv h_0(0)/h_1(0)$.

The rather simple velocity-concentration relationship (3.5) was first obtained from a simple tight-coupling picture [12, 20]. The latter picture is based on two assumptions: (i) the motor makes one step of mean step size Δx per ATP hydrolysis; and (ii) the ATP hydrolysis follows Michaelis-Menten kinetics which implies that

its rate constant ω_{hyd} is given by $\omega_{\text{hyd}}(\Gamma) = \omega_{\text{max}} \Gamma / [\Gamma_* + \Gamma]$. A combination of these two assumptions leads to

$$v(\Gamma, 0) = \Delta x \omega_{\text{max}} \Gamma / [\Gamma_* + \Gamma] \quad (3.6)$$

which is equivalent to (3.5).

The simple tight-coupling picture just described has the disadvantage that it provides no insight into its range of validity or its limitations. In contrast, the systematic theoretical approach based on the (M, K) -ratchets leads to a classification scheme which contains the simple Michaelis-Menten-type relation (3.5) as a special case. In other words, this approach characterizes both those motors for which the relationship as given by (3.5) holds exactly as well as those motors for which this relationship is replaced by more general relations as in (3.3).

The simple relationship (3.5) is always valid provided (i) the motor cycle is characterized by $Q = 1$ and (ii) this unbalanced rate follows Michaelis-Menten kinetics. If (ii) remains valid but $Q \geq 2$, one should, in general, expect to see the more general relationships as given by (3.3). There are exceptional cases, however, which arise from additional symmetries and constraints within the motor cycle. Indeed, as a result of these symmetries or constraints, some of the polynomial coefficients in (3.3) may vanish for all values of F . The latter situation has been explicitly shown to apply to some models for dimeric kinesin with $(M, K) = (2, 2)$ and $(M, K) = (3, 2)$ [9, 10].

Finally, one should note that the enzymatic reaction does not necessarily follow Michaelis-Menten kinetics. One example for a different kinetics would be provided by molecular motors with allosteric domains which bind regulatory molecules. The reaction rate Δ_q can then exhibit a sigmoidal dependence on Γ [25]. If such a sigmoidal form is inserted into the general expression (3.2) for the dependence of the motor velocity v on the unbalanced transition rates Δ_q , one will obtain a v - Γ relationships which differs from (3.3).

In summary, it is possible to classify the Γ -dependence of the motor velocity v as discussed in the present subsection. Such a classification scheme is not available for the F -dependence of v which depends on the details of the interaction potentials between the filament and the motor and, thus, reflects the specific features of the underlying molecular structure. It is, however, not difficult to calculate the F -dependence of the motor velocity if one makes some specific assumptions about the interaction potentials as shown in [9] for several types of sawtooth potentials.

4 Random Walks of Motors and Motor Traffic

Even processive motors eventually unbind from the filaments. As mentioned in the introduction, this unbinding process can be characterized by a walking time Δt_b , which is of the order of seconds, and by the corresponding walking distance Δx_b , which is of the order of micrometers. Both quantities depend on the overall ionic strength [5, 8], on the presence of certain ions such as magnesium [C. Schmidt, M. Rief, private communication], and on the molecular roughness of the filaments arising from adsorbed tau proteins [7] or from chemically altered tubulin[8].

4.1 Diffusion of unbound motors

A motor which is no longer bound to the filament, will undergo diffusive motion in the surrounding liquid. For an *in vitro* system, the corresponding diffusion coefficient D_{ub} is given by the classical Stokes-Einstein relation $D_{ub} = k_B T / (6 \pi \eta R_{hyd})$, see, e.g., [26], and depends on the thermal energy $k_B T$, on the dynamic viscosity η of the solution, and on the effective hydrodynamic radius R_{hyd} of the motor particle (here and below, the subscript *ub* means ‘unbound’ state of motor and filament). If the dynamic viscosity has a value close to $\eta \cong 0.9 \text{ mPa s}$ ($\cong \text{cP}$) as appropriate for pure water, one has $D_{ub} = 24 \text{ } \mu\text{m}^2/\text{s}$ for a motor molecule with a hydrodynamic radius of 10 nm, and $D_{ub} = 2.4 \text{ } \mu\text{m}^2/\text{s}$ for a motor with an attached bead of radius 100 nm.

It is interesting to compare these estimates for the unbound diffusion coefficient D_{ub} with the values for the bound state diffusion coefficients D_b as measured for kinesin. These latter values are of the order of $10^{-3} \text{ } \mu\text{m}^2/\text{s}$ [13, 6] and $5 \times 10^{-2} \text{ } \mu\text{m}^2/\text{s}$ [6] for two-headed and one-headed kinesin, respectively. The relatively small values of D_b reflect the additional friction arising from the binding between motor and filament. Thus, for a normal aqueous solution, the unbound diffusion coefficient D_{ub} is much larger than the bound state diffusion coefficients D_b . In principle, one could reduce D_{ub} by a factor up to 10^2 if one changes the viscosity of the aqueous solution by adding some solutes such as glycerol or sucrose.

It is more difficult to estimate the diffusive motion of an unbound motor *in vivo*. The cytosol contains macromolecules, supramolecular structures, and organelles, and the unbound motor may experience both repulsive and attractive interactions with these ‘particles’. For repulsive interactions, the particles represent additional steric barriers for the diffusive motion of the motor which will then exhibit a reduced diffusion coefficient. This reduction may be estimated by comparison with the diffusion of inert particles in fibroblasts for which the values $D_{ub} \cong 1.6 \text{ } \mu\text{m}^2/\text{s}$ and $\cong 3000 \text{ nm}^2/\text{s}$ have been measured for particle radii of 10 nm

and 80 nm, respectively [27]. Compared to water, this corresponds to a size-dependent reduction factor of 10^{-1} and 10^{-3} , respectively.

4.2 Tracer motors in open compartments

On time scales which are large compared to the walking time Δt_b , the motor undergoes random walks which consist of alternating sequences of bound and unbound motor states, i.e., of directed walks along the filaments and nondirected diffusion in the aqueous solution. When bound to a filament, the motor walks in a certain direction until it unbinds; it then undergoes nondirected diffusive motion in the surrounding aqueous solution until it encounters the same or another filament to which it can rebind and continue its directed walk.

It is intuitively clear that the relative importance of directed and diffusive motion will depend on the number and arrangement of the filaments and on the confinement of the motion by boundaries or walls. For 2- and 3-dimensional systems without boundaries, the random walks of the motors were studied by Ajdari [28] using scaling arguments. For these systems without confining walls, we have been able to obtain complete analytical solutions for the time evolution of the drift velocity and of the diffusion coefficient [29]; in addition, we also determined these quantities by Monte Carlo (MC) simulations and found very good agreement with the analytical results.

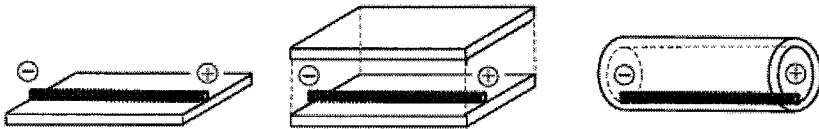


Figure 5. Various compartments with one filament attached to the confining walls: (left) half space; (middle) slab; and (right) open tube. The filament corresponds to the thick rod with its minus-end on the left and its plus-end on the right. All three compartments are open in at least one spatial direction.

We have also extended these studies to bounded geometries or compartments as shown in Fig. 5 which are accessible to *in vitro* experiments. [30] We explicitly

considered oriented filaments which are attached to the interior surfaces of three types of *open* compartments (half space, slab, open tube) as in Fig. 5. In the half space, the motor velocity v decays $\sim 1/t$ for long times t and the advancement of the motor is so slow that it will be difficult to measure. In the slab, $v \sim 1/t^{1/2}$ and the advancement should be measurable if one tracks the motor for a couple of minutes. For an open tube which resembles an axon, the velocity is reduced by a constant factor which depends on the radius of the tube.

4.3 Motor traffic in closed compartments

Finally, let us close the orifices of the open tube in Fig. 5. One now attains a compartment which confines the motors in all three spatial directions. If an ensemble of *many* motors is placed in such a compartment, the ATP hydrolysis of the bound motors on the nanometer scale generates motor concentration gradients and motor currents on the micrometer scale. These mesoscopic gradients and currents lead to novel stationary states far from equilibrium which are characterized by a subtle balance between bound currents along the filaments and diffusive currents in the aqueous solution [30].

For the open compartments discussed in the previous subsection, it was implicitly assumed that the motor concentration is relatively small and that one can safely ignore possible interactions between different motor particles. In contrast, if the compartment is closed, one must take into account the mutual exclusion of two motor particles in order to describe the motors bound to the filaments. Indeed, these filaments become easily overcrowded or 'jammed' even if the overall motor concentration is still rather small.

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References

1. B. Alberts *et al.*, *Essential cell biology: An introduction to the molecular biology of the cell* (Garland, New York, 1998).
2. J. A. Spudich, *Nature* **372**, 515 (1994).
3. K. Svoboda, C. Schmidt, B. Schnapp, and S. Block, *Nature* **365**, 721 (1993).
4. A. Mehta *et al.*, *Nature* **400**, 590 (1999).
5. R. D. Vale *et al.*, *Nature* **380**, 451 (1996).

6. Y. Okada and N. Hirokawa, *Science* **283**, 1152 (1999).
7. B. Trinczek, A. Ebnet, E.-M. Mandelkow, and E. Mandelkow, *J. Cell Sci.* **112**, 2355 (1999).
8. Z. Wang and M. Sheetz, *Biophys. J.* **78**, 1955 (2000).
9. R. Lipowsky and T. Harms, *Eur. Biophys. J.* **29**, 542 (2000).
10. R. Lipowsky, *Phys. Rev. Lett.* **85**, 4401 (2000).
11. R. Lipowsky, in *Stochastic Processes in Physics, Chemistry and Biology*, Vol. 557 of *Lecture Notes in Physics*, edited by J. A. Freund and T. Pöschel (Springer, Heidelberg, 2000), pp. 21-31.
12. J. Howard, A. J. Hudspeth, and R. D. Vale, *Nature* **342**, 154 (1989).
13. M. J. Schnitzer and S. M. Block, *Nature* **388**, 386 (1997).
14. W. Hua, E. C. Young, M. L. Fleming, and J. Gelles, *Nature* **388**, 390 (1997).
15. W. O. Hancock and J. Howard, *J. Cell Biol.* **140**, 1395 (1998).
16. M. Thormählen *et al.*, *J. Mol. Biol.* **275**, 795 (1998).
17. S. Gilbert, M. Moyer, and K. Johnson, *Biochemistry* **37**, 792 (1998).
18. K. Visscher, M. J. Schnitzer, and S. M. Block, *Nature* **400**, 184 (1999).
19. F. J. Kull *et al.*, *Nature* **380**, 550 (1996).
20. K. Svoboda and S. Block, *Cell* **77**, 773 (1994).
21. J. Prost, J.F. Chauwin, L. Peliti, and A. Ajdari, *Phys. Rev. Lett.* **72**, 2652 (1994).
22. F. Jülicher, A. Ajdari, and J. Prost, *Rev. Mod. Phys.* **69**, 1269 (1997).
23. A. Parmeggiani, F. Jülicher, A. Ajdari, and J. Prost, *Phys. Rev. E* **60**, 2127 (1999).
24. D. Keller and C. Bustamante, *Biophys. J.* **78**, 541 (2000).
25. A. Lehninger, D. Nelson, and M. Cox, *Principles of biochemistry*, 2nd ed. (Worth Publishers, New York, 1993).
26. H. C. Berg, *Random walks in biology* (Princeton University Press, Chichester, 1993).
27. K. Luby-Phelps, *Int. Rev. Cytology - A Survey of Cell Biol.* **192**, 189 (2000).
28. A. Ajdari, *Europhys. Lett.* **31**, 69 (1995).
29. T. Nieuwenhuizen, S. Klumpp, and R. Lipowsky, in preparation.
30. R. Lipowsky, S. Klumpp, and T. Nieuwenhuizen, submitted.

LIGAND MIGRATION AND BINDING IN MYOGLOBIN MUTANT L29W

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Myoglobin, a small globular heme protein that binds gaseous ligands such as O₂, CO, and NO reversibly at the heme iron, has for many years been a paradigm for studying the effects of structure and dynamics on protein reactions. Time-resolved spectroscopic measurements after photodissociation of the ligand reveal a complex ligand binding reaction with multiple kinetic intermediates, resulting from protein relaxation and movements of the ligand within the protein. To observe structural changes induced by ligand dissociation, we have investigated carbonmonooxy myoglobin (MbCO) mutant L29W using time-resolved infrared spectroscopy in combination with x-ray crystallography. The presence of two distinct infrared stretch bands of the bound CO, A_I at 1945 cm⁻¹ and A_{II} at 1955 cm⁻¹, implies that L29W MbCO assumes two different conformations at neutral pH. Low-temperature flash photolysis experiments with monitoring of the absorption changes in the individual CO lines reveal markedly different rebinding properties. While recombination in A_{II} is conceptually simple and well described by a two-state transition involving a distribution of enthalpy barriers, recombination in A_I is more complicated: Besides a fast kinetic component, a second, slower kinetic component appears; its population grows with increasing temperature. X-ray crystallography of crystals illuminated below 180 K to photodissociate the CO reveals that the slow component arises from ligands that have migrated from their initial docking site to a remote site within the distal heme pocket. This process occurs in an essentially immobilized, frozen protein. Subsequently, ligands rebound by thermal activation over a barrier that is much higher than the barrier for recombination from the initial docking site. Upon photodissociation above 180 K, ligands escape from the distal pocket, aided by protein fluctuations that transiently open exit channels. The x-ray structure shows a large proportion of ligands in a cavity on the opposite side of the heme group. The relevance of these results for the physiological binding process is discussed.

1 Introduction

Myoglobin is a small, mainly helical, monomeric oxygen storage protein that facilitates oxygen transport in vertebrate skeletal muscle and is especially abundant in sea-diving mammals. It was the first protein to have its three-dimensional structure determined. Dioxygen and other small ligands bind to the heme iron from an internal cavity, the distal heme pocket. Upon ligand binding or release, the geometry of the heme group changes. These structural alterations are communicated to the polypeptide moiety through the covalent bond of the heme iron to the

proximal histidine (H93) and nonbonded interactions, and the globin changes its conformation accordingly. In carbonmonoxy myoglobin (MbCO), these events can be triggered by light, and therefore, this protein has been employed for a long time as a model system in the study of protein dynamics^{1,2}.

Initially, the energy deposited in the heme group by absorption of a visible photon cleaves the covalent bond between the heme iron and the ligand, leading to a non-equilibrium structure, denoted Mb*CO. Subsequently, the CO either rebinds internally (geminately), without ever leaving the distal heme pocket, or it escapes from there, migrates through the protein and exits into the solvent. Ligand migration is enabled by protein fluctuations that transiently open exit channels, and protein relaxations lead to the equilibrium unligated (deoxy) Mb structure. Eventually, another ligand will enter the protein and rebound to the heme iron in a bimolecular process.

At room temperature, almost every CO ligand escapes from the protein after dissociation, and only a few percent of the molecules rebound geminately³. Lowering the temperature slows protein motions that permit ligand escape, and the relative fraction of geminate rebinding grows successively. A variety of experimental techniques have given evidence that a dynamical transition occurs around ~180 K in which large-scale protein motions become arrested^{4,5,6,7}. Below this temperature, ligands can no longer leave the distal heme pocket after dissociation and rebound only geminately^{1,2,8}.

Crystal structure analyses of the photoproduct intermediate, Mb*CO, at liquid-helium temperatures (20 – 40 K) have revealed the structural changes that occur upon photodissociation below the transition temperature^{9,10,11}: the CO moves to a docking site above heme pyrrole C, the heme group shifts by ~0.2 Å towards the distal side, and the iron shifts by ~0.3 Å out of the mean heme plane as a result of its low-to-high spin transition and coordination change. Only very minor shifts occur in the atomic positions of the frozen-in polypeptide moiety. Investigations using time-resolved infrared spectroscopy¹², x-ray crystallography¹³, and molecular dynamics simulations^{14,15,16} have suggested that the short-time photoproduct (before protein relaxation) at room temperature is essentially identical to the low-temperature photoproduct (with inhibited relaxation).

Until recently, structural information about the processes that occur subsequent to the formation of the distal pocket intermediate Mb*CO was scarce. In molecular dynamics simulations by Elber and Karplus¹⁷, ligands escaping from the distal heme pocket preferred routes involving a few internal cavities in the structure of myoglobin. Note that proteins are usually quite densely packed in their cores. Myoglobin, however, contains a few additional internal cavities in addition to the active site that are large enough to bind xenon atoms^{18,19}. They are also referred to as 'xenon holes'.

To experimentally investigate the role of the internal cavities, we have combined x-ray crystallography of photoproduct structures and time-resolved infrared spectroscopy²⁰. A substantial problem in the x-ray experiments is that the

photoproduct states are metastable, and in order to be well resolved, they need to be populated to a high extent in the crystal. In sperm whale myoglobin, no more than ~10 % of the population can be kept photolyzed under steady illumination at temperatures where large-scale protein motions are activated²¹. Consequently, we have chosen the sperm whale myoglobin mutant L29W, which has a tryptophan at sequence position 29 instead of the naturally occurring leucine. The expression system was kindly provided by Dr. John S. Olson (Rice University, Houston, TX). This mutant has an exceptionally small association rate coefficient, 130-fold smaller than that of the native protein at room temperature²². This enabled us to photolyze 100% of the molecules at moderate light intensities without damaging the crystals. The illumination protocols that yielded maximum enhancement of the intermediate states were developed using the kinetic information obtained from infrared spectroscopy.

2 Infrared Spectroscopy of L29W Carbonmonoxy Myoglobin

2.1 Taxonomic Substates

Iron-bound CO in heme proteins is characterized by a strong infrared stretch absorption near 5 μm . Frequently, multiple lines that have been associated with different conformations of the protein, sometimes called 'taxonomic substates' occur²³. For example, in native myoglobin, there exist three taxonomic substates, A_0 , A_1 , and A_3 , that rebinding CO with markedly different kinetics²⁴. Moreover, their relative populations depend on environmental variables such as temperature, pressure, and solvent composition. A detailed study of the pH dependence has shown that A_0 forms by protonation of the distal histidine, H64, imidazole side chain²⁵. The different CO stretch frequencies of the A states arise from the presence of different structural environments affecting the bond order of the bound CO ligand. Comparison of a wide variety of distal pocket mutants of myoglobin has revealed that the interaction of the CO dipole with the local electric field (Stark effect) is a major determinant causing the observed shifts^{26,27}.

FTIR spectra of L29W MbCO (20 mM, pH 7.5, in 75% glycerol / 25% potassium phosphate buffer solution), measured with a Bruker IFS 66v/S FTIR spectrometer at a number of different temperatures are plotted in Fig. 1. They contain two discrete lines, which we call A_I (1945 cm^{-1}) and A_{II} (1955 cm^{-1}). Above ~180 K, their relative populations are strongly temperature dependent; whereas A_I dominates at low temperature, essentially its entire population converts to A_{II} towards 300 K. We note that, unlike in the A_0/A_1 equilibrium in native MbCO,²⁸ the temperature-dependent changes are not governed by a protonation equilibrium of H64 shifting with temperature. Only at lower pH, when H64 side chains become protonated to a noticeable extent, an additional line similar to the A_0 band in native

MbCO appears. The ratio of A_I and A_{II} , however, is sensitively dependent on the solvent composition.

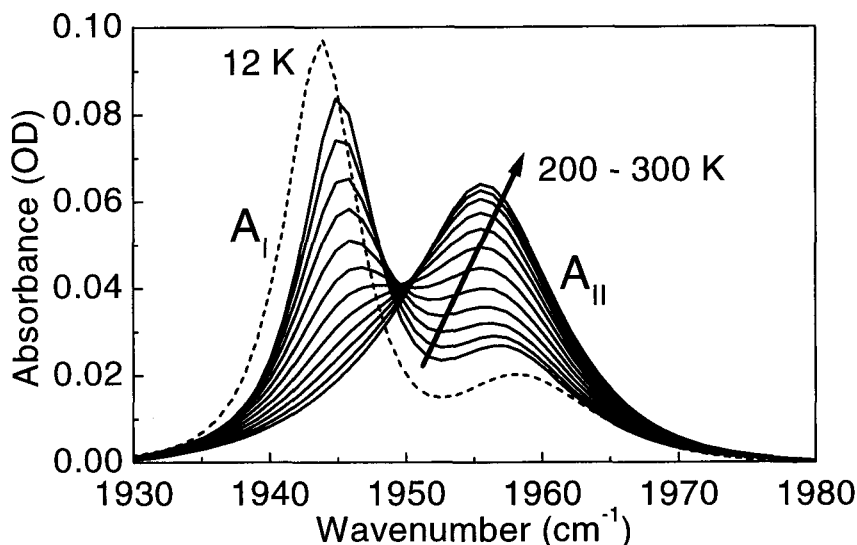


Figure 1. Infrared difference spectra of the CO stretch bands in sperm whale carbonmonoxy myoglobin mutant L29W at pH 7.5, in 75% glycerol / 25% potassium phosphate buffer solution, referenced to metmyoglobin, for 12 K (dashed) and 200 – 300 K in 10 K intervals. Two discrete lines are visible, which represent the A_I (1945 cm^{-1}) and A_{II} (1955 cm^{-1}) conformations. Between 200 and 300 K, a pronounced change in the relative populations is observed.

2.2 Ligand Binding in A_I below 180 K

We measured the kinetics of ligand binding by monitoring absorption changes in the infrared stretch band of the heme-bound CO. Photodissociation was achieved with 6 ns wide pulses (30 mJ) from a frequency-doubled Nd-YAG laser (532 nm). To measure the CO stretch absorption, infrared light from a lead-salt laser diode (Laser Photonics, Analytics Division) was focused on the sample and a photovoltaic InSb infrared detector (SAT Infrared Detectors) with off-axis paraboloid mirrors. The signal from the detector was digitized with a logarithmic time-base digitizer from 1 μs to 200 s and from 20 ns to 400 μs with a digital storage oscilloscope (Tektronix TDS 430A). A closed-cycle helium refrigerator (Helix Technology Corp., CTI Cryogenics Division) cooled the laser diode and the sample.

Flash photolysis measurements in the A_1 state were carried out with monitoring at 1944 cm^{-1} at temperatures below 180 K. The kinetic traces in Fig. 2 A and B are plotted against time in double-logarithmic plots. They are obtained by proper normalization of the observed absorbance changes and represent the fraction of proteins, $N(t)$, that have not yet rebound a ligand at time t after photolysis. Nonexponential rebinding is observed because in this temperature range the sample constitutes a static ensemble of protein molecules in many different conformational substates having different enthalpy barriers for rebinding^{29,30}. At the lowest temperatures (10 – 50 K), recombination is faster than in native myoglobin. Starting at 60 K, the curves flatten at long times, indicative of the appearance of a second kinetic species with markedly slower rebinding. The fraction of slow rebinders increases with temperature, implying that the two kinetic species can interconvert. The kinetics can be described quantitatively with the three-well model depicted in Fig. 2 C, in which rebinding to the bound state A_1 is governed by enthalpy barriers from two photoproduct states, B_2 and B_3 . We remark that we use the index 2 (instead of 1) here to allude to the B_2 substate of native MbCO,¹ characterized by a CO absorption at 2121 cm^{-1} , similar to the position at 2123 cm^{-1} of the fast-rebinding B_2 substate observed in L29W MbCO (data not shown). Time-resolved infrared spectroscopy results suggests that the difference between B_1 and B_2 is the orientation of the CO molecule in the docking site on top of pyrrole C.¹²

Photodissociation initially populates state B_2 . The subsequent rebinding kinetics are governed by the three rate coefficients k_{B_2A} , $k_{B_2B_3}$, $k_{B_3B_2}$ (k_{AB_2} can be neglected here) and given by

$$N(t) = N_f \exp[-(k_{B_2A} / N_f)t] + N_s \exp[-(k_{B_3B_2} \times N_f)t]. \quad (1)$$

Here, N_f and N_s denote the fast and the slowly rebinding fractions, respectively. Their population ratio is given by $N_f / N_s = k_{B_2A} / k_{B_2B_3}$; they are normalized such that $N_f + N_s = 1$. To describe the nonexponential kinetics, the barriers H_{B_2A} and $H_{B_3B_2}$ in Fig. 2 C are not assumed sharp, but modeled as Gaussian distributions,

$$g(H_i) = (2\pi\sigma_i^2)^{-1/2} \exp[-(H_i - H_{p,k,i})^2 / (2\sigma_i^2)]. \quad (2)$$

After including the barrier distributions $g(H_{B_2A})$ and $g(H_{B_3B_2})$, Eq. 1 becomes

$$N(t) = N_f \int g(H_{B_2A}) \exp[-(k_{B_2A} / N_f)t] dH + N_s \int g(H_{B_3B_2}) \exp[-(k_{B_3B_2} \times N_f)t] dH. \quad (3)$$

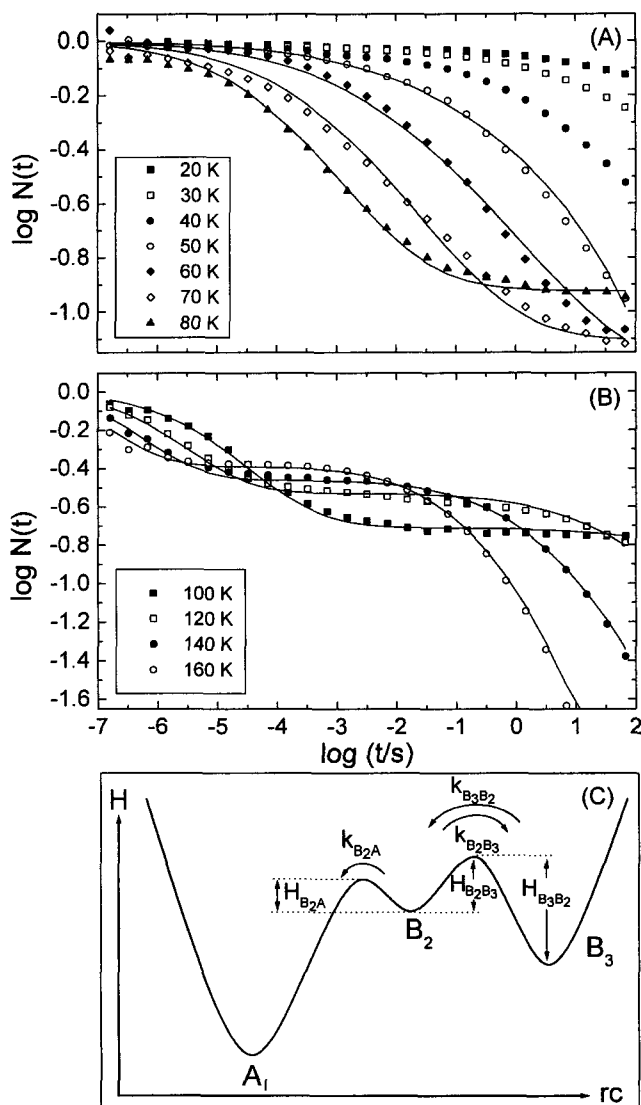


Figure 2. Flash photolysis kinetics of the A_1 conformation of L29W MbCO below 180 K, measured in the infrared band of the bound CO at 1944 cm^{-1} . Experimental data (symbols) for (A) $20\text{ K} \leq T \leq 80\text{ K}$ and (B) $100\text{ K} \leq T \leq 160\text{ K}$, including fits (solid lines) using the model sketched in panel C. (C) Schematic of the three-well model that describes the low temperature kinetics, with bound state A_1 and two photoproduct states, B_2 and B_3 . Details of the kinetic analysis are given in the text; the Arrhenius parameters governing the rate coefficients are compiled in Table 1.

If we assume thermally activated barrier crossing, the temperature dependence of the rate coefficients is governed by the Arrhenius relation,

$$k_i = A_i \frac{T}{T_0} \exp \left[-\frac{H_i}{RT} \right], \quad (4)$$

with preexponential A_i and enthalpy H_i of barrier i ; R is the gas constant, T the absolute temperature and T_0 a reference temperature set to 100 K. Below 50 K, the competing process of tunneling through the barrier becomes significant^{31,32}, and thus we omitted the data for 20 - 40 K in the non-linear least-squares fit of Eqs. 2 - 4 to the data.

The parameters $A_i, H_{pk,i}, \sigma_i$ for k_{B_2A} and $k_{B_3B_2}$ obtained from the fit are compiled in Table 1. Also included are the Arrhenius parameters for $k_{B_2B_3}$, determined from the temperature dependence of the population ratio N_f / N_s , using $k_{B_2A} (H_{pk,B_2A})$. We note that the kinetics will be governed by a rate coefficient k_{B_3A} , with Arrhenius parameters essentially identical to those of $k_{B_3B_2}$ in the case that return from B_3 avoids state B_2 . The solid lines in Fig. 2 A and B are the kinetic traces that result from fitting the model to the data.

Table 1. Preexponentials and parameters of gaussian barrier distributions for the rate coefficients in the three-well model (Fig. 2 C) describing rebinding in the A_1 conformation.

k_i	$\log (A_i/s^{-1})$	$H_{pk,i} \text{ (kJ/mol)}$	$\sigma_i \text{ (kJ/mol)}$
k_{B_2A}	8.8	6.9	1.8
$k_{B_3B_2}$	11.9	32.6	4.0
$k_{B_2B_3}$	10.4	11.1	---

2.3 Ligand Binding in A_{II} below 180 K

Ligand rebinding in the minority species A_{II} was monitored at 1959 cm^{-1} . Kinetic traces are shown in Fig. 3 A for temperatures between 40 and 160 K. Unlike the A_I state, no slow component occurs, and the kinetics can be modeled by recombination of a single photodissociated species B to the bound state A_{II} so that

$$N(t) = \int g(H_{BA}) \exp[-(k_{BA})t] dH, \quad (5)$$

with a gamma distribution of enthalpy barriers,

$$g(H_{BA}) = \frac{\alpha^{(\alpha(H_{pk} - H_{min}) + 1)}}{\Gamma[\alpha(H_{pk} - H_{min}) + 1]} (H_{BA} - H_{min})^{\alpha(H_{pk} - H_{min})} \exp[-\alpha(H_{BA} - H_{min})]. \quad (6)$$

The parameters of the distribution resulting from a non-linear least-squares fit of Eqs. 4-6 to the data are: $10^{7.8}\text{ s}^{-1}$ for the preexponential A_{BA} (Eq. 4), and $H_{min} = 0$, $H_{pk} = 6.3\text{ kJ/mol}$, $\alpha = 0.26$ characterizing the gamma distribution shown in Fig. 3 B.

2.4 Summary of the Low-Temperature Kinetic Studies

Below 180 K, large scale protein motions are arrested, and interconversions between the A states cannot occur. When monitoring in the CO stretch bands of the A_I and A_{II} substates of L29W MbCO, the two subpopulations can be studied individually, and a markedly different recombination behavior for the two A substates is observed. A_{II} exhibits simple two-state recombination, whereas a fast and a much slower kinetic component appear in A_I , the latter gaining importance at the higher temperatures. This raises questions about the structural changes responsible for the enormous slowing of the rebinding observed in the A_I state and why this effect is not observed in the A_{II} state. We have used x-ray crystallography to address these issues.

3 X-ray Structure Analysis of Low-Temperature Photoproducts

X-ray crystallography of ground state structures of small proteins is fairly straightforward, given that sufficiently good crystals are available. More difficult are crystallographic investigations of reaction intermediates as they are metastable. In some proteins, such intermediates can be created by illumination with visible light. To determine structures of photodissociated L29W MbCO, crystals were illuminated with a strong light source (cw Nd-YAG laser, 532 nm) to maintain a large fraction of molecules in the photolyzed state throughout the data acquisition.

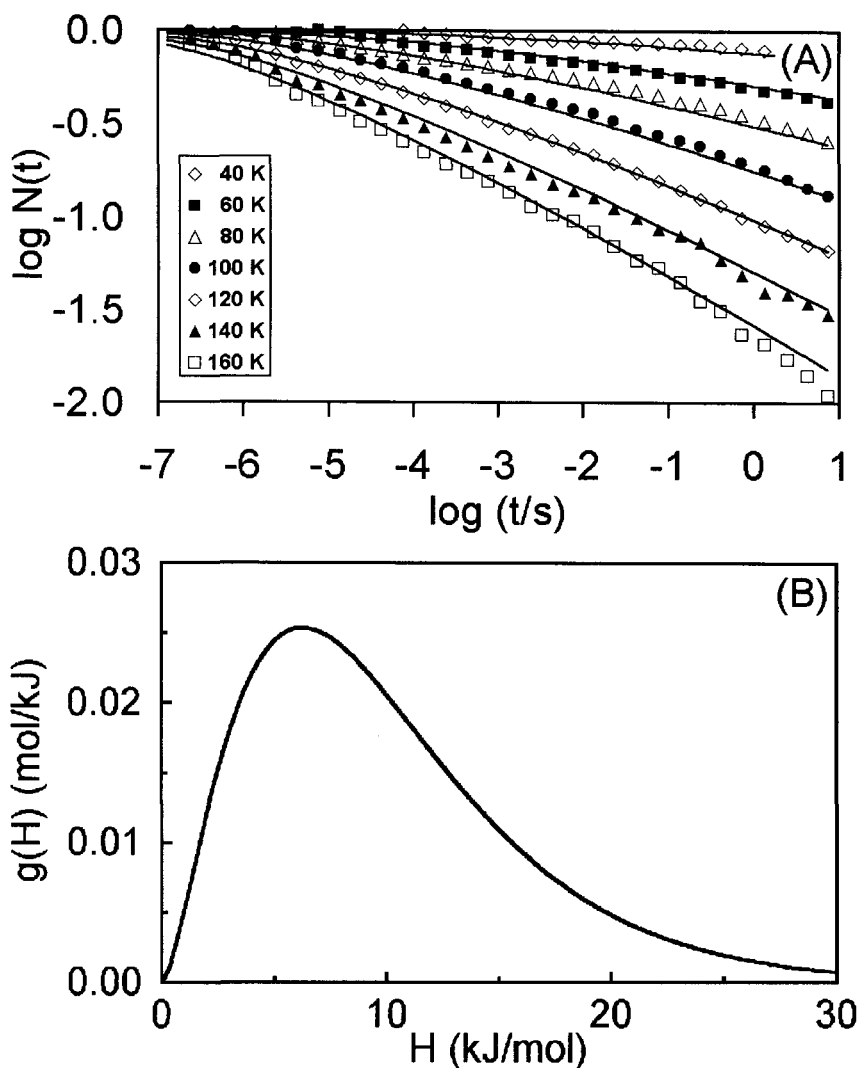


Figure 3. Flash photolysis kinetics of the A_{II} conformation of L29W MbCO below 180 K, measured in the infrared band of the bound CO at 1959 cm^{-1} . (A) Experimental data (symbols) for $40\text{ K} \leq T \leq 160\text{ K}$, including fits (solid lines) with a distribution of enthalpy barriers. (B) Normalized gamma distribution that fits the kinetic data in panel A.

Because of the slow recombination of L29W MbCO, this can be achieved easily at cryogenic temperatures. Note that only the A_I state can be studied directly as the A_{II} state is a minority species at low temperature that is not separately visible in the electron density maps.

3.1 Structural Changes in A_I upon Photodissociation at $T < 180$ K

To visualize the structural changes responsible for the drastic increase in the barrier to ligand rebinding in A_I at cryogenic temperatures, x-ray data sets were measured of both the ligand-bound and the photoproduct state of L29W MbCO crystals at 105 K. X-ray diffraction data sets were collected with a SIEMENS HISTAR multiwire proportional counter using $\text{CuK}\alpha$ radiation from an ENRAF NONIUS FR591 rotating anode generator. Laser illumination leads to significant heating of the crystals, and the photolysis was performed extremely carefully to avoid heating the crystal over 180 K during illumination, while achieving a maximum fraction of molecules trapped in the slow state B_3 ³³. More details on the experiment, data collection and refinement statistics are given in Ref. 20.

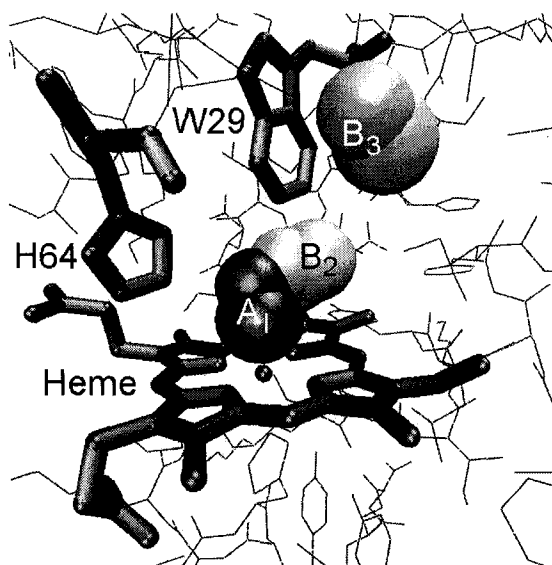


Figure 4. Model of the L29W mutant MbCO from our x-ray structure analysis at 105 K (pdb entry 1DO7), showing the heme group and the side chains of H64 and W29. Besides the bound ligand (labeled A_I), we also show the location of the ligand (B_3) from the photoproduct structure with photolysis at $T < 180$ K (pdb entry 1DO4). The protein atoms from this structure are not shown, as their locations are indistinguishable from those of the bound-state structure. The position of the CO in the 36 K photoproduct of native MbCO is also included¹¹.

Figure 4 shows a model of the active site of L29W MbCO that contains the crucial structural features. For the protein atoms, only the bound-state positions are shown, as the shifts upon photodissociation are very small. The largest change is that of the iron out of the mean heme plane by 0.2 Å, all other protein atoms shift only by a few hundredth of an angstrom. In Fig. 4, the small sphere in the middle of the large planar heme group represents the central iron atom. The side chains of H64 and W29 are emphasized, and three positions of the CO ligand are included. The CO in the bound state, denoted by A_1 , can be seen on top of the heme iron, binding at an angle of 14° to the heme normal. Upon ligand dissociation in the L29W MbCO crystal below 180 K, only very small shifts are observed for the protein atoms. The CO ligand, however, has migrated far away from the active site to a cavity in the back of the distal heme pocket, which is labeled with B_3 in Fig. 4. This location is known as the 'xenon 4' site, one of four internal cavities in myoglobin that bind xenon with high affinity¹⁸. We have also included another CO molecule, labeled B_2 , from the photoproduct structure of native MbCO at temperatures below 40 K by Hartmann *et al.*¹¹; it sits on top of the heme group above pyrrole ring C.

The kinetic analysis and the structural data yield the following interpretation of the ligand binding process after photodissociation of L29W MbCO below 180 K: after the flash, ligands settle in a location where they can rebound rather quickly. From the position of the CO stretch band near that of the B_2 substate in native MbCO and the fact that the position above pyrrole C is sterically accessible in the L29W mutant, we suggest that the fast rebinding (see Fig. 1) occurs from a CO location similar to that discovered previously in the native protein. Instead of rebinding, the CO can move over a second barrier to an alternate location, B_3 , where rebinding involves a much higher enthalpy barrier, and thus the kinetics are substantially slower. These processes occur while the polypeptide moiety is frozen in and thus unable to rearrange.

3.2 Structural Changes in A_1 upon Photodissociation at $T > 180$ K

We have determined a second photoproduct structure of an L29W MbCO crystal, but this time with photolysis above the dynamical transition at 180 K, so that we expect the protein structure to rearrange. After illumination with a laser power sufficient to ensure heating to $T > 180$ K, x-ray data were collected at 105 K. In the structure, large shifts of a few tenths of an angstrom are observed in the region of helices B, C, and E, and still substantial displacements in the region of helices F and G. The most obvious changes are seen in the structural model in Fig. 5, which also includes the bound-state structure for comparison. The H64 imidazole sidechain moves deeper into the heme pocket and occupies the space vacated by the ligand. This change is usually seen when comparing ligated and deoxy forms of various mutant myoglobins⁶. Moreover, the indole group of W29 rotates by about 45° into a position close to the one found in the room temperature structure of L29W met Mb

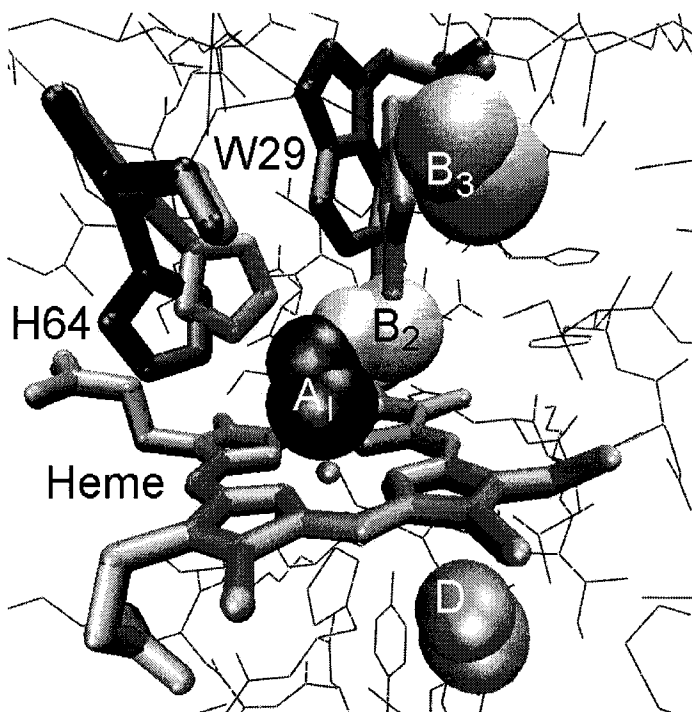


Figure 5. Model of the L29W mutant MbCO as in Fig. 4, but including the coordinates of the heme group, the side chains of H64 and W29, and the CO from our photoproduct structure with photolysis at $T > 180$ K (pdb entry 1DO3). Upon relaxation, the H64 imidazole side chain rotates further into the distal heme pocket so as to partially block the ligand binding site. The W29 indole side chain rotates and moves closer to the heme group, blocking the B_2 site and closing off access to the B_3 site.

(our unpublished data) and MbO_2 ³⁴. No electron density associated with the CO ligand is found in the distal heme cavity. Rather, 62% of the CO population is recovered on the proximal side at the 'xenon 1' site. This cavity, labeled with the letter 'D' in the figure, was the first xenon-binding site discovered¹⁸, it is the one with the highest affinity for xenon. We emphasize that no channels exist in the average structure connecting the distal and proximal cavities. Apparently, even in the crystal protein fluctuations are large enough to transiently open channels for the ligands to migrate. But where are those CO molecules that are not in the 'xenon 1' site? They may reside in the other cavities or have escaped from the protein into the solvent. In any case, their fractional occupation in any particular other site is too small to be detected in the electron density map.

3.3 Structural Properties of A_{II}

At the present time, an x-ray structure of the A_{II} substate of L29W MbCO has not yet been determined. As already mentioned, A_{II} is the dominant conformation at room temperature, and the fact that the aquomet and oxy ligated forms of L29W myoglobin both show the W29 side chain in an almost identical position to our photoproduct with illumination above 180 K strongly suggests that the A_{II} substate is similar to this photoproduct. Further support for this view comes from the lack of a slowly rebinding fraction in the A_{II} kinetics at low temperatures. With the W29 side chain rotated and pushed closer towards the heme group, the B_2 site is occluded and the pathway to the B_3 site is blocked.

4 Why Is Ligand Binding in L29W So Much Slower Than in Native MbCO?

We have combined time-resolved infrared spectroscopy and x-ray crystallography to connect structural, spectroscopic and kinetic properties of reaction intermediates in myoglobin. Experimental evidence has been given that the ligand binding process involves migration of the ligand through internal cavities in the myoglobin structure before recombination takes place at the heme iron. A key question that we have not yet addressed is how these results are related to the physiological ligand binding process, taking place above 300 K, and especially, why the bimolecular recombination is so much slower in L29W than in native MbCO. A detailed discussion of the ligand binding process is beyond the scope of this paper, but we will use the remainder of this section to give a sketch of the crucial ideas.

We start by showing in Fig. 6 an Arrhenius plot of the bimolecular reaction rate coefficient, λ_s' , in the temperature range of 190 K to 300 K. Below 200 K, the rate coefficient increases steeply. This behavior is well understood from studies on a variety of heme proteins. At low temperatures, the ligand binding rate is essentially limited by protein fluctuations that allow access to the interior¹. These are known to have a large apparent activation enthalpy ($H > 80$ kJ/mol). From 200 – 300 K, we observe simple Arrhenius behavior in L29W MbCO with an activation enthalpy of 36 kJ/mol, showing that a different barrier has taken control over the rate of ligand binding. It appears that no new physical processes occur in this temperature range; the rate coefficient just increases with temperature as more thermal energy becomes available to overcome the barrier. Incidentally, in the native protein, the activation enthalpy of λ_s' is reported to change from 79 to 12 kJ/mol; the change in slope of the Arrhenius plot is much more gradual, however, and occurs over the entire temperature interval.¹ This difference in the temperature dependence of λ_s' of the L29W mutant indicates that a crucial inner barrier exists that is much higher than in the native protein. It is somewhat amusing to find that the peak barrier heights for the final binding step in L29W MbCO, $B_2 \rightarrow A$ in the A_I conformation (6.9 kJ/mol)

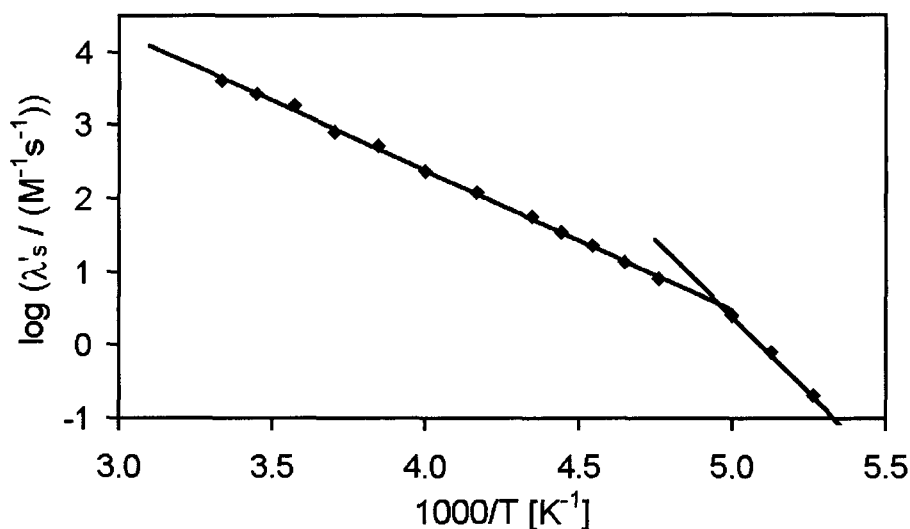


Figure 6. Arrhenius plot of the rate coefficient λ'_s for bimolecular CO binding from the solvent between 190 and 300 K. Two different temperature regions can be distinguished. Below 200 K, ligand binding is limited by the rate of protein fluctuations, giving rise to a steep temperature dependence. Above 200 K, a linear dependence indicates that ligand binding rates increase with temperature as more thermal energy becomes available to overcome an enthalpy barrier of ~ 36 kJ/mol.

and $B \rightarrow A$ in the A_{II} conformation (6.3 kJ/mol) are significantly lower than the 'canonical' 10 kJ/mol in sperm whale MbCO.^{1,2} The activation enthalpy of the barrier for ligand return from the B_3 to the B_2 site in the A_I conformation ($H_{B_3B_2} \approx 33$ kJ/mol), however, is very similar to that of λ'_s , which suggests that this barrier plays a key role as a bottleneck in the ligand binding process in L29W MbCO. Hence we conclude that ligands entering the protein become trapped in the B_3 site in the mutant. To bind, they overcome a large barrier to get to B_2 . There, they can either bind to the heme iron or escape again to B_3 (see Fig. 2 C). Whereas rebinding was more likely in our experiments below 180 K (Fig. 2), escape to B_3 becomes dominant at physiological temperature. This is a consequence of the Arrhenius parameters in Table 1. The high-temperature limit is given by the preexponentials A_i , and the rate coefficient k_{B_2A} for bond formation has a preexponential that is significantly smaller than the one for leaving the B_2 site, $k_{B_2B_3}$. By contrast, in native sperm whale myoglobin, the B_3 state traps ligands to a much lesser extent, and the crucial barrier that controls the binding reaction is associated with the bond formation step.

The scenario sketched here crucially involves trapping in the B₃ state to explain ligand binding in L29W MbCO. Clearly, the extremely slow recombination from B₃ at low temperature is the key difference in ligand binding behavior compared with native MbCO. Note that this photoproduct state is connected to the A_I bound state. We see from Fig. 1, however, that ligand-bound L29W favors the A_{II} state at 300 K. Moreover, the protein without a ligand in the distal pocket prefers to assume a structure that does not allow access to the B₂ site (essentially an A_{II}-like structure). We even have kinetic data suggesting that ligand binding from outside of the distal pocket is extremely slow or entirely impossible in A_{II}. So how can the slowing around 300 K be explained consistently involving B states belonging to the A_I conformation, whereas the protein prefers the A_{II} conformation at the higher temperature? To resolve this problem, we have to assume that the ligand induces a structural change of the distal heme pocket to an A_I-like structure upon binding in the B₃ pocket.

To summarize, by introducing the W29 side chain in the heme pocket instead of leucine, the reaction surface for ligand binding in myoglobin has been modified such that ligands migrating within the protein get trapped in the B₃ site. Compared with the native protein, the ligand has to overcome a much higher barrier to bind at the heme iron.

5 Acknowledgments

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References

1. Austin, R. H., Beeson, K. W., Eisenstein, L., Frauenfelder, H. & Gunsalus, I. C., *Biochemistry* **14**, 5355-5373 (1975).
2. Steinbach, P. J. et al., *Biochemistry* **30**, 3988-4001 (1991).
3. Henry, E. R. et al., *J. Mol. Biol.* **166**, 443 (1983).
4. Parak, F., Knapp, E. W., Kucheida, D., *J. Mol. Biol.* **161**, 177 (1982).
5. Doster, W., Cusack, S., & Petry, W., *Nature* **337**, 754 (1989).
6. Nienhaus, G. U., Heinzl, J., Huenges, E. & Parak, F., *Nature* **338**, 665 (1989).
7. Rasmussen, B. F., Stock, A. M., Ringe, D. & Petsko, G. A., *Nature* **357**, 4 (1992).
8. Ahmed, A. M. et al., *Chem. Phys.* **158**, 329 (1991).
9. Schlichting, I., Berendzen, J., Phillips, G. N., Jr. & Sweet, R. M., *Nature* **371**, 808-812 (1994).
10. Teng, T. Y., Srajer, V. & Moffat, K., *Nature Struct. Biol.* **1**, 701-705 (1994).

11. Hartmann, H. et al., *Proc. Natl. Acad. Sci. U. S. A.* **93**, 7013-7016 (1996).
12. Lim, M., Jackson, T. A. & Anfinrud, P. A., *Nature Struct. Biol.* **4**, 209 (1997).
13. Srajer, V. et al., *Science* **274**, 1726 (1996).
14. Ma, J. P., Huo, S. H., & Straub, J. E., *J. Am. Chem. Soc.* **119**, 2541 (1997).
15. Vitkup, D., Petsko, G. A. & Karplus, M., *Nature Struct. Biol.* **4**, 202 (1997).
16. Meller, J. & Elber, R., *Biophys. J.* **74**, 789 (1998).
17. Elber, R. & Karplus, M., *J. Am. Chem. Soc.* **112**, 9161-9175 (1990).
18. Schoenborn, B. P., Watson, H. C. & Kendrew, J. C., *Nature* **207**, 28-30 (1965).
19. Tilton, R. F., Kuntz, I. D. & Petsko, G. A., *Biochemistry* **23**, 2849-2857 (1984).
20. Ostermann, A., Waschipky, R., Parak, F. G. & Nienhaus, G. U., *Nature* **404**, 205-208 (2000).
21. Nienhaus, G. U., Mourant, J. R. & Frauenfelder, H., *Proc. Natl. Acad. Sci. U. S. A.* **89**, 2902-2906 (1992).
22. Springer, B. A. et al., *Chem. Rev.* **94**, 699 (1994).
23. Frauenfelder, H., Sligar, S. G. & Wolynes, P. G., *Science* **254**, 1598-1603 (1991).
24. Johnson, J. B. et al., *Biophys. J.* **71**, 1563-1573 (1996).
25. Müller, J. D., McMahon, B. H., Chien, E. Y. T., Sligar, S. G. & Nienhaus, G. U., *Biophys. J.* **77**, 1036-1051 (1999).
26. Kushkuley, B. & Stavrov, S.S., *Biophys. J.* **70**, 1214-1229 (1996).
27. Phillips, G. N., Jr., Teodoro, M. L., Li, T., Smith, B. & Olson, J. S., *J. Phys. Chem. B* **103**, 8817-8829 (1999).
28. Johnson, J. B., Lamb, D. C., Frauenfelder, H., Müller, J. D., McMahon, B., Nienhaus, G. U., & Young, R. D., *Biophys. J.* **71**, 1563-1573 (1996).
29. Frauenfelder, H., Parak, F. & Young, R. D., *Ann. Rev. Biophys. Biophys. Chem.* **17**, 451 (1988).
30. Nienhaus, G. U. & Young, R. D., in *Encyclopedia of Applied Physics* **15**, 163 (1996).
31. Alben, J. O. et al., *Phys. Rev. Lett.* **44**, 1157-1160 (1980).
32. Lamb, D. C., Kriegl, J., Kastens, K., & Nienhaus, G. U., *J. Phys. Org. Chem.* **13**, 659- 663 (2000).
33. Nienhaus, G. U. et al., *Biochemistry* **33**, 13413 (1994).
34. Hirota, S., Li, T., Phillips, G. N., Jr., Olson, J. S., Mukai, M. & Kitagawa, T., *J. Am. Chem. Soc.* **118**, 7845-7846 (1996).

THE POLYMER TRANSLOCATION AND BARRIER CROSSING

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Dynamics of polymers crossing barriers is not only a basic problem in soft matter physics but also important in biological applications such as polymer transport across membranes, DNA gel electrophoresis, etc. In this paper we review our theoretical investigations on (1) polymer translocation through a narrow pore in a membrane [1-4], and (2) dynamics of a polymer surmounting a potential barrier [5-6]. The process of polymer translocation into or across biomembrane is a ubiquitous phenomenon occurring in cells [7]. Proteins are transported across a cellular membrane and endoplasmic reticulum, while RNAs are transported across a nuclear membrane after their synthesis. There are similar macromolecular transport mechanisms in drug delivery, as well as in biotechnology of gene transfer where it is fundamental to understand how DNAs can be incorporated into cells. It is a highly complex process with specificity involving conformational changes of the translocating polymers that can occur in both *cis* and *trans* sides as well as inside of the membrane. Recently the problem has been actively investigated experimentally because it provides new prospects for direct and high-speed detection of sequences of single DNAs and RNAs [8]. In dealing with the dynamics of complex systems, the first thing to do is to determine the primary, relevant dynamical variable(s) which can effectively describe the dynamics in question. In terms of the variable, the dynamics is described by a stochastic equation of motion involving the free energy associated with the variable, which is, in the Markovian approximation, the overdamped Langevin equation. For the case of unfolded polymer translocation through a narrow pore, the variable is chosen as the number of translocated segments (n) across the membrane, and the free energy for a given n is calculated from polymer conformational entropy [1]. The translocation time, now viewed as the mean first passage time of the diffusion process crossing the free energy barrier, can be readily calculated. We considered [1] the effects of chain flexibility as well as the asymmetry caused by transmembrane chemical potential difference. With no asymmetries, the translocation times scales as N^3 irrespective of chain flexibility, where N is the total segment number. From the prefactors, however, an ideal chain translocation rate is found to be slower compared with that of a rigid chain of the same length by 23%. With a chemical potential difference per segment as small as kT/N , the translocation time shows a pronounced change; for a forward bias, the scaling behavior is changed to N^2 . Regardless of chain flexibility, this extreme sensitivity is a cooperative phenomena rising from chain connectivity; the segments respond all hand in hand rather than as individuals to a driving asymmetry. This cooperativity of the chain is also a manifestation of the first order phase transition in chain partitioning behavior between two compartments[9]. As another type of asymmetry mechanism we studied the Brownian Ratchet (BR), which was originally suggested by Simon, Peskin and Oster[10] as a nonspecific driving mechanism for biased diffusion arising from chemical binding of chaperons on the chains entering the *trans* side of the membrane. It is found that the strong ratchet (many chaperone binding) not only rectify the motion but also suppresses the chain flexibility to rigid rod like behavior. We also have investigated other types of driving asymmetries, such as curvature effect considering polymer release out of spherical vesicle [2], and polymer adsorption on the *trans* side [3]. It turns out that all of these driving forces induce the crossover on the translocation time behavior from N^3 to N^2 in the same manner as mentioned before. Since the biological membrane is subject to ceaseless non-equilibrium fluctuations from the background, the chemical potential of each segment can be modulated in

a random fashion. The effect of the fluctuations to the transport of single Brownian particle have been extensively studied [11]. An important finding is the phenomena of resonant activation, where the fastest translocation occurs when the time-scale of a non-equilibrium fluctuation matches a certain optimal time. It is found that, compared with a single particle transport, the polymer resonant translocation become much more rapid as the magnitude of the fluctuation in chemical potential difference increases, again due to chain connectivity [4]. In the second part of the paper we consider a polymer crossing the one-dimensional Kramers potential [5,6]. The potential varies over a large distance so that the polymer dimension is much smaller than the distance between the well bottoms and the barrier top. As an application, an efficient device can be suggested to separate polymers with different lengths by preparing potential barrier of a macroscopic distance. For this case, the relevant variable is the center of mass position of the chain, for which the Langevin equation is set up. The driving force in the equation is given by the gradient of the polymer free energy for a given center of mass position, which is obtained by using the Rouse model. As the barrier crossing of the center of mass is a quasi-stationary process, we adopt the Kramers rate theory of a single Brownian particle by considering both the friction and the potential to be N times that of the single particle and replacing the potential barrier in the Arrhenius factor by the free energy barrier. For small N or the well and barrier of small enough curvatures, the polymer keeps the globular shape throughout the process of barrier crossing, having essentially the same crossing dynamics of a globular Brownian particle of N segments. For the other cases, however, the crossing rate tends to be enhanced beyond the globular limit, due to chain flexibility that manifests confined and extended conformations at the well and barrier respectively. Above a critical chain length and below a critical chain spring constant the chain at the barrier top undergoes coil-stretch transition, resulting in reduction of the activation barrier and thus crossing time. The polymer in barrier crossing thus reveals its conformational flexibility and adjustment to external forces as characteristic features of soft matter dynamics.

References

1. W. Sung and P. J. Park, *Phys. Rev. Lett.* **77**, 783 (1996)
2. P. J. Park and W. Sung, *Phys. Rev. E* **57**, 730 (1998)
3. P. J. Park and W. Sung, *J. Chem. Phys.* **108**, 3013 (1998)
4. P. J. Park and W. Sung, *Int. J. Bifur. and Chaos* **8**, 927 (1998)
5. P. J. Park and W. Sung, *J. Chem. Phys.* **111**, 5259 (1999)
6. S. Lee and W. Sung, *Phys. Rev. E* **63**, 021115 (2001)
7. B. Alberts *et al.*, *Molecular Biology of the Cells*, 3rd ed., Garland, New York (1994).
8. J. J. Kasianowicz, E. Brandin, D. Branton, and D. W. Deamer, *Proc. Natl. Acad. Sci. USA* **93**, 13770 (1996).
9. E. A. DiMarzio and A. J. Mandell, *J. Chem. Phys.* **107**, 5510 (1997).
10. S. M. Simon, C. S. Peskin, G. F. Oster, *Proc. Natl. Acad. Sci. USA* **89**, 3770 (1992).
11. C. R. Doering and J. C. Gadoua, *Phys. Rev. Lett.* **69**, 2318 (1992).

LONG RANGE INTERACTION BETWEEN PROTEIN COMPLEXES IN DNA CONTROLS REPLICATION AND CELL CYCLE PROGRESSION: THE DOUBLE HELIX AND MICROTUBULES BEHAVE LIKE ELASTICALLY BRACED STRINGS

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A nonstationary interaction model, that controls the gross behaviour of DNA replication and cell cycle progression, is derived in terms of manybody physics in a chemically open T cell. The model predicts a long range force $F(\varphi) = -(\kappa/2) \varphi(1 - \varphi/N)(2 - \varphi/N)$ between the origin recognition complexes (ORCs) bound by DNA, φ being the number of ORCs, N the threshold for initiation, and κ the compressibility modulus in the lattice of ORCs which behaves like an elastically braced string. Initiation of DNA replication is induced by a switch of sign of F , from attraction (-) and assembly in the G_1 phase ($0 < \varphi < N$), to repulsion (+) and partial disassembly in the S phase ($N < \varphi < 2N$), with release of licensing factors from pre-replication complexes (pre-RCs) and prevention of re-replication. Termination of replication is due to a vanishing of F at $\varphi = 2N$, when all primed replicons have been duplicated once, and $F(0) = 0$ corresponds to a resting cell in absence of a driving force at $\varphi = 0$. The switch of sign of F at $\varphi = N$ also explains the dynamic instability in growing microtubules (MTs), as well as the switch in the interleukin-2 (IL2) interaction with its receptor in late G_1 , at the R -point, after which a T cell proceeds to replication without further exposure to IL2. Shape, slope and scale of the response curves derived agree well with data from dividing T cells and polymerizing MTs, the variable length of which is due to a nonlinear dependence on initial concentrations of guanosine-triphosphate (GTP) and tubulin dimers.

1 Initiation of DNA replication

To replicate or not to replicate that is the question, not only for a T lymphocyte extracted from Hamlet, but also for the phenomenon of life. Obviously, DNA replication cannot be initiated by the engagement of one sole receptor at the surface membrane or one origin in the DNA double helix. The activation of a resting cell requires the cooperation of many interacting molecular units at the surface membrane, in DNA, as well as in cytoskeleton. In other words, also living matter is condensed and, apart from biological and chemical aspects, must hence be described in terms of manybody physics relevant for the actual situation, a nonstationary chemically open cell system. Initiation of DNA replication at hundreds or thousands of sites along the DNA duplex requires an exact timing and spatial coordination over long distances. Such a long range coherence could not occur without nonlocal

correlations between the molecular complexes engaged. Considering the large number of reactants at work during DNA replication, another problem is to pinpoint the factor that actually plays the key role at the initiation.

Recent discoveries provide possible answers. In DNA, both from yeast and animal cells [1,2], certain openings termed origins (Fig. 1) with specific nucleotide sequences (replicators) work as receptor for an origin recognition complex (ORC), which is considered to be the best candidate for an initiator protein [1]. Only ORCs liganded with adenosine-tri-phosphate (ATP) can recognize and bind to DNA origins in the double helix [3]. The DNA binding by ORC does not require nucleotide hydrolysis and ATP remains stably associated with ORC [4], however, the actual role of ATP hydrolysis at the initiation is unknown.

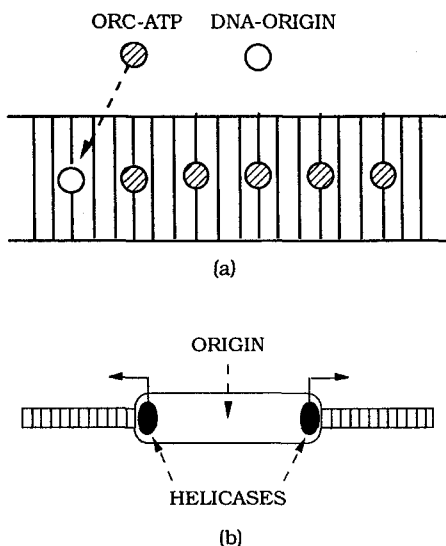


Figure 1. (a) Unfolded DNA helix schematically pictured as a lattice of origins liganded by ORC-ATP complexes. In reality the distance between the origins is much longer and the DNA structure more complex. (b) Bidirectional replication initiated at one DNA origin.

DNA replication starts by synthesis of adequate concentrations of the four nucleotide precursors and critical enzymes that carry out the replication. These precursors are regulated by a family of transcription factors (TF) [5], but TFs already bound by DNA are tonically suppressed in the interphase by the Rb proteins, termed after their discovery in Retinoblastoma cells. At a critical point in late G_1 the Rb proteins become rapidly phosphorylated and undergoes conformational changes

which induce their dissociation from the TFs, thus allowing a rapid transcription of all genes necessary for DNA synthesis and replication.

Phosphorylation of Rb proteins in turn is induced by specific kinases, cyclin-dependent kinases (CDK), that remain inactive until complexed with proteins termed cyclins. In T lymphocytes a sequential expression of these factors, cyclin D2 in mid G_1 , cyclin D3 in late G_1 , followed by cyclin E at the G_1 -S interface, is strictly dependent upon a continuous interleukin-2 (IL2) exposure until a restriction point in late G_1 , the *R*-point, beyond which the cell automatically proceeds to the initiation of replication [6]. Moreover, the degradation of another protein, p27, which inhibits the activity of the cyclin-CDK complex, is promoted by IL2 [7].

Other key regulators that must prime the DNA origins before start of replication are the licensing factors (LF), which may control the completion of replication in such a way that the genome is copied once and only once per cell cycle. For instance, in budding yeast cells the Cdc6 and Mcm2-7 proteins must first license the origins before the TFs, e.g., the Mcm1 protein, can carry out the transcription. Thus the pre-replicative protein complexes (pre-RC) include those that directly select origins in the DNA duplex, the ATP-ORC units, and others such as the LFs, that are subsequently recruited to DNA origins [1]. The ORCs remain bound by DNA also after initiation, hence, contained in the post-RCs, while the LFs like Cdc6 and Mcm2-7 proteins are released during the S phase [8].

All reactions and biochemical pathways activated in a T cell via the IL2 receptor (IL2R) are not entirely identified, but the important consequence is the transfer between the *R* point and the G_1 -S interface, from the surface membrane to the nucleus, of the molecular on-off switch mechanism, and that IL2 controls both the accumulation of D-type cyclins and the degradation of the cyclin-CDK inhibitor p27 [9]. The target for the cyclin-CDK phosphorylation is the pre-RC bound to each replicon in the DNA duplex [1]. A theory from first principles, based on all these factors, seems unobtainable but not impossible.

Recent advances in the studies suggest that DNA replication is regulated by a binary set of states: the assembly state which allows the formation and binding of pre-RCs at the origins in DNA but does not allow initiation of replication; and the replication state which allows the initiation of DNA replication with disassembly of pre-RCs but inhibits formation and binding of new such complexes (Tab. 1) [1]. However, without a definite threshold at the G_1 -S interface this "consensus" picture is not clear. The binding of arbitrary one ORC by a DNA origin could then imply two different reactions - continued assembly of pre-RCs, or partial disassembly of pre-RCs with release of LFs, and concomitant start of replication. This apparent degeneracy prevails irrespective of hydrolysis, phosphorylation, or any other factor.

To solve the problem, and ensure a complete duplication of DNA, the following alternative interpretation is suggested. An ORC with the help of LFs and other factors should be unable to initiate replication unless a definite number of origins/replicons in the DNA duplex have first been engaged by ORCs.

Table 1. Control of replication and cell cycle progression by long range interaction between protein complexes in DNA; replication is initiated when all DNA origins ($\varphi = N$) are engaged by ATP-ORC. Doubling of mass in G_2 is not critical in the sense that it normally follows if replication is initiated.

	G_1	S	G_2	M
Turnover of material	assembly R-point	disassembly	doubling of mass	chromatid separation
Characteristics	ligand-receptor.....association..... ...pre-RC binding to DNA...	DNA replication, disassembly of pre-RCs, release of LFs, prevention of re-replication		cyto- kinesis
Force between complexes, sign of force	attraction (-)	repulsion (+)		
Order-parameter intervals	$0 < \varphi < N$	$N < \varphi < 2N$		
Thresholds	$\varphi = 0$	$\varphi = N$	$\varphi = 2N$	

The cell must hence be able to count at a distance the number of ORCs bound by the DNA duplex, implying the existence of long range (nonlocal) correlations. As shown subsequently, a nonlinear manybody physics model can explain such a counting mechanism, as well as the transition into the replication state at a definite (quantal) threshold number of ORCs. The switch of interaction from assembly to partial disassembly [1,8], which can explain the initiation of DNA replication and the prevention of re-replication during the S phase, is generated by the contribution of energy from one ATP-ORC above the lattice ground state. This energy, provided by hydrolysis of ATP, makes the DNA lattice of ORCs unstable.

Obviously, the DNA duplex, the nuclear envelope, and the surface membrane alone are too liquid like and soft to explain the presence of nonlocal correlations between DNA origins or between IL2Rs engaged. However, a stabilized cytoskeletal network of microtubules (MTs) [10], the most rigid elements of the cell, can induce such correlations, within and between the membrane and nucleus subsystems. Also the histones may contribute to the DNA stabilization. The DNA replication, for which this stabilization is a requirement, is regulated by concentrations of various factors such as cyclins, kinases, ATP, ORC and origins, the interaction of which is indirectly controlled by the two surface membrane reactants, IL2 and IL2R. Such a dynamical system, regulated by concentrations, is termed lyotropic [11]. The problem is that most condensed matter theories are based on thermotropic systems, thus regulated by temperature variations and, like the Hill and Langmuir equations, based on assumption of a chemical stationary state [12,13].

However, DNA replication is initiated by the binding of ORC to origins, an unidirectional hence nonstationary process like many other functions of a living cell. On a T cell, for instance, the high affinity receptors engaged by IL2 internalize already 15 min after binding to IL2, while dissociation of IL2 from such receptors

does not start until 30 min later [14]. Therefore, biological response may occur long before a stationary state has been attained, which is mainly also the reason why the threshold and scale derived in stationary models may deviate by up to three orders from that assessed [15] (Fig. 2). In order to rescue such models the scale derived is then usually replaced by the ligand concentration for half-maximal response. However, this converts the derived response into a mere logistic equation. Obviously, that curve fits better to experimental data, but, a physical interpretation of the results in terms of the molecular interaction is then lost [16]. The discrepancy between theory and experimental data has also been ascribed to factors like transport proteins or a variable affinity [17]. Although such factors may well contribute too, they cannot compensate for the neglected nonstationary effects in the first place.

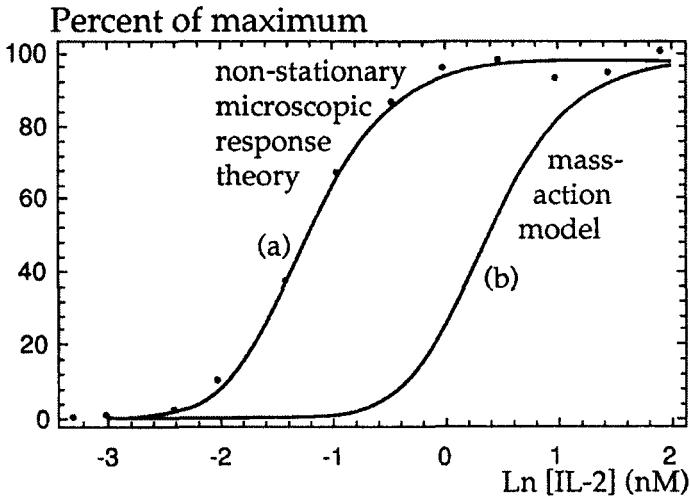


Figure 2. (a) Dose-response curve predicted by the lyotropic model (solid line) compared with proliferation data of a cell line MLA-144 (dots) from Gibbon ape with spontaneous lymphoma. Data taken from ref. [25] with permission. (b) Dose-response curve derived from mass-action model with a dissociation constant $K = 1.0 \pm 0.5$ nM.

Studies on T cell proliferation show that a definite number of IL2-receptor associations are required to initiate DNA replication [18]. This all-or-none (quantal) type response has also been observed in the T cell receptor (TCR) system [19]. Accordingly, T cells are somehow capable to count at a distance the number of interacting receptors [20], irrespective of their location on the surface membrane. Like for the origins in DNA, the indication of a definite threshold number of

receptors engagements implies the existence of nonlocal correlations between individual receptors, hence a long range interaction that controls the exact timing of DNA replication. On the contrary, a correlation between the G_1 -S transition and a critical mass threshold [21] has not been indicated.

However, as stated here before, the surface membrane and the DNA duplex alone are not rigid enough to explain the long range forces responsible for such correlations, neither within nor between the actual subsystems. A molecular complex at one end of the membrane, or at one end of the DNA duplex, would not know much about another complex at the other end of these soft condensed structures, in which only adjacent complexes are correlated. A diffusive propagation of signals would be too slow to explain the exact coordination in time and space and the simultaneous counting of interacting complexes at the *R*-point and the initiation of DNA replication. However, signal transduction, energy storage, cell motility and division, depend on the dynamics of cytoskeletal MTs, the most rigid elements in the cell.

At critical points in the cell cycle, when the cytoskeleton is stabilized, a realistic cell model could hence be expected to be more solid like. Unfortunately, a couple of hitherto unsolved problems have prevented a physical description of the dynamics and growth of MTs, the dynamic instability and the variable length of growing MTs [10,22,23]. However, the chemical driving reaction is well known: Only tubulin dimers liganded with two guanosine-tri-phosphate (GTP) molecules polymerize into MTs [10]. It is also known that a fraction of growing MTs actually attains a stabilized state at critical points in the cell cycle. The MTs and the histones are probably the only cell elements that could explain the nonlocal correlations between the membrane receptors, between DNA origins, and between the switch of interaction at the *R*-point and the initiation of replication at the G_1 -S interface.

Before investigating this possibility, a few words are in place on the energy induced by the ligand-receptor interaction. Each receptor engagement contributes an equal quantum of excitation energy [24] and, hence, there is no reason to believe that this is not true also for the pre-RC bound by DNA origins. But the total energy stored is not a linear sum of the individual excitations because the rate of cell division, as well as the dynamics that controls DNA replication and the cell cycle progression, are nonlinear functions of the ligand-receptor interaction. The rate of division of T cells is a nonlinear function determined exclusively by the concentrations of IL2 and IL2R [14].

This may appear unexpected because all time variation of the cell cycle is normally contained in the G_1 phase, and the rate of division of cells therefore approximates to the rate of cells taking a decision to initiate DNA replication, a function depending on concentrations such as of ORC, DNA origins, ATP, p27, Rb proteins, cyclins and CDK [1], albeit indirectly controlled by the two surface membrane reactants, IL2 and IL2R [6,7]. However, from a physical point of view the exclusive dependence on IL2 and IL2R [14] is not a mystery. It is a form of

enslavement, in which all other variables are damped at the actual time scale (hours), and which is also a prerequisite for modelling of the long range forces.

As demonstrated subsequently, the nonlocal correlations, corresponding to the long range interaction, have the same form in the three actual subsystems irrespective of which chemical reaction that drives the local interaction. If this was not true, dose-response data would depend on more variables than IL2 and IL2R. A slope equal to one, such as in the division rate of the T cell line MLA-144 [25], corresponds to the Langmuir response or a Hill equation with exponent $n_H = 1$ (Fig. 2). The so called Hill exponent n_H denotes the number of binding sites per receptor protein and, consequently, $n_H > 1$ is only due to short range cooperativity between adjacent receptor molecules. Consequently, a slope equal to one excludes a net contribution from short range cooperativity, however, as shown subsequently, permits nonlocal correlations.

Summarizing so far, a realistic, self-consistent but nontrivial gross interaction model of a dividing cell should thus:

- i) yield a dose-response which is a nonlinear function of the two reactant concentrations only, in compliance with the nonstationary boundary constraints,
- ii) yield the correct shape,
- iii) slope, and
- iv) scale of the growth rate,
- v) explain the initiation of DNA replication at a definite threshold number of receptor engagements and a corresponding definite number of origins engaged,
- vi) express the definite number in terms of the initial reactant concentrations,
- vii) define the constant quantum of energy induced per molecular complex engaged in each actual subsystem,
- viii) explain the prevention of re-replication during the *S* phase,
- ix) explain the dynamic instability and variable length of growing MTs,
- x) explain physically the roles of phosphorylation, and of hydrolysis at the initiation of DNA replication and in growing MTs.

A preliminary version of the model [26] could explain all these results except vii) and x). The model presented here, which has been published elsewhere [27] in a different form, also yields a constant quantum of energy contributed by each molecular complex. Moreover, the model explains the gross behaviour of DNA replication and the cell cycle progression in terms of the interacting protein complexes in DNA. It also explains the role of hydrolysis at the initiation of DNA replication and at the dynamic instability in growing MTs.

2 A nonstationary driving reaction

Let the variables ρ , r and ψ denote the concentrations of IL2, IL2R and their bound complex, or the concentrations of ATP-ORC units (ligand), DNA origins (receptor) and ATP-ORC-origin complexes (Fig. 1a). The rate of ligand-receptor association at one specific site, or recognition-binding of ATP-ORC at an origin, is given by

$$\begin{aligned} \frac{\partial \psi}{\partial t} &= k\rho r - k'\psi = k((a_0 - \psi)^2 - (a_0^2 - b_0^2)) \\ &= k(1 - \psi/a_0)^2 (a_0^2 - \frac{(a_0^2 - b_0^2)}{(1 - \psi/a_0)^2}) \end{aligned} \quad (2.1)$$

where the parameters $a_0 = 1/2(\rho_0 + r_0 + K)$ and $b_0 = \sqrt{\rho_0 r_0}$ are obtained by insertion of the initial constraints, $r + \psi = r_0$ and $\rho + \psi = \rho_0$, and zero indicates the start concentrations. $K = k'/k$ is the affinity constant, k' and k the dissociation and association constants in the actual subsystem. Integration of (2.1) then yields

$$\begin{aligned} \ln\left(\frac{\rho_K - \psi}{r_K - \psi} \frac{r_K}{\rho_K}\right) &= \ln\left(\frac{\rho'}{r'} \frac{r_K}{\rho_K}\right) \\ &= k(\rho_K - r_K)t = 2k\sqrt{a_0^2 - b_0^2} t \end{aligned} \quad (2.2)$$

where $\rho_K = a_0 + \sqrt{a_0^2 - b_0^2}$, $r_K = a_0 - \sqrt{a_0^2 - b_0^2}$, ρ' and r' play the role of dynamic concentrations, with $\rho' + \psi = \rho_K$ and $r' + \psi = r_K$. In a high affinity system, where $k' \approx K \approx 0$, the dynamic concentrations equal the initial ones, $r_K \approx r_0$, $\rho_K \approx \rho_0$, $r' \approx r$, and $\rho' \approx \rho$. Therefore no distinction will be made between dynamic and initial concentrations unless required.

The parameter t in (2.2) may be interpreted as a "physiological" time, which relates the usual time to the two continuous reactant concentrations. At the all-or-none transition to S phase, however, the dynamics suddenly becomes sharply sensitive to the number of molecular units engaged [9,18,19,20]. This switchlike behaviour, has also been observed in progesterone induced oocyte maturation [28]. Individual oocytes then showed a mitogen-activated protein kinase (MAPK) cascade equivalent to a cooperative enzyme with Hill coefficient of at least 35, but as has already been remarked [29], there is no cooperativity between individual enzymes in the MAPK cascade. It should be emphasized, however, that lack of local cooperativity between molecular units does not exclude nonlocal correlations. As demonstrated here, both the long range interaction responsible and the threshold for response are determined by such nonlocal correlations.

3 Nonlocal correlations and long range interaction

The perplexing discovery that a T cell counts the number of interacting receptors/origins before the decision to start DNA replication indicates the presence of nonlocal correlation between the receptors at the surface membrane and between the origins in DNA. The nonlocal correlation between n identical molecular complexes at x_i , $1 < i < n$, which equals the probability to find the complexes at the same sites, could be assumed to be $\varphi \approx \mu \psi(x_1) \psi(x_2) \psi(x_3) \dots \psi(x_n) / a_0^n$, where μ and a_0 are normalization constants. The reason for choosing the same a_0 as in (2.1) will soon become clear. The problem with this expression is that in nonstationary, lyotropic systems the notion of concentration could become undefined at too many or too few reactant molecules.

A well-defined concentration requires a large enough number of molecules in each macroscopically small but microscopically large volume element. At very high concentrations of vacant receptors on the surface membrane, however, the volume element around each one of them may contain less than one ligand. *Vice versa*, at the transition to S phase, where vacant receptors could become almost depleted, each volume element around a ligand may contain no receptor. Hence, the notion of concentration is not welldefined enough for a description of the actual lyotropic interaction in which $\psi(x)$ merely denotes the probability that (2.1) generates an engaged receptor at x . The probability to find a definite number of ligand-receptor complexes must therefore be extended to an average of all orders

$$\varphi = \mu \sum_{n=0}^{\infty} \left(\frac{1}{a_0} \right)^n \prod_{v=0}^n \psi(x_v) \quad (3.1)$$

In this expression the surface membrane could be regarded as an one-dimensional aperiodic lattice. The term $n = 0$ in (3.1) accounts for the possibility of having a contribution already from the start. A summation to infinite order is required because a truncation of (3.1) at any finite order could yield a condensate containing less than the interacting IL2Rs actually observed, since each factor ψ may or may not represent one engaged receptor.

The same expression (3.1) may also denote the probability to find a definite number of ORC complexes bound by origins in DNA. The first order growth of a solid type lattice (3.1) of complexes in DNA, or at the surface membrane, $\partial \psi / \partial t = k^* \psi$, k^* being the on-rate constant, should be restricted by the rate of formation (2.1) of complexes in the actual subsystem, hence, with $k' \approx 0$ one has $\partial \psi / \partial t = k^* \psi = k \rho r$. The lattice growth rate could then be obtained by combining (3.1) with (2.1). As shown subsequently, this yields a unique form of interaction that controls the cell at the initiation of DNA replication. The same form of interaction would be obtained also if (2.1) describes the rate of binding of LFs by ATP-ORC units previously bound by DNA, and (3.1) represents the probability to find a corresponding number

of pre-RCs. On a gross interaction level the ATP-ORCs bound by DNA and the pre-RCs could therefore be treated on equal footing in the assembly state.

At increasing density of receptors engaged, the area per receptor becomes successively smaller and the liquid-diffusive type motion more restricted. At a certain threshold the diffusion then changes into a harmonic-displacive type interaction, since the link between receptors through the rigid cytoskeletal network converts the liquid crystal type membrane to a more solid close-packed lattice. Thus, the critical threshold is also determined by the rigidity and elasticity of cytoskeleton and the medium. A transition into a harmonic type dynamics is expected also in the DNA duplex because a diffusive propagation in a liquid type DNA would be too slow to explain the exact timing and spatial coherence observed at the initiation of replication. The average distance between DNA origins engaged is then reduced to a critical limit, corresponding to a definite number of origins engaged.

Given such a liquid-to-solid type transition in the membrane and DNA lattices, implying nonlocal correlations between the protein complexes, as demonstrated subsequently the switch of ligand-receptor interaction at the *R*-point and the transition from assembly to replication at the *G*₁-*S* interface follows as a result of the nonstationary driving reaction (2.1). Replication does not start at all origins at once, however, the cell first makes sure that a definite number of pre-RCs, proportional to the definite number of IL2 receptors engaged by IL2, are engaged by DNA before it decides to start replication.

In the long wavelength, slow interaction limit, that characterizes the bound state, the average lattice spacing *d* between the surface receptors engaged or, between the DNA origins engaged, becomes negligible compared to the wavelength. The probability (3.1) to find a lattice (condensate) with a definite number of bound complexes in any of these subsystems then approximates to a geometric series

$$\varphi = \mu \sum_{n=0}^{\infty} \left(\frac{\psi}{a_0} \right)^n = \frac{\mu}{1 - \psi/a_0} \quad (3.2)$$

In a field theoretic perturbation expansion (3.2) corresponds to a Bethe-Salpeter like summation of ladder diagrams (Fig. 1a). Like in (2.1), *a*₀ is a function of the initial concentrations of the reactants which, as shown subsequently, determine the number of molecular complexes contained in the actual subsystem (3.2).

Combining (3.2) and (2.1), the rate of binding of complexes into a whole lattice (3.2) in any of the subsystems, then becomes

$$\frac{\partial \varphi}{\partial t} = \frac{ka_0}{\mu} \left(\mu^2 - \frac{(a_0^2 - b_0^2)}{a_0^2} \varphi^2 \right) \quad (3.3)$$

which apart from a constant has the solution

$$\varphi = \frac{\mu a_0}{\sqrt{a_0^2 - b_0^2}} \tanh(k \sqrt{a_0^2 - b_0^2} t) \quad (3.4)$$

Accordingly, the total number of membrane receptors, or DNA origins engaged, is

$$N = \varphi(\infty) - \varphi(0) = \frac{\mu a_0}{\sqrt{a_0^2 - b_0^2}} \approx \mu \frac{\rho_0 + r_0}{\rho_0 - r_0} \quad (3.5)$$

Depending on μ and the initial reactant concentrations, N can thus assume different values in the various subsystems. If N does not attain the threshold number observed, replication does not start.

4 A chemical potential energy

There are no experimental signs of replication until a definite number of receptors have been engaged [14]. Consequently, during the lag time, from start of exposure of the cell to ligands until start of replication, the chemically induced energy must be stored as a potential energy, $V(\varphi)$. In DNA this storage could take place in the form of pre-RCs bound by DNA, with an average lattice spacing d between the complexes. If the number of membrane receptors engaged, corresponding to a definite number of pre-RCs bound by the DNA duplex, attains the definite threshold number observed [18], replication is initiated. The rate of cells initiating replication thus equals the rate of cells in which the DNA lattice of bound pre-RCs attains a definite threshold length (L_0) corresponding to a definite threshold number of pre-RC. Hence, (3.3) could be interpreted as a travelling wave equation generated by a chemical potential $V(\varphi)$:

$$\frac{1}{v_N} \frac{\partial \varphi}{\partial t} = \frac{\partial \varphi}{\partial x} = \pm \sqrt{2V(\varphi)} = \frac{ka_0}{\mu v_N} (\mu^2 - g^2 \varphi^2) \quad (4.1)$$

with v_N being the traveling wave velocity in units of lattice spacings per unit time. Assuming that the binding of pre-RCs induce a corresponding elongation or compression of the DNA lattice, (4.1) could also be interpreted as a result of displacive motion due to a long range interaction between the protein complexes. A subsequent comparison with experiments shows that this assumption is realistic.

The nonlinear coupling g is defined by

$$g = \frac{\sqrt{a_0^2 - b_0^2}}{a_0} = \frac{\rho_0 - r_0}{\rho_0 + r_0} \quad (4.2)$$

and the threshold number becomes

$$N = \mu \frac{\rho_0 + r_0}{\rho_0 - r_0} = \frac{\mu}{g} \quad (4.3)$$

With a symmetric chemical potential (Fig. 3) defined by (4.1):

$$V(\varphi^2) = \frac{g^2}{2} \left(\frac{\mu^2}{g^2} - \varphi^2 \right)^2 \quad (4.4)$$

the traveling wave velocity becomes

$$v_N = k \frac{\sqrt{a_0^2 - b_0^2}}{\mu} = \frac{1}{N} v_1 \quad (4.5)$$

where v_1 is the wave velocity with only one molecular complex in the lattice:

$$v_1 = k a_0 = k \frac{(\rho_0 + r_0)}{2} \quad (4.6)$$

This relation trivially states that the time required for the lattice to grow one unit approximates to the average time it takes to form a molecular complex.

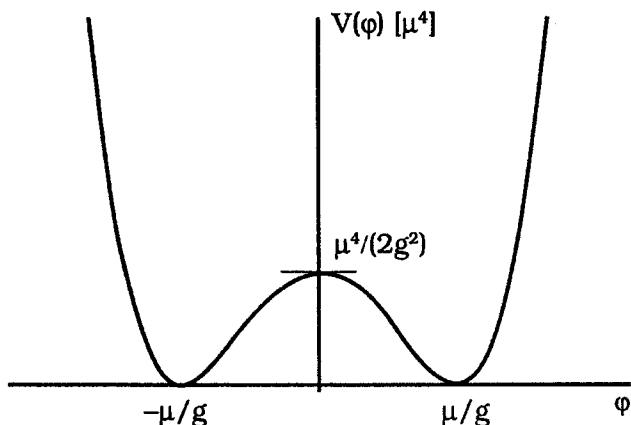


Figure 3. The symmetric, lyotropic double-well potential.

However, awaiting the ultimate binding of an ORC/pre-RC before initiation of DNA replication the lattice is confined in a bound state harmonic-displacive type dynamics. The second order equation of motion for small harmonic undulations and displacive motions corresponding to (4.1), in the close-packed lattice of pre-RCs at the threshold short before initiation, should then take the form [30,31]

$$\begin{aligned} \frac{1}{s^2} \frac{\partial^2 \phi}{\partial t^2} - \frac{\partial^2 \phi}{\partial x^2} &= -\frac{\partial V}{\partial \phi} \\ &= F_0(\phi) = 2g^2 \phi(N - \phi)(N + \phi) \end{aligned} \quad (4.7)$$

s being the velocity of sound, and $F_0(\phi)$ the nonlinear driving force. If both sides of the static part of (4.7) are multiplied by $\partial \phi / \partial x$ equation (4.1) is reobtained after integration. A diffusive type equation instead of (4.7), with a first order derivative in time corresponding to a more liquid like dynamics in the two subsystems, could not

explain the exact timing and spatial coherence observed at the initiation of DNA replication, which indicate the presence of nonlocal correlations.

The same dynamics (4.7) is obtained for the ligand-receptor interaction at the surface membrane, which has also been approximated by an one-dimensional lattice in the model. All pathways responsible for the transfer of one and the same gross interaction, from the surface membrane to the nucleus, are not entirely known, but, the important point is that it is transferred without change of form, a direct result of that the nonlocal correlations (3.2) and the driving reaction (2.1) have the same forms in the surface membrane as in DNA. Consequently, also the response, which equals the rate of cell cycle progression at the *R*-point and the *G*₁-*S* interface, is the same in the two subsystems. This also follows from the fact that the lag time between the *R*-point and the *G*₁-*S* interface is approximately constant.

This form of enslavement of the DNA dynamics under the interaction of the two surface membrane reactants, also indicated by experiments [14], crucially depends on the stabilization of cytoskeleton, without which neither the nonlocal correlations (3.2) nor the long range interaction would exist. Hence, also the cytoskeleton dynamics must obey the same gross interaction because, only one long range interaction can control the cell at a time. In other words, cytoskeleton should work together with the surface membrane dynamics before and at the *R*-point and, after a certain lag time, with the DNA dynamics at the initiation of DNA replication.

In cytoskeleton the geometrical series (3.2) yields the probability to find an MT fragment (condensate) with a definite number of GTP-tubulin dimers. The first order rate of polymerization of MTs, $\partial\psi/\partial t = k^*\psi$, is restricted by a second order rate, $\partial\psi/\partial t = k^*\psi = k\rho r$ (with $k' \approx 0$), and in this subsystem (2.1) represents the association of GTP with tubulin dimers. Hence, also the driving of cytoskeleton dynamics, and the attachment or detachment of energetic GTP-tubulin dimers to all orders of (3.2), is provided by reaction (2.1). When combining (2.1) with (3.2) the same equation of motion (4.7), driven by the nonlinear potential (4.4), is obtained also for the cytoskeletal network of MTs. Accordingly, in cytoskeleton the threshold *N* represents the content of bound GTP-tubulin dimers, i. e., the turbidity [10], which is determined by the initial concentrations of GTP and tubulin dimers.

5 Comparison with data

By inspection of (4.4) one finds that the potential energy stored at $\varphi = 0$, corresponding to zero pre-RCs, is nonzero, and the long range intermolecular force F_0 in (4.7) turns out to be repulsive (+) for $0 < \varphi < N$ when the system is expected to be in the assembly state. However, this looks worse than it actually is. The equilibrium of *V* in (4.4) at $\varphi = 0$ (Fig. 3) is unstable, and as is usually the case in Nature, the system spontaneously goes to one of its lowest energy states of *V*(φ):

$$\varphi \rightarrow -\frac{\mu}{g} + \varphi = -\mu \frac{\rho_0 + r_0}{\rho_0 - r_0} + \varphi \quad (5.1)$$

The reflection symmetry of $V(\varphi)$ (Fig. 3) is then spontaneously broken and the total probability (3.4) becomes positive definite:

$$\varphi = \mu \frac{\rho_0 + r_0}{\rho_0 - r_0} \{1 + \tanh(k\sqrt{a_0^2 - b_0^2}t)\} \quad (5.2)$$

Expressing the time parameter in (5.2) in terms of the actual reactant concentrations, through (2.2), the response of the lyotropic system is given by

$$\varphi = \mu \frac{\rho_0 + r_0}{\rho_0 - r_0} \{1 + \tanh(1/2 \ln(\frac{\rho}{\rho_{50}}))\} \quad (5.2)$$

Instead of the constant scale $\rho_{50} = K^{1/n}$, usually obtained in stationary type theories [12,13], the scale is here given by $\rho_{50} = r\rho_0/(Er_0)$, a function of the initial concentrations of the two actual reactants and that of vacant receptors. The integration constant E could be related to efficacy. At suitable initial reactant concentrations the quantal number N can attain the definite number of ligand-receptor engagements observed, and (5.3) then yields the rate of transition into the S phase, which approximates to the rate of cell division

$$R(\rho) = CN \{1 + \tanh(1/2 \ln(\frac{\rho}{\rho_{50}}))\} = 2CN \frac{\rho / \rho_{50}}{1 + \rho / \rho_{50}} \quad (5.4)$$

The constant C is here proportional to the number of cells in the start population. Shape and slope ($n_H=1$) of (5.4) agree almost exactly with data from a cell line MLA-144 [25], and contrary to stationary models [12,13] an exact agreement, between the derived scale and that assessed, could be obtained for realistic values of the initial and vacant receptor concentrations (Fig. 2).

The same enslaving interaction that yields the rate of transition into the S phase (5.4), induced by the membrane reactants at the cell surface, also controls the cytoskeleton dynamics with assembling-disassembling of MTs. Although laboratory conditions are markedly different from those of a living cell, and recalling that cytoskeleton is not one-dimensional and MTs are not equally long in all spatial directions, on the gross interaction level (5.4) should still be proportional to the average length of an MT, i. e., to the content of tubulin dimers [10]. Let C denote the number of MTs and N that of GTP-tubulin dimers per MT, (5.4) then yields the turbidity as a function of the concentrations of GTP and tubulin dimers, and (5.2) should be proportional to the variable length of a growing MT as a function of time. However, at the actual GTP concentration of 500 μM GTP, a large majority of reacting GTP molecules do not have the same chance to bind to a tubulin dimer as those in the vicinity of the less abundant dimers. Consequently, the effective concentration of GTP at work in the second order rate equation (2.1), that controls the growth of MTs, is much lower, and the on and off rate constants [22,23] are

correspondingly changed by a factor f , which is also due to that the growth of MTs is markedly nonstationary.

With an effective GTP concentration of $\rho_0 = 32 \mu\text{M}$, an association constant $k = 3.80f (\mu\text{Ms})^{-1}$, a dissociation constant here modified by a contribution from hydrolysis (h), $k' \rightarrow k' + k_h = 1.14f \text{ s}^{-1}$, where $f = 0.028/60$, hence, with $K = k'/k = 0.3 \mu\text{M}$ being left unchanged of f , a satisfactory agreement of the actual one-dimensional model (5.2) with data from growing MTs (Tab. 2) is obtained at the three highest

Table 2. Turbidity time data at different tubulin dimer concentrations; (a) 19.0 μM , (b) 17.0 μM and (c) 13.8 μM . Data taken from ref. [10] with permission.

Time (min)	Turbidity (a)	Time (min)	Turbidity (b)	Time (min)	Turbidity (c)
0.90	2.10	1.15	1.70	0.70	0.10
1.05	2.50	1.30	2.40	1.40	0.20
1.15	3.40	1.50	3.40	2.00	0.90
1.55	5.50	1.80	4.80	2.50	2.00
2.00	9.00	2.00	6.50	3.00	3.90
2.60	12.40	2.50	9.60	3.50	5.80
3.00	13.75	3.00	11.50	4.10	7.70
4.00	14.65	4.00	13.00	4.70	8.90
4.90	15.10	5.00	13.50	5.20	9.50
5.80	15.40	6.00	13.85	6.20	10.40
6.60	15.50	7.00	14.00	7.00	10.60
7.50	15.65	8.00	14.00	7.40	10.70

tubulin concentrations, $r_0 = 19.0$, 17.0, and 13.8 μM (Fig. 4) [10], if the essential start of polymerization occurs at $t_0 = 1.85$, $t_0 = 2.1$ and $t_0 = 3.45$ min, respectively. At lower tubulin concentrations, the system responds much slower and laboratory conditions could therefore be expected to distort the results. The good agreement of the derived response (5.4) with the three concentration dependent MT amplitudes assessed (Fig. 4), which explains the variable length of growing MTs, and with the growth data of dividing cells (Fig. 2), lend strong support to the proposed long range gross interaction (4.7). It should be observed that not only the shape, slope and scale of the response must fit data (Fig. 4); the amplitude, which is as a function of the initial reactant concentrations, must now also explain the variable length of growing MTs (Tab. 2).

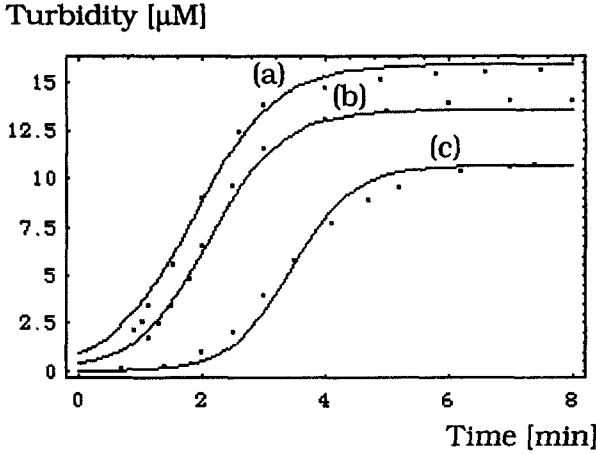


Figure 4. The variable length of growing microtubules, as a function of time, at an effective initial GTP concentration of 32 μM , and initial tubulin dimer concentrations at; (a) 19.0 μM , (b) 17.0 μM and (c) 13.8 μM . Data taken from ref. [10] with permission.

6 Long range interaction control of DNA replication

The displacement (5.1) yields an assymetric potential (Fig. 5):

$$V_A(\varphi^2) = \frac{g^2}{2} \varphi^2 \left(\varphi - \frac{2\mu}{g} \right)^2 \quad (6.1)$$

and the equation of motion becomes

$$\begin{aligned} \frac{1}{s^2} \frac{\partial^2 \varphi}{\partial t^2} - \frac{\partial^2 \varphi}{\partial x^2} &= -\frac{\partial V_A}{\partial \varphi} \\ &= F_A(\varphi) = -2g^2 \varphi \left(\frac{\mu}{g} - \varphi \right) \left(\frac{2\mu}{g} - \varphi \right) \end{aligned} \quad (6.2)$$

in which $F_A(\varphi)$ is the long range force between the interacting molecular complexes in the actual lattice. By insertion of the definite number (4.3) the chemically induced driving force becomes

$$F_A(\varphi) = -2g^2 \varphi (N - \varphi) (2N - \varphi) \quad (6.3)$$

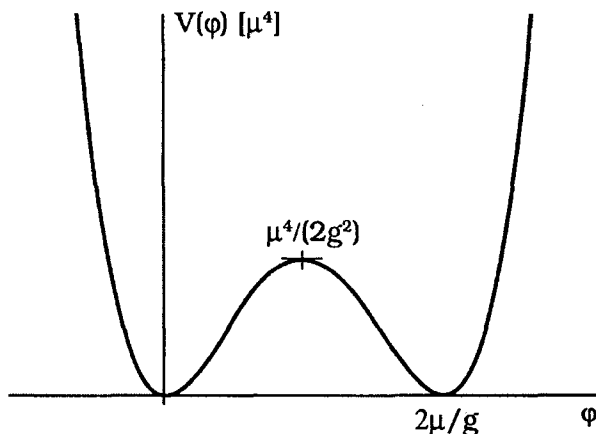


Figure 5. The asymmetric, lyotropic double-well potential.

which vanishes at $\varphi = 0$, $\varphi = N$, and $\varphi = 2N$. The force F is hence attractive (-) during the whole G_1 , that is, in the assembly state ($0 < \varphi < N$), which allows the formation and binding of pre-RCs by the DNA duplex but not initiation of replication (Tab. 1). Initiation of DNA replication is induced by a switch of interaction at $\varphi = N$, from attraction (-) to repulsion (+). During replication, for $N < \varphi < 2N$, the force is hence repulsive (+) implying a partial disassembly of pre-RCs with release of LFs and prevention of re-replication in the entire S -phase. Termination of replication is due to a vanishing of the force at $\varphi = 2N$, at which DNA replication is completed, with all replicons at the N primed origins duplicated once. $F(0) = 0$ corresponds to a resting cell, in the absence of driving source. Without knowledge of the dependence on hydrolysis and phosphorylation, and without inclusion of polarization and electromagnetic effects, at this stage of development the model is unable to describe the bidirectional character of DNA replication (Fig. 1b) and to predict the value of η , to mention just a couple of remaining unsolved problems.

Due to a vanishing force (6.3) at $\varphi = N$, however, the entire DNA system stays at rest and the lattice of pre-RCs awaits the event of excitation into the S phase [1]. Until this ultimate signal has been elicited, the DNA lattice obeys the harmonic-displacive type dynamics (4.7). Before examining the physical mechanism for initiation of DNA replication, the principles of which also explain the dynamic instability of growing MTs, a useful classical interpretation of the one-dimensional lattices of molecular complexes in the DNA duplex and the MTs is made.

7 Microtubules and DNA duplex work like elastically braced strings

After the spontaneous symmetry breakdown (5.1), and multiplication by the linear elastic modulus $\varepsilon = \rho^* s^2$, ρ^* being the linear (gravitational) mass density, (6.2) could be interpreted as the equation of motion of an elastically braced string [32]

$$\rho^* \frac{\partial^2 \varphi}{\partial t^2} - \varepsilon \frac{\partial^2 \varphi}{\partial x^2} + \kappa \varphi = F_D(\varphi) \quad (7.1)$$

driven chemically by the nonlinear force $F_D(\varphi)$ defined by

$$\begin{aligned} -\varepsilon \frac{\partial V_A}{\partial \varphi} &= F(\varphi) = -\kappa \varphi + F_D(\varphi) \\ &= -\frac{\kappa}{2} \varphi \left(1 - \frac{\varphi}{N}\right) \left(2 - \frac{\varphi}{N}\right) \end{aligned} \quad (7.2)$$

with a compressibility modulus $\kappa = 4\mu^2 \varepsilon$, φ/N being the (continuous) relative deformation of the embedded lattice, and $-\kappa \varphi$ the linear part of a long range restoring force $F(\varphi) = \varepsilon F_A(\varphi)$ on the same stringlike lattice. The force (7.2), derived from the model with spontaneously broken symmetry, thus assures that replication is initiated at $\varphi = N$, when all DNA origins are primed, and turned off at $\varphi = 2N$ such that the DNA content of G_2 cells is exactly twice that of G_1 cells.

The N excitations induce an energy density

$$\begin{aligned} H(\varphi) &= \frac{\rho^*}{2} \left(\frac{\partial \varphi}{\partial t}\right)^2 + \frac{\varepsilon}{2} \left(\frac{\partial \varphi}{\partial x}\right)^2 + \frac{\kappa}{2} \varphi^2 - \Phi(\varphi) \\ &= \varepsilon \left\{ \frac{1}{2s^2} \left(\frac{\partial \varphi}{\partial t}\right)^2 + \frac{1}{2} \left(\frac{\partial \varphi}{\partial x}\right)^2 + V_A(\varphi) \right\} \end{aligned} \quad (7.3)$$

where $1/2\kappa\varphi^2$ is proportional to the energy stored in the medium per unit length of the lattice. Rewriting (4.1) on the form $1/2(\partial\varphi/\partial x)^2 = V(\varphi)$, which yields a factor 2 to the potential in the static part of (7.3), for small φ -values, $\varphi^2 \approx 1 \ll N^2$, κ can thus be interpreted as the quantum of energy stored per ligand-receptor engagement, or the energy per pre-RC, or per GTP-tubulin dimer [24]. In the continuum limit the one-dimensional lattice models, of ligand-receptor complexes at the surface membrane, of pre-RCs in the DNA duplex, and of GTP-tubulin dimers in the cytoskeletal MTs, approximate to elastically braced strings. The parameters κ , ε and the coefficients of $F(\varphi)$, which like N , s , ρ^* , μ and the initial reactant concentrations may differ from one subsystem to another, depend on the elastic properties of the actual subsystem the string is embedded in, and on each other via the cytoskeleton which connects different subsystems. The interlinked parameters thus depend particularly on the rigidity and elasticity of the network of MTs and the MT associated proteins (MAPs). A comparison of the terms in (7.3) yields

$$\Phi(\varphi) = 2\varepsilon\mu^2\varphi^2 - \varepsilon V_A(\varphi) \quad (7.4)$$

and from (7.2) one has

$$\begin{aligned}
 F_D(\varphi) &= 6\mu^2 \varepsilon \frac{\varphi^2}{N} - 2\mu^2 \varepsilon \frac{\varphi^3}{N^2} \\
 &= \frac{\kappa}{2} \varphi \left(3 \frac{\varphi}{N} - \frac{\varphi^2}{N^2} \right)
 \end{aligned} \tag{7.5}$$

A change in the chemical potential, through (7.2), due to a compression of MTs could then be quantitatively related by (4.3) to a lowering of the critical tubulin concentration, at the threshold for depolymerization, as previously proposed in the tensegrity model [33]. By use of $2V(\varphi) = (\partial\varphi/\partial x)^2$, $\partial\varphi/\partial t = 0$, and the static wave solution (Fig. 6a)

$$\varphi = \frac{\mu}{g} \tanh(\mu x) \tag{7.6}$$

the localized energy density (7.3) becomes (Fig. 6b)

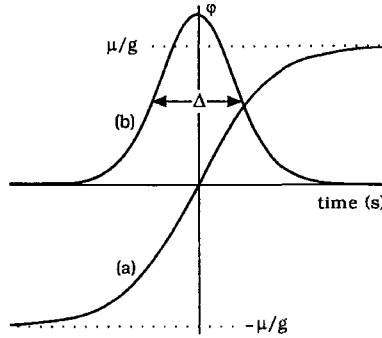


Figure 6. (a) The symmetric kink-solution. (b) The corresponding energy density $H = \varepsilon\mu^4/[g^2 \cosh^4(\mu x)]$ of width $\Delta = 1/\sqrt{2} \mu$.

$$H(\varphi) = \varepsilon \left(\frac{\partial\varphi}{\partial x} \right)^2 = \frac{\mu^2}{g^2} \frac{\varepsilon}{\cosh^4(x)} \tag{7.7}$$

The total energy stored by the N (complexes) excitations in the membrane, in the MTs, or in the DNA helix, is given by [30]

$$E \approx \varepsilon \int_0^{+\infty} \left(\frac{\partial\varphi}{\partial x} \right)^2 dx = \frac{1}{2} \varepsilon \int_{-\infty}^{+\infty} \left(\frac{\partial\varphi}{\partial x} \right)^2 dx = \frac{2}{3} \varepsilon \frac{\mu^3}{g^2} = \frac{1}{6} \frac{\kappa}{g} N \tag{7.8}$$

Hence, the total energy is not a direct sum of the N excitations, each of which contributes an equal quantum of energy κ . Through the coupling g (7.8) is a nonlinear function of the initial reactant concentrations.

The ligand-receptor induced energy, can thus be stored as quanta in the form of pre-RCs in DNA, until the cell takes the unanimous, irrevocable decision to initiate DNA replication. Interestingly, the mechanism responsible for this all-or-none type transition into the S phase, i. e., the switch of interaction from attraction to repulsion which also explains the prevention of re-replication during the S phase by repulsion of the LFs, also explains the dynamic instability in growing MTs. As will be shown by analysis of a time-dependent perturbation, the initiation of DNA replication and the dynamic instability in growing MTs are both due to the contribution of energy from one molecular complex above the lattice ground state.

8 Initiation of DNA replication and dynamic instability in MTs

By inserting small time-dependent excitations in the dynamics (4.7) of the surface membrane, the DNA, or the MT lattices

$$\varphi(x, t) = \varphi(x) + e^{i\omega qt} \psi(x) \quad (8.1)$$

and keeping only the linear terms, a Schrödinger like, exactly soluble equation is obtained [32]:

$$\begin{aligned} \left(\frac{d^2}{dx^2} + V''(\varphi) \right) \psi_v(x) = \\ \left(\frac{d^2}{dx^2} + 4\mu^2 - 6\mu^2 \frac{6\mu^2}{\cosh^2(\mu x)} \right) \psi_v(x) = \frac{\omega_v^2}{s^2} \psi_v(x) \end{aligned} \quad (8.2)$$

This equation has two discrete eigenmodes

$$\begin{aligned} \frac{\omega_v^2}{s^2} = \mu^2 v(4-v); v = 0, 1 \\ \psi_0(x) = A_0 \frac{1}{\cosh^2(\mu x)}; \psi_1(x) = A_1 \frac{\sinh(\mu x)}{\cosh^2(\mu x)} \end{aligned} \quad (8.3)$$

and a continuum ($q \geq 0$) starting at 2μ ($v = 2$):

$$\frac{\omega_v^2}{s^2} = 4\mu^2 + q^2 \quad (8.4)$$

$$\psi_q(x) = e^{i\omega_q x} \left(3 \tanh(\mu x) - 3iq \frac{\tanh(\mu x)}{\mu} - 1 - \frac{q^2}{\mu^2} \right)$$

Discrete eigenvibrations: When comparing the expansion $\phi(x + \Delta x) \approx \phi(x) + \Delta x d\phi/dx$ with (8.1), which shows that $d\phi/dx$ equals ψ_0 (8.3), one finds that $\omega_0 = 0$ is a so called translation mode of the unperturbed wave (7.6) of the N molecular complexes. The translation mode implies that all the N membrane receptor engagements, and the corresponding definite number of pre-RCs in DNA, can respond coherently as one unit in the all-or-none type transition into the S phase. This zero frequency mode, which is a coherence effect corresponding to the infinite wavelength approximation in (3.2), is a result of the rigidity of the system, a prerequisite for the long range correlations and the cell's ability to count the number of molecular complexes irrespective of their localization. The cell first counts, becomes "aware", and then takes the unanimous all-or-none type decision to initiate DNA replication. The nonzero mode $\omega_1 = \sqrt{3}\mu s$ (at $v = 1$) is a stable eigenvibration determined by the values of μ and s in the different subsystems. The model thus automatically provides a frequency selective mechanism [34] that could play an essential role in the storage of energy and signal transduction.

Scattering threshold: At vanishing momentum, $q = 0$, the scattering (continuum) energy threshold is given by $\omega_c^2 = 4\mu^2 s^2$ ($v = 2$), or $\omega_c^2 = 4\mu^2 s^2 / d^2$ if the usual length scale, d , of the lattice unit is used. Multiplying by $\varepsilon = \rho^* s^2$ and inserting $4\mu^2 = \kappa/\varepsilon$, this threshold energy equals $\rho^* \omega_c^2 d^2 = \kappa = 1/2 \rho^* \omega_c^2 d_m^2$, which corresponds to a "melting" displacement $d_m \approx \sqrt{2} d$. An energy contribution by hydrolysis of ATP from one pre-RC, κ_{DNA} , above the ground state energy of the lattice of N pre-RCs is hence sufficient to excite the DNA system to the scattering threshold. The system then switches from a condensating state (8.3) with attractive force to a scattering state (8.4) with repulsive force. This explains the role of ATP hydrolysis at the initiation of DNA replication.

In the DNA duplex this switch of interaction explains the transition from the assembly state in the G_1 phase to partial disassembly in the S phase, with initiation of replication and release of LFs from the pre-RCs, after degradation of the cyclin-CDK inhibitors, phosphorylation of Rb proteins and other preceding steps. However, a description of the bidirectional character of DNA replication (Fig. 1b), including enzymatic reactions, also requires the inclusion of electromagnetic effects with gauge invariant transfer (scattering) of correlated pairs of widely separated electrons. For instance, this should be required to control at a distance the ATPase activity in the pairs of helicases that start from η of the N primed origins.

It is anticipated here that the quantum of energy per origin engaged in the DNA lattice, κ_{DNA} , should be associated with the binding of one complete pre-RC. Consequently, after being engaged by the ultimate ATP-ORC unit, in order to reach the scattering threshold, the corresponding DNA origin should be primed by a complete set of licensing factors, before the hydrolysis and phosphorylation of various factors, transfer of electrons, and initiation of DNA replication could finally take place. Alternatively, the mere attempt to bind to DNA by one ultimate ATP-ORC unit could turn out to be sufficient to induce initiation of DNA replication. Only experiments can show what actually takes place.

Similarly, the switch of interaction, from condensation to repulsion, at the energy contributed by hydrolysis of one ultimate GTP-tubulin dimer, κ_{TD} , above the threshold at $\varphi = N$, can also explain the dynamic instability in a growing MT. The MT filament then suddenly starts to depolymerize. Given the different (gravitational) mass densities (ρ^*) and the velocities of sound (s), of the hydrolyzed (h) and nonhydrolyzed (n) parts of a MT or DNA string, the acoustic impedance, $Z = \rho^*s$, becomes known like the amplitude ratios for reflection, $A_r/A_{in} = \tanh(1/2 \ln(Z_n/Z_h))$, and transmission, $A_t/A_{in} = 1 + \tanh(1/2 \ln(Z_n/Z_h))$, which could then be related to the traveling waves before and after the symmetry breakdown.

9 Summary

The gross behaviour of DNA replication and cell cycle progression is explained here by a nonstationary manybody physics model, in which the cell cycle is controlled by a long range force, $F(\varphi) = -(\kappa/2)\varphi(1 - \varphi/N)(2 - \varphi/N)$, alternatively $F(\varphi) = -(\kappa^*/2)(\varphi/N)(1 - \varphi/N)(2 - \varphi/N)$ with ρ^* , ε , and κ in (7.1) and (7.2) renormalized by a factor N , between the ORCs bound by DNA, φ being the number of protein complexes, N the threshold number at which DNA replication is initiated and κ the compressibility modulus in the lattice of complexes which behaves like an elastically braced string. The so obtained force possesses the desired properties needed to understand the gross regulation of DNA replication and the cell cycle progression in terms of the basic molecular interactions (Tab. 1).

Initiation of replication is thus explained by a switch of sign of the long range force $F(\varphi)$, from attraction (−) and assembly in the G_1 phase ($0 < \varphi < N$), to repulsion (+) and partial disassembly in the S phase ($N < \varphi < 2N$), with concomitant release of LFs from pre-RCs and, hence, also prevention of re-replication. Termination of DNA replication is due to a vanishing of the long range force $F(\varphi)$ at $\varphi = 2N$, at which all primed replicons in DNA have been duplicated once, and $F(0) = 0$ corresponds to a resting cell in the absence of a driving force at $\varphi = 0$. Thus the model makes sure that the DNA content of G_2 cells is exactly twice that of G_1 cells.

The switch of sign at $\varphi = N$ similarly explains the dynamic instability in growing MTs as well as the switch in the interaction at the surface membrane of IL2 with its

receptor in late G_1 , at the R -point, after which the T cell proceeds to DNA replication without further exposure to IL2. Shape, slope and scale, $\rho_{50} = r\rho_0/(Er_0)$, of the response curves derived agree well with experimental data from dividing T cells (Fig. 2a) and polymerizing MTs, the variable length of which is explained by the model through a nonlinear dependence of the growth amplitude on the initial concentrations of GTP and tubulin dimers (Fig. 4).

The model also provides a nonlinear expression for the definite threshold number (4.3) of molecular complexes engaged, $N = \mu(\rho_0 + r_0)/(\rho_0 - r_0)$, at which DNA replication is initiated, and at which threshold also the dynamic instability in growing MTs occurs, ρ_0 and r_0 being the initial reactant concentrations in the actual subsystem. Given the explicit forms of the threshold and scale, in terms of the initial reactant concentrations, apart from the functions of a normal cell, it is now also possible to examine certain forms of cancer, in which the cell line is unable to internalize receptors and therefore becomes hypersensitive to ligands at such low concentrations where normal cells remain inactive. In addition the model also defines positive definite and constant quanta of energy, $\kappa = 4\mu^2\varepsilon$, per ligand-receptor association [24], per pre-RC, and per GTP-tubulin dimer, which was not the case in the previous version [26]. A property like that may look formal, however, is nontrivial too because, had one tried to define $\kappa = 4\mu^2\varepsilon$ through the symmetric equation (4.7) instead of the assymetric one (6.2), then μ^2 and hence κ would have had the wrong sign.

The model is derived as if DNA replication is initiated simultaneously with the switch of the IL2-IL2R interaction at the R -point, beyond which the cell automatically proceeds to replication. However, also with a nonzero but constant lag time, between the R -point and the G_1 - S interface (Tab. 1), the rate of cell cycle progression at these two critical points of the cell cycle are approximately equal. The important point is that the same form of interaction, responsible for the molecular switch mechanism, is actually transferred from the surface membrane to the nucleus [9]. This in turn is due to the fact that the nonlocal correlations (3.2), and the driving reaction (2.1), have the same forms in each subsystem irrespective of if the pair of reactants are IL2 and IL2R, or ATP-ORC units and vacant DNA origins.

Much about DNA replication is obtained from yeasts, and it can therefore be asked whether such information could be applied to T lymphocytes from mammals. There is no general answer to this but for particular questions. Analogues of yeast ORC genes and proteins have been described in a wide range of eukaryotic organisms, suggesting that ORC is a conserved component of eukaryotic DNA replication [1]. Experiments [2] also provide solid evidence for replication origins, with specific DNA sequences (replicators), in animal cells [35]. Other components of eukaryotic DNA replication may turn out to be non-conserved, however, this should have little influence on the model proposed here as long as ORC is considered to be the best candidate for an initiator protein [1]. Moreover the switch

of sign of the long range interaction (6.3), crucial for the phase-transition between the assembly state and the replication state, is determined solely by the number of origins engaged by ORCs.

In the actual one-dimensional version the model cannot yet account for spatio-temporal dependence of the initial reactant concentrations. Cytoskeletal MTs therefore grow identically in all directions. A more realistic response theory, that also explains the hydrolysis of GTP and ATP, the increased cyclin-CDK activity, the rapid phosphorylation of Rb proteins, and which predicts the value of η , should obviously be derived in three spatial dimensions and include electromagnetic interaction and polarization. The elastically braced string dynamics (7.1) accounts for the leading order interaction, and the remaining dissipative effects (8.1) could then be treated as perturbations [32].

However, the explanations of the variable length and dynamic instability of self-assembling MTs, the counting of receptors and origins engaged, the mechanisms for initiation and termination of DNA replication and for the assembly and disassembly of pre-RCs with prevention of re-replication during the *S* phase, and the good agreement with growth data of dividing cells and polymerizing MTs, are results obtained through one and the same model. This has encouraged me to propose that already the one-dimensional model proposed could serve as a guide for further studies of both transformed and normal cells, as well as of their functions and dysfunctions in the organism. Through charge conservation, which implies a symmetry termed gauge invariance that links the electromagnetic field to the matter field, the lyotropic one-dimensional model obtained could also work as platform for generalizations into three spatial dimensions. I also propose that the spatio-temporal coherence, induced by stabilization of the cytoskeleton, could be taken as a classical definition of a form of consciousness of the cell, a state of "mind" developed prior to its unanimous decision to initiate replication. In the infinite wavelength approximation such a definition could work also for a system of cells connected through a rigid cytoskeletal network.

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References

1. Dutta A. and Bell S.P., Initiation of DNA replication in eukaryotic cells, *Annu. Rev. Cell Dev. Biol.* **13** (1997) pp. 293-332.

2. Aladjem M. I., Rodewald L. W., Kolman J. L. and Wahl G. M., Genetic Dissection of a Mammalian Replicator in the Human β -Globin Locus, *Science* **281** (1998) pp. 1005-1009.
3. Bell S.P. and Stillman B., ATP-dependent recognition of eukaryotic origins of DNA replication by a multiprotein complex, *Nature* **357** (1992) pp. 128-134.
4. Klemm R.D., Austin R.J. and Bell S.P., Coordinate binding of ATP and origin DNA regulates the ATPase activity of the origin recognition complex, *Cell* **88** (1997) pp. 493-502.
5. Cress W.D. and Nevins J.R., Use of the E2F Transcription Factor by DNA Tumor Virus Regulatory Proteins, *Curr. Topics Microbiol. Immunol.* **208** (1996) pp. 63-78.
6. Turner J.M., IL-2-dependent induction of G₁ cyclins in primary T cells is not blocked by rapamycin or cyclosporin A, *Int. Immunol.* **5** (1993) pp. 1199-1209.
7. Nourse J., Firpo E., Flanagan W.M., Coats S., Polyak K., Lee Mong-Hong, Massague J., Crabtree G.R., and Roberts J.M., Interleukin-2-mediated elimination of the p27^{Kip1} cyclin-dependent kinase inhibitor prevented by rapamycin, *Nature* **372** (1994) pp. 570-573.
8. Zou Lee and Stillman B., Formation of a Preinitiation Complex by S-phase Cyclin CDK-Dependent Loading of Cdc45p onto Chromatin, *Science* **280** (1998) pp. 593-596.
9. Smith K. A. Why do cells count? In *Nonlinear Cooperative Phenomena in Biological Systems*, ed. by Matsson L. (World Scientific, Singapore, 1998) pp. 13-19.
10. Voter W.A. and Erickson H.P., The Kinetics of Microtubule Assembly, *J. Biol. Chem.* **259** (1984) pp. 10430-10438
11. Collings P.J., *Liquid Crystals: Nature's Delicate Phase of Matter*. (Adam Hilger, Bristol, UK, 1990) pp. 147-216.
12. Hill A.V., The combinations of Hemoglobin with Oxygen and with Carbon Monoxide I, *Biochem. J.* **7** (1913) pp. 471-480.
13. Langmuir I., The adsorption of gases on plane surfaces of glass, mica and platinum, *J. Am. Chem. Soc.* **40** (1918) pp. 1361-1403.
14. Smith K.A., The Interleukin-2 Receptor, *Annu. Rev. Cell Biol.* **5** (1989) pp. 397-425.
15. Bevan J.A., Oriowo M.A. and Bevan R.D., Physiological Variation in α -Adrenoceptor-Mediated Arterial Sensitivity: Relation to Agonist Affinity, *Science* **234** (1986) pp. 196-197.
16. Barlow R. and Blake J.F., Hill coefficients and the logistic equation, *Trends Pharmacol. Sci.* (Nov.) **10** (1989) pp. 440-441.
17. Bevan J.A., Bevan R.D., Kite K., and Oriowo M.A., Species differences in sensitivity of aortae to norepinephrine are related to α -adrenoceptor affinity, *Trends Pharmacol. Sci.* **9** (1988) pp. 87-89.

18. Cantrell D.A. and Smith K.A., The interleukin-2 T-cell system: A new cell growth model, *Science* **224** (1984) pp. 1312-1316.
19. Viola A. and Lanzavecchia A., T Cell Activation Determined by T Cell Receptor Number and Tunable Thresholds, *Science* **273** (1996) pp. 104-106.
20. Rothenberg E.V., How T Cells Count, *Science* **273** (1996) pp. 78-79.
21. Chen K.C., Csikasz-Nagy A., Gyorffy B., Val J., Novak B. and Tyson J.J., Kinetic Analysis of a Molecular Model of the Budding Yeast Cell Cycle, *Mol. Biol. Cell* **11** (2000) pp. 369-391.
22. Mitchison T. and Kirschner M., Microtubule assembly nucleated by isolated centrosomes, *Nature (London)*, **312** (1984) pp. 232-237.
23. Mitchison T. and Kirschner M., Dynamic instability of microtubule growth, *Nature (London)*, **312** (1984) pp. 237-242.
24. Paton W.D.M., A theory of drug action based on the rate of drug-receptor combination, *Proc. R. Soc. London Ser. B* **154** (1961) pp. 21-69.
25. Smith K.A., T-cell growth factor and glucocorticoids: Opposing regulatory hormones in neoplastic T-cell growth, *Immunobiology*. **161** (1982) pp. 157-173.
26. Matsson L., Response Theory for Non-Stationary Ligand-Receptor Interaction and a Solution to the Growth Signal Firing Problem, *J. Theor. Biol.* **180** (1996) pp. 93-104.
27. Matsson L., Long Range Interaction between Protein Complexes in DNA Controls Replication and Cell Cycle Progression, *J. Biol. Syst.* **9** No. 1 (2001) pp. 41-65. (On page 52, 4th line above (5.1) in this reference work the upper limit of the interval should be N, not 5.)
28. Ferrell J. E. and Machleder E.M., The Biochemical Basis of an All-or-None Cell Fate Switch in *Xenopus* Oocytes, *Science* **280** (1998) pp. 895-898.
29. Koshland Jr. D.E., The Era of Pathway Quantification, *Science* **273** (1998) pp. 852-853.
30. Rajaraman R., *Solitons and Instantons*. (North Holland, Amsterdam, 1982) pp. 1-83.
31. Ziman J.M., *Principles of the theory of solids*. (Cambridge University Press, Cambridge, 1964) pp. 324-346.
32. Morse P. and Feshbach H., *Methods of Mathematical Physics Part I*. (McGraw-Hill, New-York, 1953) pp. 139, 256, 305, 729-736.
33. Ingber D., Cellular tensegrity, defining new rules of biological design that govern the cytoskeleton, *J. Cell. Sci.* **104** (1993) pp. 613-627.
34. Fröhlich H., Theoretical Physics and Biology. In *Biological Coherence and Response to external Stimuli*. (Springer Verlag, Berlin, 1988) pp. 1-24.
35. Huberman J.A., Choosing a Place to Begin, *Science* **281** (1998) pp. 929-930.

FROM THE BIOCHEMISTRY OF TUBULIN TO THE BIOPHYSICS OF MICROTUBULES

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Microtubules (MTs) are protein polymers of the cytoskeleton that once fully understood will provide a deeper understanding of many cell functions. Assembly dynamics with the characteristic dynamic instability phenomenon has been intensively investigated over the past two decades and several models have been developed which adequately describe this phenomenon. Since the tubulin structure was imaged by Nogales and Downing, the dipole has been calculated and also the charge distribution on the surface of the protein together with a hydrophobicity plot. However, it still remains to be seen how the dipole changes upon the conformational change due to GTP hydrolysis. Furthermore, the contribution of the carboxyl terminus to the dipolar and electrostatic properties has not been accounted for. Using the crystallographic data of Nogales and Downing, some properties of the new structure of tubulin were examined. The so called multi-tubulin hypothesis seems to be explained by the differences in the electrostatic potentials produced by various tubulin isotypes produced by only several amino-acid substitutions. Such small changes in the tubulin structure may render the MTs less susceptible to naturally occurring agents which would otherwise bind them and impair their function. The hypothesis of electrostatic binding between protofilaments seems to be well founded. The MT structure has been compared with the previous work, to comment on models of motor protein movement and to consider how isotype changes affect the electrostatic potential surrounding the MT. The nature of binding between the MT and motor proteins also seems to be electrostatic and can be used to explain the stepping of these motors along the MT surface. The overall picture emerging from these studies is that the tubulin's molecular structure and the ensuing microtubular architecture can provide a microscopic-level understanding of the biological function in the cell.

1 Introduction—Microtubules

Of the three known types of the filaments comprising the cytoskeleton, microtubules (MTs) have the largest diameter, 25 nm. They are found in nearly all eukaryotic cells and are polymers of tubulin protein. MTs serve as tracks on which motor proteins may carry materials about the cell and serve as scaffolding to maintain the cell shape since they are among the most rigid structures within a typical cell. They also form the core of cilia and flagella which beat in a coordinated manner to either move objects along the cell membrane or to propel the cell through its environment. Within the cell body, the majority of the MTs emanate from a centriole. The negative ends of MTs are anchored at these microtubule organizing centres. The MTs *in situ* are interconnected and intraconnected by microtubule associated proteins (MAPs). MAPs have a stabilizing effect on the dynamics of MTs.

Ledbetter and Porter [1] were the first to describe these tubules found within the cytoplasm and dubbed them both cytotubules and microtubules. The name microtubules has stuck and the general structure of the MT has since been well-established by experiment [2,13]. MTs are polymers formed from two largely homologous globular proteins, α -tubulin and β -tubulin. These two proteins are very closely structurally related and they bind together to form a heterodimer known as $\alpha\beta$ -tubulin. This dimer is the basic subunit which polymerizes to form the MT. The MT is a hollow tube with an outer diameter of 25 nm and an inner diameter of 15 nm (see Fig. 1). The tube is composed of strongly bound linear polymers, known as protofilaments, that are connected via weaker lateral bonds to form a sheet that is wrapped up to form a tube.

The electron crystallography analysis of Nogales *et al.* [4] has indeed shown that the α and β monomers are nearly identical. However, this small difference on the monomer level allows the possibility of several lattice types, in particular, the so-called MT A and B lattices [5]. Moving around the MT in a left-handed sense, protofilaments of the A lattice have a vertical shift of 4.92 nm upwards relative to their neighbours. In the B lattice, this offset is only 0.92 nm because the α and β monomers have switched roles in alternating protofilaments. This change results in the development of a structural discontinuity in the B lattice known as the seam.

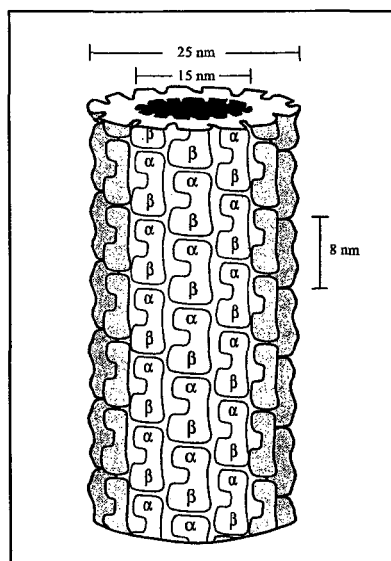


Figure 1. A section of a typical microtubule demonstrating the hollow interior which is filled with cytoplasm, the helical nature of its construction. Each vertical column is known as a protofilament and the typical MT has 13 protofilaments.

2 Tubulin—The Building Block of a Microtubule

Tubulin is an important and an interesting globular protein. The tubulin that polymerizes to form MTs is actually a heterodimer of α -tubulin and β -tubulin. These two proteins are highly homologous and have 3D structures which are nearly identical. Although the similarity of α -tubulin and β -tubulin had long been suspected, the fact that tubulin resisted crystallization for about 20 years prevented confirmation of this hypothesis until very recently. Nogales *et al.* [4] were able to perform cryo-electron crystallography on sheets of tubulin formed in the presence of zinc ion. Figure 2 produced from the Nogales data, obtained from the protein data bank (PDB entry: 1tub), using MOLSCRIPT [6] makes clear the similarities between the two proteins. Each is composed of a peptide sequence more than 400 members long which is highly conserved between species. The amino acid sequences for these proteins may be compared given the data in Table 1 which lists the conventional one and three letter codes for the 20 naturally-occurring amino acids. Codes should be read from left to right and spaces are inserted every 10 residues for clarity. The sequences of a few representative samples of tubulin have been retrieved from the Swiss-Prot protein sequence database [7] and are shown in Table 2.



Figure 2. A diagram of the tubulin molecule produced from the Nogales *et al.* electron crystallography data [4] shows the similarity between the α -subunit (upper half) and β -subunit (lower half). The stick outlines near the base of each subunit indicate the location of GTP when bound.

Table 1. The twenty naturally occurring amino acids are listed along with their one- and three-letter codes as well as whether they have a polar character and whether they are charged. *Histidine has a pKa of 6.5 and consequently will be protonated and positively charged should the pH of the cytoplasm dip below this value.

1 Letter	3 Letter	Amino Acid	Polar	Charge
A	Ala	Alanine		
C	Cys	Cysteine	yes	
D	Asp	Aspartic Acid		-e
E	Glu	Glutamic Acid		-e
F	Phe	Phenylalanine		
G	Gly	Glycine		
H	His	Histidine		+e*
I	Ile	Isoleucine		
K	Lys	Lysine		+e
L	Leu	Leucine		
M	Met	Methionine		
N	Asn	Asparagine	yes	
P	Pro	Proline		
Q	Gln	Glutamine	yes	
R	Arg	Arginine		+e
S	Ser	Serine	yes	
T	Thr	Threonine	yes	
V	Val	Valine		
W	Trp	Tryptophan		
Y	Tyr	Tyrosine	yes	

Based on their charge, the amino acids may be classified into 3 groups: those with a positive charge, those with a negative charge and the neutral residues. The size of the residue and its ability to react with other amino acids will affect protein folding and consequently function. Mutations involve changes to the sequence of amino acid residues. This may occur by substitution, addition or deletion of one or more of the residues. If the change is for a residue with similar steric or electrostatic properties, the mutated tubulin protein will likely fold properly and retain its function. However, a change that substitutes a residue whose properties differ substantially will likely result in a non-functional protein. Nevertheless, several different versions of the tubulin protein exist today and are known as isotypes when they exist within the same species. Both α -tubulin and β -tubulin appear in several isotypes. The isotypes are versions of these proteins which differ to a smaller degree than between the α and β variants. The different isotypes are expressed to varying degrees in specific cells types. For example, in humans, the β_2 isotype is found

predominately within neurons. Due to this localization of isotypes, there is a suspicion that tubulin has adapted for specific functional reasons. There is also a third isoform family known as γ -tubulin. The γ species is found within the MT organizing centers and is important in the nucleation of new microtubules.

Table 2. The amino acid sequences of human α_1 , β_1 and β_2 tubulin show a high degree of homology. The sequence is given along with the total number of amino acids and the molecular weight of the molecule.

Human tubulin α_1 amino-acid sequence (451 amino acids, 50157 Da)

MRECSIHVG	QAGVQIGNAC	WELYCLEHGI	QPDGQMPSDK	TIGGGDDSEN	TFSETGAGK
HVPRAVFVDL	EPTVIDEVRT	GTYRQLFHPE	QLITGKEDAA	NNYARGHYTI	GKEIIDLVLD
RIRKLADQCT	RLQGFLVFHS	FGGGTSGSGT	SLLMERLSVD	YGKKSKLEFS	IYPAPQVSTA
VVEPYNSILT	THTTLEHSDC	AFMVDNEAIY	DICRRNLDIE	RPITYNLNRL	IGQIVSSITA
SLRFDGALNV	DLTEFQTNLV	PYPRIHFPLA	TYAPVISA EK	AYHEQLSVAE	ITNACFEFAN
QMVKCDPGHG	KYMACCLLYR	GDVVPKDVNA	AJATIKTKRT	IQFVDWCPTG	FKVGINYQPP
TVVPGDLAK	VQRAVCMLSN	TTAIAEAWAR	LDHKFDLMYA	KRAFVHWYVG	EGMEEGEFSE
AREDMAALEK	DYEEVGHVSH	EGEGEEEGEE	Y		

Human tubulin β_1 amino-acid sequence (444 amino acids, 49759 Da)

MREIVHIQAG	QCGNQIGAKF	WEVISDEHGI	DPTGTYHGDS	DLQLDRISVY	YNEATGGKYV
PRAILVDLEP	GTMDSCRS GP	FGQIFRPDNF	VFGQSGAGNN	WAKGHYTEGA	ELVDSVLDVV
RKEAESDCDL	QGQFLTHSLG	GGTGSGMGTL	LISKIREEYP	DRIMNTFSVV	PSPKVS DTVV
EPYNATLSVH	QLVENTDETY	CIDNEALYDI	CFRTLRLTTP	TYGDLNLHVS	GTMECVTTCL
RFPQQLNADL	RKLAVNMVFP	PRLHFFMPGE	APLTSRGSQQ	YRALTVPDLT	QQVFDKNNMM
AACDPRHGRY	LTVAAVFRGR	MSMKEVDEQM	LNQVNKNSSY	FVEWIPNNVK	TAVCDIPPRG
LKMAVTFIGN	STAIQELFKR	ISEQFTAMFR	RKAFLHWYTG	EGMDEMEFTE	AESNMNDLVS
EYQQYQDATA	EEEEDFGEEA	EEEE			

Human tubulin β_2 amino-acid sequence (445 amino acids, 49831Da)

MREIVHLQAG	QCGNQIGAKF	WEVISDEHGI	DPTGTYHGDS	DLQLERINVV	YNEATGGKYV
PRAVLVDLEP	GTMDSVRS GP	FGQIFRPDNF	VFGQSGAGNN	WAKGHYTEGA	ELVDSVLDVV
RKEAESDCDL	QGQFLTHSLF	GGTGSGMGTL	LISKIREEYP	DRIMNTFSVV	PSPKVS DTVV
EPYNATLSVH	QLVENTDETY	CIDNEALYDI	CFRTLRLTTP	TYGDLNLHVS	ATMSGVTTCL
RFPQQLNADL	RKLAVNMVFP	PRLHFFMPGE	APLTSRGSQQ	YRALTVPELT	QQMFDKNNMM
AACDPRHGRY	LTVAAVFRGR	MSMKEVDEQM	LNQVNKNSSY	FVEWIPNNVK	TAVCDIPPRG
LKMSATFIGN	STAIQELFKR	ISEQFTAMFR	RKAFLHWYTG	EGMDEMEFTE	AESNMNDLVS
EYQQYQDATA	EEEEGEFEEA	EEEEA			

In humans, six α isotypes and seven β isotypes of tubulin are found. Although the sequence of amino acids is highly conserved overall, certain regions of α_1 tubulin show divergence from α_2 tubulin and so on. Recent studies have shown that the differences in α -tubulin are more subtle than those in β -tubulin [8]. Table 3 gives a comparison between the main β -tubulin isotypes in cows. The location of cells expressing that particular variant of tubulin are given along with the homology in percent with β_1 which is derived from a comparison of primary sequences. Finally, the abundance of each tubulin isotype in the bovine brain is given.

Table 3. Localization and Homology of Bovine β -Tubulin

Isotype	Localization	Homology	Abundance in Brain (%)
β_1	everywhere, thymus	100.0	3
β_2	brain	95.0	58
β_3	brain, testis, tumours	91.4	25
β_4	brain, retina, trachea	97.0	11

There are some differences between MTs assembled from the various β -tubulins in terms of their assembly properties, cross-linking behaviours and drug interactions. Although MTs incorporate without difficulty more than one isotype of tubulin, we can consider MTs with primarily a single β isotype in order to distinguish their respective properties. MTs composed of β_2 and β_3 assemble more easily in the presence of MAP τ and MAP2; and these two MAPs have the same localization as the β_2 and β_3 isotypes *in vivo*. MTs polymerized from β_2 and β_4 tubulin may be connected by crosslinking proteins, but β_3 MTs may not. This may be a result of the conformations of the isotypes and the cross-linker length. MTs formed from β_3 tubulin are the primary MTs found in tumours but it is also the isotype which cannot be bound by colchicine, an anti-cancer drug. Therefore, one of the main differences between the β -tubulin isotypes is thought to be the available microtubule-associated protein binding sites on the outside of the protein's surface. Additional differences between the tubulin isotypes are found when 'tubulin decay' is studied. Specifically, sulf-hydryl groups become exposed over time and the β_2 isoform of tubulin seems to decay more quickly than β_3 . Finally, there is the issue of localization within the cell. The β_1 and β_4 isotypes are not found in cell nuclei but are present along with β_2 in the mitotic spindle. Wilson and Borisy pointed out that the function of the β_4 isotype of tubulin in axonemes is suggestive that the interaction of tubulin with extrinsic proteins may direct the architecture and organization of MTs [9].

Post-translational modifications are those changes to the tubulin molecule that occur after the protein has already been produced. These changes such as detyrosination, acetylation, γ -glutamylation and phosphorylation are limited to the exposed portions of the protein molecule but may still affect some of the properties of the protein. Detyrosination of α -tubulin is an enzymatic process that removes the final amino-acid residue of the carboxy-terminal [10]. Behind this tyrosine residue lie several charged glutamic acid residues, thus the removal of the tyrosine makes the extended carboxy-terminal tail much more electro-negative. Acetylation may occur at lysine-44 of α -tubulin and the addition of an acetyl group to lysine neutralizes its positive charge. Similarly, γ -glutamylation can result in the addition of up to six glutamic acid residues to the already highly negatively charged carboxyl tail of tubulin [10]. Consequently, all of these post-translational modifications

change the electrostatic properties of tubulin and hence its interaction with other molecules of tubulin. Phosphorylation is conversion of an alcohol group (OH^-) to a phosphate group (PO_4^{3-}). In this case, the addition is quite bulky and is often used to regulate enzymatic processes. The steric hindrance resulting from these modifications alters the binding affinities between the tubulin molecule and certain substrates such as GTP, MAPs and drugs such as colchicine. It is also believed that MT stability is affected by post-translational modification as well as membrane affinity of the MTs.

3 The Electrostatic Properties

The process of describing electric fields about molecules is one of assigning partial charges to atomic positions based on the electro-negativity of the atoms. It also involves the association of dipoles to molecular bonds. The resultant fields are then compared to empirical measurements. After years of theoretical and experimental work, this practice has been refined and parameters adjusted to best reflect reality. This has meant the inclusion of effects due to atoms which are separated by more than a single bond and ultimately to functional groups such as entire amino-acid groups when proteins are studied. Simply placing a partial charge on each atomic site does not describe the electromagnetic field well compared to *ab initio* calculations on small molecular systems with fewer than 100 electrons. One may instead attempt to represent the electrostatic potential and field of a molecule by placing a sequence of multipoles at its center of mass. However, the use of distributed multipole analysis (DMA) provides a much more accurate representation of the electrostatic field about a molecule. Due to the size of small groups, diatomics, triatomics and tetra-atoms may each be described to high precision using only monopoles, dipoles and quadrupoles. Even for large systems, the DMA does a reasonable job of representing the electrostatic potential since dipoles and quadrupoles are the most important terms for description of molecular bonding.

In the case of the MT, each monomer is comprised of approximately 450 amino acids and has close to 7000 atoms. The good news is that beyond a certain distance, the so-called Bjerrum length, we can ignore electrostatic effects. In our case, beyond 2.0 nm charge-charge, charge-dipole and Van der Waals interactions are neglected. The potential is gradually switched off in the calculation so there are no discontinuities in the electrostatic potential, ϕ . This is close to the true situation since ions in the surrounding solution will screen any surface charges. The results presented for the electrostatic potential in this section represent 'vacuum' results given that the solvent is not explicitly taken into account. If the surrounding mixture of ions is considered, then the potential due to a point charge does not fall off simply as $1/r$, but instead as

$$\phi \propto \frac{1}{r} \exp(-Kr) \quad (1)$$

where K^{-1} is the Debye length, typically 0.6 nm under physiological conditions. Since we consider locations within 1.0 nm of the MT surface, they are not screened by the ions of the solution as there is not sufficient room for even water to be located in the intervening space.

As was mentioned earlier, Nogales *et al.* published the structure of α - and β -tubulin which were co-crystallized in the heterodimeric form [4]. Imaging was completed in the form of zinc sheets. The presence of zinc(II) ion, causes the tubulin heterodimers to form anti-parallel protofilaments. These sheets do not curl up to form the familiar MT but rather remain flat and are therefore suitable for electron crystallography. The work establishes that the structures of α - and β -tubulin are nearly identical and confirms the consensus speculation. A detailed examination shows that each monomer is formed by a core of two β -sheets that are surrounded by α -helices. The monomer structure is very compact, but can be divided into three functional domains: the amino-terminal domain containing the nucleotide-binding region, an intermediate domain containing the taxol-binding site, and the carboxy-terminal domain, which probably constitutes the binding surface for motor proteins [4].

Calculations of the potential energy were done with the aid of TINKER [4]. This computer program serves as a platform for molecular dynamics simulations and includes a facility to use protein specific force-fields. The first thing studied using TINKER was the overall charge, and dipole on the tubulin molecule (see Table 4). It turns out that tubulin is highly negatively charged at physiological pH but that much of the charge is concentrated on the C-terminus. This is the one portion of the tubulin dimer which was not imaged due to its freedom to move following formation of the tubulin sheet. This tail of the molecule extends outward away from the MT and into the cytoplasm and has been described by Sackett [10]. At neutral pH, the negative charge on the carboxy-terminus causes it to remain extended due to the electrostatic repulsion within the tail. Under more acidic conditions, the negative charge of the carboxy-terminal region is reduced by associated hydrogen ions. The effect is to allow the tail to acquire a more compact form by folding (see Fig. 3). Although, this is probably the largest structural change which occurs due to changes in the cell's pH, we shall see that other structural changes, the results of post-translational modification, can similarly affect the electrostatics of the tubulin dimer.

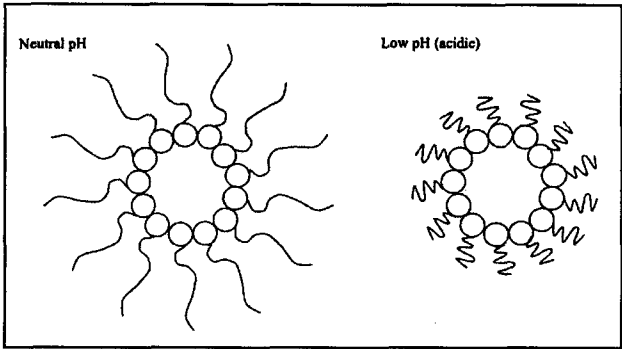


Figure 3. Cross-section of a MT including the carboxyl-terminus of the tubulin subunits. Folding of the carboxyl-terminus of the tubulin dimer demonstrates the change of the geometry of the molecule with pH. Neutral pH is shown on the left, the tail folds at lower pH as the negative charges are screened.

Table 4. Tubulin's Electrostatic Properties (tail region excluded)

Tubulin Properties	Calculated Value
charge (electronic charges)	-10
dipole (Debye)	1714
components: $\begin{Bmatrix} p_x \\ p_y \\ p_z \end{Bmatrix}$	$\begin{Bmatrix} 337 \\ -1669 \\ 198 \end{Bmatrix}$

In Table 4, the x-direction coincides with the protofilament axis. The α monomer is in the direction of increasing x values relative to the β monomer. This is opposite to the usual identification of the β monomer as the 'plus' end of the MT, but all this identifies is whether the MT is pointed towards or away from the cell body.

An important result that may be derived from the electrostatic potential are those regions of the MT's outer surface that are negatively charged and which may attract hydrogen ions (see Fig. 4). If electronic conduction occurs by proton ferrying, then the locations where protons would bind can be thus clarified. Finally, it also identifies locations on the MT where motor proteins may bind as in the case of at least one motor protein, kinesin; its attachment has been shown to be primarily electrostatic [12]. In calculating the electrostatic potential, 2.0 nm was selected as the cutoff distance for charge, dipole and Van der Waal interactions. The electrostatic potential was calculated for a 12.0 nm segment of the line, thereby including an additional 2.0 nm above and below each tubulin molecule. Periodic boundary conditions were then applied in the direction of the protofilament because this is the configuration of the tubulin dimers within a MT. The resulting profiles of

some of the electrostatic potential are shown in Figure 5 and are located about the tubulin dimer as shown in Figure 4. The lateral boundary conditions were not considered in the calculation of the potential.

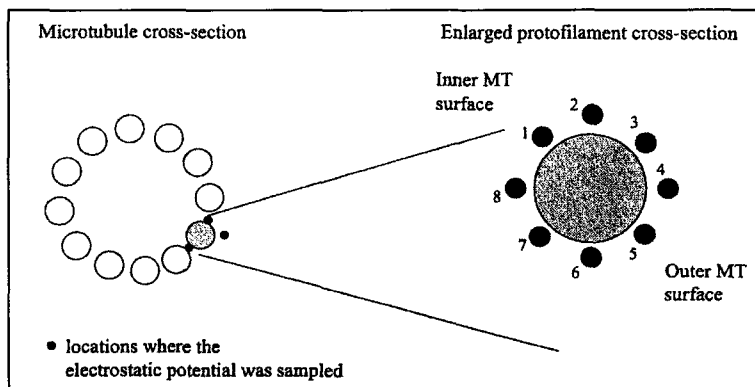
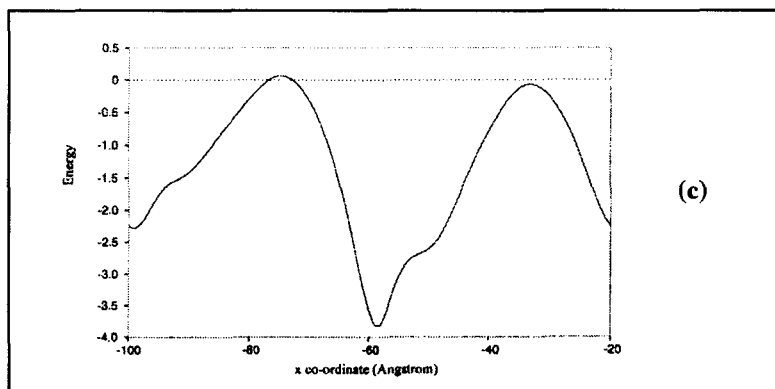
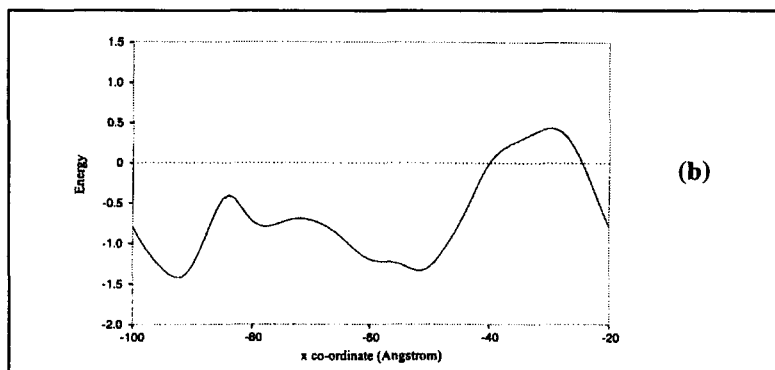
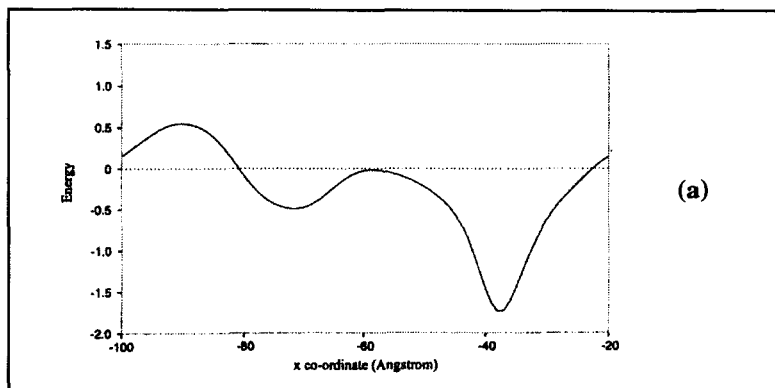


Figure 4. A MT cross-section illustrates where the electrostatic potential was examined along lines parallel to the protofilament axis (a line perpendicular to the plane of the page).

Consider the profile of the electrostatic potential in Figures 5a and 5b and compare them with the profiles in Figures 5e and 5f. These are the left and right sides respectively, of the tubulin molecule, which interact laterally to hold one protofilament together with neighbouring protofilaments. In these figures, each unit of energy represents 14.4 kcal/mol or 0.62 eV. This is roughly the energy available from the hydrolysis of two to three molecules of GTP or just a little more than the hydrolysis of one molecule of ATP. What is interesting is that the electrostatic potential is largely negative on the left side and positive on the right side. Thus there is a net electrostatic attraction between tubulin dimers with parallel alignment when their opposite sides face each other. In fact, if the minima in the left side's profile are aligned with the maxima in the electrostatic potential of the right side, we find that the neighbouring tubulin dimer will be shifted by 1.4 nm or 5.4 nm which compares reasonably well to the observed 0.9 nm or 4.9 nm offsets that depend on the lattice type [5]. The simple change of a residue on the surface offers the possibility of specifying one shift and locking the resulting MT into either the A or B type lattice. Hence post-translational modification or more likely the expression of a particular isotype over another could select a specific lattice [14].



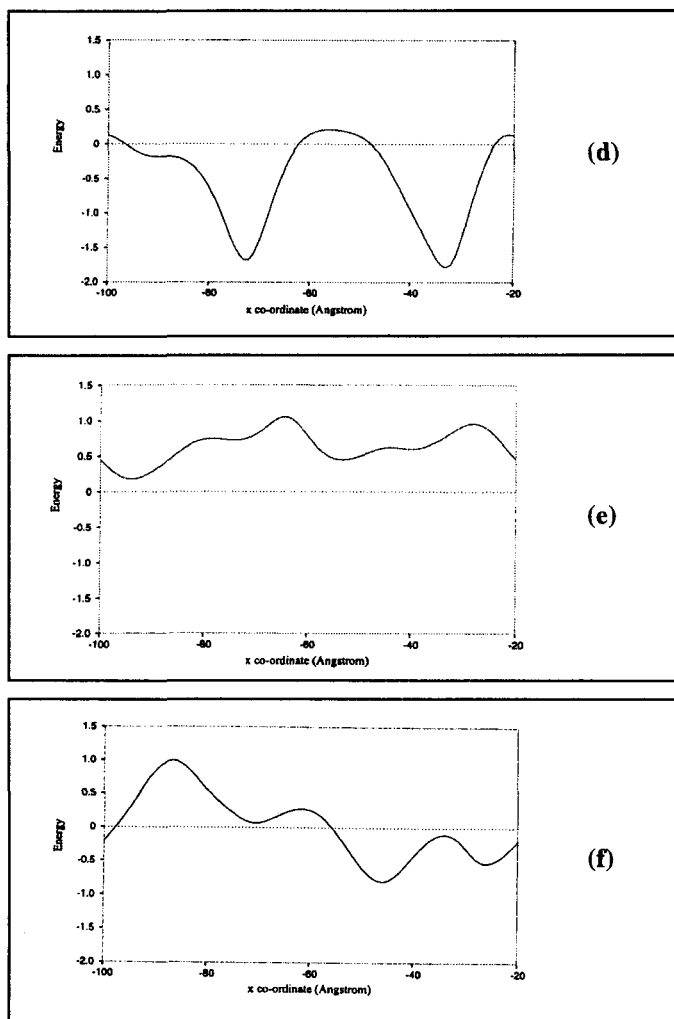


Figure 5. Electrostatic profiles: (a) along line 3 of tubulin's exterior that is on the A side of the protofilament-protofilament interface. (b) Along line 4 of tubulin's exterior that is on the A side of the protofilament-protofilament interface. The profile is largely negative indicating the surface is negatively charged. (c) Along line 5 of tubulin's exterior that is on the outside of the MT. (d) Along line 6 of tubulin's exterior that is on the outside of the MT. The large negative surface charges help to keep the carboxy-tail away from the MT surface. (e) Along line 7 of tubulin's exterior that is on the B side of the protofilament-protofilament interface. The largely positive surface charge is complementary to the opposite side of the dimer and contributes to protofilament-protofilament binding. (f) Along line 8 of tubulin's exterior that is on the B side of the protofilament-protofilament interface.

Along the outer surface of the MT, the profiles in Figures 5c and 5d of the electrostatic potential must be considered. Overall, the surface is either neutral or negatively charged. It is particularly interesting that here again, there are two deep wells which are locations favorable for positively charged protein surfaces such as the head domain of motor proteins. In fact, these wells with a depth of 10-20 kcal/mol and a width of about 1 nm represent a localized electric field of between 10^4 and 10^5 V/m which is not uncommon on atomic scales. In fact, it is this knowledge of the structure of the electrostatic potential that is the basis for an improved model of motor protein motion along MT surfaces but which goes beyond the scope of this thesis. The important features are simply its periodicity in that the α -monomer looks very similar to the β -monomer electrostatically and the presence of binding regions for positively charged substrates.

4 Variations by Tubulin Isotype

An interesting possibility to study involves the comparison of the various tubulin isotypes by considering their electrostatic properties. Since the tubulin structure is now available, it is possible to consider making changes to the structure on a computer and then to calculate the resulting changes in the electrostatic potential surrounding the tubulin dimer. It should be noted that substitution of one amino acid for another may be conservative or non-conservative. The former occurs when the substituted amino acid has similar charge and steric characteristics. However, the interesting changes are those which are non-conservative. It is in this light that we have examined three different substitutions to the β_1 -tubulin structure. In one case, we have simply exchanged one known tubulin isotype for another, and secondly we have made two targeted substitutions based on the discussion of Burns and Surridge [13] who explain that the methylation of α -Lys394 prevents MT assembly and that the substitution of alanine for β -Pro287 specifies 13 protofilaments.

We have used the tubulin structure from the β_2 isotype and made the appropriate substitutions to arrive at the β_1 isotype. We have then looked in the regions where structural differences exist and examined the changes in the electrostatic potential. The changes which are examined here concern the region of β -tubulin from residues 231-235. This region was selected because in a sequence of five amino acid residues, three of them change. The location of these residues is near the inner surface of the MT. The residues ATMSG of the Nogales data are changed for GTMEC from β_2 to β_1 tubulin. The change at position 234 is particularly significant because the serine residue, S, is exchanged for glutamic acid, E, which carries a negative charge. One therefore expects some region to become attractive to protons. In order to interpret the following set of figures, all distances are quoted in Angstroms and energies in kcal/mol. The more darkly colored regions represent locations where the potential is negative and that attract protons or

positively charged molecules. The lightly colored regions are those regions which exclude protons. Figure 6 depicts a contour plot of the electrostatic potential in this region and what is interesting in this case is the channel in the lower left-hand corner of each picture. In the upper figure that represents the β_2 -tubulin isotype, a channel appears to exist that is open for proton movement, while in the lower figure which represents the β_1 -tubulin isotype, a barrier seems to prevent proton movement through this part of the protein.

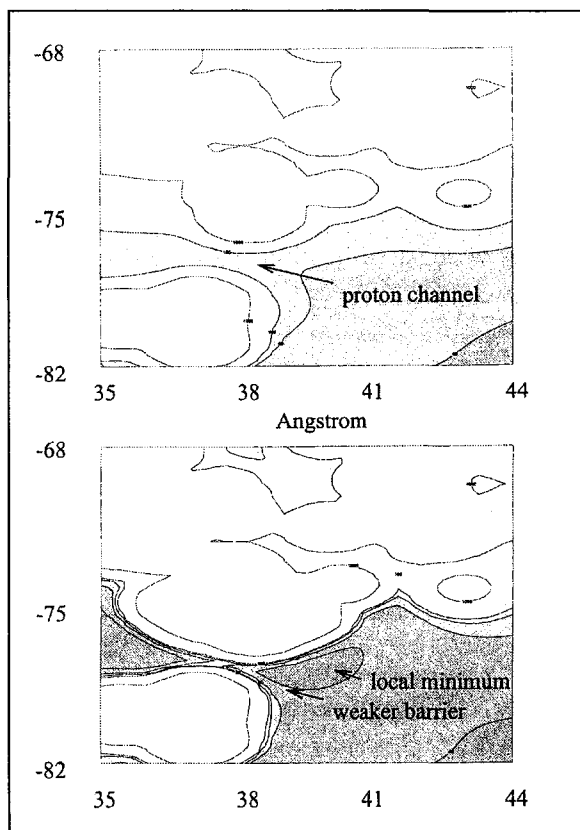


Figure 6. The energy landscape of β_2 -tubulin (above) is compared with that of β_1 -tubulin (below) in the neighbourhood of residues 231-235.

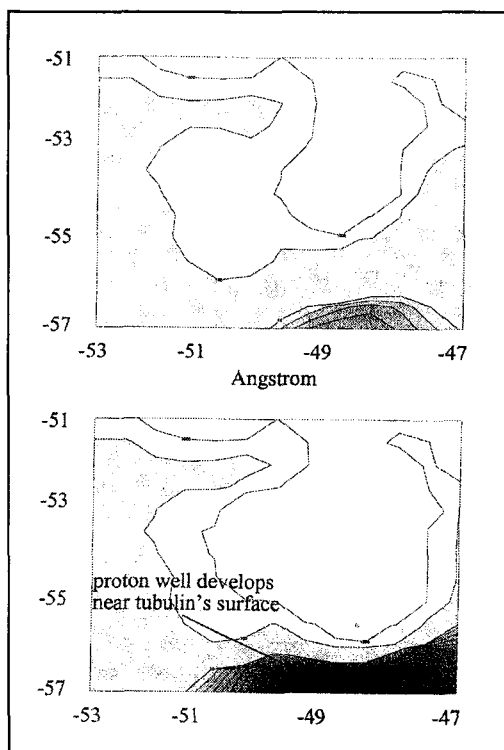


Figure 7. Methylation of α -Lys394 demonstrates the large proton well that develops once the lysine residue is shielded.

Upon closer inspection, we see how the glutamic acid has reduced the potential of the region. In fact, a local minimum exists at the mouth of this channel that will facilitate proton movement through the narrow channel since it may be able to trap them briefly. Thus this one example shows how the nervous system's β_2 -tubulin isotype may trap protons that are to be used to ferry electrons about. The β_1 -tubulin would be more likely to allow the proton to escape. Should a proton become associated with an electron, the narrowness of the channel in the case of β_2 -tubulin would also aid in trapping the proton.

The methylation of α -Lys394 involves the addition of a methyl group to the existing amino group at the end of the lysine residue (see Fig. 7). The location of the α -Lys394 is on the surface of the α -monomer but facing the MT exterior. This change serves to screen this charge from the surroundings since the methyl group has somewhat more bulk than a single hydrogen atom. Consequently, the local electrostatic properties are changed. It is clear that locally the electrostatic potential

is such that protons feel a slight repulsion from the MT surface which agrees with the charge on the lysine residue. However, once the residue has been methylated, a well for positively charged objects develops further up along the MT's outer surface. Given the location of this change which is a long way from the dimer-dimer interaction site which is responsible for the polymerization of protofilaments and also the fact that this is not near the protofilament-protofilament interface, it is difficult to understand how this prevents MT polymerization. The solution to this problem must be by one of two mechanisms. The first would hypothesize the interaction of tail with this negatively charged region of the MT surface and that this interaction impairs MT assembly. Since the tail is negatively charged itself, it would be driven to extend into the cytoplasm away from the MT surface and seems unlikely to interfere with MT polymerization. Consequently, the more likely mechanism is that the tubulin adopts a different configuration that is not conducive to polymerization. In particular, this lysine residue is in the region of the GTP binding region of the α -tubulin monomer. While this molecule of GTP is not hydrolyzed upon MT polymerization, it remains essential for this process. Thus it seems likely that this change prevents GTP association with the α -monomer of the tubulin dimer and thereby prevents MT assembly. This can be rationalized when one considers that the phosphate group carries a negative charge and that consequently, it will be repelled as the GTP molecule attempts to bind the α -monomer.

The substitution of alanine for β -Pro287 was a simple change to make as the β and δ carbon atoms are exchanged for hydrogen atoms and the γ carbon atom deleted. The β -Pro287 residue is compact and its location is on the surface of the β -monomer. It is close to the α -monomer in the axial direction and at the protofilament-protofilament interface in the circumferential direction around the tubulin molecule. The result of the alanine for proline substitution is apparent in Figure 8. Not only does the smaller alanine residue allow protons to come closer to the surface of the tubulin molecule, but in fact, a large binding pocket for a positively charged molecule is developed. Since this pocket becomes so much more prominent after the proline has been substituted by alanine, the binding to a neighbouring protofilament at this location becomes much stronger relative to the unsubstituted tubulin dimer. Provided this location has a corresponding surface with an angular position favouring 13 protofilaments, strengthening this contact will tend to fix the number of protofilaments. By controlling the location of corresponding contacts, the protofilament number could theoretically be adjusted to be most any number. The fact that the contact areas are almost exactly opposite each other on the tubulin dimer gives a MT flexibility to choose a protofilament number that is close to the ideal 13. If these regions are much closer, in angular terms, about the circumference of the tubulin dimer then a lower number of protofilaments would be expected.

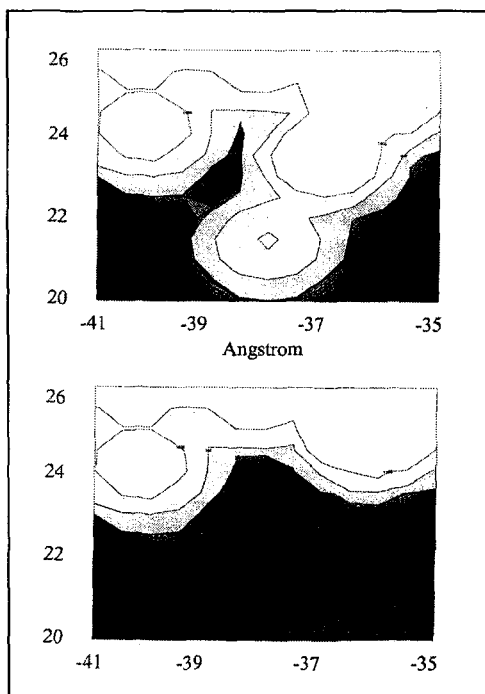


Figure 8. Substitution of alanine for β -Pro287.

This rather cursory look at the electrostatic potential about tubulin has revealed glimpses of some fundamental questions. A periodic potential with a depth comparable to the free energy of ATP hydrolysis has been observed on the outer surface of the MT. This seems to explain the tight binding of motor proteins to MTs and to explain the ATP activation of these molecules. We have also seen how changing between tubulin isotypes changes the electrostatic potential, in our specific case, to modulate proton mobility targeted substitution and post-translational modifications with known physical consequences demonstrate that indeed electrostatics govern tubulin's assembly properties. It is apparent how one side of a tubulin dimer attracts the opposite side of another dimer, how the vertical offset between protofilaments arises and even how the number of protofilaments is specified in the structure of tubulin. Beyond this, the gross structure of tubulin sheets was predicted and then verified. All that remains is to explain dimer-dimer interactions along the protofilament. This will likely have to wait until additional structural information is available on the α - and β -tubulin monomers, and on the free $\alpha\beta$ -dimer. This additional information is required to understand the

dimerization process, that presumably includes the formation of a covalent bond, and to distinguish it from the dimer-dimer association which is much weaker. Since GTP is present near the interface for the formation of both of these bonds and the fact that GTP hydrolysis is concomitant with dimer-dimer association, an understanding of the interaction may be indeed gleaned from electrostatics but one must also consider the GTP molecule and associated water molecules. In the case where a covalent bond forms, the purely electrostatic picture may not be complete since a quantum mechanical interpretation is required.

5 Conclusions

The challenge in the theoretical work in this area is to link concepts of MT structural dynamics and function which are macroscopic properties with the microscopic properties of tubulin's structure, its electrostatic properties, individual MT dynamics, flexural rigidity, post-translational modifications and so forth. The picture which emerges is one in which the MT is perhaps the most multi-functional cellular component due to the microscopic characteristics of the tubulin dimer, which was studied in this paper using atomic resolution data. The conclusion is that the structure's polymerization characteristics will be better understood once the molecule is successfully imaged in additional conformations.

6 Acknowledgments

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References

1. M. C. Ledbetter and K. R. Potter. *A Microtubule in Plant Cell Fine Structure*. J. Cell Biol. **19**, 239-250 (1963).
2. L. A. Amos and W. B. Amos. *Molecules of the Cytoskeleton*. Macmillan Press, London (1991).
3. D. Chretien and R. H. Wade. *New data on the Microtubule Surface Lattice*. Bio. Cell. **71**, 161-174 (1991).
4. E. Nogales, S. G. Wolf, and K. H. Downing. *Structure of the alpha-beta Tubulin Dimer by Electron Crystallography*. Nature (London). **391**, 199-203 (1998).
5. L. A. Amos. *The Microtubule Lattice—20 Years On*. Trends Cell. Biol. **5**, 48-51 (1995).

6. Per J. Kraulis. Molscript: A Program to Produce *Both Detailed and Schematic Plots of Protein Structures*. Journal of Applied Crystallography. **24**, 946-950 (1991).
7. A. Bairoch and R. Apweiler. The SWISS-PROT Protein Sequence Data Bank and Its Supplement TrEMBL in 1998. Nucleic Acids Res. **26**, 38-42 (1998).
8. R. Luduena. *Function and Distribution of Tubulin Isoforms*, Banff Workshop: Molecular Biophysics of the Cytoskeleton August (1997). Presentation.
9. P. G. Wilson and G. G. Borisy. *Evolution of the Multi-tubulin Hypothesis*. Bioessays. **19**, 451-454 (1997).
10. D. L. Sackett. *Subcellular Biochemistry* volume 24 of *Proteins: Structure, Function and Engineering* chapter Structure and Function in the Tubulin Dimer and the Role of the Acidic Carboxyl Terminus. Plenum Press New York (1995).
11. M. J. Dudek and J. W. Ponder. J. Comput. Chem. **16**, 791 (1995).
12. G. Woehlke, A. K. Ruby, C. L. Hart, B. Ly, N. Hom-Booher and R. D. Vale. *Microtubule Interaction Site of the Kinesin Motor*. Cell. **90**, 207-216 (1997).
13. R. G. Burns and C. D. Surridge. *Tubulin: Conservation and Structure*, p. 3-31, John Wiley and Sons, New York, NY (1994).
14. R. D. Vale, C. M. Coppin, F. Malik, F. J. Kull and R. A. Milligan. *Tubulin GTP Hydrolysis Influences the Structure, Mechanical Properties, and Kinesin-driven Transport of Microtubules*. J. Biol. Chem. **269**, 23769-23775 (1994).

PATH INTEGRAL APPROACH TO REACTION IN COMPLEX ENVIRONMENT: A BOTTLENECK PROBLEM

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The path integral method for handling the polaron problem, as developed by Feynman [1], is applied to the problem of the rate of reaction of a system coupled to a complex environment consisting of an infinite set of oscillators. After eliminating the harmonic oscillator degrees of freedom, an effective action containing a reaction coordinate coupled to the non-local harmonic oscillator, is obtained. This non-local behavior represents the complex heat bath of the system and can be expressed in terms of a spectral function. For a simple system with a single dominant frequency, and simple reaction coordinates containing quadratic terms, the path integral can be calculated exactly. In this simple model there are three parameters: two represent the frequency ω , the amplitude of the environment κ , and the third, α , represents the strength of the bottleneck potential.

This paper derives the survival path, the effective rate coefficient, the correlation function and the survival probability from the generating functional associated with the effective action. These results are compared with the work of Wang and Wolynes [2].

1 Introduction

The transport process of complex systems such as liquids, glasses and biomolecules over a barrier has been the subject of many studies. For molecules in a complex environment, the overall barrier-crossing reaction rate can be treated as classical phenomenological chemical kinetics. Numerous treatments of a molecule reaction dynamic using the reaction diffusion equation approach have been reported. However this approach is not applicable to reactions in a highly viscous environment. For example, Frauenfelder and Wolynes [3] show that, in the case of carbon monoxide, recombination of myoglobin needs a higher-barrier relax equation of a highly non-exponential property.

Using Feynman's Path Integration, the problem of survival paths for reaction dynamics in fluctuating environments has been investigated extensively by Wang and Wolynes [2,4] beginning with a consideration of a simple model of the rate process in general Gaussian fluctuating environments. This approach assumes that the fluctuations relax exponentially according to the stretched exponential law given by

$$\langle r(\tau) r(\sigma) \rangle = \theta \exp[-(\lambda |\tau - \sigma|)^\beta] \quad (1)$$

where θ is the amplitude of equal time correlation, λ is the frequency or relaxation rate and β denotes the stretched parameters.

The $\beta=1$ case corresponds to exponentially relaxing fluctuations, while $\beta<1$ corresponds to the more general fluctuating environment, with stretched exponential relaxation encountered in glasses and biomolecules. To obtain an analytic result, Wang and Wolynes [4] assume that the reaction terms are linear and quadratic; α^2 , where, α is another parameter denoting the strength of the reaction oscillator. These linear and quadratic reaction terms, together with stretched exponential fluctuation in an analytical survival path with an effective rate as a constant, are obtained. In order to treat a more general effective rate coefficient, Wang and Wolynes [5] have generalized the calculation of the reaction in a complex environment by using the instanton method to calculate the rate coefficient.

This paper follows the idea of Wang and Wolynes [4] by using a path integral method for handling the reaction rate within a complex environment. However instead of assuming that the fluctuation decays as a stretched exponent, the rate constant with the reaction coordinate coupled to the heat bath or an infinite set of oscillators is used. This microscopic Hamiltonian model to describe a dissipative system was first introduced by Ullesma (1966) and became popular after Caldeira and Leggett [7] applied it to the tunneling problem. It is also assumed that the rate coefficient has a Gaussian dependence on its environment coordinates. Adopting the Caldeira and Leggett model, and after eliminating the oscillators, an effective action was obtained. To simplify the discussion only a single dominant oscillator is considered. Within this limitation, the effective action constraint of non-local action can be solved exactly. Further this paper develops a Lagrangian model in section 2; discusses the bottleneck problem in section 3; and the survival path, the correlation function, the effective rate coefficient and the survival probability in sections 4 and 5, respectively. Several limiting cases as well as numerical results are given in section 6 and the results are compared with the work of Wang and Wolynes. Finally section 7 follow discussion conclusion.

2 The Lagrangian Model

2.1 Wang and Wolynes Model

In this section, the Wang and Wolynes [2,4] mathematical formulae for reactions in fluctuating environment are used. The survival probability associated with a reaction whose environment is described by a generalized Langevin or Fokker-Plank equation can be presented as a functional integral over paths, each exponentially

weighted according to the time integral of an appropriate Lagrangian along that path. Wang and Wolynes illustrate this by starting with the path probabilities in the absence of reaction and write:

$$P(r_f, r_i; T) = \exp\left[-\int_0^T L(r, \dot{r}, \tau) d\tau\right]. \quad (2)$$

This gives the probability for observing a particular path $r(\tau)$ with the boundary value r_f for the stochastic variable r at time T and the boundary value $r_i=r_i$. This path probability gives rise to a generalized Langevin equation

$$dr + h(r) d\tau = dw(\tau), \quad (3)$$

where dw is the Gaussian white-noise term; and $h(r)$ is the mean force as a function of r . The path probability then is:

$$P(r_f, r_i; T) = \exp\left[-\frac{1}{2} \int_0^T [\dot{r} + h(r)]^2 d\tau + \alpha \int_0^T h'(r) d\tau\right], \quad (4)$$

where, $h'(r)$ is the derivative of $h(r)$ with respect to r . (Note that this model is local in time.)

If $h(r)$ is quadratic, then the model is a Gaussian one. If $h(r)$ is not quadratic, then, in general, the model is non-Gaussian. From the complex systems Wang and Wolynes studied, it is necessary to consider cases where there is correlation between fluctuations of the coordinates r 's at different times; that is, memory effects need to be taken into account. In this case Wang and Wolynes extended the formulation for a general Gaussian memory form:

$$P(r_f, r_i; T) = \exp\left[-\int_0^T \int_0^T r(\tau) A(\tau, \tau') r(\tau') d\tau d\tau'\right], \quad (5)$$

where A is defined as

$$\int d\tau' A(\tau, \tau') \langle r(\tau'), r(\tau') \rangle = \delta(\tau, \tau') \quad (6)$$

Wang and Wolynes define A^{-1} as the function inverse of A , thus, $A^{-1}(\tau, \tau') = \langle r(\tau), r(\tau') \rangle$ is the correlation function between the variable r 's at different times labelled by τ and τ' . (The brackets refer to an average over noise.)

In general, the correlation function can have many forms of time dependence. In complex systems such as proteins, glasses or complex structured fluids, non-exponential decay of the correlation function, which can be fitted to a stretched exponential law as Eq. (1), are often encountered, taking the path probability given in Eq. (2) to be valid when there is no reaction. The reaction, by recognizing that the survival probability decays along any given trajectory by the first-order kinetic equation, can also be taken into account. For simplicity, the back-reaction can be ignored:

$$dP/d\tau = -K(r)P \quad (7)$$

where $K(r)$ is the rate coefficient which depends on the environment fluctuation coordinate r . By combining the Eqs. (2) to (7) Wang and Wolynes obtain a path integral expression for the calculation of the survival probability:

$$P(r_f, r_i; T) = \frac{\int Dr(\tau) \exp \left[- \int_0^T K(r) d\tau - \frac{1}{2} \int_0^T \int_0^T r(\tau) A(\tau, \tau') r(\tau') d\tau d\tau' \right]}{\int Dr(\tau) \exp \left[- \frac{1}{2} \int_0^T \int_0^T r(\tau) A(\tau, \tau') r(\tau') d\tau d\tau' \right]} \quad (8)$$

When the surviving population seeks out path $r(\tau)$, it is because the path probability is a local maximum. When variation of the exponential of the path probability with respect to $r(\tau)$ is undertaken, a nonlinear integral equation is obtained:

$$r(\tau) = - \int_0^T \frac{dK(r)}{dr} A^{-1}(\tau, \tau') d\tau' \quad (9)$$

where τ and τ' are within the range of 0 and T , the variation equation for the general Gaussian fluctuating environment. The survival probability can easily be calculated by substituting the dominant path solution into the exponential of the path integral formulation. The rate coefficient is weakly dependent on the environment variable, the dominant survival path following the ordinary relaxation to equilibrium as in the Onsager [6] regression hypothesis. When the rate coefficient strongly depends on the environmental variable, the dominant survival path exhibits behavior very distinct from ordinary relaxation, including reflection off rapid variations in the rate constant, as well as refraction, giving paths very different from equilibrium relaxation.

2.2 Present Model

By considering the Wang and Wolynes [2,4] path probability of the surviving path along a given trajectory the first order kinetic equation can be written as Eq. (7). Next the reaction coordinate is coupled to the environment, and, in this case, as a set of an infinite number of oscillators as discussed by Caldeira and Leggett [7] and Poulter and Sa-yakanit [10] were introduced. Therefore, the Lagrangian model is:

$$L = \frac{m}{2} \dot{r}^2 - K(r) + \frac{1}{2} \sum_j m_j [\dot{x}_j^2 - \kappa_j (r - x_j)^2] \quad (10)$$

where r is the reaction coordinate with mass m moving in a potential $K(r)$ and x_j , m_j , κ_j are the coordinates, mass and coupling constant of the environment oscillators, respectively. By eliminating the environmental degrees of freedom, an effective action is obtained:

$$S_{\text{eff}} = \int_0^T d\tau \left[\frac{m}{2} \dot{r}^2(\tau) - K(r) \right] - \frac{1}{2} \int_0^T \int_0^T d\tau d\sigma g(\tau - \sigma) |r(\tau) - r(\sigma)|^2. \quad (11)$$

Here, $g(\tau - \sigma)$ is the Green function

$$g(\tau - \sigma) = \frac{1}{2\pi} \int_0^\infty d\omega J(\omega) \left\{ \frac{\cosh[\omega(|\tau - \sigma| - T/2)]}{\sinh[\omega T/2]} \right\} \quad (12)$$

and J denotes the spectral function

$$J(\omega) = \frac{\pi}{2} \sum_j m_j \kappa_j \omega_j \delta(\omega - \omega_j) \quad (13)$$

with $\omega_j = (\kappa_j/m_j)^{1/2}$.

This spectral function represents the heat bath of the system. In general, this spectral function is very complicated. Physically, it must be terminated by a certain cut-off frequency such as the Debye cut-off in the lattice dynamic problem and the electron-plasmon interaction employed in the electron gas problem. In the dissipation system there is a well known empirical expression [11]:

$$J(\omega) = \eta \omega^s e^{-\omega/\omega_c}, \quad (14)$$

where η is the friction constant, s is the power of the ω , and ω_c is the oscillator cut-off frequency. Further it is shown that if $s=1$ this expression can lead to ohmic

friction. The case $0 < s < 1$ and $s > 1$ are known as sub-ohmic and super-ohmic, respectively. It is also assumed that there exists a single oscillator that dominates the spectral function and is identified as ω and κ with $\omega = (\kappa/m)^{1/2}$ with m equal to the fictitious mass.

3 Bottleneck Problem

The action from Eq. (11) is obviously a translation invariant and therefore cannot lead to the equilibrium path. In order to obtain the equilibrium path, the action is rewritten with explicit symmetry breaking. Then the action becomes:

$$S_{eff} = \int_0^T d\tau \left[\frac{m}{2} \dot{r}^2(\tau) - K_R(r) \right] + \frac{m\kappa\omega}{4} \int_0^T \int_0^T d\tau d\sigma \left\{ \frac{\cosh[\omega(|\tau - \sigma| - T/2)]}{\sinh[\omega T/2]} \right\} r(\tau)r(\sigma) \quad (15)$$

where $K_R(r)$ is the renormalized rate coefficient as,

$$K_R(r) = K(r) + \frac{m}{2} \kappa r^2. \quad (16)$$

$K_R(r) = (m/2)\alpha r^2$ is related to the Wang and Wolynes geometrical bottleneck problem, where α is the strength of the bottleneck rate coefficient. This model is used by many authors for calculation of the *CO* in myoglobin or the transport through a bottleneck. Then, a bottleneck reaction is obtained:

$$S_B = \int_0^T d\tau \left[\frac{m}{2} \dot{r}^2(\tau) - \alpha r^2(\tau) \right] + \frac{m\kappa\omega}{4} \int_0^T \int_0^T d\tau d\sigma \left\{ \frac{\cosh[\omega(|\tau - \sigma| - T/2)]}{\sinh[\omega T/2]} \right\} r(\tau)r(\sigma). \quad (17)$$

Since this action is again quadratic the classical action can be calculated exactly. The result is:

$$S_{cl, B} = \frac{m}{2\omega} \ddot{\Delta}(0) (r_f - r_i)^2 + \frac{m\omega}{8\Delta(0)} (r_f + r_i)^2 \quad (18)$$

where

$$\Delta(\tau) = \frac{\omega}{2} \left(\frac{\Omega^2 - \omega^2}{\Omega^2 - \psi^2} \right) \frac{\cosh[\Omega(\tau - T/2)]}{\Omega \sinh[\Omega T/2]} + \frac{\omega}{2} \left(\frac{\psi^2 - \omega^2}{\psi^2 - \Omega^2} \right) \frac{\cosh[\psi(\tau - T/2)]}{\psi \sinh[\psi T/2]} \quad (19)$$

with

$$\Omega^2 = \frac{1}{2}(\omega^2 - \alpha) + \frac{1}{2}\sqrt{(\omega^2 - \alpha)^2 - 4\omega^2(\kappa - \alpha)} \quad (20)$$

and

$$\psi^2 = \frac{1}{2}(\omega^2 - \alpha) - \frac{1}{2}\sqrt{(\omega^2 - \alpha)^2 - 4\omega^2(\kappa - \alpha)} \quad (21)$$

In order to calculate the survival path and correlation function, a generating function that introduces the driving force $f(\tau)$ must be constructed. Then the general action is:

$$\begin{aligned} S[f] = & \int_0^T d\tau \left\{ \frac{m}{2} [\dot{r}^2(\tau) - \alpha r^2(\tau)] + f(\tau) r(\tau) \right\} \\ & + \frac{m\kappa\omega}{4} \int_0^T \int_0^T d\tau d\sigma \left\{ \frac{\cosh[\omega(|\tau - \sigma| - T/2)]}{\sinh[\omega T/2]} \right\} r(\tau) r(\sigma) \end{aligned} \quad (22)$$

with the classical action $S_{cl}[f]$,

$$\begin{aligned} S_{cl}[f] = & \frac{m}{2\omega} \ddot{\Delta}(0) (r_f - r_i)^2 + \frac{m\omega}{8\Delta(0)} (r_f + r_i)^2 + \int_0^T d\tau f(\tau) \left[\frac{\dot{\Delta}(\tau)}{\omega} (r_f - r_i) + \frac{\Delta(\tau)}{2\Delta(0)} (r_f + r_i) \right] \\ & + \frac{1}{2m\omega} \int_0^T \int_0^T d\tau d\sigma f(\tau) f(\sigma) \left[\frac{\Delta(\tau)\Delta(\sigma)}{\Delta(0)} - \Delta(|\tau - \sigma|) \right]. \end{aligned} \quad (23)$$

Then the propagator becomes

$$P(r_f, r_i; T) = F(T) e^{-S_{cl}} \quad (24)$$

where the prefactor is given by

$$F(T) = \left(\frac{m\omega}{8\pi\Delta(0)} \right)^{1/2} \frac{\sinh[\omega T/2]}{\sinh[\Omega T/2] \sinh[\psi T/2]} \quad (25)$$

From Eq. (23) the generating function allowing for the carrying out of the calculations for all physical quantities can be calculated. The end points can be obtained by differentiating S_{cl} with respect to r_f and expressing r_f in terms of r_i as

$$r_f = \left(\frac{4\Delta(0)\ddot{\Delta}(0) - \omega^2}{4\Delta(0)\dot{\Delta}(0) + \omega^2} \right) r_i + \frac{1}{\frac{m}{\omega}\ddot{\Delta}(0) + \frac{m\omega}{4\Delta(0)}} \int_0^T d\tau f(\tau) \left[\frac{\dot{\Delta}(\tau)}{\omega} + \frac{\Delta(\tau)}{2\Delta(0)} \right]. \quad (26)$$

Further, the average classical path $R(\tau)$ is shown by differentiating Eq. (23) once with respect to $f(\tau)$:

$$\begin{aligned} R(\tau) = \langle r(\tau) \rangle = & \left[\frac{\dot{\Delta}(\tau)}{\omega} + \frac{\Delta(\tau)}{2\Delta(0)} \right] r_f - \left[\frac{\dot{\Delta}(\tau)}{\omega} - \frac{\Delta(\tau)}{2\Delta(0)} \right] r_i \\ & + \frac{1}{2m\omega} \int_0^T d\sigma f(\sigma) \left[\frac{\Delta(\tau)\Delta(\sigma)}{\Delta(0)} - \Delta(|\tau - \sigma|) \right]. \end{aligned} \quad (27)$$

The correlation function $C(\langle r(\tau)r(\sigma) \rangle)$ can be obtained from the generating function by differentiating Eq. (23) twice with respect to $f(\tau)$:

$$\begin{aligned} C(\langle r(\tau)r(\sigma) \rangle) &= \langle r(\tau) r(\sigma) \rangle - \langle r(\tau) \rangle \langle r(\sigma) \rangle \\ &= -\frac{1}{2m\omega} \left[\frac{\Delta(\tau)\Delta(\sigma)}{\Delta(0)} - \Delta(|\tau - \sigma|) \right] \end{aligned} \quad (28)$$

The effective rate K_{eff} , can be obtained by taking the trace of the propagator for large T :

$$K_{eff} = \frac{1}{T} \ln \text{Tr} P(r_f, r_i; T) \quad (29)$$

and to obtain the survival probability $P(T)$, the propagator in Eq. (24) was considered by setting $f(\tau)=0$ and $r_f=r_i=0$,

$$P(T) = P(0,0;T) = F(T). \quad (30)$$

The above results will be discussed in details in the next section.

4 Survival Paths and Correlation Function

In this section the survival path and the correlation function is discussed. The survival path can be obtained from Eq. (27) by setting $f(\tau)=0$. The results are given in Fig. 1 for $\alpha=0$, the equilibrium path, and in Fig. 3 for $\alpha > \kappa$, the unstable path. For $\alpha < \kappa$, all paths decay exponentially as shown in Fig. 2. The correlation function can be obtained from Eq. (28) and it represents the same behavior to the survival path as shown in Fig. 4.

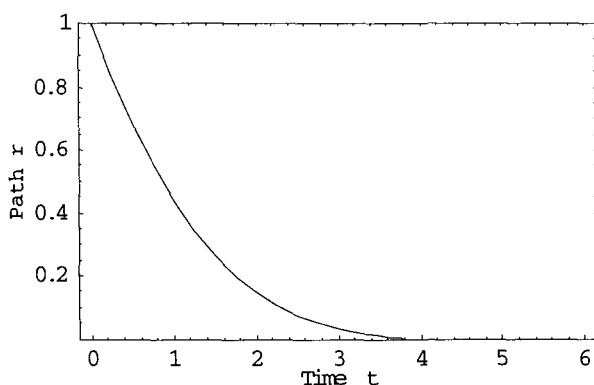


Figure 1. The equilibrium path for $\alpha=0$ with $\omega=\kappa=1$.

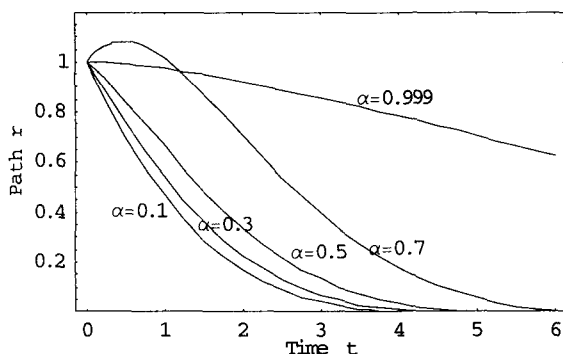


Figure 2. The survival paths $\alpha < \kappa$ for any α with $\omega=\kappa=1$.

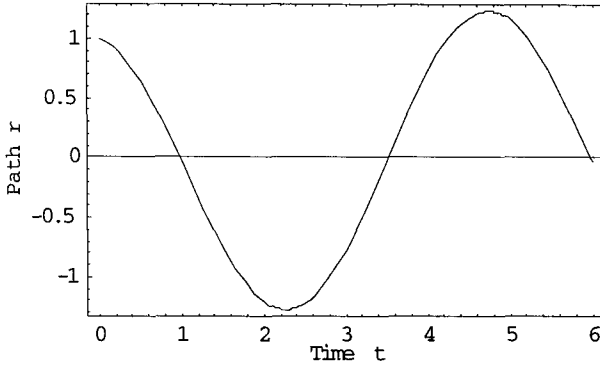


Figure 3. The unstable path for $\alpha > \kappa$; $\alpha=2$ with $\omega=\kappa=1$.

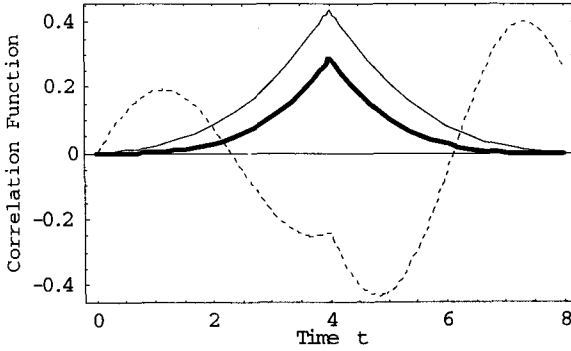


Figure 4. The correlation function; thick line for $\alpha=0$, light line for $\alpha=0.5 < \kappa$ and dashed line for $\alpha=2 > \kappa$ setting $\omega=\kappa=1$ and $\sigma=4$.

5 Effective Rate Coefficient and Survival Probability

The rate coefficient can be obtained from Eq. (29) for large T . The rate approach K_{eff} result is shown in Fig. 5. The survival probability can be obtained from Eq. (30) and the result is presented in Fig. 7. This result can be compared with the experiments given in Fig. 6.

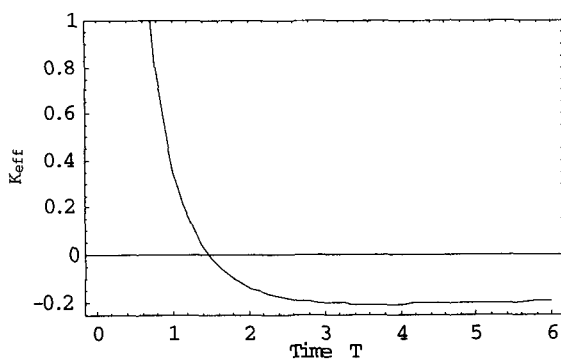


Figure 5. The effective rate coefficient; $\alpha=0.5$ setting $\omega=\kappa=1$.

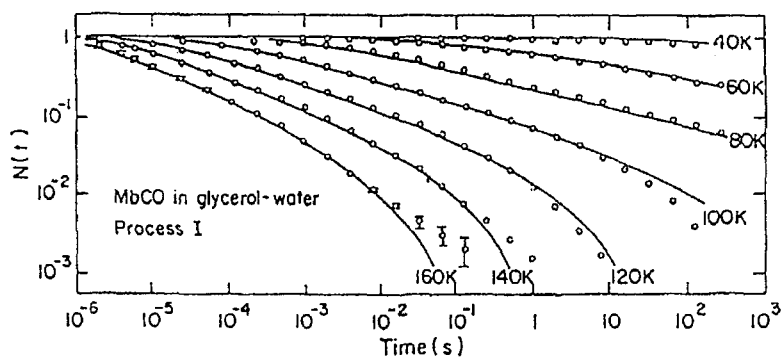


Figure 6. Survival probability adapted from H. Frauenfelder [15].

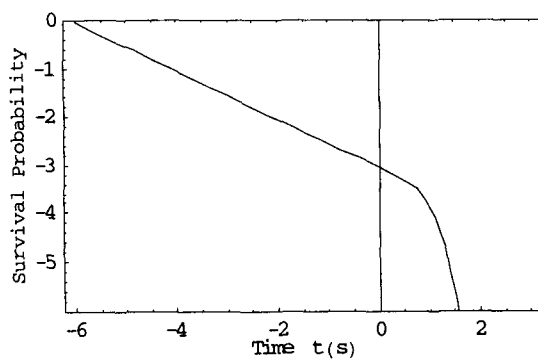


Figure 7. Log-Log plot of the survival probability vs. time setting $m=0.000005$, $\omega=\kappa=1$ and $\alpha=0.5$.

6 Limiting Cases

This section discusses the survival path using the results of the previous section in comparison with Wang and Wolynes [2,4] and showing that their result can be derived from the present result by further constraint the parameters. Two limiting cases are considered: $\Omega = \psi$ and $\psi \rightarrow 0$

6.1 Case $\Omega = \psi$

This allows for a reduced Eqs. (20) and (21) to

$$\Omega^2 = \frac{1}{2}(\omega^2 - \alpha) \quad (31)$$

with the condition $\kappa = \alpha + \omega^2 - \alpha^2/(4\omega^2)$ implying that $\kappa > \alpha$. Therefore the path will decay exponentially. Then Eq. (19) is simplified to

$$\begin{aligned} \Delta_1(\tau) = & \frac{\omega}{4\Omega^3 \sinh[\Omega T/2]} \left[(\Omega^2 + \omega^2) \sinh[\Omega T/2] \cosh[\Omega(\tau - T/2)] \right. \\ & \left. - \frac{T}{2} \Omega(\Omega^2 - \omega^2) \cos[\Omega \tau] + \Omega(\Omega^2 - \omega^2) \sinh[\Omega T/2] \sinh[\Omega(\tau - T/2)] \right] \end{aligned} \quad (32)$$

Substitute the above two equations in the generating function setting $f(\tau)=0$, the relation with end-points in Eq. (26) then becomes

$$r_f = \left(\frac{4\Omega^2 \Delta_1^2(0) - \omega^2}{4\Omega^2 \Delta_1^2(0) + \omega^2} \right) r_i. \quad (33)$$

The average classical path can be obtained from Eq. (27):

$$R(\tau) = \left[\frac{\dot{\Delta}_1(\tau)}{\omega} + \frac{\Delta_1(\tau)}{2\Delta_1(0)} \right] r_f - \left[\frac{\dot{\Delta}_1(\tau)}{\omega} - \frac{\Delta_1(\tau)}{2\Delta_1(0)} \right] r_i. \quad (34)$$

The correlation function can be obtained from Eq. (29). The result is

$$C(< r(\tau)r(\sigma) >) = -\frac{1}{2m\omega} \left[\frac{\Delta_1(\tau)\Delta_1(\sigma)}{\Delta_1(0)} - \Delta_1(|\tau - \sigma|) \right]. \quad (35)$$

This expression can be compared with the expression given by Wang and Wolynes in Eq. (1) where the above result corresponds to $\beta=1$. However in this study the correlation function was derived from the first principle microscopic model system. The rate coefficient can be derived from the pre-factor of the propagator given in Eq. (25). The pre-factor becomes

$$F(T) = \left(\frac{m\omega}{8\pi\Delta_1(0)} \right)^{1/2} \frac{\sinh[\omega T/2]}{\sinh^2[\Omega T/2]} \quad (36)$$

The effective rate coefficient from Eq. (29) is

$$K_{eff} = \frac{1}{T} \ln \left[\frac{1}{2} \frac{\sinh[\omega T/2]}{\sinh^2[\Omega T/2]} \right] \quad (37)$$

The asymptotic of the propagator was considered at the large time T , thus:

$$Tr P \approx e^{-K_{eff}T} = \exp[(\omega/2 - \Omega)T] \quad (38)$$

This expression can be rearranged by expressing it in terms of Ω . Then the effective rate coefficient becomes

$$K_{eff} = \frac{\omega}{2} [2^{1/2} (1 - (\omega/\omega^2))^{1/2} - 1] \quad (39)$$

This expression can be compared with Wang and Wolynes [4] as

$$K_{eff}^{WW} = \frac{\lambda}{2} [(1 + (4\alpha\theta)/\lambda)^{1/2} - 1] \quad (40)$$

This result differs from Wang and Wolynes by a factor of $2^{1/2}$. Note that both results are identical: $\theta \approx 1/(4m\Omega)$ and is evident in the following result on the correlation function.

Wang and Wolynes case

For comparison, we present Wang and Wolynes expression:

$$R^{WW}(\tau) = \frac{1}{2\sinh[\lambda T]} \left[(r_f - r_i e^{-\lambda\tau}) e^{\lambda\tau} + (r_f e^{\lambda\tau} - r_i) e^{-\lambda\tau} \right] \quad (41)$$

One shows that for case $\omega = \Omega$ the survival path in Eq. (34) approaches to Wang and Wolynes result. In our research the classical survival path from Eq. (34) is

$$R(\tau) = \frac{1}{\sinh[\Omega T]} (\sinh[\Omega \tau] r_f + \sinh[\Omega(T - \tau)] r_i) \quad (42)$$

and the relation of the end-points from Eq. (26) is

$$r_f = \frac{1}{\cosh[\Omega T]} r_i \quad (43)$$

The correlation function from Eq. (35) for $\tau > \sigma$ is

$$C(< r(\tau) r(\sigma) >) = \frac{1}{2m\Omega \sinh[\Omega T]} \sinh[\Omega(T - \tau)] \sinh[\Omega \sigma] \quad (44)$$

For long-time, it becomes

$$C(< r(\tau) r(\sigma) >) = \frac{1}{4m\Omega} e^{-\Omega \tau - \sigma} \quad (45)$$

This result confirms the previous value of θ as $\theta = 1/4m\Omega$. The rate coefficient from Eq. (29) is:

$$K_{eff} = \frac{1}{T} \ln \left[\frac{1}{2 \sinh[\Omega T / 2]} \right] \quad (46)$$

The survival probability from Eq. (30) becomes

$$P(T) = F(T) = \sqrt{\frac{m\Omega}{2\pi \sinh[\Omega T]}} \quad (47)$$

These results for the classical path, correlation function, rate coefficient and survival probability are presented in Figs. 8, 9, 10 and 11, respectively.

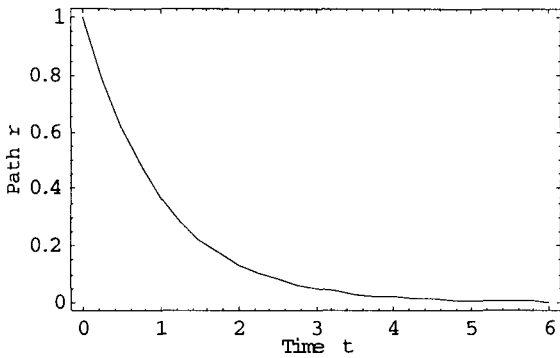


Figure 8. The survival path for $\Omega=\psi$ with $\Omega=\omega$ compared with the result of Wang and Wolynes in survival path for $\Omega=1$ and large T limit.

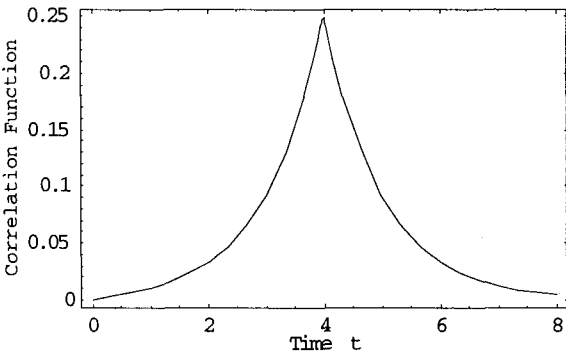


Figure 9. The correlation function of Eq. (44) with $\Omega=\omega$ setting $\Omega=1$, $\sigma=4$ and large T limit.

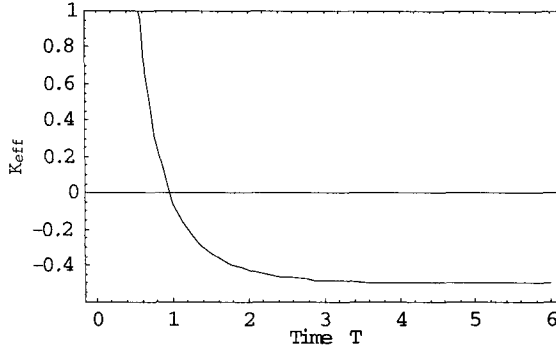


Figure 10. The effective rate coefficient in $\Omega=\psi$ case with $\Omega=\omega$ and setting $\Omega=1$.

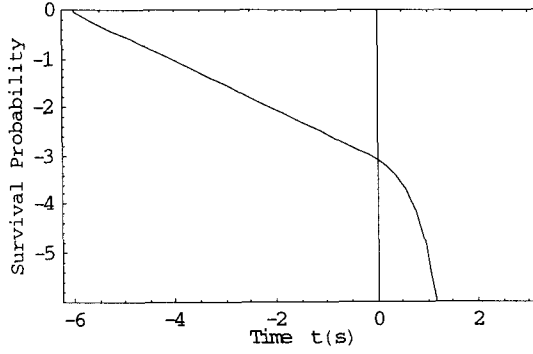


Figure 11. Log-Log plot of the survival probability for $\Omega=\psi$ with $\Omega=\omega$ setting $\Omega=1$ and $m=0.000005$.

6.2 $\psi \rightarrow 0$

This case corresponds to $\kappa-\alpha \rightarrow 0$, or $\psi \rightarrow 0$ with

$$\Omega^2 = \omega^2 - \alpha . \quad (48)$$

Then, $\Delta(\tau)$ given in Eq. (19) is reduced to $\Delta_2(\tau)$,

$$\Delta_2(\tau) = \frac{\omega}{2} \left(\frac{\omega}{\Omega} \right)^2 \frac{\cosh[\psi(\tau - T/2)]}{\psi \sinh[\psi T/2]} . \quad (49)$$

To express r_f in terms of r_i , Eq. (23) is used to obtain

$$r_f = r_i . \quad (50)$$

The survival path is obtained from Eq. (27) by setting $f(\tau)=0$,

$$R(\tau) = \left(\frac{\omega}{\Omega} \right)^2 \left[-\frac{1}{2} + \frac{\tau}{T} - \frac{\alpha}{\omega^2} \frac{\sinh[\Omega(\tau - T/2)]}{2 \sinh[\Omega T/2]} \right] (r_f - r_i) + \frac{1}{2} (r_f + r_i) . \quad (51)$$

This can be expressed in terms of end points using Eq. (50) as

$$R(\tau) = r_i . \quad (52)$$

This result shows that there is no exponential decay path. Next the correlation function was considered to obtain from [13] in the case where $\tau > \sigma$ as

$$C(\langle r(\tau)r(\sigma) \rangle) = - \left(\frac{\alpha}{m\Omega^3} \frac{\sinh[\Omega(T-\tau)/2] \sinh[\Omega\sigma/2] \cosh[\Omega(T-\sigma)/2]}{\sinh[\Omega T/2]} \right) + \frac{1}{2m} \left(\frac{\omega}{\Omega} \right)^2 \left(\frac{\tau\sigma}{T} - \sigma \right) \quad (53)$$

For large time, $T \rightarrow \infty$, the correlation in time was determined as

$$C(\langle r(\tau)r(\sigma) \rangle) = - \frac{1}{2m} \left(\frac{\omega}{\Omega} \right)^2 \sigma . \quad (54)$$

This correlation function approaches a constant confirming of the previous result that the correlation function diversifies.

The effective coefficient rate for the case $\psi \rightarrow 0$ from Eq. (29) is

$$K_{eff} = \frac{1}{T} \ln \left[\frac{1}{2} \frac{\sinh[\omega T/2]}{\sinh[\Omega T/2] \sinh[\psi T/2]} \right] . \quad (55)$$

For large T then

$$Tr P = \exp [(\omega - \Omega) T/2] . \quad (56)$$

This expression can show the effective rate as

$$K_{eff} = \frac{\omega}{2} [(1 - (\alpha/\omega^2))^{1/2} - 1] \quad (57)$$

and the survival probability becomes

$$F(T) = \left(\frac{m}{2\pi T} \right)^{1/2} \frac{\Omega \sinh[\omega T / 2]}{\omega \sinh[\Omega T / 2]} . \quad (58)$$

For this subsection $\psi \rightarrow 0$, the effective coefficient rate and the survival probability are presented in Figs. 12 and 13, respectively.

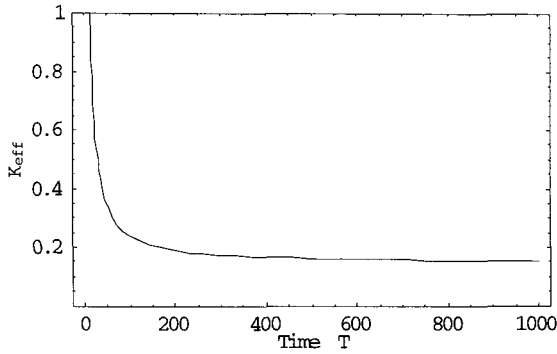


Figure 12. The effective rate coefficient case $\psi=0.000001 \rightarrow 0$ setting $\omega=1$ and $\alpha=0.5$.

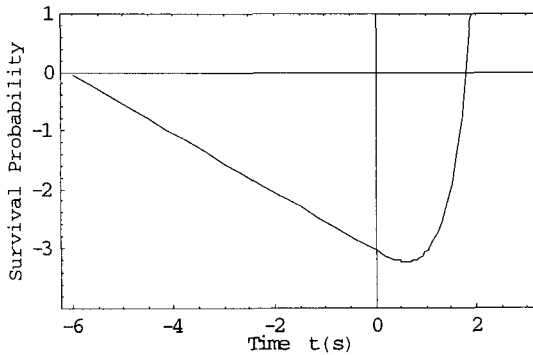


Figure 13. Log-Log plot of the survival probability for $\psi \rightarrow 0$ setting $m=0.000005$, $\Omega=1$ and $\alpha=0.5$.

7 Discussion and Conclusion

In this paper, the Feynman path integral method was applied to the rate reaction coupled to a complex environment, the model consisting of the reaction coefficient coupled to the microscopic heat bath with an infinite set of oscillators. This heat bath is assumed to behave as in Eq. (14) and consists of two adjustable parameters s and ω_c . The s represents the deviation from the ohmic friction and ω_c is the frequency cut-off.

This empirical spectral function suggests that there exists a single dominant frequency occurring at $\omega \approx s\omega_c$. This can be seen by maximizing the spectral function Eq. (14). However with this single oscillator there are still two parameters in the model: κ representing the amplitude of fluctuation and ω representing the frequency of the oscillator. How these two parameters allow a discussion of a wide range of physical quantities is shown.

The bottleneck problem introduced by Wang and Wolynes[4] corresponds to adding additional quadratic potential with α representing the amplitude of the bottleneck. Because the bottleneck models are quadratic all path integrals can be performed exactly. The generating function associated with the effective action is obtained and used to calculate the survival path given in Eq. (27), the correlation function in Eq. (28), with the effective rate and the survival probability given in Eqs. (29) and (30), respectively.

This paper shows that for $\alpha=0$ the equilibrium path is obtained. When $\alpha > \kappa$ the survival path is unstable and starts to oscillate instead of decaying exponentially. This behavior also appears in the correlation function for $\alpha > \kappa$. For $\alpha < \kappa$ all survival paths are stable and decay exponentially. The correlation function is calculated and shows similar in behavior as the survival path. The effective rate is also derived from the pre-factor of the propagator limit of large T . Finally it is shown that for the survival probability can be obtained from the pre-factor. In order to compare the results with those of Wang and Wolynes, several limiting cases were considered. It is shown that for the special case of $\Omega = \psi$ exactly the same equilibrium result as Wang and Wolynes was obtained.

Finally, the method developed in the paper can be generalized to include the full spectral function with a complete and complex environment as well as a more complicated coefficient reactions such as Gaussian or exponential.

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References

1. Feynman R. P., Slow Electrons in a Polar Crystal, *Phys. Rev.* **97** (1955) pp. 660-665.
2. Wang J. and Wolynes P., Survival paths for reaction dynamics in fluctuating environments., *Chem. Phys.* **180** (1994) pp. 141-156.
3. Frauenfelder H. and Wolynes P. G., Rate theories and Puzzles of Hemoprotein Kinetics., *Science* **229** (1985) pp. 337-345.
4. Wang J. and Wolynes P., Passage through fluctuating geometrical bottlenecks. The generation Gaussian fluctuating case., *Chem. Phys. Lett*, **212** (1993) pp. 427-433.
5. Wang J. and Wolynes P., Instantons and the Fluctuating Path Description of Reactions in Complex Environments., *J. Phys. Chem.* **100** (1996) pp. 1129-1136.
6. Onsager L. and Machulp S., Fluctuations and Irreversible Processes., *Phys. Rev.* **91** (1953) pp.1505-1515.
7. Caldeira A. O. and Leggett A. J., Quantum Tunneling in a Dissipative System., *Ann. Phys.* **149** (1983) pp. 374-456.
8. Sa-yakanit V., Electron density of states in a Gaussian random potential: Path-integral approach, *Phys. Rev.* **19** (1979) pp. 2266-2275.
9. Sa-yakanit V., The Feynman effective mass of the polaron, *Phys. Rev.* **19** (1979) pp. 2377-2380.
10. Poulter J. and Sa-yakanit V., A Complete expression for the propagator corresponding to a model quadratic action, *J. Phys. A Math. Gen.* **25** (1992) pp.-1539-1547.
11. Leggett A. J., Chakravarty S., Dorey A. T., Fishes M. A., Gong A. and Zwenger W., Dynamics of the dissipative two-state system. *Rev. Mod. Phys.* **59** (1987) pp.1-85.
12. Sa-yakanit V., Path-integral theory of a model disordered system. *J. Phys. C* **7** (1974) pp. 2849-2876.
13. Castrigiano D. P. L. and Kokiantonis N., Classical paths for a quadratic action with memory and exact evaluation of the path integral. *Phys. Lett. A* **96** (1983) pp. 55-60.
14. Eizenberg N. and Klafter J., Molecular motion through fluctuating bottlenecks. *J. Chem. Phys.* **104** (1996) pp. 6796-6806.
15. Frauenfelder H., Biomolecules. In *Emerging Syntheses in Science*, ed. By Pines D. (Addison-Wesley Publishing Company, Inc, 1988) pp. 155-165.

PULSED RADIOBIOLOGY: POSSIBILITIES AND PERSPECTIVES

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Problems arising in the irradiation of biological objects by ionizing radiation (mainly by X-ray photons) which has low dose but high dose power (high power flux density of radiation within the object) are discussed. A survey of modern sources of hard radiation generating ultrashort pulses (equal or below nanosecond time intervals) is presented. Applications of the sources to radio-enzymology (a branch of "pulsed radiobiology") and its possible use in X-ray microscopy of living bio-objects are analyzed.

1 Problems of Flash Radiobiology and Medicine

The illumination of biological objects by radiation, which is able to ionize atoms or molecules in them, is used in life sciences for diagnostics (including microscopy and tomography), medical treatment, as well as for directed changes of structure, composition and properties of the objects. The last procedure is aimed both to *investigate* the role of different elements and systems of a biological object which are connected with each other by very complicated non-linear tights and mechanisms [1], and to *change* its role, e.g., for the goals of gene engineering. "Ionizing radiation" can be fast electrons, soft and hard X-rays and γ -radiation, neutron and ion beams, α -particles, nuclear fission fragments and nuclear fusion products. In any of these cases the minimal energy of an individual particle or photon of the beam must be of the order of or higher than the ionization energy of an atom or the dissociation energy of a molecule (~ 10 eV) of the biological object under irradiation. Another type of radiation—very powerful beams of low energy particles and especially photons (those as, e.g., in solid state lasers)—also can produce ionization, but the corresponding mechanisms and consequences in this last case will be different. Because usually ionization of matter in this second case occurs side by side with the process of its ablation (evaporation) we shall concentrate our attention only on the first type of radiation.

Classical radiobiology as well as radiation treatment of patients [2] deals with "weak" beams of radiation and operates in terms of "doses D," i.e., how many ionizing particles or photons interact with the object without paying attention to the time duration of the interaction? It presumes that it works in the conditions where each fast particle or hard photon interacts with the object *individually*.

In principle the term "flash radiobiology" in a wide sense could be applicable to all particular acts of interaction of ionizing radiation beams with objects provided that the time of this interaction is short compared with the duration of a

corresponding specific biological process. So generally speaking pulsed radiobiology is operated with "instant" flashes of radiation. But due to various physical restrictions (e.g. because of finiteness of the speed of light) it is impossible to produce *mathematically* "instant" pulses. Thus we shall examine here *physically* instant flashes and shall operate with "dose power P." We shall specify the time range of applicability of the term "flash radiobiology" in any particular case by comparison of the beam pulse duration with the biological process under discussion.

Because the time interval of the interaction process plays an important role in flash radiobiology we shall organize our analysis of the problem in both terms—dose and dose power. In particular we shall operate with the power flux density of the irradiating beam on the target. Or, taking into consideration the mean free path of fast particles of the beam within a biological tissue, we shall discuss the problem eventually in terms of a *concentration* of the fast particles within the biological object *during* the irradiation process.

Moreover, we shall see that under certain conditions the effect of the action of short-pulsed radiation on an object will have essentially non-diffusive character and could be accompanied by synergetic (collective) effects. These conditions can be formulated more accurately in the following way:

During an irradiation pulse a concentration of effective spheres having radius equal to a mean free path of the primary ionizing particles of the irradiating beam or secondary particles produced by the beam ("spurs" and "bubbles" in terms of radiobiology) should be so high that these spheres must overlap each other. And this condition should be realized during a time interval (radiation pulse duration) lasting less than the corresponding biochemical process.

However, the concentration of energy of the ionizing radiation must be lower than the ablation threshold for a particular tissue.

The other feature of flash radiobiology—non-stationary development of biochemical processes—is closely connected to this idea. Usually when a long-lasting pulse of low intensity is used for irradiation, all processes are ruled by diffusion of reactants. The reaction speed at any moment and during the whole process in this case is determined by a collision rate (concentration, temperature, ...) of reactants and reaction products. Contrary to this scenario when reactants are produced "instantly" and at short distances from each other the reaction will not have a diffusive character anymore, and we shall come to a non-stationary process. Moreover, compression in time and space of the same number of photons or fast particles over a certain limit will result in their mutual action and may have catastrophic consequences for the object in spite of the fact that the overall dose is quite far from the upper allowable threshold.

There are four other points, which become of special importance when short powerful pulses are used for irradiation of biological objects. The first one is the non-threshold biological effect of radiation [3], which means that even doses on

the level of cosmic radiation or earth emanation can produce radiation effects on living tissues. The second point is the so-called 'problem of low doses' [4], i.e., doses of a certain level—higher than the natural one but much lower than the doses allowable by radiation hygiene. It appeared that doses in a certain niche could produce much stronger influence on biological objects than higher or lower doses. The third effect is the selective interaction of quasi-monoenergetic radiation with an object having an energy (wavelength) resonance in the absorption coefficient for the particular radiation. And finally the fourth problem is coming from the statement of N. Bohr [5] on complementarity of life phenomenon and its investigation process. One of the facets of the problem can be formulated in the following way.

As many researchers have shown, the dose needed to make an X-ray picture of a cell with proper resolution and contrast is much higher than cell's survival (even vaporization) level. But probably during a very short period of time (at the 'inertial confinement' of an object) it will be possible to take the picture of still 'living' tissue in spite of a subsequent 'death' (evaporation) of it.

Discussion of all the above aspects in the frame of pulsed radiobiology is the main aim of the paper. Let's point out these issues in more details by listing some of the basic directions in which radiation biology and medicine are developing.

1.1 Diagnostics

Diagnostics in medicine and in fundamental biology is the most important way to get information on biological objects. It can be done either on living tissue or (in experimental biology) on the objects, which have been dried beforehand and then covered by metallic film enhance the contrast in the irradiating beam. Generally speaking there are three types of diagnostics:

1. Passive ones, when organs and tissues are investigated by characterization of its own activity (blood pressure, temperature, etc.) and emanations (liquids, gases, light, etc.).
2. Non-destructive active diagnostics, when characteristics of an object are tested by external probes (X-rays, electric probes, etc.).
3. Active influence on tissues with the goal of acting on a certain parameter or mechanism to clarify its role in the whole non-linear coupled complex.

As for diagnostics of the second type the permanently existing issue is whether the particular method is really 'non-destructive', especially in the case of implementation of a new procedure.

1.1.1 X-Ray Diagnostics (to let the beam through)

Here the main problems are as follows:

- a) *Portable apparatus* (emergency cars and military surgery). The idea is to equip modern medicine with X-ray diagnostic techniques which are light enough to be carried, say to a certain floor of a building, and to be supplied with electricity from apartment's mains.
- b) *Soft X-rays* (angiography, mammography, dentistry and pediatrics). The main aim is to elaborate a low dose source of a low energy (soft) X-rays fitted to visualize 'soft' tissues with a high contrast and resolution, but at the same time having high efficiency, low cost and long life-time.
- c) *High-Resolution Computer Tomography* (HRCT). The key problem here is to ensure a small focal spot of the e-beam of a tube and a very high reproducibility of the X-ray yield from shot to shot.
- d) *X-ray Microscopy* (XRM) for applications in medicine and fundamental biology. Most important issue is the development of imaging with high resolution and contrast. Other problems are to provide elemental and chemical mapping, and X-ray excited fluorescent technique, all of them probably for living tissues.
- e) *Miniature sources*, ready for application, e.g., in laparoscopy.
- f) *High sensitivity high-resolution detectors*—at the moment investigations are mainly concentrated on the increase of sensitivity and the miniaturization of elements of CCD matrixes.

1.1.2 One- and Two-photon Emission Tomography (including the Positron Emission Tomography—PET)

The main issues in these types of diagnostics are the following:

- a) Elaboration of *non-isotope sources of X-rays and γ -rays* of a special kind suited for diagnostics, which can be positioned *inside* the body of a patient.
- b) Production of short-lived γ - and β^+ -radioactive isotopes; the main problems here are:
 - The isotope production process should be fast and cheap.
 - Whether it is possible to site the production factory (must be ecologically clean) directly in clinics to have a possibility to prepare the isotopes *in vitro* in a close vicinity to patients?
 - Is it achievable to prepare the isotopes *in vivo*, i.e., inside the patient's body?

1.2 Therapy

During radiation treatment the main problems are: how to destroy 'bad' cells, how to destroy all of them, and how to give the lowest possible dose to surrounding 'good' cells and tissues? But at the same time it should be taken into account that

the relative number of 'bad' cells destroyed directly *during* radiation treatment procedure is about 10^{-6} —all others would die later on.

1.2.1 X-Ray and γ -Ray Therapy

- a) *Low-dose high power sources*. The issue is whether it is possible to decrease the dose of radiation on a patient by increasing its power flux density with the same therapeutic effect.
- b) *Multi-beam systems*. Present day sources of radiation are cumbersome accelerators. At the same time a rotation of a patient with respect to the beam is unacceptable in many cases. Thus the only possibility to resolve the problem of *localization of hard radiation inside the patient's body* with minimal influence on surrounding tissues is a multi-beam source which should be of small size and adjustable.
- c) *Miniature ecologically clean powerful sources*. The intriguing problem here is the following: will it be possible to develop a suitable source of hard X-ray radiation to be positioned for a short period of time directly inside the patient's body?

1.2.2 Fast Electron Therapy

Because the main part of the energy loss by fast electrons takes place near the surface of the object where the e-beam starts to penetrate inside a tissue this type of therapy is used mainly for surface tumors and concerns about 10% of all cancer patients. The main goals of fast electron therapy are as follows:

- a) *Same as above*
- b) *Outpatient service*
- c) *Adjustable electron energy spectrum*

1.2.3 Neutron Therapy (tumors and arthritis)

- a) *Same as above*
- b) *Monochromatic neutron spectrum*. The problem under investigation is formulated in the following way. Present day neutron therapy is executed with the help of neutron beams from fission reactors or isotopes having a broad energy spectrum. In this connection it will be interesting to learn whether the monochromatic neutron beam (available, e.g., from nuclear fusion sources of neutrons) has any advantages.
- c) *Spectroscopic investigations*. It will also be interesting to study the difference of the therapeutic effect of neutrons depending on their energy and energy spectrum distribution.

1.2.4 Proton and Fast Ion Therapy

- a) *Same as above*
- b) "*Coulomb explosion*"(?) This question is connected with the idea of having certain experimental evidence in the field of registration of high-Z ions by means of a nuclear emulsion method. It was found that when a high-Z ion penetrates into an emulsion it attracts initially for its neutralization many more electrons than the magnitude of its charge, thus producing a local 'explosion' of the substance.

1.3 Tests, Simulations (radiobiology)

1.3.1 Cell Survival

- a) *Low-dose high-power pulses?* The problems are: whether low doses but compressed in time and space are dangerous for living cells or, contrary to this issue, could be more effective in the destruction of tumor cells.
- b) *High repetition rate pulsed device.* This source parameter is important for two tasks: to complete a patient treatment in a short period of time and at the same time to investigate the influence of periodicity of the pulses on cell survival.
- c) *Resonance frequencies?* This requires the investigation of possible coincidence of frequencies of, e.g., the irradiating X-ray photons with the characteristic absorption frequencies of the molecules of an irradiated object.
- d) *Combined illumination* of tumor cells by different types of radiation as well as investigation of the consequences of *mutual implementation* of radiation therapy + chemotherapy.

1.3.2 Biochemical Activity

- a) *Same as above.*
- b) *Resonance frequencies for repetition rate of radiation pulses.*
- c) *Effect of low doses* (so-called 'resonance' with a whole cell, a system or an organism).
- d) *Spectrally selective absorption.*

So the general aims in the development of pulsed sources of ionizing radiation for use in medical diagnostics and basic biology as well as for therapy are:

- a) Availability of hard radiation flashes with *different pulse duration and repetition rate* comprising a very wide range of time intervals both for the flashes and for the intervals between them. They cover atto-, femto-, pico-, nano-, micro-, milliseconds, seconds, minutes, hours, ..., years, ...—more than 25 orders of magnitude. These time intervals and repetition rates refer to various processes taking place within biological objects—from very simple

chemical reactions (e.g. production of photoelectrons) to the organism's growth and development.

- b) To have sources producing *different types* of pulsed ionizing radiation (X-ray photons, neutrons, etc.) *during one run of the apparatus*.
- c) To tune the *parameters of sources*: pulse duration; coherence and spectrum; source's size or beam's divergence; dose and/or power flux density.

2 Modern Short Pulse X-Ray Sources

At the present time several sources of hard radiation of different types which can produce short pulses of ionizing radiation are available, and at the same time are ecologically clean in comparison with fission reactors, isotopes and classical accelerators. We shall discuss here only devices, which are able to provide during a short pulse the necessary concentration of ionized particles within the bio-object or having good perspectives for realizing this task in the future.

2.1 High Harmonics of IR (Nd-GLASS) Laser Pulses

The source of this type [6] can produce very powerful pulses (up to several kJ) due to transformation of infrared radiation in nonlinear crystals into ultraviolet radiation. Also, the pulse duration covers the range from nanoseconds to femtoseconds. The main disadvantages of these sources are:

long wavelengths, i.e., low photon energy, usually not enough to ionize molecules (in fact, belonging to UV range of radiation because of the optical properties of nonlinear crystals)

low efficiency

cumbersome devices

2.2 X-Ray Lasers [7 - 9]

- Advantages:

high spatial and temporal coherence

good divergence

table-top dimensions

pico- and nano-second pulses ($10^{-12} - 10^{-7}$ s)

- Disadvantages:

very low overall energy (less than 1 mJ) achieved at the moment

VUV region and only a restricted selective number of wavelengths (15...150 Å)

low efficiency (much less than 10^{-4})

2.3 *Laser Produced Plasma [10]*

- Advantages:
- small size of the source ($\geq 10 \mu$)
- medium energy (up to 10% of heating IR laser energy)
- nano- and picosecond pulses
- different wavelengths
- Disadvantages:
- low efficiency (less than 10^{-4})
- not very high overall energy
- cumbersome and expensive devices
-

2.4 *Laser Synchrotron Sources [8]*

This device consists of a synchrotron generating a relativistic electron beam and a Nd-glass laser. X-ray photons appear during the scattering of a pulsed IR laser beam with a tightly focused electron beam.

- Advantages:
- femtosecond pulses (~ 300 fs)
- short wavelength ($0.4 \text{ \AA}$)
- Disadvantages:
- low output (10^5 photons per pulse at present time)
- poor efficiency
- cumbersome and ecologically dangerous

3 **Dense Plasma Focus–Pulsed Powerful Source of Hard Radiation of Different Types**

Dense Plasma Focus (DPF) is a sort of a pulsed Z-pinch [11]. It produces hard radiation at the discharge of a capacitor bank of medium voltage (~ 20 kV), inductive storage or explosive generator through various gases. During plasma compression by a magnetic field pressure it may generate soft X-rays of different wavelength depending on the working gas used. After this ‘pinching’ process magnetic energy is converted into the energy of beams of fast electrons and ions because of a number of turbulent phenomena. Interaction of the beams with the anode and plasma produces hard X-ray flashes and neutron radiation. Because at the present time this source is the most convenient one for various applications in pulsed technologies (and has been used in our experiments) we shall list here its most important advantages:

- a) Generation of many types of radiation and the possibility of tuning within a wide spectral range—fast electrons and X-rays (100 eV... 1.0 MeV), fast

ions (up till 100 MeV), neutrons (monochromatic–2.45 or 14.0 MeV) and fast plasma jets.

- b) High efficiency (10% for soft X-rays and fast particles), high brightness, and high repetition rate of the source.
- c) Wide range of feeding energy and relatively compact size of the device (at the moment 100 J through 1 MJ; it can be portable at low energies and transportable at medium ones).
- d) Small size of radiating zones of the source (1 cm...1 μm).
- e) Relatively low charging voltage of the capacitor bank used (~ 10 kV).
- f) In comparison with sources based on fission materials and classical accelerators it is ecologically clean, safe and cheap.
- g) Possibility to generate nanosecond pulses (with picosecond substructure).

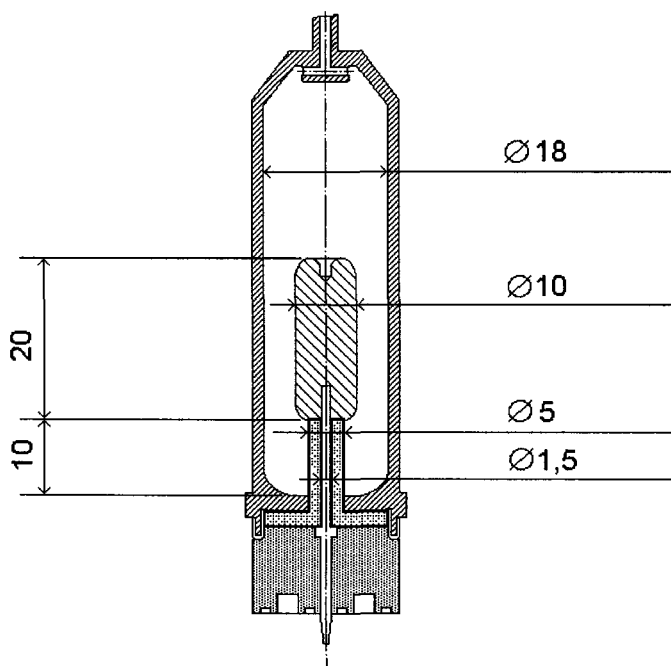


Figure 1. Construction of the chamber of the device PF-0.2 [20]. Primary energy storage $E_{el} = 100$ J, $E_{\text{soft X-rays}} = 10.0$ J, $E_{\text{HV}} = 9$ keV ≤ 1.0 J, $E_{\text{hard X-rays}} \geq 0.1$ J, $\tau_{\text{pulse}} = 4$ ns. All dimensions are in mm.

Based on our 30 years-experience of investigation of physical phenomena taking place in the discharge [12-14] and working on the improvement of technology of these installations [15], we have developed several devices of this

type (Fig. 1, 2, 3) suited to applications. They demonstrate high efficiency, high repetition rate and long lifetime, and they are designed specifically for particular assignments (see, e.g. [16]).

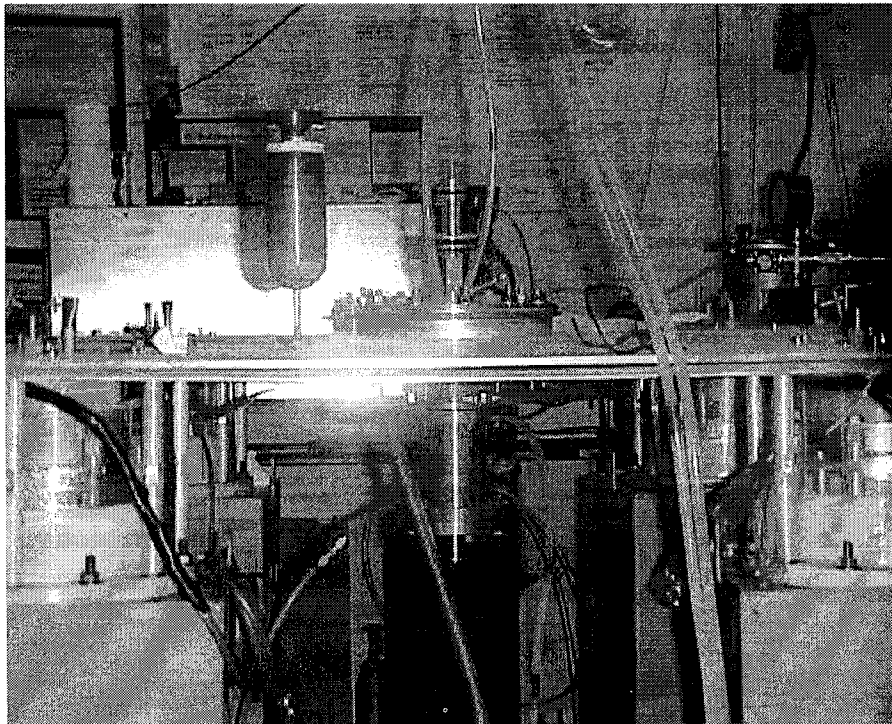


Figure 2. DPF device NX1. Primary energy storage $E_{el} = 2 \text{ kJ}$, $E_{\text{soft X-ray}} = 100 \text{ J}$, $E_{hv=9 \text{ keV}} \leq 10.0 \text{ J}$, $E_{\text{hard X-rays}} \geq 1.0 \text{ J}$, $\tau_{\text{pulse}} = 1 \dots 10 \text{ ns}$, repetition rate $\sim 3.5 \text{ Hz}$

Figure 1 shows a finger-like chamber of a portable DPF. The weight of whole device including capacitor and control panel is about 15 kg, and it can be supplied from the usual mains. It has been used as an X-ray and neutron source for various aims in medicine, biology, and oil industry, for the calibration of detectors and characterization of materials and premises.

Figure 2 and 3 show transportable DPF, developed for soft X-ray generation for uses in nanoelectronics and micromachining. In this field we have shown by a proximity lithography technique the possibility of producing an image on a photoresist with elements having dimensions about 50 nm [15, 20b]. It also may be used (with different working gas filling) for dynamical fault detection in industry as well as in radiobiology.

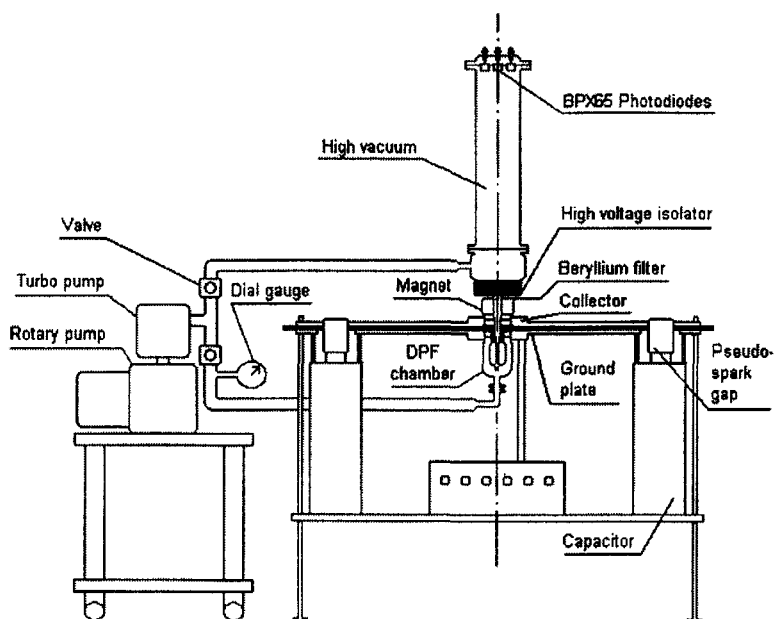


Figure 3. Diagram of the DPF NX1 device

4 Portable Dense Plasma Focus for X-Ray Diagnostics in Medicine?

Because of the above-mentioned characteristics it was proposed to use a small size DPF as a portable pulsed X-ray apparatus for various applications [12, 13, 14-18]. One of the fields of its use which as the most attractive one is in medical diagnostics (to let beam through).

Indeed, there are several advantages of a DPF device in comparison with the conventional X-ray tubes used in present day medicine. First, this device (e.g. PF-0.2, Fig. 1) may have low weight (about 10 kg). Another advantage is that it has a capacitor as an intermediate part between the mains and the discharge chamber (as shown on Fig. 3). This means that it is possible to use normal low power apartment mains for its power supply whereas classical X-ray apparatus used in clinics needs several kW. Moreover, acceleration of electrons in DPF is provided by collective plasma mechanisms. Because of this fact, normal charging voltage at the power supply output in DPF (about 10 kV) is 10 times less than in classical tubes. These features give an opportunity to discuss possible use of this portable device in emergency cars, military field surgery, etc.

Second, the ultrashort X-ray pulse of a DPF, irradiated by a practically point source, has high brightness. Therefore it is possible to take X-ray picture of any organ of a patient practically instantly, not having to be afraid of any movement of the object during the exposure time (which is of the order of one nanosecond).

Third, due to self-focusing of the electron beam inside the pinch plasma of a DPF [19] the e-beam focal diameter (X-ray source diameter) at the anode is of the order of 100 μm . It gives very high spatial resolution of an X-ray picture. Such a resolution can be reached with vacuum X-ray tubes only during many thousands of flashes and with the use of a diamond needle with special cooling as an anode. This system is costly and has a very low lifetime. So taking into consideration the small size of a DPF chamber, it seems that it would be very convenient to use a miniature Dense Plasma Focus in dentistry.

Next, as X-rays of the DPF are irradiated by a very small source and its spectrum is highly enriched by the low energy photons [12], its use for examination of soft tissues with high resolution (dentistry, mammography, pediatric diseases, angiography, etc.) is of great importance. Moreover, because of the miniature size of the DPF chamber it is possible to position the chamber inside the body (e.g. within a mouth of a patient) thus making irradiation (e.g. of teeth) *from* within the patient's body. It provides a much lower dose on a patient during the production of a panoramic picture of tissues and is accompanied by the irradiation of non-sensitive organs.

The lower dose associated with the use of this short-pulse point source positioned inside a body arises because of geometrical factors. Such geometry results in a magnification of the image at some distance out of the body and thus gives a possibility to use an intensifier, e.g., fluorescent screen, without losing spatial resolution of the image. More important fact is that the dose decrease takes place also due to the fact that a short (about nanoseconds and less) pulse produces a much stronger photographic effect on detectors (e.g. X-ray film) than a long (about few seconds) pulse having the same number of X-ray photons (dose). Our experiments [20] have shown that the dose needed for production of the same optical density of the image on the X-ray film in case of a nanosecond pulse is several times lower than in the case of a conventional X-ray tube used in clinics. A possible explanation of this phenomenon [21] is based on a synergetic effect of X-ray photons when their high concentration in the case of X-ray pulse compression in time and space is reached within a sensitive layer of the film.

Thus in view of the above mentioned characteristics of the DPF-based X-ray source the problem can be formulated in the following way. Is it really favorable to use in medicine this short-pulse, ecologically clean, low-dose, portable X-ray apparatus having a better balance in its spectrum, which is very convenient for "instant" visualization of both hard and soft tissues? And in particular, being formulated more specifically, is it indeed safer to use this pulsed apparatus which produces the same image at doses several times less than the conventional ones used in clinics at the present moment?

At this point one has to realize that *decreasing the dose* with this device by several *times* will *increase power flux density (dose power)* by several *orders of magnitude*. Is it dangerous for potential patients or not? To prove the tremendous importance of the above question, let us give an example.

During half an hour of tanning under the sunshine, a man receives 1 MJ of UV radiation (energy of the same “quality”!) upon his body. And he experiences nothing harmful, just pleasure. However, an energy of 1 MJ is enough to produce work against the Earth’s gravity in lifting up to 1.5 km a 100-kg body. It’s a lot of energy! Moreover, being compressed in time within a microsecond interval, this energy is equivalent to that released in the explosion of four grenades. So from this point of view a shortening of a radiation pulse must increase the probability of radiation damage. This example shows that the problem is very serious and should be examined more closely.

5 Dose Versus Dose Power (power flux density)

Although flash radiography has been known for more than half a century, it should be stressed that sources of ionizing radiation having ultrashort pulses of sub-nanosecond range and high brightness became available for laboratory experiments only during the last two to three decades [11]. These installations have appeared side by side with the progress of high current electronics. The main applications of these devices from the very beginning have been concentrated in military electronics, namely in simulations and testing of electronic devices under flashes of ionizing radiation [22 - 24]. Only a few reports on the interaction of short high brightness pulses of ionizing radiation with matter having fundamental interest can be found in the current literature. The overall picture looks as follows.

Let’s fix a dose of ionizing radiation received by a sample from isotopes or a fission reactor during a relatively long period of time (seconds through hours) which is already high enough but still does not produce within the sample any measurable effects. We shall now irradiate the same samples of solids (e.g. crystals), organic materials and living tissues with the same dose but compressed in space and in time (to about nanoseconds). It is clear that this “instant” energy release within the samples must produce certain effects, at least because of a simple mechanism—fast heating of it (we have to compare here the heating interval with the characteristic relaxation cooling time).

Real experiments show that in crystals we shall have an appearance of *irreversible damage of its structure and formation of defects* within the sample [24]—contrary to the previous case of prolonged illumination. As for organic materials there is some evidence that side by side with the formation of the defects certain *reconstruction mechanisms* are taking place. For instance, the author of the paper about 20 years ago has observed the effect of opaqueness of plexiglass windows of DPF chamber under the action of a 60-ns very bright X-ray flash,

which relaxed (the windows became transparent) after about 2 microseconds. As for living tissues we may expect *not only reconstruction but also certain reparation (rehabilitation)* mechanisms. The main specific feature of biological tissue is a high water content in it. Unfortunately there is not enough experimental material on this point within the frame of flash radiobiology. So the problem can be formulated as follows:

What *dose power P* is critical for living tissues under their irradiation by very short and very bright flashes of ionizing radiation *at the low absolute dose D*?

Let's compare now data on experimentally investigated effects produced by pulses of hard radiation in two different spheres—in military electronic tests and in the irradiation of living objects. The author believes that there is a certain analogy between the two objects and their behavior (functioning) during and after the irradiation. The data taking from the available literature [25] are collected in the table presented at the beginning of the next paragraph.

6 Military Electronics Tests Versus Radiation Disease

The table below represents effects produced by X-ray photons of different energy on various substances subjected to irradiation by soft and hard X-rays in electronic devices based on semiconductors and in biological objects.

Table 1. Consequences of irradiation of chips and biological objects

≤ 1 keV continuous and pulsed source	9 keV pulsed source	Dose	≥ 60 keV continuous and pulsed source
10 J/cm ² Morphological destruction [28] DPF-300 [26]		5•10 ⁷ Gy	Fission reactors Not available in a short pulse of laboratory devices
Structural changes [28] NXI [15]		5•10 ³⁺⁴ Gy	1÷10 cal/cm ² [26] Morphological destruction [25]
10 ⁻³ J/cm ² Sensitivity of RCA		5•10 ¹⁺² Gy	10 ⁻²⁺¹ cal/cm ² [25] DPF-300, 100 J Structural changes
		5 Gy	10 ⁻³ cal/cm ² Information damage [17, 25]
	~10 ⁻⁵ Gy Enzyme activation /inactivation [17]	$\leq 5 \cdot 10^{-2}$ Gy 5 rem, "Safe" upper limit	10 ⁻⁵ cal/cm ² No detectable influence

Let's examine the situation produced in these cases by not paying attention to the duration of the irradiation process.

All types of damage can be divided roughly into 3 classes:

- Morphological destruction, when samples are melted and evaporated.
- Structural changes, when state of an object becomes different from the previous one.
- Information damage, when everything looks as before, but because of currents flown through the "circuit" *during* the radiation pulse the "program of an object" for a future is changed.

The most important difference between the two types of irradiated objects is that electronics consists of solid matter (semiconductors) whereas the main substance in biological tissues is water.

Morphological changes in both spheres need no extra comments. But "structural changes" mean from a medical point of view immediate radiation disease, whereas "information damage" (DNA damage, gene modification) probably will result in future cancer. In electronics those structural changes correspond to such phenomena as, e.g., change of state of a flip-flop circuit, whereas information effects during irradiation of a chip may influence its executive mechanisms, thus having distant consequences.

At the present time it is admitted that in radiation disease the main role is taken by biochemical reactions with the assistance of free radicals. So in discussing the difference between ultrashort pulses and continuous radiation we have to compare the radiation pulse duration with the time intervals characteristic of various physicochemical processes. This is so in spite of the fact that many of them, especially those taking place during the very initial period of irradiation (10^{-11} s and less), are not well investigated and understood. There are several stages of response of matter to a radiation action on it (R. L. Platzmann, 1953).

During the first one—*the physical stage*—primary processes of water radiolysis take place. Between them photoelectrons production, their free travelling and diffusion, inelastic collisions and energy transfer ("electronic activation"), creation of δ -electrons, their solvation, production of recoil atoms and radicals, their migration, scattering and relaxation, very complicated dynamics and eventually its recombination. Also, of no less important are the time intervals between these processes, i.e., their repetition rate (e.g. collision rates).

The physicochemical stage consists of processes of transformation (spontaneous or due to collisions) of initial radiolysis products into intermediate ones, such as ions and free radicals. Chemical reactions of the latter products with each other and with the environment; its dynamics and production of new substances occupy a more prolonged period of time and form the third stage—*the chemical one*. These processes take place when a system reaches thermal equilibrium. The forth stage—*biological*— is occupied by reaction of the whole organism and its systems with the chemical products created. We have to correlate

all these processes with life cycles of the organism and its organs (e.g. with the frequency of creation of new cells).

The time duration of the most important processes *and the intervals between them* relevant to radiation damage in living objects [27] are presented in Table 2.

Table 2. Time scales of the most important processes

Time scale (s)	Processes
$10^{-17} - 10^{-14}$	Photoelectric absorption of X-rays, atomic transitions, Auger processes, Compton effect, pairs and free radicals production
$10^{-13} - 10^{-11}$	Free electrons collide, thermalize and hydrate; relaxation of rotational/vibrational energy of molecules
$10^{-11} - 10^{-9}$	Beginning of chemical processes, Brownian movement, diffusion of 10 nm particles for distances ~ 0.2 nm
$10^{-6} - 10^0$	Mean distances of diffusion 6 nm ... 3 μ m for particles of the diameter of 10...50 nm; cell's creation rate; end of radical reactions, organ's movement (e.g. winking or hart beating)
$10^{-6} - 10^3$	Biochemical reactions, stages of mitosis and meiosis, <i>intervals between the stages</i>
$10^2 - 10^8 \dots$	Human's life duration; radiation disease development, different consequences of gene mutations

It can be seen that the present day pulsed radiation sources can have pulse duration comparable with all the above processes except for the first line. DPF can produce pulses of 10^{-9} s probably with picosecond structure and with a very high brightness. Let's discuss this fact in connection with one of the possible problems of radiation biology to make the next step in understanding whether it is dangerous to compress a low dose of radiation in time and space.

One of the most fundamental issues is the problem of radiation damage of living cells of an organism. The scheme of a cell's life cycle is presented in Fig. 4.

The most sensitive stages when the influence of radiation on a cell is of great importance are "M" and "S." The total number of cells contained in a human body is well known. Taking into consideration the total possible number of acts of their division during the organism's life (50) it is easy to estimate (on the average) *an interval between divisions of two different cells*. It cannot be shorter than 1 microsecond. This fact means that pulses of nanosecond range may irradiate only those cells which are already in the stages M or S.

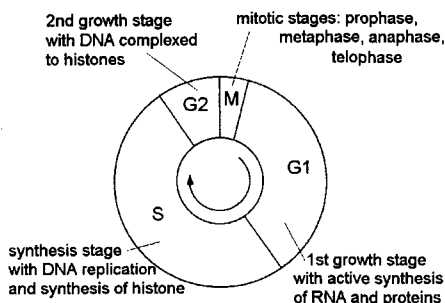


Figure 4. Diagram of a cell's life cycle

Contrary to this situation a process of irradiation executed during seconds (e.g. as in the case of clinic's examination of patients by classical X-ray tubes) spreads onto many more cells (up to 1 million times more!) newly entering into the above mentioned sensitive stages of mitosis within the irradiation pulse. So from this point of view the shorter the pulse the better! Thus the question is whether the compression of a dose in space and time is more dangerous or less dangerous for living cells.

From a general point of view it is clear that working below the absolute upper limit of power flux density of radiation which may result in an evaporation of living tissue (morphological damage) may have two possibilities:

- 1) Same dose at low dose power (long pulse) versus same dose at high power (short pulse) should result correspondingly in safe or destructive consequences, and
- 2) High dose power at low dosage (short pulse) versus same dose power at higher dosage (long pulse) should be safe or destructive correspondingly in contrast to the previous case.

The difference lies in concrete figures. This dilemma may be elucidated practically by experiments.

7 Radiobiology Versus X-Ray Microscopy

As was mentioned above, there is another interesting possible application of pulsed radiation sources besides radiobiology and medicine: microscopy. For certain tasks the Dense Plasma Focus may turn out to be important because of its possible use as one source of several types of radiation generated in the same short interval of time—e.g. of X-rays and neutrons. Images of an object received in the two beams may give different information enriching the whole picture of it. But we shall discuss here only some possibilities of a pure X-ray flash microscopy of living (wet) tissues.

At present, resists with chemical amplification (RCA) have a sensitivity about 1 mJ/cm^2 . Being illuminated from a proper source of soft X-rays, it may give an image of an object with spatial resolution about 20 nm [29]. This figure is at the same time a diffraction limit for a conceivable geometry of the experiment and very close to the diffusion length of Auger electrons produced within the resist by X-ray photons. Such resolution makes possible the investigation of structure and elemental distribution of molecules in protein, enzymes, as well as large bio-objects (e.g. the cell's elements). It can be provided only with a source having a small size of radiating zone, which has been reached in the DPF device NX1 [15, 16].

Short pulses (in the nanosecond to picosecond range) of X-ray photons of proper energy (within the "water window"—0.5...0.3 keV, i.e. 24...43 Å— and higher, up to 4000 eV) and dosage (10^4 - 10^8 Gy) are needed for the goals of X-ray flash microscopy [28]. This is now available with present-day DPF devices. The above mentioned figures must be ensured for the following tasks.

First, we have to have X-ray pulse duration in sub-nanosecond range because of two reasons:

- a) Diffusive and Brownian movements of bio-objects of 20-50 nm size cannot blur its own image taken during the above period of time.
- b) It is very likely that during this time interval no phase transition of water will occur.

The above dosage is necessary for the visualization of a 20-nm object with signal/noise ratio being 3. And here the most promising opportunity of flash X-ray microscopy is represented. Indeed usually with the use of the long exposure times (e.g. from a synchrotron source) the cell survival at these dosages is not an important question. Only the "structural stability," i.e., the number of atoms in a resolution element and the position of the element within the wet specimens under irradiation is considered [28]. But if so high a dosage is absorbed during the "confinement time" of an object (i.e. during the time interval when this object "start to be evaporated" and its shape and structure are not changing considerably) we shall receive probably an image of a "living tissue."

However, at this point we have to come back to our Tables 1 and 2. It can be seen that certain physicochemical processes even during sub-nanosecond time will take place already. Also some structural changes will be executed by radiation during this inertial confinement time. In this connection, contrary to the question put by L. Matsson [30]—"What is life?"—we have to formulate the main issue of X-ray microscopy of *living* tissues as "What is death?" To resolve this problem many experiments supported by theoretical investigation should be performed.

In contrast to X-ray microscopy, radiobiology focuses on radiation damage of biological objects. It operates in different X-ray photon energy range (1 through 500 keV) and dosage (0.1 through 100 Gy). Photon energy, being much higher than that for microscopy, is determined by the absorption coefficient and thickness of the layers (usually several mm and cm) of the biological objects under

irradiation. Lower dosage in comparison with microscopy result from the fact that usually initial changes in biological activity can be found namely at this level of irradiation (see e.g. [17]). However, it should be taken into consideration also that recently a certain niche between negligible and allowable doses where some effects of activity change was found was discovered [18]. It appears that not only the low dosage of radiation, but even pure chemical action executed by a low dose of a reagent can produce certain effects similar to radiation damage. There is no complete understanding of the phenomena at the present time. So any experiments performed showing the effects of low dosage are of great importance.

8 Enzyme Activation and Inactivation Induced by Low Dose/High Dose Power Radiation

A search for the optimal ratio between dosage and dose power in the field of radiation biology is important because of the possible applications of DPF in medicine. Indeed if it appears that radiation with low doses but of high power flux density is safe for living objects it will be possible to develop *a portable low-dose apparatus for X-ray diagnostics*. In the opposite case, if low-dose high power radiation destroy cells with higher efficiency than long pulses of the same absolute dosage, such a device can be used for *a low-dose therapy* of cancer.

To reach a conclusion on this issue we have made experiments on irradiation of several types of enzymes *in vitro* using various sources of hard radiation with different dosage, dose power and spectrum [17].

8.1 Materials and Methods

8.1.1 ACE

We have used for irradiation an isolated and purified electrophoretically homogeneous ACE (Mw 180 kDa) from bovine lungs. The enzyme contained about 98% of active molecules as determined by stoichiometric titration with a specific competitive inhibitor.

8.1.2 Peroxidases

Native horseradish peroxidase C (HPR, Mw 44 kDa) purchased from Biozyme, and recombinant wild-type HRP (Mw 34 kDa) produced from *Escherichia coli* inclusion bodies was under irradiation. Tobacco anionic peroxidase (TOP, Mw 36 kDa) from *Nicotiana glauca* transgenic plants also has been used.

8.1.3 Measurements

ACE enzymatic activity was determined with 10^{-5} M carbobenzoxy-L-phenylalanyl-L-histidyl-L-leucine (Cbz-Phe-His-Leu) (Serva, Germany) as a substrate in a 0.05 M phosphate buffer, pH 7.5, containing 0.15 M NaCl, 25°C, using o-phthalaldehyde modification of His-Leu as a reaction product.

Peroxidase activity was measured with 2,2'-azino-bis (3-ethylbenzothiazoline-6-sulfonate) (ABTS) and guaiacol (Sigma) as substrates using a Shimadzu UV 120-02 spectrophotometer (Japan) at 25°C as follows:

1. 0.05 mL of ABTS solution (8 mg/mL) and an aliquot of the enzyme were added to 2 mL of 0.1 M Na-acetate buffer (pH 5.0). The reaction was initiated by the addition of 0.1 mL of hydrogen peroxide (100 mM). A molar absorptivity of ABTS oxidation product was taken equal to 36800 L/mol/cm at $\lambda = 405$ nm.
2. 0.15 mL of guaiacol water solution (1 mg/mL) and an aliquot of the enzyme were added to 0.1 M Na-acetate buffer (pH 5.0, 0.2 mL). The reaction was initiated by the addition of 0.1 mL of hydrogen peroxide (100 mM). A molar absorptivity of guaiacol oxidation product was taken equal to 25500 L/mol/cm at $\lambda = 436$ nm.

The accuracy of activation/inactivation measurements is estimated as about $\pm 10\%$.

8.2 Irradiation

We have used several sources of X-rays:

- A standard γ -source ^{137}Cs : $E_{\text{hv, max}} = 662$ keV, $P_D = 5 \cdot 10^{-2}$ Gy/s
- A standard X-ray tube used in clinics for patient body examination working in 3 regimes: $U = 50.0, 60.0$ and 90.0 keV, $E_{\text{hv, max}} \approx 35.0, 40.0$ and 60 keV correspondingly, pulse duration ~ 1 s, $D \approx 10^{-5} - 10^{-3}$ Gy
- Miniature DPF "PF-0.2": energy storage—100 J, $E_{\text{hv}} \approx 8 \dots 100$ keV, X-ray pulse duration—4 ns, X-ray yield $\sim 0.1 \dots 1.0$ J/shot, $D \approx 10^{-5} - 10^{-4}$ Gy/shot, neutron yield $\sim 10^6$ n/pulse ($E_n = 2.45$ MeV)
- Medium-size DPF "PF-2": energy storage—2.0 kJ, $E_{\text{hv}} \approx 1 \dots 200$ keV, X-ray pulse duration—1...10 ns, X-ray yield $\sim 1.0 \dots 10.0$ J/shot, $D \approx 10^{-5} - 10^{-3}$ Gy/shot, neutron yield $\sim 10^8$ n/pulse ($E_n = 2.45$ MeV)
-

The irradiation was performed at 18-20°C. The absorbed dose was determined by thermoluminescent detectors based on LiF activated with Mg, Cu, and P with the help of analyzer Harshaw TLD system-4000. X-ray spectrum of sources and its pulse shape in case of DPF were monitored by the filter method [12, 24]. As detectors in these measurements we used Roentgen- γ dosimeter 27040 (Germany), calibrated X-ray films, photomultipliers SNFT (Russia) with plastic scintillators (of 2 ns time resolution) and individual dosimeters (gas-ionizing chambers).

All types of enzymes have been irradiated in the following regimes covering dosage range from 10^{-6} to 10^2 Gy and dose power from 10^{-2} up to 10^6 Gy/s:

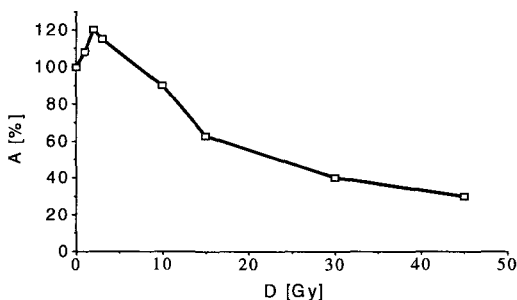
- By ^{137}Cs isotope continuous source—at various distances and during different time periods lasting from seconds up to many hours
- By classical X-ray tubes—in the above mentioned regimes with 1...10 irradiation séances ('shots') for a specimen
- By both DPF sources – at various distances and with 1...100 irradiation séances for a specimen
- By both DPF sources with two types of filters for X-rays—Al and Cu foils
- By both DPF sources with two types of filters—transparent for X-rays and opaque for neutrons or transparent for neutrons and opaque for X-rays.

8.3 Results

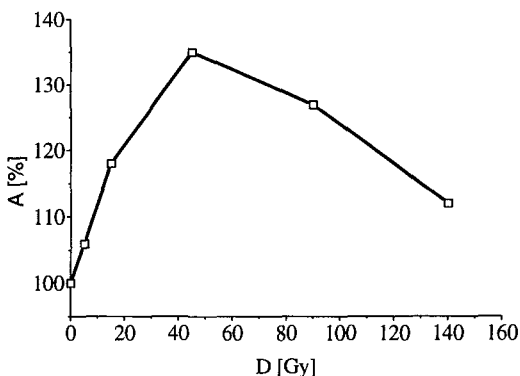
Typical results on radiation-induced changes in the activity of enzymes found in our experiments are illustrated in Fig. 5. In both cases—irradiation by isotope source and by DPF-based source—we have registered activation and inactivation of enzymes. In all investigated regimes these changes appeared to be irreversible ones—at least for several days. As can be seen the main differences between changes in activity of the enzymes irradiated by isotope sources (*a* and *b*) and ones illuminated by DPF (*c*; 1, 2, and 3) are as follows:

- In spite of the fact that the amplitudes of the activity changes are about the same in both cases, the changes start to be measurable in the case of an isotope source only at dosage about 0.1 Gy. Contrary to it in the case of DPF irradiation they are registered at the *dosage lower* than in the case of isotope by *5-6 orders of magnitude*. *Inverse difference in dose power* in both cases was *4-5 orders of magnitude*
- The peaks of the activity changes are very narrow in a dosage scale for DPF case whereas quite wide for the isotope irradiation case
- At a *decrease of dosage* in both cases below the above level (isotope: lower than 10^{-1} Gy, and DPF: lower than 10^{-5} Gy) no changes in the enzyme activity have been detected
- However, an *increase of doses* produces quite different consequences. In the case of isotope irradiation it results in consecutive enzyme activity changes, so that eventually the activity of specimens receiving high dosage fall down to a very low figures. Contrary to this situation a *dosage increase* under the DPF irradiation *during one shot* (e.g. by distance shortening from the source) results in the situation where the *change* of the activity of a specimen becomes *lower*. Thus in the DPF case at doses higher by an order of magnitude than those where the changes was found to be high, no appreciable changes in the enzyme activity (within the limits of the method's accuracy) were found as in the case of a very low dosage.

To clarify details of the situation we have undertaken several additional experiments. First, we have checked for possible influence of neutrons on the results achieved. By positioning between the source and the specimens special filters, which cut the X-rays completely but are transparent for neutrons, we have shown that no any appreciable effect can be detected in this case. Analysis of shots with high neutron/low X-rays and low neutron/high X-rays gave the same conclusion. So at this stage of our research it should be admitted that very likely neutrons have not played any role in the phenomenon (unless we have any compensation effects).



(a)



(b)

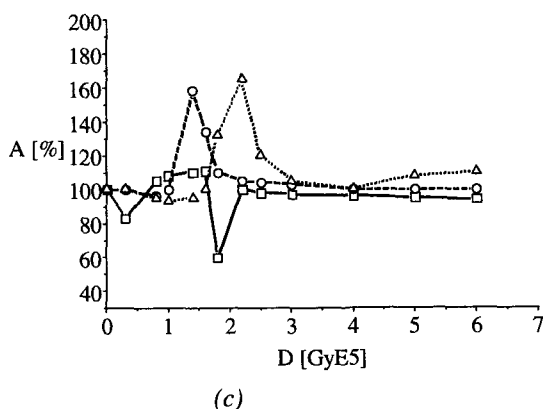


Figure 5. Changes in catalytic activity of (a) γ -irradiated by ^{137}Cs source native horseradish peroxidase (10^{-7} M enzyme solution in acetate buffer, pH 6.0, substrate guaiacol), (b) γ -irradiated by ^{137}Cs source tobacco peroxidase (10^{-7} M enzyme solution in acetate buffer, in the presence of $5 \cdot 10^{-2}$ M CaCl_2 , substrate guaiacol), and (c) DPF-irradiated: 1) - - native HPR (10^{-7} M enzyme solution in phosphate buffer, pH 6.0, substrate ABTS), 2) Δ - ACE (10^{-8} M, phosphate-borate buffer, pH 6.5, substrate Cbz-Phe-His-Leu), and 3) α - ACE (10^{-8} M, phosphate-borate buffer, pH 7.5, substrate Cbz-Phe-His-Leu).

Second, we have used classical X-ray tubes for irradiation of our objects in the range of dosage 10^{-5} through 10^{-1} Gy. No appreciable changes in the enzyme activity throughout the whole range of dosage under investigation were found. The only exception was when we used the lowest voltage at the X-ray tube (50 kV). In this case in one experiment we have found also the narrow peak of the same character (activity increase and then decrease with the dosage increase). But in comparison with the DPF experiment it was found at the dose 10^{-2} Gy and it has been about 12-15% by amplitude (therefore very close to the accuracy of the measurements).

Third, we have changed by 10 times the X-ray dose power of the DPF source by increasing the distance from the source to the specimen. We produced at this dose power various numbers of shots on different specimens, and at a certain seance we have reached a confirmation of the above effect. It appeared that in this case we might see the same narrow peak in the dosage scale, as was the case in Fig. 5c. But again the amplitude was about 10-15% only, and appeared at a dosage of 100 times higher (at 10^{-3} Gy) than in the case of Fig. 5c.

Finally we have changed the filter in our DPF window from Cu foil into Al. All attempts to repeat the above results failed in this case.

Let's discuss the results received taking into consideration our above control experiments as well as the following facts,

1. The overall spectrum of X-rays generated by DPF consists of two parts—X-ray photons of thermal nature peaked roughly around 1-2 keV, exponentially decreasing to higher photon energy and having a production efficiency of about 10%, and X-ray photons radiated by accelerated electrons. The last spectrum has a power-like decreasing to the high-energy wing and possesses a much lower efficiency.
2. Our DPF has an anode (where the relativistic electron beam generates X-ray photons) made from copper. So in the vicinity of a photon energy of about 9 keV the fast electrons generated in DPF produce several very bright lines, the highest brilliance of which belongs to Cu K_{α} line [31].
3. The difference in the radiation spectrum in two cases—Al-foil filter and Cu-foil filter (Fig. 6a)—is only that in case of Cu foil we have a “window of transparency”—K-edge—near ≤ 9 keV in a scale of photon energies. The total X-ray energy yield of our two DPF devices (about 1.0 and 10 J), working with the copper foil, mainly composed of a Cu K_{α} line, a tail of the thermal plasma radiation of the pinch and bremsstrahlung radiation from accelerated electrons plus high energy photons in the range above 20 keV. The first three types of radiation are concentrated namely within this “9-keV-window” (Fig. 6b). The Al filter gave about the same total energy in the range above 20 keV, but nothing in the low-energy wing (Fig. 6c).
4. The enzyme with which we’ve seen strong activation/inactivation effects had in its structure an atom of Zn. The spectral absorption curve of Zn atom overlaps with the Cu K-edge curve because these elements are neighbors in Mendeleev’s periodic table of the elements.
5. The thickness of the enzyme solution layer under irradiation (3 mm) was about the mean free path of the 9 keV photons (~ 1 mm). But it is considerably lower than the mean free path of 30 keV photons (peak of the high-energy component of the DPF spectrum) and much lower than that for the 662 X-ray photons irradiated by the ^{137}Cs source.

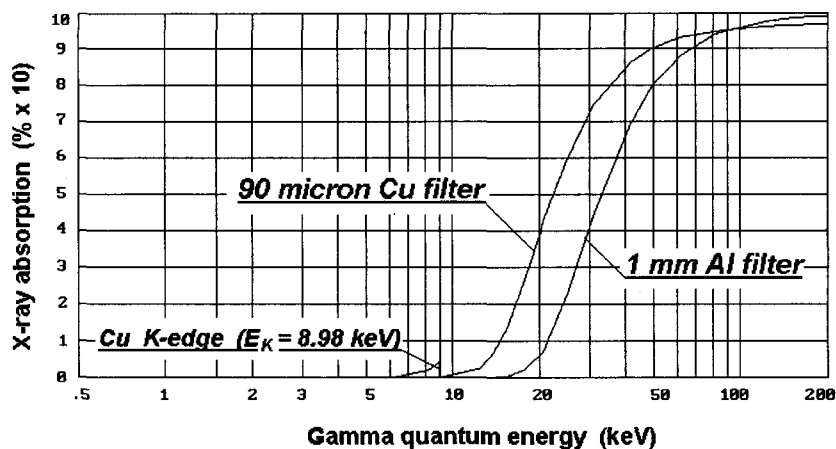
As is well known [32] for X-ray quanta having an energy below 1 MeV almost all the energy is used in the creation of spurs (size about 10 nm) and bubbles. Moreover, for X-ray photons below about the 5-keV limit, bubbles are mainly produced. Their characteristic size is about 100 nm. Our fast electrons in the cases of isotope source (662 keV) and DPF’s hard component (30-100 keV) are of Compton nature. As for 9 keV photons, they are primarily photoelectrons.

When all these electrons penetrate water solutions they never produce δ -electrons with energy more than 1 keV (the main part of them have an energy of the order of the ionization potential of the water molecules). It is easy to estimate roughly the mean distance between bubbles (in the case of 9 keV photons) and spurs (in the cases of hard component of DPF and isotope γ -quanta) produced in the water solutions of enzymes. It appears that in the case when the DPF is

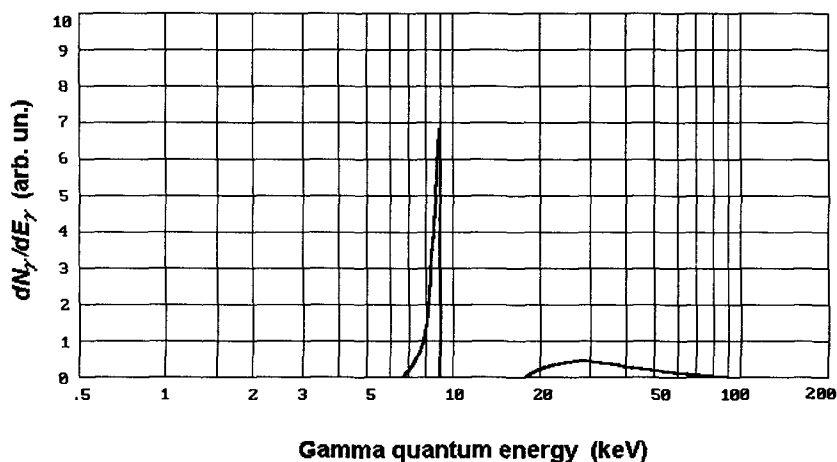
positioned 10 cm apart from the specimen and equipped with a copper foil (i.e. 9 keV photons) this distance is equal or less than the bubble size as is shown in Fig. 7a. It means that our ionization zones are overlapped. And what's more this overlapping takes place during the time interval (few nanoseconds) which is compared with the time intervals ($10^{-12} - 10^{-8}$ [32c], p. 38]) of creation of molecules H_2 , H_2O_2 , radicals HO_2 , atoms O , ions OH^- . Together with the primary ('instant') products of the physical stage of radiolysis escaping from a recombination— e^-_{aq} , H , OH , H^+ —they begin to diffuse and interact with enzyme molecules (*not* with each other). The last process takes place even at low concentration of enzyme (even less than $10^{-6} M$).

It should be mentioned that certain synergetic effects as was mentioned in the beginning of the paper might accompany this short-lasting process of mutual creation of the above intermediate radiolysis products. One of the types of such an effect is presented in Fig. 7c. If the absorption of radiation locally overheats the substance in many points at the same time, a multiple shock wave production throughout the volume is possible. At oblique collisions of the spherical shock waves micro-cumulative streams may appear [33]. These streams have velocities 4 times higher than those of the primary micro-shock waves. They may provide an evident turbulence and intensification of the chemical reactions (as in the case of 'sonochemistry') within the volume.

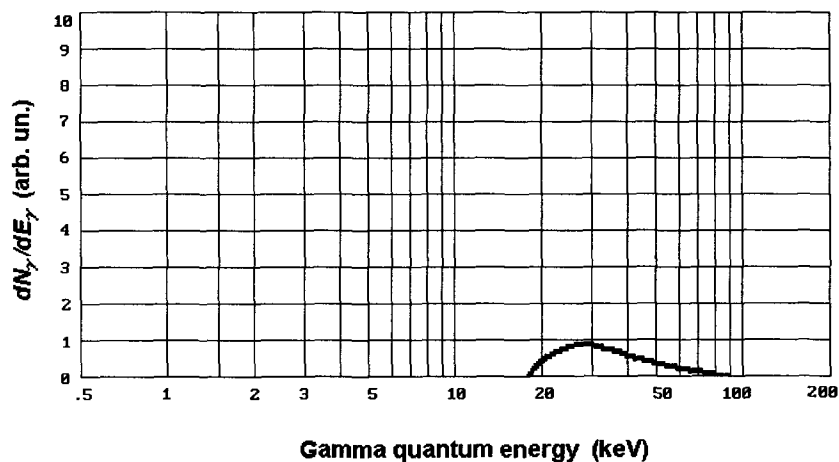
Another possible and probably more important consequence of this 'instant' event is the *simultaneous* creation of the intermediate products and excitation of the Zn atoms of the enzyme as is shown in Fig. 7c. Indeed the interaction of the above products with excited Zn atoms very likely might produce stronger (or the same but much easier) conformational changes in the enzyme molecule.



(a)



(b)



(c)

Figure 6. Spectral characteristics of the X-ray filters and X-ray radiation generated by DPF: (a) Spectral curves of Al and Cu foils transparencies plotted versus X-ray photon energy. (b) Spectrum of the X-ray radiation of DPF coming out through the Cu foil. (c) The same spectrum coming out through the Al foil.

Contrary to this situation the distances between spurs created by photons of the hard component of the DPF spectrum and by γ -quanta from the isotope source are much larger than their own size. It is so because of the smaller size of spurs in comparison with bubbles and due to their non-continuous creation along the track of the fast particle. Thus in the last case the picture will look as is presented in Fig. 7b—only some of the spurs and bubbles can overlap each other during low-dose short pulse or during very high-dose long pulse of irradiation. And evidently in these cases there is no spectrally selective influence on Zn atoms of the enzyme. We may suppose that a very long pulse only in a very low probability case can produce such an excitation when an individual track of a fast electron will cross the site of an enzyme molecule.

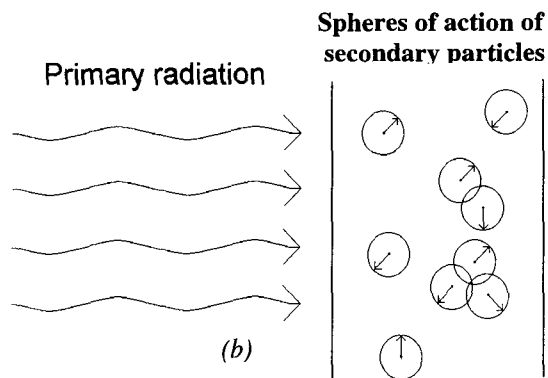
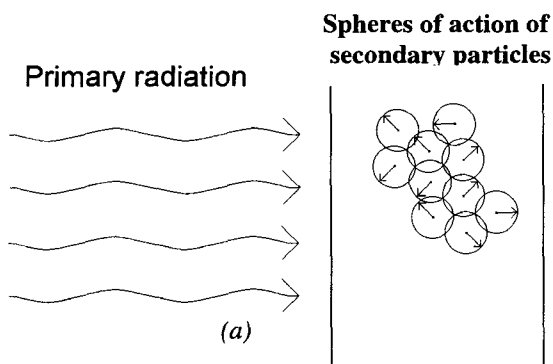
So at this time we propose three reasons for a possible non-contradictory interpretation of these experiments:

- Low-dose high-power X-ray pulses produce overlapping spurs and bubbles during a time period equal to the time of reaction of secondary radiolysis products with the enzyme molecules. Because of this fact the concentration of the reactants are greatly increased thus increasing the probability and the rate of subsequent reactions.
- It is important that the radiation spectrum of DPF in the case of a Cu foil produces a selective action onto the Zn atoms contained in the enzyme molecules (Cu K-edge window plus Cu K_{α} line and thermal X-ray spectrum within the window $\longleftrightarrow$ Zn K_{α} line). And it produces a synergetic effect of *simultaneous excitation* of the Zn atoms and *production* in the vicinity of it many secondary water radiolysis products. This fact ensures the interaction of them during each short pulse.
- Most difficult to understand is the disappearance of the effect on the increasing of the dose at the same high dose power. For sure, it means that the third reason—namely the effects of low doses [4]—was displayed in the experiments.

At the present time a concept to explain the low-dose effect which is accepted by all the community is absent. Usually speculations on the possibility to understand this phenomenon on the basis of the reaction of the macro-systems or even the whole organism with the secondary products of water radiolysis are under discussions (e.g. in a style of the traditional homeopathy). In this situation the author will risk to propose a hypothesis using a certain analogy from laser physics.

It seems to be quite possible that the effects presented in figures 5a and b from one side and 5c from the other side are due to different mechanisms. The first one is a real denaturation (destruction) of an enzyme with an irreversible change in its structure, whereas the second one is just a conformational change in its molecule, impossible at low concentration of the above secondary products. If so it is quite possible that any dose increase above a certain level will be resulted in a saturation effect. This effect must be similar to those in the case of saturation of a two-level

laser system at high photon pumping. Analogous to the balance between absorption and induced radiation in a two-level system which results in the transparency of a previously opaque substance, here we probably have a situation of a balance between activation-inactivation processes resulting in the saturation of conformational changes and eventually in the insensitivity to low-dose radiation. In principle such a situation can be reached by many methods, and in particular by a low-dose high-power radiation. It is clear that to verify or refute this hypothesis many experiments and, in particular, investigations of the conformational changes in enzyme molecules should be done.



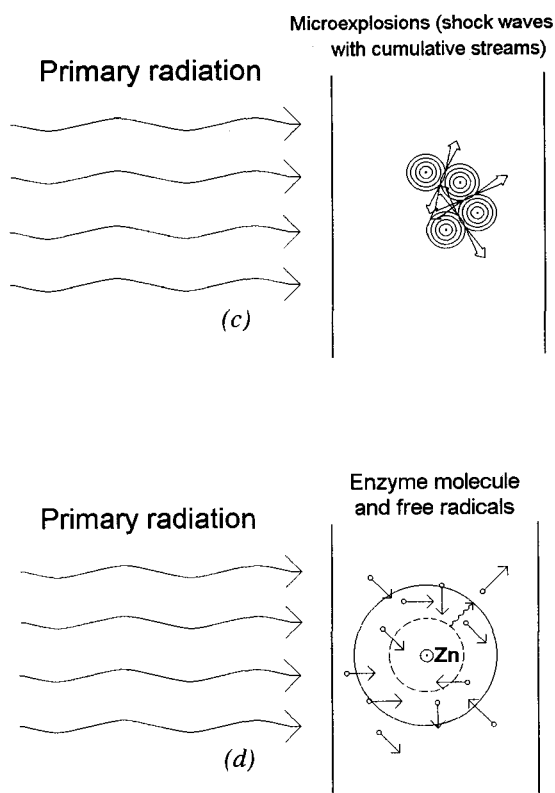


Figure 7. Effects of the X-ray interaction with the specimens: (a) Production of the overlapping bubbles within the irradiated specimens at low-dose high power short pulse of X-ray photons from DPF at the use of Cu foils (9 keV X-ray radiation). (b) Production of the partly overlapping spurs and bubbles within the irradiated specimens at low-dose high power short pulse of X-ray photons from DPF at the use of Al foils (~30 keV-peaked X-ray radiation) or at high-dose very low dose power of γ -radiation from ^{137}Cs source. (c) Synergetic effect of local absorption of high power X-ray radiation and production of volumetric multiple micro-shock waves with cumulative streams. (d) Synergetic effect of the simultaneous selective excitation of Zn atom inside the enzyme molecule and the creation of secondary products of the water radiolysis.

9 Acknowledgments

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References

1. H. Frauenfelder, P. G. Wolynes, R. H. Austin, Biological Physics, *Reviews of Modern Physics*, Vol. 71, No. 2, Centenary 1999, S419-S430
2. a) N. V. Timofeev-Resovskij, A. V. Savich, M. I. Shal'nov, Vvedenie v moleculjarnuju radiobiologiju (Introduction into Molecular Radiobiology), Meditsina, Moscow (1981) *in Russian*
b) A. V. Agafonov, Primenenie uskoritelej v medicine (Application of accelerators in medicine), *Priroda*, No. 12 (1996) 65-77, *in Russian*
3. A. D. Sakharov, in book "Radioaktivnyj uglevod jadernyh vzryvov i besporogovye biologicheskiye efekty (Radioactive carbon of nuclear explosions and non-threshold biological effects)," Atomizdat, Moscow (1959) *in Russian*
4. a) *Radiation Biology*, No. 1 (1998)
b) "Consequences of the Chernobyl Catastrophe on Human Health", ed. by E. B. Burlakova, Nova Scientific Publishers, Inc., New York (1999)
a) "Low Doses of Radiation: Are They Dangerous?", ed. by E. B. Burlakova, New York (2000)
5. N. Bohr, Atomic Physics and Human Knowledge, London (1957)—Atomnaya fizika i chelovecheskoje poznanie, Izd. Inostr. Lit., Moscow (1961) *in Russian*
6. P. W. Milonni, J. H. Eberly, Lasers, Wiley, NY (1991)
7. a) C. Steden and H. J. Kunze, Observation of gain at 18.22 nm in carbon plasma of a capillary discharge, *Phys. Lett.*, Vol. 151 (1990) 534-537
b) H. -J. Kunze, K. N. Koshelev, C. Steden, D. Uskov, H. T. Weischebrink, Lasing mechanism in a capillary discharge, *Phys. Lett. A*, Vol. 193 (1994) 183-187
8. P. Eisenberger and S. Suckewer, Subpicosecond X-ray pulses. *Science*, Vol. 274 (1996) 201-202
9. R. C. Elton, X-ray Lasers, Academic Press, Inc. (1994)

10. a) N. G. Basov, V. A. Gribkov, O. N. Krokhin, G. V. Sklizkov, Investigation of High Temperature Phenomena Taking Place under the Action of Powerful Laser Radiation on the Solid Target, *ZETP*, Vol. 54, No. 4 (1968) 268-276
 b) N. G. Basov, V. A. Boiko, V. A. Gribkov *et al.*, Investigation of dynamics of laser plasma temperature by X-ray radiation, *Pis'ma ZhETP (ZhETP Letters)*, Vol. 9 (1969) 520-524
11. a) V. A. Burtsev, V. A. Gribkov, T. I. Filippova, High Temperature Pinch Formations, in book "Fizika Plazmy", ed. by V.D. Shafranov, VINITI, Moscow (1981) *in Russian*
 b) D. D. Ryutov, M. S. Derzon, M. K. Matzen, The physics of fast Z pinches, *Reviews of Modern Physics*, Vol. 72, No. 1 (2000) 167-222
12. V. A. Gribkov, Physical processes in high-current discharges of "plasma focus" type. Doctor of Phys-Math Sci dissertation, Lebedev Physical Institute (1989) *in Russian*
13. N. V. Filippov *et al.*, Experimental and Theoretical Investigation of the Pinch Discharge of the Plasma Focus Type, *Plasma Phys. and Contr. Nuclear Fus. Research*, IAEA-CN 28/D-6 (1971)
14. V. A. Gribkov, P. Lee, S. Lee, M. Liu, A. Srivastava, Pinch Dynamics with Argon Filled Dense Plasma Focus Radiation Source, ICPP-2000. International Congress on Plasma Physics, 42nd Annual Meeting of the Division of Plasma Physics of the American Physical Society, October 23 - 27, 2000, Quebec City, Canada
15. S. Lee, P. Lee, G. Zhang, X. Feng, V. A. Gribkov, M. Liu, A. Serban, and T. K. S. Wong, "High Rep Rate High Performance Plasma Focus as a Powerful Radiation Source", *IEEE Transactions on PLASMA SCIENCE*, Vol. 26, No. 4 (1998) 1119-1126
16. V. A. Gribkov, P. Lee, S. Lee, M. Liu, A. Srivastava, Dense Plasma Focus Radiation Source for Microlithography & Micromachining, ISMA-2000: International Symposium on Microelectronics and Assembly, 27 November - 2 December 2000, Singapore
17. M. A. Orlova, O. A. Kost, V. A. Gribkov, I. G. Gazaryan, A. V. Dubrovsky, V. A. Egorov, "Enzyme Activation and Inactivation Induced by Low Doses of Irradiation", *Applied Biochemistry and Biotechnology*, Vol. 88 (2000) 243-255
18. V. A. Zuckerman, Z.M. Azarkh, People and explosions, *Arzamas-16* (1994) *in Russian*
19. V. A. Gribkov, Application of the Relativistic Electron Beams, Originating in the Discharge of DPF-Type for the Combined Laser-REB Plasma Heating, Energy Storage, Compression, and Switching, ed. by W. Bostik, V. Nardi, and O. Zuker, Plenum Press, N-Y (1976)
20. a) A. V. Dubrovsky, P. A. Silin, V. A. Gribkov, I. V. Volobuev, DPF device application in material characterization, *Nanoleonika*, Vol. 43, No. 3 (2000) 185-187

- b) V. A. Gribkov, E. P. Bogoljubov, A. V. Dubrovsky, Yu. P. Ivanov, P. Lee, S. Lee, M. Liu, V. A. Samarin, Wide Pressure Range Deuterium and Neon Operated DPF as Soft and Hard X-Ray Source for Radiobiology and Microlithography, Proceedings of the 1st International Workshop on Plasma Applications, Chengdu, PR of China, October 2000, to be published
21. V. A. Gribkov, Pulsed Radiochemistry, *Nukleonika*, to be published (2001)
 22. F. Jamet, G. Thomer, Flash Radiography, Elsevier Scientific Publisher Company, Amsterdam, Oxford, New York (1976)
 23. Proceedings of the IEEE International Pulsed Power Conferences (e.g. Proc. of the 12th IEEE International Pulsed Power Conf., Monterey, California USA, June 27-30, 1999)
 24. E. N. Astvatsatur'yan, P. G. Bobyr', V. A. Gribkov, *et al.*, Methods of investigation of X-ray pulses of DPF devices, *Pribory i tekhnika experimenta*, No. 5 (1982) 183-185, *in Russian*
 25. a) See SPIE Proceedings, e.g. Vol. 1140 (1989), 1741 and 2015 (1993), and later.
b) T. M. Agakhanjan, E. R. Astvazatur'jan, P.K. Skorobogatov, Radiazionnyye efekty v integral'nyh mikroskhemah (Radiation effects in integrated chips), Energoatomizdat, Moscow (1989) *in Russian*
 26. V. A. Gribkov, A. V. Dubrovsky, Yu. V. Igonin *et al.*, Experimental Investigations on "PLAMYA" Installation, *Sov. J. of Plasma Phys.*, v. 14, No.8 (1988) 987-992
 27. J. L. Magee, A. Chattejee, Theoretical Aspects of Radiation Chemistry, in Radiation Chemistry. Farhatatiz and Rodgers eds., VCH (1987)
 28. G. Schneider, Investigation of soft X-radiation induced structural changes in wet biological objects, Proc. IV Int. Conf. On X-Ray Microscopy, ed. by V.V. Aristov and A.I. Erko, September 1993, Chernogolovka, Bogordsky pechatnik, 181-195
 29. L. E. Ocola, F. Serrina, Parametric modeling of photoelectron effects in X-ray lithography, *J. Vac. Sci. Technol. B* 11 (1993) 2839
 30. L. Matsson, this volume.
 31. G. W. C. Kaye, T. H. Laby, "Tables of Physical and Chemical Constants and some Mathematical Functions," Longman, London and New York (1986)
 32. a) R. L. Platzmann, Physical and chemical aspects of basic mechanisms in radiobiology, ed. by J. L. Magee *et al.*, Washington, National Academy of Sciences, NRC Publication, No. 305 (1953)
b) V. M. Byakov, F. G. Nichiporov, *Vnutritrekovye khimicheskie processy* (Internal chemical processes in the track), Moscow, Energoatomizdat (1985) *in Russian*
c) V. M. Byakov, F. G. Nichiporov, *Radioliz vody* (Radiolysis of water), Moscow, Energoatomizdat (1990) *in Russian*

33. V. A. Gribkov, O. N. Krokhin, G. V. Sklizkov, *et al.*, Experimental Study of Cumulative Plasma Phenomena, Proc. of the 5th Europ. Conf. on Contr. Fusion and Plasma Phys., Grenoble, France (1972)

NONLINEAR APPROACH IN DNA SCIENCE

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We describe a new approach where DNA is considered as a physical dynamical system where many types of internal motions are possible. We focus our attention on the motions of large amplitudes the description of which requires the nonlinear technique. The history of the nonlinear approach, main results, and perspectives are discussed. To be concrete, we consider in details one of possible large-amplitude motions, namely, local unwinding of the double helix. We derive new nonlinear equations describing the motions. We show that the equations have solitary wave solutions that can be interpreted as a boundary between wound and unwound regions. We discuss new mathematical and physical problems that arise due to interaction of nonlinear mathematics and physics with DNA science.

1 Introduction

Nonlinear physics and mathematics are well known as rapidly developing fields of science with many interesting applications. One of the applications, namely, the application to DNA science, is the theme of this article. We describe a new approach where DNA is considered as a physical dynamical system where many types of internal motions are possible. We focus our attention on the motions of large amplitudes the description of which requires the nonlinear technique.

The history of the approach began with the work of Englander and co-authors¹ who presented the first nonlinear hamiltonian of DNA. This work gave a powerful impulse for investigations of the nonlinear DNA dynamics by physicists. A large group of authors, including Yomosa², Takeno³, Krumhansl⁴⁻⁵, Fedyanin⁶⁻⁸, Yakushevich⁹, Zhang¹⁰⁻¹¹, Prohofsky¹², Muto¹³⁻¹⁵, van Zandt¹⁶, Peyrard¹⁷, Zhou¹⁸, Dauxois¹⁹⁻²⁰, Gaeta²¹⁻²⁴, Salerno²⁵, Bogolubskaya²⁶, Hai²⁷, Gonzalez²⁸, Barbi²⁹, and Campa³⁰, developed the idea by improving the model hamiltonian, suggesting new models, investigating corresponding nonlinear differential equations and their soliton-like solutions, consideration of statistics of DNA solitons and calculations of corresponding correlation functions.

Beside that many attempts to explain experimental data in the frameworks of the nonlinear approach were made. Explanations of the data on hydrogen-tritium exchange¹, resonant microwave absorption^{11, 13, 31-33}, neutron scattering by DNA³⁴ were among them. Moreover investigators tried to use the nonlinear approach to explain the dynamical mechanisms of DNA functioning. The works on dynamical mechanisms of transitions between different DNA forms^{11, 35-36}, long-range effects³⁷⁻³⁹, regulation of transcription⁴⁰, DNA denaturation¹⁷, protein synthesis (namely, insulin production)⁴¹, carcinogenesis⁴² were only some of the examples. It is

important also to mention the work of Selvin and co-authors⁴³, where the torsional rigidity of positively and negatively supercoiled DNA was measured in the wide range of the DNA parameters. The results obtained gave rather reliable evidence that the DNA molecule can exhibit the nonlinear behavior. And this is a short description of the history. A more detailed description of its different stages can be found in the reviews^{18, 23, 44-45} and books⁴⁶⁻⁴⁷. In section 2 we present modern point of view on the internal DNA dynamics. We show that nonlinear features naturally appear if we consider DNA as a physical dynamical system with many types of internal motions (including large-amplitude motions). In section 3, we describe in details the algorithm of modeling one of the internal motions, namely, local unwinding of the double helix. This motion plays an important role in the processes of transcription, DNA-protein binding, DNA denaturation, DNA destruction due to radiation and so on. We derive new mathematical equations describing nonlinear DNA dynamics and show that the equations have solitary wave solutions that are interpreted as unwound regions. In section 4, we describe applications of the approach to DNA functioning. As examples, we discuss the problem of long-range effects in DNA and the problem of the direction of the process of transcription. In section 5, we discuss shortly applications of the approach to physics and mathematics. We describe shortly the problem of interaction of DNA with the environment, statistics of DNA solitons, scattering of light and thermal neutrons by DNA, the role of inhomogeneity in DNA dynamics.

2 DNA as A Physical Dynamical System

From the point of view of physicists the DNA molecule is nothing but a system consisting of many atoms interacting with one another and organized in a special way in space (Fig. 1). Under usual external conditions (temperature, pH, humidity, etc) this space organization has the form of the double helix, which is rather stable but moveable system. The thermal bath where the DNA molecule is usually immersed is one of the reasons of the DNA internal mobility. Collisions with the molecules of the solution which surrounds DNA, local interactions with proteins, drugs or with some other ligands also lead to internal mobility. As a result, different structural elements of the DNA molecule such as individual atoms, groups of atoms (bases, sugar rings, phosphates), fragments of the double chain including several base pairs, are in constant movement. Several examples of internal motions occurred in DNA, are shown in Fig. 2. They are:

- usual displacements of individual atoms from their equilibrium positions (Fig. 2a),
- displacements of atomic groups (Fig. 2b),
- rotations of atomic groups around single bonds (Fig. 2c),
- rotations of bases around sugar-phosphate chain (Fig. 2d),

local unwinding of the double helix (Fig. 2e),
transitions between different DNA forms (Fig. 2f).

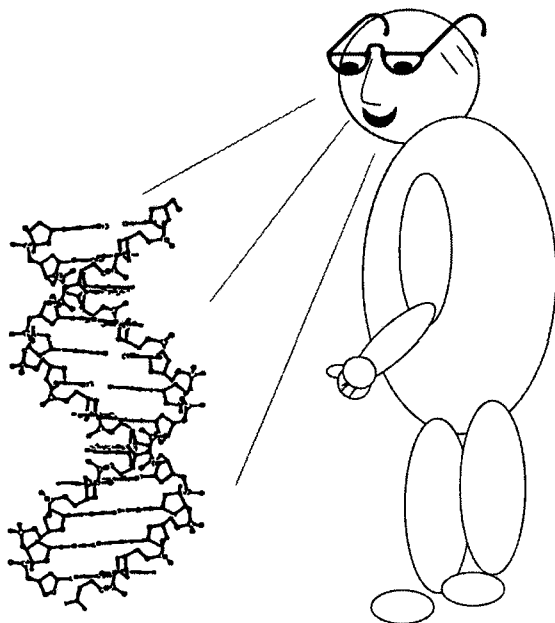


Figure 1. DNA from the point of view of physicists.

A more detailed list of internal motions and of their dynamical characteristics can be found in the works of Fritzsche⁴⁸, Keepers and co-authors⁴⁹, McClure⁵⁰, McCammon and co-authors⁵¹ and Yakushevich^{45, 52}. These lists show that the general picture of the internal DNA mobility is very complex: many types of internal motions with different amplitudes, energies of activation and characteristic times.

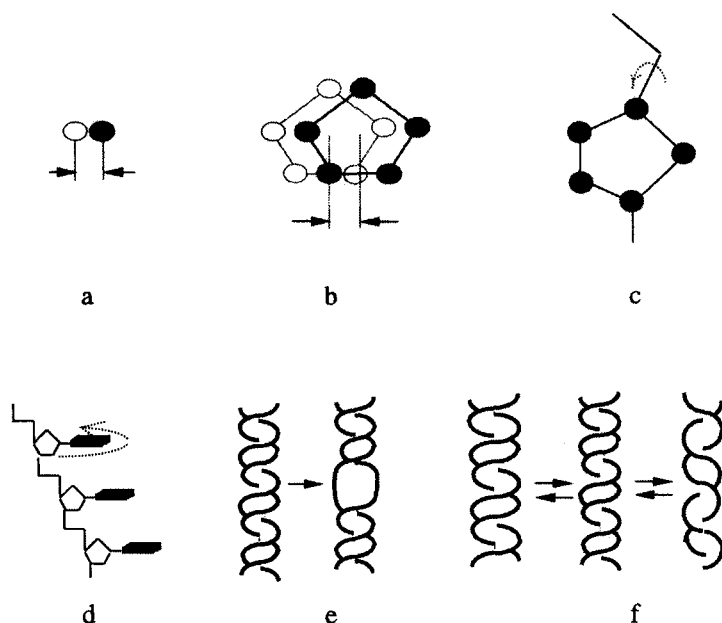


Figure 2. Some examples of possible internal motions in DNA. Displacements of individual atoms from their equilibrium positions (a), displacements of atomic groups (b), rotations of atomic groups around single bonds (c), rotations of bases around sugar-phosphate chain (d), local unwinding of the double helix (e), transitions between different DNA forms (f).

2.1 Small- and Large- Amplitude Internal Motions

Internal motions occurred in DNA can be divided conditionally into two groups: the motions of small and large amplitudes. Small displacements of atoms or atomic groups from their equilibrium positions shown in Fig. 2a, 2b, are the examples of small-amplitude motions. Local unwinding of the double helix (Fig. 2e) and transitions between different conformation states (Fig. 2f) are the examples of large-amplitude motions

To model mathematically internal motions of small and large amplitudes, investigators use different approximations: to model small-amplitude motions, they use harmonic (or linear) approximation, and to model large-amplitude motions, anharmonic (or nonlinear) approximation is usually used, because linear approximation becomes incorrect when the amplitudes of the motions are not small. So, modeling large-amplitude motions naturally lead us to nonlinear approach in

DNA science, which can be considered as a new interesting application of nonlinear mathematics and physics to DNA.

3 Mathematical Modeling of the Internal DNA Motions

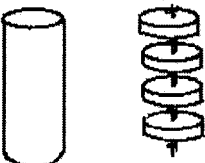
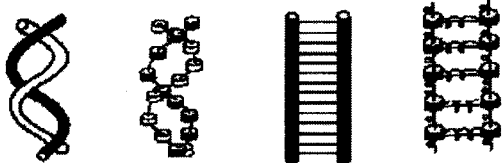
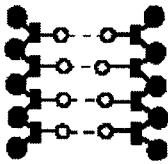
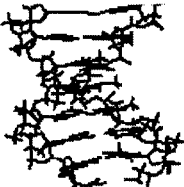
Mathematical modeling is known as one of the most effective tool of studying internal DNA motions. In the DNA molecule we have a large number of internal motions. To model all of them we need to write too large number of coupled differential equations to deal with. Fortunately, in practice, investigators deal only with limited number of motions. The choice of the motions and the number of them depend on the problem considered. Usually, investigators include to the model the motions with dynamical parameters close to the characteristics of the biological processes considered. So, the first step of the algorithm consists in the choice of the limited group of motions. To make this step it is convenient to use approximate DNA models. In the Table 1 the main of the models used are presented. For convenience they are arranged in the order of increasing complexity and each new level of the complexity is presented as a new line in the table.

In the first line of the table, the simplest models of DNA, namely, the model of elastic thread and its discreet version, are shown. To describe mathematically the internal dynamics of elastic rod, it is enough to write only three coupled differential equations: one for longitudinal motions, one for torsional motions and one for transverse motions. To describe the discreet version we need to write $3N$ equations.

In the second line of the table, more complex models of the internal DNA dynamics are shown. They take into account that the DNA molecule consists of two polynucleotide chains. The first of the models consists of two elastic threads weakly interacting with one another and being wound around each other to produce the double helix. The discreet version of the model is nearby. The next two models in the line are simplified versions of the previous two models, which are often used by investigators. In these models the helicity of the DNA structure is neglected. To describe mathematically internal dynamics of the models consisting of two weakly interacting elastic threads, we need to write six coupled differential equations: two equations for longitudinal motions, two equations for torsional motions and two equations for transverse motions in both threads. And the mathematical description of the discreet versions consists of $6N$ coupled equations.

In the third line a more complex model of the DNA internal dynamics is shown. It takes into account that each of the chains consists of three types of atomic groups (bases, sugar rings, phosphates). In the Table 1 different groups are shown schematically by different geometrical forms, and, for simplicity, the helicity of the

Table 1. Approximate models of DNA structure and dynamics.

1	
2	
3	
	
II	

structure is omitted. It is obvious; that the number of mathematical equations required to model internal motions is substantially increased in comparison with two previous cases.

The list of approximate models could be continued and new lines with more and more complex models of DNA structure and dynamics could be added till the most

accurate model which takes into account all atoms, motions and interactions, will be reached. Unfortunately, the process of improving the DNA model is accompanied by increasing the number of equations. In this paper we shall limit ourselves by considering only the continuous models of the second line, which can be completely described by six equations.

3.1 Mathematical Modeling of Large-Amplitude Motions

We present here, as an example, mathematical description of local unwinding of the double helix. Some authors name this motion “the formation of open state”. It is widely accepted that this motion plays an important role in DNA functioning. Indeed, the process of DNA-protein recognition includes the formation of open state to have a possibility to “recognize” the sequence of bases. Local unwinding is an important element of binding RNA polymerase with promoter regions at the beginning of transcription. Formation of unwounded regions is known also as an important part of the process of DNA melting.

We begin the procedure of modeling with the choosing of appropriate model. To find the model it is convenient to use the Table 1. Let us begin with the models of the first level. It is obvious, that these models can not be used to model local unwinding. Indeed, the models do not take into account the existence of two threads in the DNA structure, which are necessary to organize unwinding. The models of the second line are more appropriate, and they are the simplest models that can describe unwinding (Fig. 3).

To obtain six coupled differential equations which are enough to model internal DNA mobility in the frameworks of the continuous models of the second line, we can use the method developed recently in one of our previous works⁵³. According to the method, let us begin with the discrete version of the model and write corresponding hamiltonian in the vector form

$$H_{\text{general}} = \sum_n [m(dU_{n,1}/dt)^2 + m(dU_{n,2}/dt)^2]/2 + \sum_n K[|U_{n,1} - U_{n-1,1}|^2/2 + |U_{n,2} - U_{n-1,2}|^2/2] + \sum_n V(|U_{n,1} - U_{n,2}|), \quad (1)$$

where $U_{n,i}(t)$ is the vector which describes torsional, transverse and longitudinal displacements of the n -th nucleotide in the i -th polynucleotide chain:

$$\begin{aligned} U_{n,1} &= \{R_0(1 - \cos\Theta_{n,1}) + u_{n,1} \cos\Theta_{n,1} ; -R_0 \sin\Theta_{n,1} + u_{n,1} \sin\Theta_{n,1} ; z_{n,1}\}, \\ U_{n,2} &= \{-R_0(1 - \cos\Theta_{n,2}) + u_{n,2} \cos\Theta_{n,2} ; R_0 \sin\Theta_{n,2} + u_{n,2} \sin\Theta_{n,2} ; z_{n,2}\}. \end{aligned} \quad (2)$$

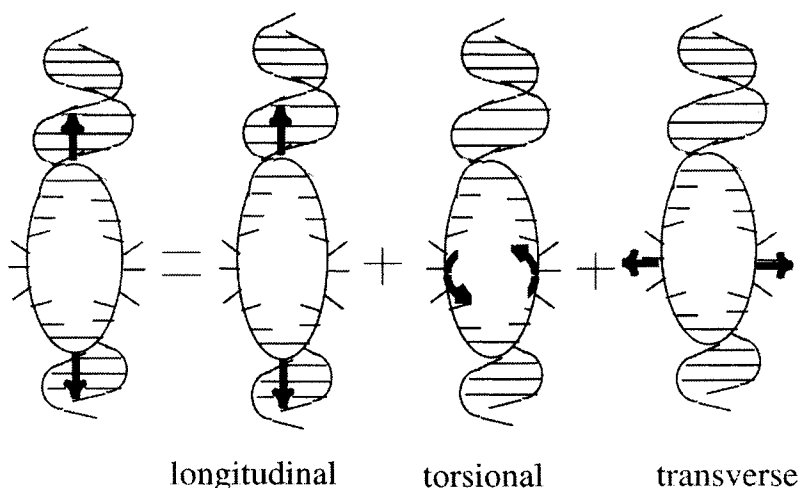


Figure 3. Local unwinding of the double helix is presented here as a sum of six more simple internal motions: longitudinal, torsional, and transverse motions of both threads.

Here $\Theta_{n,i}$ describes angular displacement of the n -th structural unit of the i -th chain; $u_{n,i}$ describes the transverse displacement; $z_{n,i}$ describes the longitudinal displacement ($i = 1, 2$); m is a common mass of nucleotides; K is the coupling constant along each strand; R_0 is the radius of DNA; a is the distance between bases along the chains; and V is the potential function describing interaction between bases in pairs. Hamiltonian (1) can be considered as a generalization of two well-known particular nonlinear models of the DNA internal dynamics: the model of Peyrard⁴⁶, which describes transverse DNA dynamics, and the model of Yomosa², which describes torsional DNA dynamics.

To obtain the explicit form of the model hamiltonian, it is enough to insert (2) into (1). To simplify calculations, we suggest a simple form for potential function,

$$V(|U_{n,1} - U_{n,2}|) = \sum_n k |U_{n,1} - U_{n,2}|^2 / 2, \quad (3)$$

and omit the terms describing the helicity of the DNA structure (the helicity can be taken into account at the final stage of the calculations^{21-22, 53}). As a result of calculations, we obtain the discrete version of the model hamiltonian,

$$\begin{aligned}
H = & (m/2) \sum_n \{ [(du_{n,1}/dt)^2 + (R_0 - u_{n,1})^2 (d\Theta_{n,1}/dt)^2 + m(dz_{n,1}/dt)^2] \\
& + [(du_{n,2}/dt)^2 + (R_0 + u_{n,2})^2 (d\Theta_{n,2}/dt)^2 + m(dz_{n,2}/dt)^2] \} \\
& + (K/2) \sum_n \{ [2R_0^2 [1 - \cos(\Theta_{n,1} - \Theta_{n-1,1})] + u_{n,1}^2 + u_{n-1,1}^2 \\
& - 2u_{n,1} u_{n-1,1} \cos(\Theta_{n,1} - \Theta_{n-1,1}) - 2R_0 u_{n,1} [1 - \cos(\Theta_{n,1} - \Theta_{n-1,1})] \\
& - 2R_0 u_{n-1,1} [1 - \cos(\Theta_{n,1} - \Theta_{n-1,1})] + |z_{n,1} - z_{n-1,1}|^2 \\
& + |z_{n,2} - z_{n-1,2}|^2] + [2R_0^2 [1 - \cos(\Theta_{n,2} - \Theta_{n-1,2})] + u_{n,2}^2 + u_{n-1,2}^2 \\
& - 2u_{n,2} u_{n-1,2} \cos(\Theta_{n,2} - \Theta_{n-1,2}) + 2R_0 u_{n,2} [1 - \cos(\Theta_{n,2} - \Theta_{n-1,2})] \\
& + 2R_0 u_{n-1,2} [1 - \cos(\Theta_{n,2} - \Theta_{n-1,2})]] \} + (k/2) \sum_n \{ [2R_0^2 \{ (1 - 2\cos\Theta_{n,1}) \\
& + (1 - 2\cos\Theta_{n,2}) + [1 + \cos(\Theta_{n,1} - \Theta_{n,2})] \} - 2R_0 u_{n,1} (1 - 2\cos\Theta_{n,1}) \\
& + 2R_0 u_{n,2} (1 - 2\cos\Theta_{n,2}) + u_{n,1}^2 + u_{n,2}^2 - 2u_{n,1} u_{n,2} \cos(\Theta_{n,1} - \Theta_{n,2}) \\
& - 2R_0 u_{n,1} \cos(\Theta_{n,1} - \Theta_{n,2}) + 2R_0 u_{n,2} \cos(\Theta_{n,1} - \Theta_{n,2})] + k |z_{n,1} - z_{n,2}|^2 \},
\end{aligned} \tag{4}$$

which can be written in a more convenient form as

$$H = H(f) + H(\Psi) + H(g) + H(\text{interact.}), \tag{5}$$

where

$$\begin{aligned}
H(f) = & (m R_0^2/2) \sum_n (df_{n,1}/dt)^2 + (m R_0^2/2) \sum_n (df_{n,2}/dt)^2 + (K R_0^2/2) \sum_n (f_{n,1} - f_{n-1,1})^2 \\
& + (K R_0^2/2) \sum_n (f_{n,2} - f_{n-1,2})^2 + (k R_0^2/2) \sum_n (f_{n,1} + f_{n,2})^2,
\end{aligned} \tag{6}$$

$$\begin{aligned}
H(\Psi) = & (m R_0^2/2) \sum_n (d\Psi_{n,1}/dt)^2 + (m R_0^2/2) \sum_n (d\Psi_{n,2}/dt)^2 \\
& + (KR_0^2) \sum_n [1 - \cos(\Psi_{n,1} - \Psi_{n-1,1})] + (KR_0^2) \sum_n [1 - \cos(\Psi_{n,2} - \Psi_{n-1,2})] \\
& + (kR_0^2) \sum_n \{ 2(1 - \cos\Psi_{n,1}) + 2(1 - \cos\Psi_{n,2}) - [1 - \cos(\Psi_{n,1} + \Psi_{n,2})] \},
\end{aligned} \tag{7}$$

$$\begin{aligned}
H(g) = & (m R_0^2/2) \sum_n (dg_{n,1}/dt)^2 + (m R_0^2/2) \sum_n (dg_{n,2}/dt)^2 + (K R_0^2/2) \sum_n (g_{n,1} - g_{n-1,1})^2 \\
& + (K R_0^2/2) \sum_n (g_{n,2} - g_{n-1,2})^2 + (k R_0^2/2) \sum_n (g_{n,1} + g_{n,2})^2,
\end{aligned} \tag{8}$$

$$\begin{aligned}
H(\text{interact.}) = & (m R_0^2/2) \sum_n (-2f_{n,1} + f_{n,1}^2) (d\Psi_{n,1}/dt)^2 \\
& + (m R_0^2/2) \sum_n (-2f_{n,2} + f_{n,2}^2) (d\Psi_{n,2}/dt)^2 \\
& + (K R_0^2) \sum_n [1 - \cos(\Psi_{n,1} - \Psi_{n-1,1})] [f_{n,1} f_{n-1,1} - f_{n-1,1} - f_{n,1}] \\
& + (K R_0^2) \sum_n [1 - \cos(\Psi_{n,2} - \Psi_{n-1,2})] [f_{n,2} f_{n-1,2} - f_{n,2} - f_{n-1,2}] \\
& - (2k R_0^2) \sum_n (f_{n,1}) (1 - \cos\Psi_{n,1}) - (2k R_0^2) \sum_n (f_{n,2}) (1 - \cos\Psi_{n,2}) \\
& + (k R_0^2) \sum_n (-f_{n,1} f_{n,2} + f_{n,1} + f_{n,2}) [1 - \cos(\Psi_{n,1} + \Psi_{n,2})],
\end{aligned}$$

and new variables

$$\begin{aligned}
f_{n,1} &= u_{n,1}/R_0, & f_{n,2} &= -u_{n,2}/R_0, \\
\Psi_{n,1} &= \Theta_{n,1}, & \Psi_{n,2} &= -\Theta_{n,2}, \\
g_{n,1} &= z_{n,1}/R_0, & g_{n,2} &= -z_{n,2}/R_0.
\end{aligned}
\tag{10}$$

are used. Here $H(f)$ describes transverse motions; $H(\Psi)$ describes torsional motions; $H(g)$ describes longitudinal motions; $H(\text{interact.})$ describes interactions between the motions.

Usually it is suggested that the solutions are rather smooth functions (that is the functions $f_1, f_2, g_1, g_2, \Psi_1, \Psi_2$ change substantially only at the distances which are much more than the distance between neighboring base pairs), and continuous approximation is used. In the continuous approximation the model hamiltonian takes the form

$$\begin{aligned}
H_{\text{cont.}} = & (\rho_m R_0^2/2) \int dz [(\partial f_1/\partial t)^2 + (\partial f_2/\partial t)^2] + (Y R_0^2/2) \int dz [(\partial f_1/\partial z)^2 + (\partial f_2/\partial z)^2] \\
& + (y R_0^2/2) \int dz (f_1 + f_2)^2 + (\rho_m R_0^2/2) \int dz [(\partial g_1/\partial t)^2 + (\partial g_2/\partial t)^2] \\
& + (Y R_0^2/2) \int dz [(\partial g_1/\partial z)^2 + (\partial g_2/\partial z)^2] + (y R_0^2/2) \int dz (g_1 + g_2)^2 \\
& + (\rho_m R_0^2/2) \int dz [(1 - f_1)^2 (\partial \Psi_1/\partial t)^2 + (1 - f_2)^2 (\partial \Psi_2/\partial t)^2] \\
& + (Y R_0^2/2) \int dz [(1 - f_1)^2 (\partial \Psi_1/\partial z)^2 + (1 - f_2)^2 (\partial \Psi_2/\partial z)^2] \\
& + (y R_0^2) \int dz \{ 2 (1 - f_1) (1 - \cos \Psi_1) + 2 (1 - f_2) (1 - \cos \Psi_2) \\
& + (-f_1 f_2 + f_1 + f_2 - 1) [1 - \cos(\Psi_1 + \Psi_2)] \},
\end{aligned}
\tag{11}$$

where $m/a = \rho_m$; $Ka = Y$; $k/a = y$. And the dynamical equations which correspond to the model hamiltonian (11), can be easily obtained from the general theory of hamiltonian systems:

$$\begin{aligned}
\rho_m (d^2 f_1/dt^2) + \rho_m (1 - f_1) (d\Psi_1/dt)^2 = & Y \partial^2 f_1/\partial z^2 + Y (\partial \Psi_1/\partial z)^2 (1 - f_1) \\
& - y (f_1 + f_2) + 2y (1 - \cos \Psi_1) \\
& - y (1 - f_1) [1 - \cos(\Psi_1 + \Psi_2)],
\end{aligned}
\tag{12}$$

$$\begin{aligned}
\rho_m (d^2 f_2/dt^2) + \rho_m (1 - f_2) (d\Psi_2/dt)^2 = & Y \partial^2 f_2/\partial z^2 + Y (\partial \Psi_2/\partial z)^2 (1 - f_2) \\
& - y (f_1 + f_2) + 2y (1 - \cos \Psi_2) \\
& - y (1 - f_2) [1 - \cos(\Psi_1 + \Psi_2)],
\end{aligned}
\tag{13}$$

$$\begin{aligned}
\rho_m (1 - f_1) (d^2 \Psi_1/dt^2) - 2\rho_m (df_1/dt) (d\Psi_1/dt) = & Y (\partial^2 \Psi_1/\partial z^2) (1 - f_1) - \\
& - 2Y (\partial \Psi_1/\partial z) [\partial f_1/\partial z] - 2y [(\sin \Psi_1)] \\
& + y(1 - f_1) [\sin(\Psi_1 + \Psi_2)],
\end{aligned}
\tag{14}$$

$$\begin{aligned} \rho_m (1 - f_2) (d^2 \Psi_2 / dt^2) - 2\rho_m (df_2 / dt) (d\Psi_2 / dt) = & Y (\partial^2 \Psi_2 / \partial z^2) (1 - f_2) \\ & - 2Y (\partial \Psi_2 / \partial z) [\partial f_2 / \partial z] \\ & - 2y [(\sin \Psi_2)] \\ & + y(1 - f_1) [\sin(\Psi_1 + \Psi_2)], \end{aligned} \quad (15)$$

$$\rho_m (d^2 g_1 / dt^2) = Y \partial^2 g_1 / \partial z^2 - y (g_1 + g_2), \quad (16)$$

$$\rho_m (d^2 g_2 / dt^2) = Y \partial^2 g_2 / \partial z^2 - y (g_1 + g_2). \quad (17)$$

These are the sought dynamical equations. In the next section we show that among the solutions of the equations there are soliton-like solutions describing local unwinding of the double helix.

3.2 Solutions of the System Equations (12) - (17)

To find the solutions of equations (12) - (17), let us divide them into two subsystems. The first subsystem consists of equations (12) - (15) which describe transverse and torsional internal motions in DNA. The second subsystem consists of equations (16) - (17) which describe longitudinal motions. We should note, however, that the division of the system into two independent parts becomes possible only because we chose simplified form of the potential function $V(|U_{n,1} - U_{n,2}|)$ (see formula (3)). In the general case, when the formula for potential function is

$$V = D\{\exp[-A(|U_{n,1} - U_{n,2}|)] - 1\}^2, \quad (18)$$

the division is not possible. But here, in this paper, we limited ourselves by simplified formula (3), which can be considered as a first term in the expansion

$$V(|U_{n,1} - U_{n,2}|) = D\{\exp[-A|U_{n,1} - U_{n,2}|] - 1\}^2 = 2DA^2|U_{n,1} - U_{n,2}|^2/2 + \dots \quad (19)$$

3.2.1 Solutions of Subsystem (16) - (17): Longitudinal DNA Dynamics

In the approximation described above, longitudinal motions can be considered independently. Corresponding equations (16) - (17) are well known linear partial differential equations having the solutions in the form of usual plane waves

$$g_1(z,t) = g_{01} \exp[i(qz - w_1^{\text{long}} t)]; g_2(z,t) = g_{02} \exp[i(qz - w_2^{\text{long}} t)], \quad (20)$$

where g_{01} , g_{02} are the amplitudes of the waves and q is the wave vector which lies in the first Brillouin zone. Inserting (20) into (16) - (17) we can easily find the frequencies of longitudinal waves in DNA,

$$w_1^{\text{long}} = [(Y/\rho_m) q^2 + 2(y/\rho_m)]^{1/2}; w_2^{\text{long}} = [(Y/\rho_m)]^{1/2} q. \quad (21)$$

As follows from (21), the spectrum of longitudinal oscillations of DNA consists of two branches: one optical branch ($w_1^{\text{long}}(q)$) and one acoustic branch ($w_2^{\text{long}}(q)$).

3.2.2 Solutions of the Subsystem of Equations (12) - (15)

Let us consider in this section the other subsystem of the system of dynamical equations (12) - (17). It consists of four equations (12) - (15) which describes transverse and torsional motions. If we assume, however, that transverse motions (variables f_1 and f_2) are much faster than the torsional motions (variables Ψ_1 and Ψ_2), adiabatic approximation can be applied, and the subsystem of equations (12) - (15) can be divided into two independent parts. The first part describes torsional DNA dynamics and the second part will describe transverse DNA dynamics.

a) Transverse Dynamics

Let us consider equations (12) - (13) describing transverse dynamics, and suggest that the slow variables Ψ_1 and Ψ_2 are constants (C_1 and C_2 , relatively). Then the dynamical equations take the form

$$\rho_m (d^2 f_1 / dt^2) = Y \partial^2 f_1 / \partial z^2 - y(f_1 + f_2) + 2y(1 - \cos C_1) - y(1 - f_1)[1 - \cos(C_1 + C_2)], \quad (22)$$

$$\rho_m (d^2 f_2 / dt^2) = Y \partial^2 f_2 / \partial z^2 - y(f_1 + f_2) + 2y(1 - \cos C_2) - y(1 - f_2)[1 - \cos(C_1 + C_2)], \quad (23)$$

and corresponding model hamiltonian is

$$H_{\text{adiab.}}(f) = (\rho_m R_0^2 / 2) \int dz [(\partial f_1 / \partial t)^2 + (\partial f_2 / \partial t)^2] + (Y R_0^2 / 2) \int dz [(\partial f_1 / \partial z)^2 + (\partial f_2 / \partial z)^2] + (y R_0^2 / 2) \int dz (f_1 + f_2)^2 + (y R_0^2) \int dz \{ 2(1 - f_1)(1 - \cos C_1) + 2(1 - f_2)(1 - \cos C_2) + (-f_1 f_2 + f_1 + f_2 - 1)[1 - \cos(C_1 + C_2)] \}. \quad (24)$$

If we suggest boundary conditions in the form

$$\cos\Psi_1 \rightarrow 1, \text{ when } \Psi_1 \rightarrow \pm\infty, \quad \text{and} \quad \cos\Psi_2 \rightarrow 1, \text{ when } \Psi_2 \rightarrow \pm\infty, \quad (25)$$

then the following relations are valid:

$$(1 - \cos C_1) = (1 - \cos C_2) = [1 - \cos C_1 + C_2] = 0, \quad (26)$$

and the final formula for hamiltonian describing transverse subsystem takes the form

$$H_{\text{adiab.}}(f) = (\rho_m R^2/2) \int dz [(\partial f_1/\partial t)^2 + (\partial f_2/\partial t)^2] + (Y R^2/2) \int dz [(\partial f_1/\partial z)^2 + (\partial f_2/\partial z)^2] + (y R_0^2/2) \int dz (f_1 + f_2)^2. \quad (27)$$

So, we can conclude that in the adiabatic approximation and for simple model of potential function (3) any influence of torsional subsystem on transverse subsystem is absent. Equations (22) - (23) then take the form

$$\rho_m (d^2 f_1/dt^2) = Y \partial^2 f_1/\partial z^2 - y(f_1 + f_2), \quad (28)$$

$$\rho_m (d^2 f_2/dt^2) = Y \partial^2 f_2/\partial z^2 - y(f_1 + f_2), \quad (29)$$

and their solutions are

$$f_1(z, t) = f_{01} \exp[i(qz - w^{\text{tr}} t)]; \quad f_2(z, t) = f_{02} \exp[i(qz - w^{\text{tr}} t)], \quad (30)$$

where f_{01} , f_{02} are the amplitudes of the waves and q is the wave vector which lies in the first Brillouin zone. Inserting (30) into (28) - (29) we find the frequencies of transverse oscillations in DNA,

$$w_1^{\text{tr}} = [(Y/\rho_m) q^2 + 2(y/\rho_m)]^{1/2}; \quad w_2^{\text{tr}} = [(Y/\rho_m)]^{1/2} q. \quad (31)$$

So, the spectrum of transverse oscillations of DNA also consists of two branches: one optical branch ($w_1^{\text{tr}}(q)$) and one acoustic branch ($w_2^{\text{tr}}(q)$).

b) Torsional Dynamics

Let us consider now equations (14) - (15) describing torsional DNA dynamics. In accordance with adiabatic approximation, let us write all terms of hamiltonian (11), which contain variables Ψ_1 , Ψ_2 , and average the values depending on the variables f_1 and f_2 . As a result, we obtain the hamiltonian describing torsional motions in the adiabatic approximation,

$$\begin{aligned}
H_{\text{adiab.}}(\Psi) = E_0 &+ [\rho_m \langle (1-f_1)^2 \rangle R_0^2/2] \int dz (\partial \Psi_1 / \partial t)^2 + [\rho_m \langle (1-f_2)^2 \rangle R_0^2/2] \int dz (\partial \Psi_2 / \partial t)^2 \\
&+ [Y \langle (1-f_1)^2 \rangle R_0^2/2] \int dz (\partial \Psi_1 / \partial z)^2 + [Y \langle (1-f_2)^2 \rangle R_0^2/2] \int dz (\partial \Psi_2 / \partial z)^2 \\
&+ [y \langle (1-f_1) \rangle R_0^2] \int dz [2(1-\cos \Psi_1)] \\
&+ [y \langle (1-f_2) \rangle R_0^2] \int dz [2(1-\cos \Psi_2)] \\
&+ [y \langle (-f_1 f_2 + f_1 + f_2 - 1) \rangle R_0^2] \int dz [1 - \cos(\Psi_1 + \Psi_2)]. \quad (32)
\end{aligned}$$

Here E_0 is an average energy of transverse oscillations and $\langle \dots \rangle$ means averaging over the states of oscillating transverse subsystem.

From the general view of formula (32) we can conclude that the influence of the transverse subsystem on the torsional subsystem is reduced to a simple renormalization of the coefficients of the torsional hamiltonian. This gives us a possibility to use the results on the torsional DNA dynamics, obtained in one of our previous works⁵⁴. Indeed, the dynamical equations corresponding to hamiltonian (32) take the form

$$\rho_m \langle (1-f_1)^2 \rangle (d^2 \Psi_1 / dt^2) = Y \langle (1-f_1)^2 \rangle (\partial^2 \Psi_1 / \partial z^2) + y [2 \sin \Psi_1 - \sin(\Psi_1 + \Psi_2)]; \quad (33)$$

$$\rho_m \langle (1-f_2)^2 \rangle (d^2 \Psi_2 / dt^2) = Y \langle (1-f_2)^2 \rangle (\partial^2 \Psi_2 / \partial z^2) + y [2 \sin \Psi_2 - \sin(\Psi_1 + \Psi_2)]. \quad (34)$$

And we showed earlier⁵⁴, that equations of that type have among others the soliton-like solution

$$\Psi_1(z-vt) = -\Psi_2(z-vt) = 4 \arctan\{\exp[\gamma(\xi-\xi_0)/d]\}, \quad (35)$$

where $\gamma = [1 - (\rho_m/Y) v^2]^{-1/2}$, $\xi = z-vt$, and v is the velocity of propagation of the soliton. The values of mass (M), energy (E) and size (d) of the soliton are a little increased in comparison with the corresponding values obtained for torsional dynamical system where interactions with transverse and longitudinal motions are not taken into account:

$$E = 8\{2Yy\}^{1/2} \{\langle (1-f)^2 \rangle\}^{1/2}, \quad (36)$$

$$M = 8\rho_m \{2y/Y\}^{1/2} \{\langle (1-f)^2 \rangle\}^{1/2}, \quad (37)$$

$$d = Y \langle (1-f_1)^2 \rangle / 2y. \quad (38)$$

Graphic representation of the solution is shown in Fig. 4a, and qualitative picture which corresponds to this solution is shown schematically in Fig. 4b. So, the solitary wave solution can be really interpreted as that describing unwound region

(or open state). Or to say more exactly, it describes the boundary between wound and unwound regions.

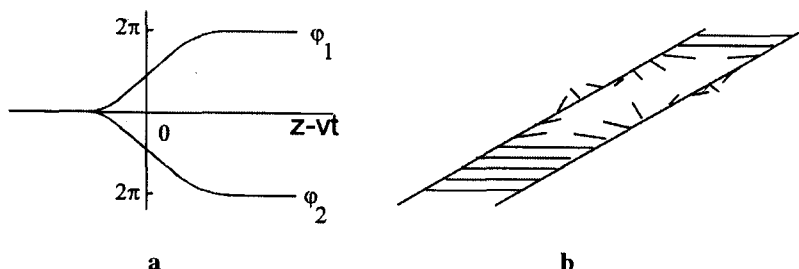


Figure 4. (a) Soliton-like solution of equations (33) - (34) and (b) corresponding qualitative picture. Long lines are used to show schematically sugar-phosphate chains, and short lines - to show bases.

4 Applications to Biology

In the previous section we described the algorithm of mathematical modeling of large-amplitude motions in DNA. To be concrete, we considered, as an example, only one type of internal motions, namely, local unwinding of the double helix. According to the algorithm, we obtained nonlinear differential equations and soliton-like solutions describing boundary between wound and unwound regions. We should note, however, that the approach described above is rather general and it can be used to model another types of large-amplitude motions. For example, it can be successfully used to describe transitions between different forms of the double helix. In this case, soliton-like solutions of corresponding nonlinear differential equations will describe the boundary between the regions with different structural forms (A- and B-forms, for example).

In this section, we discuss applications of the approach to biology. We present two examples to illustrate possible connection between nonlinear dynamics and DNA functioning. The first of the examples concerns the problem of long-range effects in DNA, and the second one concerns the problem of direction of the process of transcription.

4.1 Long-Range Effects in DNA

During the 70's and 80's, a great deal of experimental work was done on long-range effects⁵⁵⁻⁶⁸. A simple scheme illustrating the effects is shown in Fig. 5. The scheme consists of two protein molecules (1 and 2) and one DNA molecule with two sites of specific binding with the proteins. It is assumed that the first protein molecule can bind with site 1 and the other protein molecule can bind with the other site: site 2. The effect is that the binding of the first protein with site 1 influences the binding of the second protein molecule with site 2 in spite that the distance between the sites can reach hundreds or thousands (as in the case of enhancers) of base pairs.

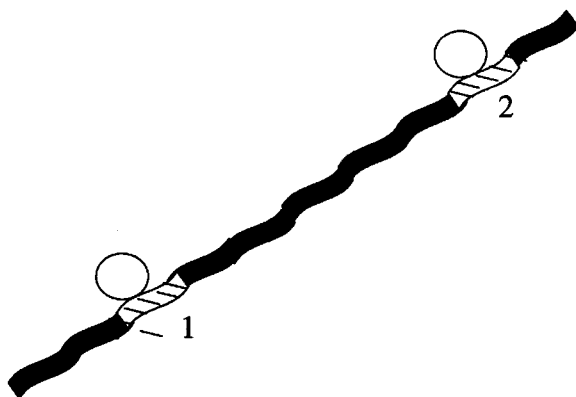


Figure 5. A simple scheme illustrating long-range interactions in DNA. The DNA molecule is presented by black band; the sites (1 and 2) interacting with proteins are shaded; protein molecules are presented by small circles.

Among different explanations of the effect (see reviews⁶⁰⁻⁶²) there is one which is of most interest. According to it, the effect of the binding of the first protein molecule to site 1 is accompanied by a local distortion of the DNA structure, which can then propagate along the double DNA chain. When reaching site 2, it changes the conformational structure of the site which, in turn, changes the binding constants of the second protein with the site.

This mechanism can be easily interpreted in terms of the nonlinear approach. Indeed, if the local distortion is associated with substantial structural changes of the DNA site (for example, with the transitions from B-form to A-form), we can consider the formation of the local distortion as a complex internal motion of large

amplitude. Then we can choose appropriate dynamical model and write corresponding nonlinear differential equations. As a result, we could expect that among the solutions of the equations there should be soliton-like solutions, which describe local distortion moving along the DNA.

4.2 *Direction of Transcription Process*

In general, transcription can be characterized as a complex multistage process which proceeds in a system of many components, including DNA, RNA polymerase, regulatory proteins, hormones, ions, water, etc. The simplest scheme of the beginning of the process includes three stages: initiation, elongation and termination.

At the first stage a special protein named RNA-polymerase binds with a special DNA site named a promoter region which can be defined as a point of initiation (Fig. 6a). When RNA polymerase takes a correct position and forms several phosphodietheric bonds, the second stage of the process begins. At this stage

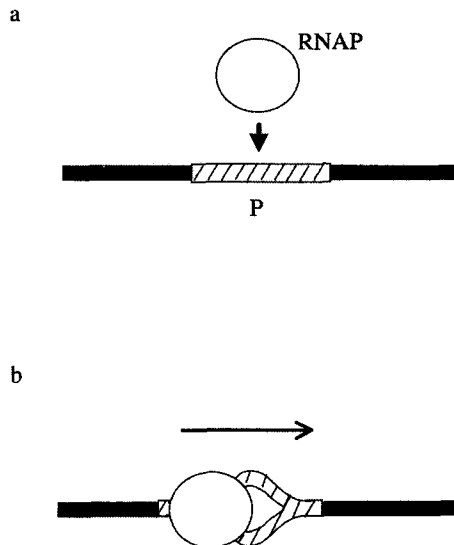


Figure 6. RNA-polymerase (RNAP) and DNA before (a) and after (b) binding and formation of unwound region.

a small fragment (subunit σ) is released from the RNA polymerase and the rest of the molecule (core ferment) moves along the DNA and elongates step by step the RNA molecule (Fig. 6b). In the third stage, the process finishes and RNA polymerase is released from the DNA molecule. A special region of DNA named terminator gives a signal to stop the process.

In this section we shall deal with the first stage of the transcription process when RNA-polymerase binds with promoter region. Details of the process of binding are now under discussion⁶³. But it is well known that the process is accompanied by considerable local distortion of the DNA conformation including local unwinding of the double helix⁶⁴⁻⁶⁸(Fig. 6b). After that the disturbed region begins to move along the DNA molecule. What is the direction of the movement? Can we use the model described above to predict the direction of the movement? The answer is "no," because the model is homogeneous (all bases in DNA are assumed to be the same). But if we modify the model by taking into account differences of the bases, that is, if we consider inhomogeneous model, the answer changes to "yes". Mario Salerno²⁵ was the first who solved the problem. He suggested that the direction of the movement depended on the sequence of bases of promoter region. To prove the suggestion, he used mathematical model of DNA in the discreet form with coefficients, which were dependent on the sequence of bases, and with initial condition in the form of soliton-like solution of usual sine-Gordon equation. As a result, he obtained good correlation of computer experiment with the available experimental data on the direction of transcription^{25, 69-71}.

5 Applications to Nonlinear Mathematics and Physics

In this section we discuss what biology can give to nonlinear mathematics and physics. We describe shortly new interesting mathematical and physical problems that appeared in connection with the internal DNA dynamics.

5.1 New Nonlinear Mathematical Problems

Interaction of DNA science with nonlinear mathematics looks very promising because it leads to new mathematical equations with new interesting solutions. System of equations (12) - (16) is only one of possible examples. If we consider subsystem (33) - (34) which is a simpler version of the system of equations (12) - (16), we should state that even this simple subsystem has not been studied completely. Particular solution (35) mentioned above is only one of possible solutions of the system. And we do not know anything about other solutions of the equations. This problem remains for future investigations.

5.2 New Types of Inhomogeneous Models

Many different models of inhomogeneity are used in physics. The most popular of them are:

- point inhomogeneity,
- boundary between two neighboring homogeneous ranges and
- random inhomogeneity.

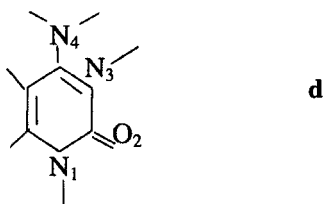
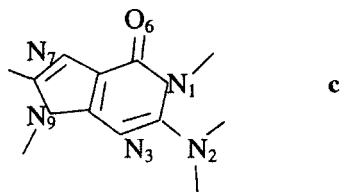
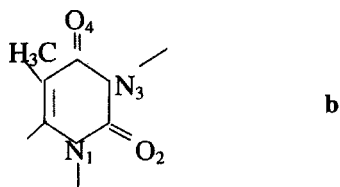
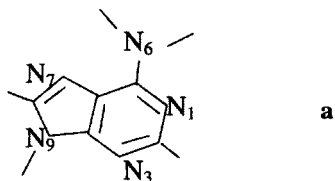


Figure 7. Four types of bases in DNA: adenine (a), thymine (b), guanine (c), and cytosine (d).

DNA gives us a new type of inhomogeneity, which is appeared due to the sequence of bases. Four types of bases (A, T, G and C) which are shown in Fig. 7 form the sequence. An example of a fragment of the sequence is shown in Fig. 8a. The bases of one of the two DNA chains interact with the bases of the other chain through hydrogen bonds, two types of base pairs (AT or GC) being possible. They are shown in Fig. 8b, 8c.

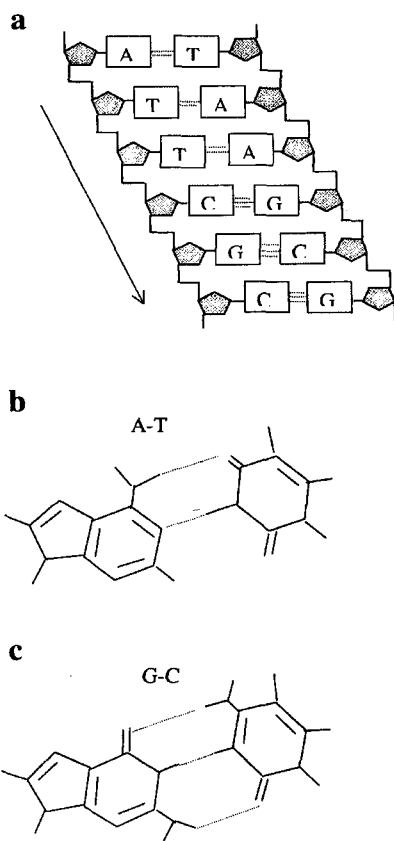


Figure 8. Fragment of the DNA double chain with the sequence : ATTCGC (a); A-T base pair (b); G-C base pair (c). Hydrogen bonds are shown by dotted lines.

In the models considered above we ignored the difference between the bases. That is, we considered the models like this



But model (39) is not correct even for homogeneous fragment of DNA (for example for synthetic poly A-poly T). A more correct variant is



From the point of view of physics, we could say that the fragment of DNA (40) looks like a quasi-one dimensional crystal with two "atoms" (nucleotides) in the cell. In real DNA, we have, however, the sequence which looks, for example, like this



This type of inhomogeneity is very unusual from physical point of view. Now, only a few works devoted to studies of the nonlinear internal dynamics of inhomogeneous DNA are known^{25, 69-72}, and the general solution of the problem remains for future investigations.

5.3 *Interaction with the Surrounding*

Till now we discussed DNA models where interactions with the surrounding were not taken into account. In the general case, the modeling of the DNA-environment interaction is a rather complex problem. In the first approximation, it can be reduced to consideration of only two effects:

- the effect of dissipation and
- the effect of interaction with an external field.

In the language of mathematics it means that we should add two additional terms to each dynamical equations:

- one term to describe dissipation, and
- the other term to describe interaction with external fields.

If we assumed that the DNA-environment interaction leads to small perturbations of the solutions of the ideal model dynamical equations, linear

perturbation technique can be used to solve corresponding equations⁷². The results obtained in this approximation show that the behavior of soliton-like solutions is very similar to the behavior of usual classical particles. And this gives us a possibility to solve other interesting physical problems: the problem of statistics of solitons in DNA and the problem of scattering by DNA solitons.

5.4 Statistics of Solitons in DNA

In the dynamical models discussed above it was suggested that only one soliton could be excited in DNA. But DNA is a rather long molecule, and we can expect that several solitons can be excited simultaneously. In this case, we should consider a problem of ensemble of solitons and discuss their statistics. Consideration of ensemble of solitons becomes very important when we try to interpret experimental data on scattering (neutrons or light) by DNA or the data on DNA denaturation.

Different approaches to study ensemble of DNA solitons and their statistics are known. Peyrard and co-authors¹⁷ developed one of the approaches based on the method of transfer operators. This approach gives a possibility to calculate the classical partition function, free energy, the specific heat and other characteristics of the nonlinear DNA system. The results of calculations were successfully applied to the problem of DNA denaturation.

Fedyanin and Yakushevich⁶ have proposed another approach. It was based on the similarity between the dynamical behavior of solitons and usual classical particles. In this approach, we can 1) ascribe mass m , velocity v and energy E to DNA soliton, and 2) consider ensemble of ordinary classical particle (instead of consideration of ensemble of solitons) with the same physical characteristics. To simplify calculations, it is suggested also that the number of solitons is not large and ensemble of particles can be described as an ideal gas. After all these suggestions it is not difficult to calculate different macroscopic characteristics of the ensemble such as large statistical sum, thermodynamical potential, correlation particles, density of the particles and others. All these results were obtained, however, only for homogeneous model of DNA. We have not any idea about inhomogeneous case.

5.5 Scattering of Neutrons and Light

To solve the problem of scattering, it is enough to calculate the dynamical form-factor. According to the general theory, the dynamical factor is completely determined by correlation functions. The functions in turn can be easily calculated in the approximation of an ideal gas^{6, 73}. The results of calculations predict the existence of the central peak in the spectrum of scattering. It was shown that the parameters of the peak such as the integral intensity and the width depend on the temperature and on the wave vector which is a difference between the final and initial wave vectors of neutron scattered by DNA. It was shown also that taking into

account the helical structure of DNA leads to the splitting of the peak into two components, and this prediction can be checked experimentally.

All these results were obtained, however, for simplified model of ensemble of DNA solitons and for homogeneous case. Improving this approach remains for future investigations.

6 Conclusions

We described here DNA as a nonlinear dynamical system To be concrete, we considered in details the algorithm of mathematical modeling one of possible internal motions, namely, local unwinding of the double helix. It was stressed, however, that the algorithm is rather general and it can be used to model other large-amplitude motions. We derived new mathematical equations describing nonlinear DNA dynamics. We showed that the equations have solitary wave solutions that can be interpreted as a boundary between wound and unwound regions. We tried to show interesting perspectives of the approach. We discussed how the approach could be applied to explain some elements of DNA functioning. We described also new mathematical and physical problems that arose due to interaction of nonlinear physics and mathematics with DNA science.

References

1. S. W. Englander, N. R. Kallenbach, A. J. Heeger, J. A. Krumhansl and A. Litwin, *Proc. Natl. Acad. Sci. USA* **77**, 7222 (1980).
2. S. Yomosa; *Phys. Rev. A* **27**, 2120 (1983).
3. S. Takeno and S. Homma, *Prog. Theor. Phys.* **70**, 308 (1983).
4. J. A. Krumhansl and D. M. Alexander; in *Structure and dynamics: nucleic acids and proteins*, eds. E. Clementi and R. H. Sarma (Adenine Press, New York, 1983, 61).
5. J. A. Krumhansl, G. M. Wysin, D. M. Alexander, A. Garcia, P. S. Lomdahl and S. P. Layne in *Structure and motion: membranes, nucleic acids and proteins*, eds. E. Clementi, G. Corongiu, M. H. Sarma and R. H. Sarma (Adenine Press, New York, 1985, 407).
6. V. K. Fedyanin and L. V. Yakushevich, *Stud. biophys.* **103**, 171 (1984).
7. V. K. Fedyanin, I. Gochev and V. Lisy, *Stud. biophys.* **116**, 59 (1986).
8. V. K. Fedyanin and V. Lisy, *Stud. biophys.* **116**, 65 (1986).
9. L. V. Yakushevich, *Phys. Lett. A*-**136**, 413 (1989).
10. Ch.-T. Zhang, *Phys. Rev. A*-**35**, 886 (1987).
11. Ch.-T. Zhang, *Phys. Rev. A*-**40**, 2148 (1989).

12. E. W. Prohofsky, *Phys. Rev. A* **-38**, 1538 (1988).
13. V. Muto, J. Holding, P. L. Christiansen and A. C. Scott, *J. Biomol. Struct. Dyn.* **5** 873 (1988).
14. V. Muto, A. S. Scott and P. L. Christiansen, *Phys. Lett. A* **-136**, 33 (1989).
15. V. Muto, P. S. Lomdahl and P. L. Christiansen, *Phys. Rev. A* **-42**, 7452 (1990).
16. L. L. Van Zandt, *Phys. Rev. A* **-40**, 6134 (1989).
17. Peyrard M. and A. R Bishop, *Phys. Rev. Lett.* **62**, 2755 (1989).
18. G.-F. Zhou and Ch.-T. Zhang, *Phys. Scripta* **43**, 347 (1991).
19. T. Dauxois, M. Peyrard and C. R. Willis, *Phys. D* **-57**, 267 (1992).
20. T. Dauxois, *Phys. Lett. A* **-159**, 390 (1991).
21. G. Gaeta, *Phys. Lett. A* **-143**, 227 (1990).
22. G. Gaeta, *Phys. Lett. A* **-168**, 383 (1992).
23. G. Gaeta, C. Reiss, M. Peyrard and T. Dauxois, **17**, 1 (1994).
24. G. Gaeta, *J. Biol. Phys.* **24**, 81 (1999).
25. M. Salerno, *Phys. Rev. A* **-44**, 5292 (1991).
26. A. A. Bogolubskaya and I.L. Bogolubsky, *Phys. Lett. A* **-192**, 239 (1994).
27. W. Hai, *Phys. Lett. A* **-186**, 309 (1994).
28. J. A. Gonzalez and M. Martin-Landrove, *Phys. Lett. A* **-191**, 409 (1994).
29. M. Barbi, S. Cocco, M. Peyrard and S. Ruffo, **24**, 97 (1999).
30. A. Campa and A. Giansanti, *J. Biol. Phys.* **24**, 141 (1999).
31. S. J. Webb and A. D. Booth, *Nature* **222**, 1199 (1969).
32. M. L. Swicord and C. C. Davis, *Biopolymers* **21**, 2453 (1982).
33. M. L. Swicord and C. C. Davis, *Bioelectromagnetics* **4**, 21 (1983).
34. K. F. Baverstock and R. D. Cundal, *Int. J. Radiat. Biol.* **55**, 152 (1989).
35. A. Khan, D. Bhaumic and B. Dutta-Roy, *Bull. Math. Biol.* **47**, 783 (1985).
36. H. M. Sobell in: *Biological macromolecules and assemblies*, eds. F.A. Jurnak and A. McPherson (John Wiley and Sons, New York, 1984, 172).
37. L. V. Yakushevich, *Mol. Biol. (Russian J.)* **23**, 652, (1989).
38. L. V. Yakushevich, *Nanobiology* **1**, 343 (1992).
39. S. N. Volkov, *J. Theor. Biol.* **143**, 485 (1990).
40. R. V. Polozov and L. V. Yakushevich, *J. Theor. Biol.* **130**, 423 (1988).
41. E. Balanovski and P. Beaconsfield, *Phys. Rev.* **32**, 3059 (1985).
42. J. J. Ladik, S. Suhai and M. Seel, *Int. J. Quant. Chem. QBS Suppl.* **5**, 35 (1978).
43. P. R. Selvin, D. N. Cook, N. G. Pon, W. R. Bauer, M. P. Klein and J. E. Hearst, *Science* **255**, (1992).
44. A. C. Scott, *Comments Mol. Cell. Biol.* **3** 5 (1985).
45. L. V. Yakushevich, *Quart. Rev. Biophys.* **26**, 201 (1993).
46. M. Peyrard, ed. *Nonlinear Excitations in Biomolecules* (Springer, Berlin, 1995).
47. L. V. Yakushevich, *Nonlinear physics of DNA* (Wiley, New York, 1998).
48. H. Fritzsche, *Comm. Mol. Biophys.* **1**, 325 (1982).
49. J. W. Keepers and Th L. James, *J. Am. Chem. Soc.* **104** 929 (1982).
50. W. R. McClure, *Ann. Rev. Biochem.* **54**, 171 (1982).

51. J. A. McCammon and S. C. Harvey, *Dynamics of proteins and nucleic acids* (Cambridge University Press, Cambridge, 1987).
52. L. V. Yakushevich and V. M. Komarov, *Mathematics. Computer. Education (in Russian)* **5**, 310, (1998).
53. L. V. Yakushevich in *Mathematical models of non-linear excitations, transfer, dynamics, and control in condensed systems and other media*, eds. L. V. Uvarova, A. E. Arinstein and A. V. Latyshev (Plenum, New York, 2000, 93).
54. L.V. Yakushevich, *Russian J. Phys. Chemistry* **69**, 1277 (1995).
55. F. M. Pohl, T. M. Jovin, W. Baehr and J. J. Holbrook, *Proc. Natl. Acad. Sci. USA* **69**, 3805 (1972).
56. G. B. Kolata, *Science* **198**, 41 (1977).
57. M. Hogan, N. Dattagupta and D. M. Crothers, *Nature* **278**, 52 (1979).
58. D. M. Crothers and M. Fried, *Cold Spring Harbor Symposia on Quantitative Biology* **47**, 263 (1983).
59. R. D. Wells, T. C. Goodman, W. Hillen, G. T. R. D. Horn Klein, J. E. Larson U. R. Müller, S. R. Nevendorf, N. Panayatatos and S. M. Stirdivant in *Progress in Nucleic Acid Research and Molecular Biology* (Academic Press, New York, **24**, 167, 1980).
60. M. Ptashne, *Nature* **322**, 687 (1986).
61. J. C. Wang. and G. N. Giaever *Science* **240**, 300. (1988).
62. K. Adzuma and K. Miznuchi, *Cell* **57**, 41 (1989).
63. R. L. Gourse, W. Ross and T. Gaal, *Mol. Microbiol.* **37**, ..., 2000.
64. D. Pribnow in *Biological Regulation and Development. I. Gene Expression*. ed. R. F. Golderberger (Plenum, New York, 1979, 219).
65. P. H. von Hippel in *Biological Regulation and Development. I. Gene Expression*. ed. R. F. Golderberger (Plenum, New York, 1979, 279).
66. U. Scienbenlist, *Nature* **279**, 651 (1979).
67. U. Scienbenlist, *Cell* **20**, 269 (1980).
68. M. Sluyer, *Trends in Biochemical Sciences* **8**, 236 (1983).
69. M. Salerno, *Phys. Lett. A*-**167**, 49 (1992).
70. M. Salerno and Yu. S. Kivshar, *Phys. Lett. A*-**193**, 263 (1994).
71. M. Salerno in: *Nonlinear Excitations in Biomolecules*, ed. M. Peyrard (Springer, Berlin, 1995, 147).
72. L. V. Yakushevich, *Stud. biophys.* **121**, 201 (1987).
73. L. V. Yakushevich, *J. Biol. Phys.* **24**, 131 (1999).

STRATEGIES TO STUDY DISTRIBUTION AND FUNCTION OF MINIGENES IN MICROORGANISMS

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The genomes of bacteria and their viruses harbor nucleotide sequences that encode very short peptides. Although these small "genes," named minigenes, have been dismissed as genomic curiosities, we have shown that they are functionally recognizable. Minigenes are transcribed into messenger molecules, or mRNAs, which are translated on ribosomes, the cellular organelles where the synthesis takes place. However, contrary to most messengers which are translated into mature proteins, some minigene mRNAs provoke peptidyl-tRNA release at high frequency. Under limiting peptidyl-tRNA hydrolase activity, tRNA remains sequestered and translation stops. Most of the above conclusions have been derived from minigene over-expression. It is possible that the role of minigenes in the cell may be more complex than just to inhibit protein synthesis. Nevertheless, fewer than expected lethal minigenes were found in a survey of a random library constructed from lambda DNA small fragments. Lethality by over expression of foreign genes, usually rich in rare codons, is common in *E. coli*. We think that at least some of those events may be explained by an inhibition mechanism similar to that of minigenes.

1 Introduction

1.1 The Central Dogma

The so-called "central dogma of molecular biology" states that the genetic information in the cells flows from DNA to protein through RNA (Fig. 1). Deoxyribonucleic acid (DNA) is a helix of two strands formed by a sequence of many chemically linked subunits named nucleotides. Each nucleotide is formed by one of four nitrogenous bases, a deoxyribose sugar and a phosphate group. The four bases in DNA are adenine, thymine, guanine and cytosine (A, T, G and C). The double helix is kept together by specific interactions between A and T and between G and C pairs. It is the particular sequence of the four different nucleotides in a gene that codes for the sequence of amino acids that form the corresponding proteins.

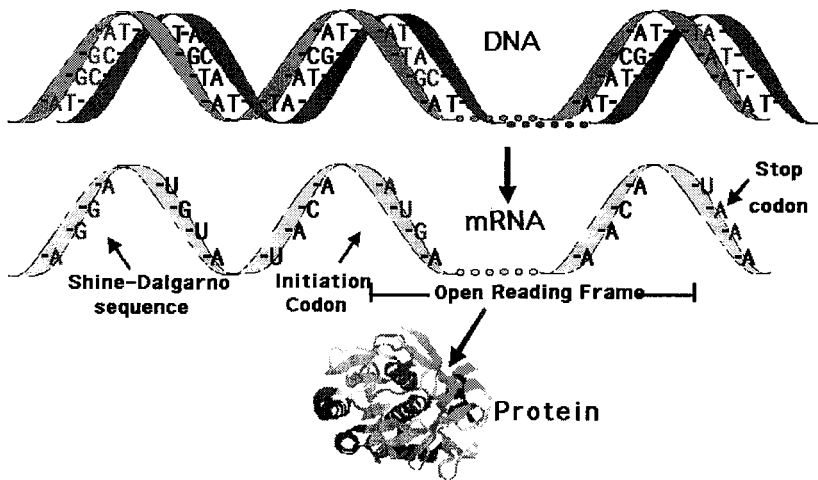


Figure 1. Central dogma of molecular biology. The translational signals in mRNA are indicated (see text).

DNA is transcribed into single-stranded ribonucleic acid molecules, or RNA, identical in sequence with one of the strands of the DNA but for possessing uracil (U) instead of T and the nucleotide subunits containing a ribose instead of deoxyribose. Three different types of RNA are involved in the synthesis of proteins: messenger RNA (mRNA), ribosomal RNA (rRNA), and transfer RNA (tRNA).

1.2 Translation

Messenger RNAs are translated into proteins. Transfer RNA (tRNA) and ribosomal (rRNA) provide other components of the protein synthesis machinery. The region of mRNA translated, named open reading frame or simply ORF, is related to a protein sequence by the genetic code. Each triplet of neighboring nucleotides, termed codon, specifies one of the 20 common amino acids in proteins. In addition to the ORF to be translatable mRNA requires a four to six nucleotide sequence named Shine-Dalgarno (SD) that pairs with ribosomal RNA, an initiator codon, usually AUG, and any of three terminator codons.

Proteins are chains of many amino acid residues (hold together by peptide bonds) synthesized on subcellular organelles named ribosomes. The codons in mRNA interact with triplets, named anticodons, in tRNA molecules, which carry specific amino acids (Fig. 2). The amino acids are then transferred to the elongating protein, one at a time, at the ribosome active center.

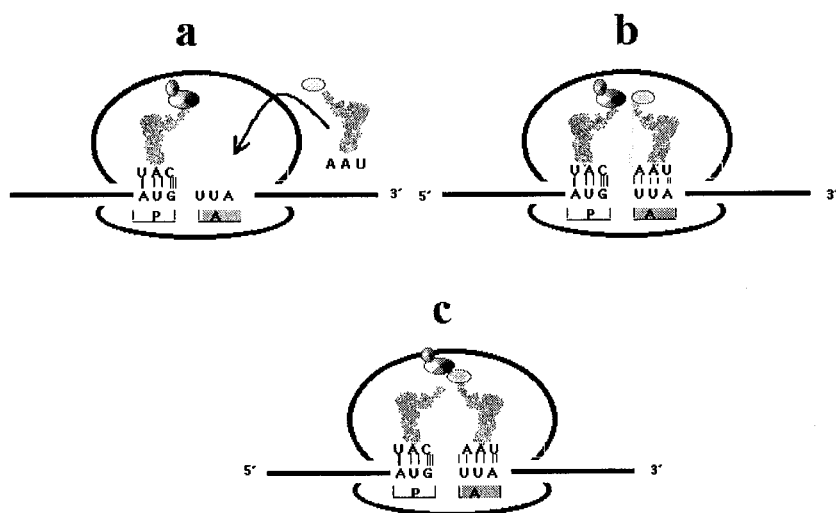


Figure 2. Translation initiation and elongation. a) Initiator fMet-tRNA harboring the anti-codon 3'-UAC pairs with initiation codon 5'-AUG at the ribosomal P-site. b) The next codon in the mRNA, 5'-UUC in this example, positioned at the ribosomal A-site is recognized by the amino acyl-tRNA bearing anti-codon 3'-AAG. c) The peptide bond forms by transfer of f-Met to the amino acyl-tRNA in the A-site.

Translation termination occurs when a terminator codon aligns on the ribosome. Then, instead of an amino acyl-tRNA, a release factor (RF1 or RF2) associates with the ribosome provoking hydrolysis of the peptidyl-tRNA linkage and release the mature protein (Fig. 3). Interference with termination process may cause release of the peptidyl-tRNAs from the ribosomes.

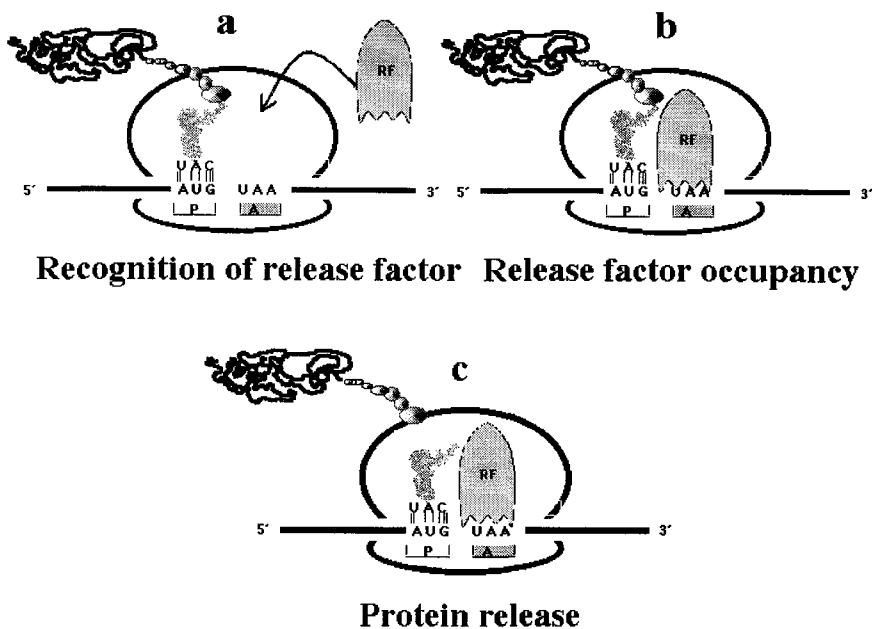


Figure 3. Translation termination. a) A termination codon, UAA in this example, at the end of the ORF in an mRNA aligns on the ribosomal A site. b) The termination codon is recognized by a release factor protein (RF1 or RF2). c) The hydrolysis of the last peptidyl-tRNA occurs releasing the mature protein.

1.3 Minigenes

Most genes in bacteria residues are transcribed in mRNAs that encode between one hundred and one thousand amino acid. We name minigenes the sequences that resemble genes in every respect but carrying ORFs containing only two to six codons. Although the presence of minigenes in bacterial genomes has been known for a long time, it was only recently that they have been associated with a phenotype.

Bacteriophage lambda is a bacterial virus that grows on *Escherichia coli*, a common bacteria in the intestine of vertebrates. However, lambda is unable to grow in bacterial mutants partly defective for peptidyl-tRNA hydrolase (Pth) an enzyme essential for the cell viability. The growth defect was associated to the expression of minigenes in the phage genome because mutations within these regions, named *bar*, allow phage growth on Pth-defective cells (Fig. 4).

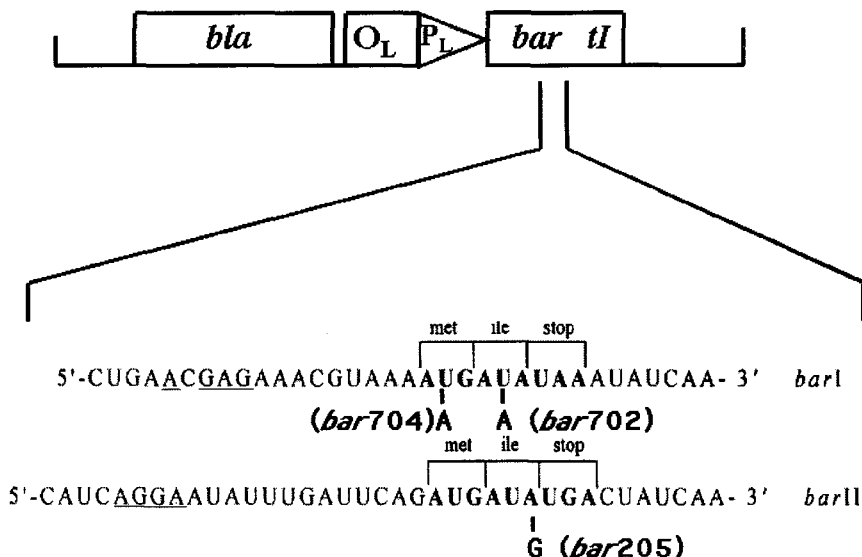


Figure 4. Minigenes in bacteriophage lambda. a) Minigenes *bar* mRNAs including the Shine-Dalgarno (underlined) and mini-ORF sequences. The encoded amino acids, termination (stop) codons, and base substitutions, which inactivate the *barI* and *barII*-mediated lethality, are indicated. b) Plasmid constructs to express minigenes. The constructs are derivatives of pBR322, a plasmid that harbors gene *bla*, which encodes beta-lactamase protein and O_L - P_L , a regulable transcriptional operator-promoter sequence. The *bar* region is a lambda DNA segment of about 500bp harboring *barI* minigene and a transcriptional terminator *tI*.

2 Results and Models

2.1 Aberrant Translation of Minigene mRNAs

We developed a model system in which minigenes are expressed from multicopy constructs (Fig. 4).

Minigene expression is lethal to Pth-defective cells where it prevents protein, but not mRNA, synthesis (Fig. 5) [11]. This result suggests that minigene expression inhibits translation but not transcription. Paradoxically, inhibition requires minigene mRNA translation; base substitutions at the SD sequence or at either of the two codons in the translatable region abolished lethality [14].

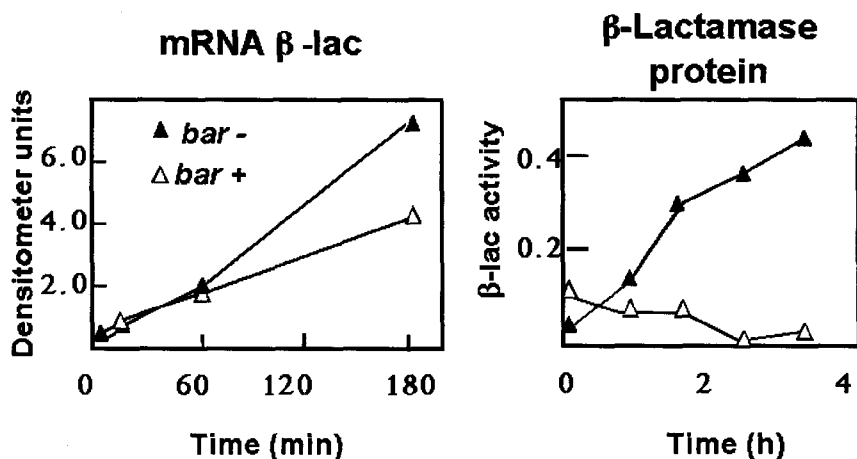


Figure 5. Minigene expression prevents translation. Upon transcriptional induction of *barI* minigene (*bar*+) or the inactive mutant *bar704* (*bar*-) in a *pth*-defective strain, *bla* mRNA concentration or beta-lactamase activity were determined at the indicated times (modified from Pérez-Morga and Guarneros, 1990 [11]).

In addition minigene mRNA and ribosomes interact in the same fashion as *bona fide* mRNAs interact with ribosomes (Fig. 6) [10].

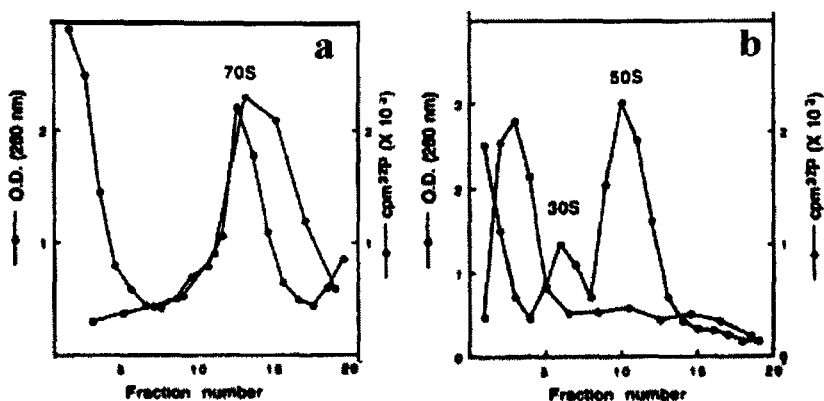


Figure 6. Transcripts from the *bar* region interact with ribosomes as authentic mRNA. a) *bar* transcripts on a ribosome profile in a sucrose gradient were revealed by a [^{32}P]-labeled probe under conditions that maintain the ribosome structure (70S) or b) under low Mg^{++} concentration, which dissociates 70S particles into 30S and 50S subunits. The ribosomal peaks were detected by absorbance (O.D.) at 260 nm. The transcripts banded in a slightly heavier fraction than the ribosomal peak in a) but remained at the top of the gradient in b) as typical mRNAs do (modified from Ontiveros *et al.*, 1997 [10]).

Why does the Pth defect aggravate minigene-mediated lethality? The function of Pth activity is to hydrolyze peptidyl-tRNAs (pep-tRNAs) released from ribosomes in abortive events of protein synthesis [9].

The expression of minigenes *in vitro* inhibits protein synthesis (Fig. 7). As additions of a purified preparation of Pth, or of specific tRNA, prevent the inhibition, the simplest explanation for this result is that minigene expression generates accumulation of pep-tRNA that sequesters tRNA essential for synthesis [6].

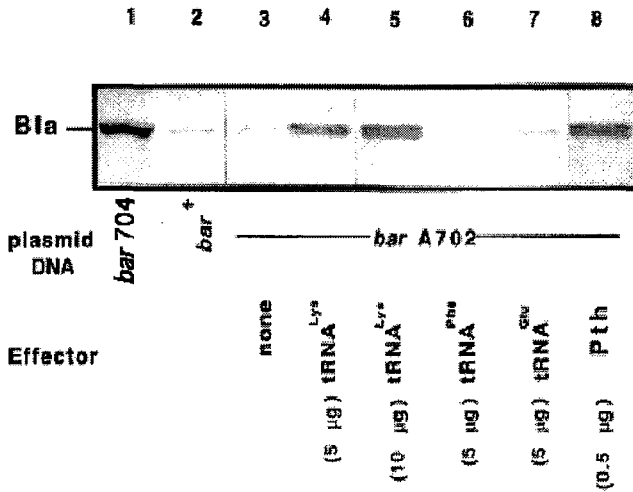


Figure 7. Starvation of tRNA by minigene expression. Bands correspond to *in vitro* labeled [35 S]-Methionine beta-lactamase protein (Bla) that were resolved by SDS-PAGE and revealed by autoradiography. The *in vitro* synthesis was primed with plasmid DNAs harboring the indicated minigenes: bar⁺ is the *barI* wild type minigene, bar704 is an inactive mutant and barA702 is an active substitution encoding Lys in *barI*-minigene second position (Fig. 4). Other effector additions in each reaction are indicated (see text). (Modified from Hernández-Sánchez *et al.*, 1998 [6]).

Recent studies by other authors have confirmed and extended these results [5]. They have shown that minigene toxicity is influenced by the rates of pep-tRNA drop-off and pep-tRNA hydrolysis mediated by Pth.

2.2 Many Possible Minigenes

The two lambda minigenes studied in depth have the same AUG AUA sequence (Fig. 4). We addressed the question whether other codon combinations behaved as lambda minigenes. In collaboration with Alexander Mankin (University of Illinois) who had isolated minigenes from a random library [12]. The library contained minigenes with fixed initiation (AUG) and termination (UAA) codons and a random sequence of four codons in between. A sample of eight minigenes selected by toxicity shows that there is a bias in codon composition (Arg, Ile and His) and in

size (less than five sense codons) (Fig. 8)[13]. However, no definitive conclusions could be drawn from such a small sample. However, latter work by other groups confirmed the size inference [7].

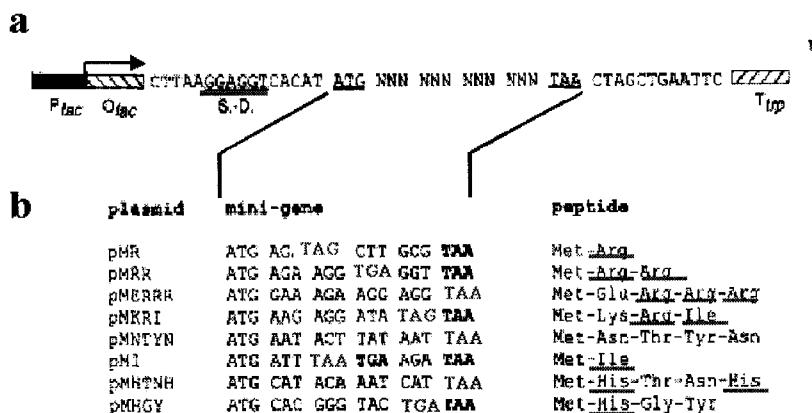


Figure 8. Toxic minigenes from a random library. a) Map of a segment in the plasmid constructs used to express the randomly generated minigenes. P_{tac} - O_{tac} is a promoter-operator regulable sequence; SD is the Shine-Dalgarno sequence that allows translation at the neighboring initiation codon AUG in the transcripts; T_{top} is a terminator that prevents transcription into the adjacent vector sequences. b) A sample of lethal minigenes isolated (Modified from Tenson *et al.*, 1999 [13]).

Interestingly, the expression of deleterious minigenes in Pth-deficient cells causes accumulation of pep-tRNA specific to the tRNA cognate to the last sense codon in the minigene. Also, the inhibitory effect of the minigene expression is suppressed by overexpression of the same tRNA. These results are consistent with the notion that minigene mediated inhibition of protein synthesis is due to starvation of the specific tRNA by sequestration into pep-tRNA.

2.3 Codon Composition in Lethal Minigenes

To investigate whether toxicity depended on the codons in the mRNA or the amino acids in the peptidyl-tRNA, we conducted a survey of minigenes encoding peptides of only two amino acids. In these constructs the initiation and termination codons were AUG and UAA, respectively, and the second codon was any of the 64 possibilities. The results (Cruz-Vera *et al.* in preparation), show that; i) the termination codons were harmless, as expected from the implication of pep-tRNA in toxicity, ii) The range of toxicity was very broad, from low (Gly, Ala and Cys) to high toxicity (Lys, Arg and Ile) and iii) Most of the minigenes which encode the same second amino acid through different but synonymous codons which are decoded by the same tRNA, confer comparable degree of toxicity (i.e., Phe, UUU and UUC; Leu, CUU and CUC; Ile AUU and AUC). However, there are exceptions that differ greatly in lethality degree, i.e., Arg, CGU, CGC and CGA and Ser, AGU and AGC. Since the amino acid and tRNA are identical for the dipeptidyl-tRNA, lethality differences may be ascribed to variations in termination efficiency due to codon anti-codon interactions [2].

Upon expression, the lethal *bar* minigenes of lambda accumulated minigene mRNA [14]. Furthermore, using several of the two-codon minigenes we observed a correlation between lethality and mRNA concentration. In general, the higher the lethality the more elevated the minigene mRNA levels in the cell (Fig. 9; Cruz-Vera *et al.* in preparation). This effect may be related to mRNA protection by ribosome pausing upon limitation of specific tRNA. The non-lethal minigenes accumulate less mRNA, however, the mRNA levels increased in the presence of pactamycin, an antibiotic known to stop translation on the ribosomes soon after initiation has taken place [8]. This result suggests that the non-lethal minigene transcripts are translated but they are not stabilized because the dipeptides are readily terminated without a significant ribosome pause.

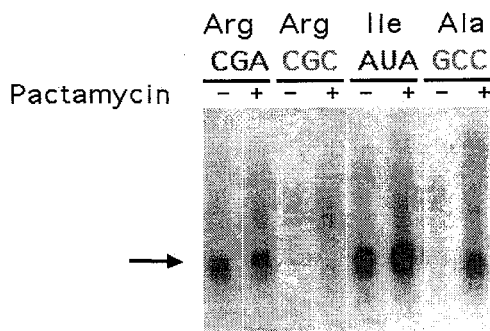


Figure 9. Two-codon minigene mRNAs concentrations correlate to the degree of lethality. Northern blot assay of minigene mRNAs (arrow) from the indicated minigenes. Highly lethal minigenes contain the codons in red at the second position and non-lethal minigene constructs harbor the codons in green. Minigene expression was monitored in the presence (+) or in the absence (-) of pactamycin. The amino acids encoded by the second codon in the minigenes are indicated.

In these experiments Pth overproduction reduces the concentration of mRNA from toxic minigenes (Fig. 10, Cruz-Vera *et al.* in preparation). As Pth hydrolyzes free pep-tRNA, we infer that the enzyme restores the tRNA levels to favor minigene dipeptide termination but not mRNA release. In the absence of ribosome pausing it would not be minigene mRNA protection. Therefore, tRNA sequestering as pep-tRNA must be the key event in mRNA stabilization.

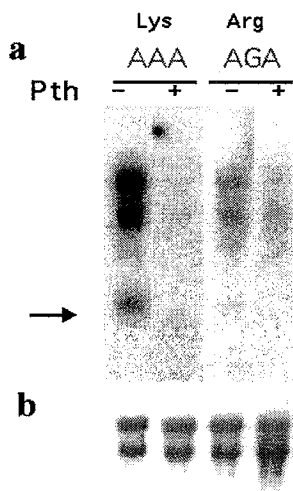


Figure 10. Over-production of peptidyl-tRNA hydrolase reduces minigene mRNA concentrations. a) Northern blot assay of lethal minigene mRNAs (arrow) in the presence (+) or in the absence (-) of wild type Pth overproduction. b) Control of RNA concentration used in a) revealing stained ribosomal RNAs at comparable concentrations

2.4 Search for Minigenes in Bacteriophage Lambda Genome

We have identified two minigenes, by genetic mapping and sequencing, which prevent phage to grow on Pth defective bacteria. However, the wide diversity in codon composition showed by synthetic minigenes prompted us to conduct a search for other minigenes in the phage genome. We used computer-assisted or “in silico” and “wet lab” approaches.

In collaboration with Julio Collado (UNAM, Cuernavaca) we developed a tandem of computer programs to quickly locate putative minigenes in any complete genomic sequence as long as it follows *E. coli* translation rules. The program, named “minigene.pl” (Fig. 11) first locates any of three initiation codons (ATG, GTG or TTG) and an in phase termination codon (TAA, TGA or TAG) within the 18bp downstream. Then, another program module searches for a SD sequence in the 18bp preceding the initiation codon. This was done by weight-matrix assessing after alignment of the sequences with a catalog of 52 *bona fide E. coli* SD regions. Using this program on lambda genome we identified 68 possible minigenes. However a SD relaxed search, allowing one base mismatch in any of six positions, yielded 533

prospects. We take these two estimates as the limits for the number of minigenes in the complete lambda sequence.

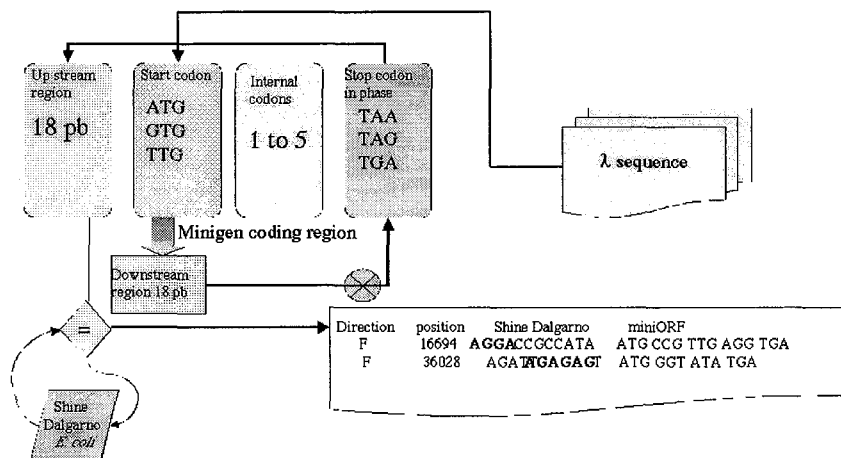


Figure 11. Strategy for the search of minigenes in *silico* (see text).

To investigate whether the predicted minigenes actually corresponded with lethal minigenes we constructed a genomic library to select for lethal clones in Pth-defective bacteria. Lambda DNA was fragmented to an average size of 100bp and cloned into a plasmid vector next to an inducible transcriptional promoter.

This design favors expression of short ORFs, carrying their own translation initiation SD region upon transcription induction. Among 10,500 transformants only 18 clones were lethal for a Pth defective mutant like previous lambda minigenes. The inserts were sequenced; eleven carried minigenes including six new candidates and four with the two known minigenes (Fig. 12, Oviedo *et al.*, in preparation). As the program predicted many more minigenes than those actually found, either most of the predicted minigenes are not lethal or not expressed.

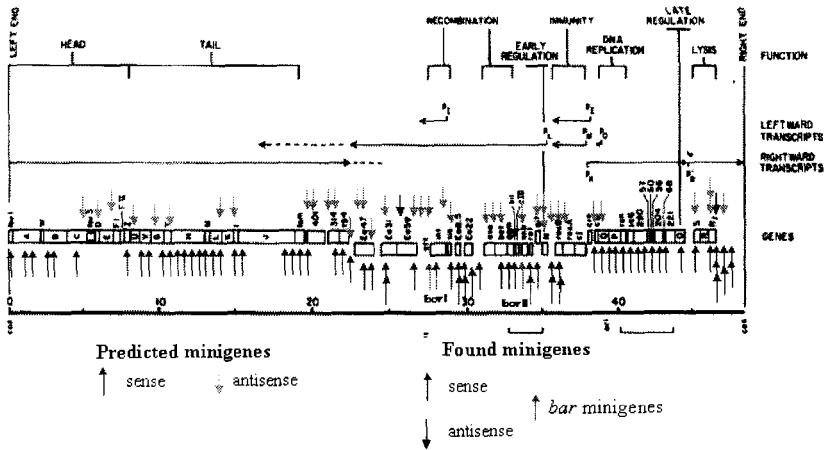


Figure 12. Minigenes in lambda DNA. The position of predicted minigenes in silico is indicated by upward green arrows. Downward green arrows correspond to predicted minigenes in the antisense DNA chain. Minigenes actually found in lethal constructs, in the sense or antisense chains, are indicated by red arrows. Map modified from Daniels *et al.* 1983 [3]).

Among the lethal clones selected above, five contained a common but unpredicted feature; they all harbored SDs from *bona fide* lambda genes and variable lengths within the corresponding ORF (Oviedo *et al.*, in preparation). We have not examined the nature of this phenomenon but it may be related to the defective expression of genes I address next.

2.5 Over Expression of Some Genes May Act as Minigenes.

It often happens that foreign genes cannot over-express from multicopy vectors in *E. coli*. In some cases the genes contain a high proportion of codons rarely used in the bacteria and the defect can be suppressed by over-expression of the cognate tRNAs [1,4]. Gene *int* of lambda, that encodes a protein catalyzing integration of phage DNA into the bacterial genome, contains a high proportion of rare codons including three tandems of the rare Arg codons AGA and AGG at positions 3 and 4, 108 and 109 and 176 and 177 [15].

We reasoned that depletion of the $\text{tRNA}^{\text{Arg}}_{\text{UCU}}$ cognate for AGA and AGG by an excess of *int* mRNA translation may provoke pausing of ribosomes before positions 3 and 4, favoring a condition analogous to minigene inhibition, i.e., Nascent peptidyl-tRNA drop-off. To test this hypothesis we repeated in a *pth* defective strain observations made by Zahn in a wild type strain. Since overproduction of $\text{tRNA}^{\text{Arg}}_{\text{UCU}}$, specific for AGG AGA, suppresses the defective Int protein yield, it is likely that this tRNA is depleted during *int* over-expression [15].

Of all rare Arg codons gene *int*, the first tandem, at positions 3 and 4, seems the critical one for tRNA depletion because it suffices to substitute it for two CGT codons to overcome the inhibition (Fig. 13, Olivares *et al.*, in preparation). CGT is a frequently used Arg codon in *E. coli* recognized by an abundant tRNA different from $\text{tRNA}^{\text{Arg}}_{\text{UCU}}$. Substitutions of the same two codons for a tandem of the rare codon AUA (Ile) did not improve Int yield but a tandem of the common AAA codon (Lys) did. Both AUA and AAA in two-sense codon minigenes determine toxicity but the AAA is recognized by the abundant tRNA^{Lys} whereas AUA is cognate to the scarce tRNA^{Ile} . Therefore, it is likely that low usage codons at early positions in an ORF and the scarce cognated tRNAs determine low yields of protein. This hypothesis needs confirmation by a pep-tRNA accumulation assay.

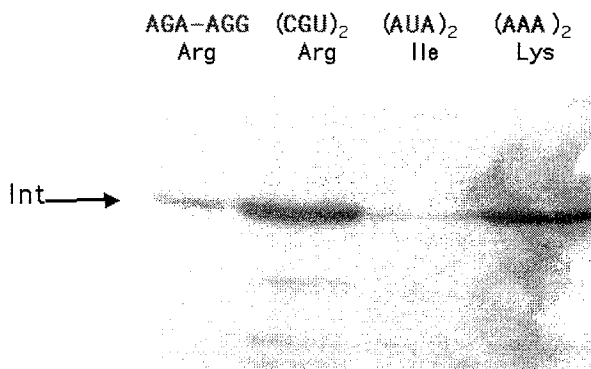


Figure 13. Early codon substitutions in lambda *int* gene affect Int protein yield. Codons AGA and AGG at *int* positions 3 and 4 were substituted by pairs of the indicated codons and the mutant *int* genes were over-expressed from plasmid constructs. Int protein was detected by immunoblot assay.

3 Conclusions

The results presented above, and those from other groups, have allowed to understand the peculiarities of minigene expression in a system grossly amplified. However, It is likely that the role of minigenes in the cell may be subtler than just to inhibit protein synthesis and kill the cells. The search for minigenes in the complete genome of bacteriophage lambda yielded very few lethal minigenes. This result may imply that lethal minigenes are "avoided" in the genome and that harmless minigenes may be the dominant class. We hope to elucidate minigene function analyzing their effects on lambda gene translation. The mechanism of minigene on protein synthesis inhibition may explain some cases of inhibition by foreign gene over-expression in bacteria.

4 Acknowledgements

Parts of the results shown were done in collaboration with Richard Buckingham (Paris) and Shura Mankin (Chicago). This work was supported by grants from the Consejo Nacional de Ciencia y Tecnología (Mexico), Consejo del Sistema Nacional de Educación Tecnológica (México) and Howard Hughes Medical Institute (U. S. A.).

References

1. Chen, G.F.T. and Inouye, M. (1994) Role of the AGA/AGG codons, the rarest codons in global gene expression in *Escherichia coli*. *Genes and Develop.* 8:2641-2652.
2. Curran, J. (1995) Decodin Oviedo *et al.*, in preparation g with the A:I wobble pair is inefficient. *Nuc. Acid Res.* 23:683-688.
3. Daniels, D.L., Schroeder, J.L., Szybalski, W., Sanger, F. and Blattner, F.R. A molecular map of coliphage lambda. In Hendrix, R.W., Roberts, J.W., Stahl, F.W. and Weisberg, R.A. (ed.). *Lambda II*. Cold Spring Harbor Laboratory, 1983. pp. 469-517.
4. Del Tito, Jr., B.J., Ward, J.M., Hodgson, J., Gershater, C.J.L., Edwards, H., Wysocki, L.A., Watson, F.A., Sathe, G. and Kane, J.F. (1995) Effects of a minor isoleucyl tRNA on heterologous protein translation in *Escherichia coli*. *J. Bacteriol.* 177:7086-7091.
5. Dinçbas, V., Hergué-Hamard, V., Buckingham, R. K. and Ehrenberg, M. (1999) Shutdown in protein synthesis due to the expression of mini-genes in bacteria. *J.Mol. Biol.*, 291: 745 - 759.

6. Hernández-Sánchez, J., Valadez, J. G., Vega-Herrera, J., Ontiveros, C. and Guarneros, G. (1998) Lambda *bar* minigene-mediated inhibition of protein synthesis involves accumulation of peptidyl-tRNA and starvation for tRNA. *EMBO J.*, 17: 3758 - 3765.
7. Heurgué-Hamard, V., Dinçbas, V., Buckingham, R.H., and Ehrenberg, M. (2000) Origins of minigene-dependent growth inhibition in bacterial cells. *EMBO J.* 19:2701-2707.
8. Mankin, A.S. (1997) Pactamycin resistance mutations in functional sites of 16S rRNA. *J. Mol. Biol.* 274: 8-15.
9. Menninger, J. R. (1976) Peptidyl tranfer RNA dissociates during protein synthesis from ribosomes of *Escherichia coli*. *J. Biol. Chem.* 25:3392 - 3398.
10. Ontiveros, C., Valadez, J. G., Hernández, J. and Guarneros, G. (1997) Inhibition of *Escherichia coli* protein synthesis by abortive translation pf phage I minigenes. *J. Mol. Biol.*, 269:167 - 175.
11. Pérez-Morga, D. and Guarneros, G. (1990) A short DNA sequence from phage I inhibits protein synthesis in *Escherichia coli*. *J. Mol. Biol.* 172: 243 - 250.
12. Tenson, T., Xiong, L., Kloss, P. and Mankin, A. S. (1997) Erythromycin resistance peptides selected from ramdon peptide libraries. *J. Biol. Chem.*, 272: 17425 - 17430.
13. Tenson, Oviedo *et al.*, in preparation T., Vega-Herrera, J., Kloss, P., Guarneros, G. and Mankin, A. S. (1999) Inhibition of translation and cell growth by minigene expression. *J. Bacteriol.* 181: 1617 - 1622.
14. Valadez-Sánchez, J.G., Hernández-Sánchez, J., Magos, M.A., Ontiveros, C. and Guarneros, G. (2000) Increased *bar* minigene mRNA stability during cell growth inhibition. In press.
15. Zahn, K. and Landy, A. (1996) Modulation of lambda integrase synthesis by rare arginine tRNA. *Mol. Microbiol.* 21:69-76.

FRACTIONAL BROWNIAN MOTION: THEORY AND APPLICATION TO DNA WALK

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This paper briefly reviews the theory of fractional Brownian motion (FBM) and its generalization to multifractional Brownian motion (MBM). FBM and MBM are applied to a biological system namely the DNA sequence. By considering a DNA sequence as a fractal random walk, it is possible to model the noncoding sequence of human retinoblastoma DNA as a discrete version of FBM. The average scaling exponent or Hurst exponent of the DNA walk is estimated to be $H = 0.60 \pm 0.05$ using the monofractal R/S analysis. This implies that the mean square fluctuation of DNA walk belongs to anomalous superdiffusion type. We also show that the DNA landscape is not monofractal, instead one has multifractal DNA landscape. The empirical estimates of the Hurst exponent falls approximately within the range $H \sim 0.62 - 0.72$. We propose two multifractal models, namely the MBM and multiscale FBM to describe the existence of different Hurst exponents in DNA walk.

1 Introduction

In biology, there exist many phenomena that exhibit fractal and multifractal structures. Examples include physiological time series such as the ECG [1], human gait [2], morphogenesis of cancer cells and tissues [3-4], complex molecule landscapes [5] and biological transport processes.

Recall that for normal diffusion, the mean square displacement varies linearly with time $\langle X(t)^2 \rangle \sim t$. However, there exist many natural phenomena which satisfy $\langle X(t)^2 \rangle \sim t^\alpha$, $0 < \alpha < 2$. Such diffusion processes with α different from 1 are known as anomalous diffusion. For $\alpha > 1$, one gets the enhanced diffusion or superdiffusion, and $\alpha < 1$ gives suppressed diffusion or subdiffusion. In biological systems enhanced diffusion occurs long-range correlation of DNA sequences [6] and, whereas suppressed diffusion can be found in transport processes in living cells [7], heart rate variability is produced by cardiac control [8] and local viscoelasticity in filamentous actin network [9]. Many of these fractal biological phenomena can be modeled by fractal stochastic processes. One of the most widely used stochastic process for modeling scale invariant long-range correlated phenomena is the fractional Brownian motion [10].

In this paper, we shall first review the theory of fractional Brownian motion (FBM). FBM can also be generalized to multifractional Brownian motion (MBM) in order to describe phenomena that exhibit multifractal characters. Finally we discuss the application of FBM and MBM to DNA walk.

2 Fractional Brownian Motion

FBM was popularized by Mandelbrot and van Ness through their seminal paper [11]. They studied the basic properties of FBM and stressed its applications in the modeling scaling phenomena with power spectra of power-law type, $1/\omega^\alpha$, with frequency ω and spectral exponent $1 < \alpha < 3$. The standard FBM employed in most applications was introduced by Mandelbrot and van Ness using the following definition [11]:

$$B_H(t) = \frac{1}{\Gamma(H + 1/2)} \left[\int_{-\infty}^0 [(t-u)^{H-1/2} - (-u)^{H-1/2}] dB(u) + \int_0^t (t-u)^{H-1/2} dB(u) \right], \quad (1)$$

where $B(t)$ is the standard Brownian motion, Γ is the Gamma function and Hurst index or Hurst exponent H lies in the range $0 < H < 1$. Equation (1) is also known as the moving average representation of FBM. Note that $B_H(t)$ is continuous, everywhere nondifferentiable with a unique scaling exponent H for all t , which reflects the monofractal or homogeneous fractal character of the process. $B_H(t)$ is a Gaussian process with zero mean and its variance and covariance are respectively

$$\langle (B_H(t))^2 \rangle = \sigma_H^2 |t|^{2H}, \quad (2a)$$

$$\langle (B_H(t)B_H(s)) \rangle = \frac{\sigma_H^2}{2} [|t|^{2H} + |s|^{2H} - |t-s|^{2H}], \quad (2b)$$

with

$$\sigma_H^2 = \langle (B_H(1))^2 \rangle = \frac{\Gamma(1-2H) \cos(\pi H)}{\pi H}. \quad (2c)$$

The standard FBM has the following desirable properties. It is a self-similar process with the scaling exponent H :

$$B_H(at) = a^H B_H(t), \quad \forall a > 0, t \in \mathbb{R}, \quad (3)$$

where the equality is in the sense of finite joint distribution. Though B_H is itself non-stationary, its increment process

$$\Delta B_H(t; \tau) = B_H(t+\tau) - B_H(t), \quad \tau > 0 \quad (4)$$

is stationary with covariance

$$R_{\tau}(s) = \langle \Delta B_H(t; \tau) \Delta B_H(t-s; \tau) \rangle \\ = \frac{\sigma_H^2}{2} \left[|s - \tau|^{2H} + |s + \tau|^{2H} - 2|\tau|^{2H} \right]. \quad (5)$$

The stationary property of ΔB_H makes it possible to define a stationary fractional Gaussian noise as derivative of B_H (in the sense of distribution or generalized stochastic process), which allows a generalized power spectrum to be associated with B_H . One note that the Fourier transform of the distribution $f(s) = |s|^{2H}$ is

$$\tilde{f}(\omega) = -\frac{2\Gamma(2H+1)\sin(H\pi)}{|\omega|^{2H+1}}, \quad (6)$$

which implies the Fourier transform of $R_{\tau}(s)$ is

$$S(\omega) = \frac{4\sigma_H^2\Gamma(2H+1)\sin(\frac{H\pi}{2})}{|\omega|^{2H+1}}. \quad (7)$$

In order to find a generalized power spectrum for $B_H(t)$, we express the increment process as a convolution

$$\Delta B_H(t; \tau) = B_H(t) * h_{\tau}(t), \quad (8)$$

where $h_{\tau}(t) = \delta(t) - \delta(t-\tau)$. Now recall that if $X(t)$ is a stationary process, then $Y(t) = X(t) * h_{\tau}(t)$ is also stationary and the power spectra of X and Y are related by

$$S_X(\omega) = \frac{S_Y(\omega)}{|\tilde{h}(\omega)|^2}. \quad (9)$$

Though B_H is nonstationary, it is tempting to use (9) in order to obtain a generalized power spectrum for B_H

$$S_{B_H}(\omega) = \frac{S_{\Delta B_H}(\omega)}{|\tilde{h}_{\tau}(\omega)|^{2H+1}} = \frac{\sigma_H^2}{|\omega|^{2H+1}}. \quad (10)$$

Alternatively, the "one-sided" FBM introduced by Levy [12] based on the following Riemann-Liouville fractional integral (see also Barnes and Allan [13])

$$X_H(t) = \frac{1}{\Gamma(H+1/2)} \int_0^t (t-u)^{H-1/2} dB(u) \quad (11)$$

which represents a linear system driven by white noise $\eta(t)$, with the impulse response function $t^{H-1/2}(\Gamma(H+1/2))^{-1}$. (Note that the white noise is formally related to

the Brownian motion by $dB(t) = \eta(t)dt$. The Riemann-Liouville (RL) FBM $X_H(t)$ is a zero mean Gaussian process with a rather complicated covariance [14]:

$$\langle X_H(t)X_H(s) \rangle = \frac{t^{H-1/2}s^{H+1/2}}{(H+\frac{1}{2})(\Gamma(H+\frac{1}{2}))^2} {}_2F_1\left(1, \frac{1}{2}-H, \frac{3}{2}+H, s/t\right), \quad (12)$$

where $s < t$ and ${}_2F_1$ is the Gauss hypergeometric function. However the variance of X_H has the same time-dependence as B_H :

$$\langle (X_H(t))^2 \rangle = \frac{t^{2H}}{2H(\Gamma(H+1/2))^2}. \quad (13)$$

Despite its complex covariance, X_H shares with B_H many properties, which includes self-similarity, regularity of sample paths, etc. The usual reason given for the absence of stationarity in the increments of RL-FBM is the over-emphasis on the time origin in X_H . The lack of stationary property in its increments implies X_H cannot be associated with a generalized spectrum of power-law type as in the case of B_H . This is the main reason that RL-FBM is seldom used in modeling phenomena with $1/\omega^\alpha$ type power spectrum.

In order to see whether there exists some kind of stationary property in the increment property ΔX_H of the RL-FBM, it suffices to consider its variance since for a real process, the covariance of its increment process can be obtained from the variance using the following identity:

$$\begin{aligned} \langle \Delta X_H(t, v) \Delta X_H(s, u) \rangle &= \frac{1}{2} \left[\langle \Delta X_H(t, u)^2 \rangle + \langle \Delta X_H(s, v)^2 \rangle \right. \\ &\quad \left. - \langle \Delta X_H(u, v)^2 \rangle - \langle \Delta X_H(s, t)^2 \rangle \right]. \end{aligned} \quad (14)$$

The conditions for ΔX_H to be stationary can be obtained by considering the conditions for the variance of $\Delta X_H(t, \tau)$ to be independent of t . With some changes of variables, one gets

$$\langle (\Delta X_H(t, \tau))^2 \rangle = \frac{\tau^{2H}}{(\Gamma(H+1/2))^2} \left(I + \frac{1}{2H} \right), \quad (15)$$

where

$$I = \int_0^{t/\tau} \left[(1+u)^{H-1/2} - u^{H-1/2} \right]^2 du \quad (16)$$

is independent of t provided (a) $t/\tau \rightarrow 0$, or (b) $t/\tau \rightarrow \infty$.

Condition (a) gives $I = 0$. As this is satisfied for very large time lag τ , such a condition is of little physical interest. When condition (b) is satisfied, one gets

$$(\Gamma(H + 1/2))^2 I = \frac{\Gamma(1 - 2H) \cos(\pi H)}{\pi H} = \sigma_H^2. \quad (17)$$

There are two possible ways to fulfill condition (b): either $t \rightarrow \infty$ for all τ , or $\tau \rightarrow 0$ for all t . The requirement that $t \rightarrow \infty$ means that the increment process of the large-time asymptotic RL-FBM is stationary. On the other hand, the condition $\tau \rightarrow 0$ implies the increment process of RL-FBM is locally asymptotically stationary with

$$\langle (\Delta X_H(t, \tau))^2 \rangle = D_H \tau^{2H}, \quad \tau \rightarrow 0, \quad (18)$$

where

$$D_H = \sigma_H^2 + \frac{1}{2H(\Gamma(H + 1/2))^2}. \quad (19)$$

For most practical applications, both the conditions for stationary increments (i.e., $t \rightarrow \infty$ and $\tau \rightarrow 0$) are rather too stringent. It would be useful if the conditions can be suitably relaxed. This can be achieved by assuming τ small enough such that those terms of $O(\tau^2)$ and higher powers can be neglected. A change in variable and evaluating the integral I up to order $O(\tau^2)$ gives

$$\begin{aligned} I &= t^{2H} \int_0^1 \left[\left(u + \frac{\tau}{t}\right)^{H-1/2} - u^{H-1/2} \right]^2 du \\ &\approx (H-1/2)^2 t^{2(H-1)} \tau^2 \int_0^1 u^{2H-3} du \\ &= \frac{(H-1)^2}{8(H+1)} t^{2(H-1)} \tau^2. \end{aligned} \quad (20)$$

Thus the increment process of X_H is stationary for τ sufficiently small so that $O(\tau^2) = 0$. Since $(H-1) < 0$, $t^{2(H-1)}$ decreases as t increases. Therefore τ can take larger values progressively as t increases such that $t^{2(H-1)} \tau^2 \approx 0$ still holds.

In other words, the size of the interval of stationarity for the increment process $\Delta X(t, \tau)$ is time-dependent. Between the two extreme stationary conditions $t \rightarrow \infty$ and $\tau \rightarrow 0$, there exist "intermediate" conditions which allow the interplay of the values of t and τ . The interval of approximate stationarity for the increments of RL-FBM is very much smaller at the beginning than at large t . We shall call this latter property as the local stationary property since it is the consequence of the local assumption $\tau \ll t$. Local stationary increments provide the flexibility necessary for practical applications. By using (20) and omitting $O(\tau^2)$ term, one notes that locally X_H approaches B_H since the local covariance of X_H and B_H are the same (up to a multiplicative constant):

$$\begin{aligned} \langle B_H(t)B_H(t+\tau) \rangle &\equiv \langle X_H(t)X_H(t+\tau) \rangle \\ &\sim |t|^{2H} + |t+\tau|^{2H} - |\tau|^{2H}, \quad \tau \ll t. \end{aligned} \quad (21)$$

From the above discussion, one sees that the increment process of RL-FBM is not totally lacking of stationary property. Instead, it satisfies some weaker forms of stationarity. Thus, one can regard RL-FBM as the same as standard FBM for $t \rightarrow \infty$. For most physical applications, the process involved begins at finite time (which can be chosen as $t = 0$), and usually one is interested in the asymptotic state (for example, in anomalous diffusion). Thus RL-FBM turns out to be a more suitable FBM for modeling phenomena that behave asymptotically as self-similar Gaussian processes with stationary increments [14]. In such cases, all the nice properties of standard FBM are applicable.

3 Multifractional Brownian Motion

For some complicated systems, FBM turns out to be inadequate since it can only be used in modeling phenomena that have same irregularities globally or monofractal structure due to the constant Hurst exponent. In order to study phenomena that have more intricate structures with variations in irregularities, it is necessary to allow the Hurst exponent to vary as a function of time (or position).

A direct way of extending the monofractal FBM to a multifractal FBM or multifractional Brownian motion (MBM) is to replace the constant Hurst exponent by $H(t)$: $[0, \infty) \rightarrow (0, 1)$ with Holder regularity, r such that $r > \sup H(t)$. This time-varying Hurst exponent $H(t)$ describes the local variation of the irregularity of the MBM process. Note that in general $H(t)$ can be a deterministic or random function, and it needs not be a continuous function. In this paper, however, we shall restrict $H(t)$ to be a smooth deterministic function of time.

Generalization of the standard FBM B_H to the standard MBM $B_{H(t)}$ was first carried out independently by Peltier and Levy Vehel [15] and by Benassi, Jaffard and Roux [16]. Following [15], one defines the standard MBM as

$$\begin{aligned} B_{H(t)}(t) = \frac{1}{\Gamma(H(t) + 1/2)} &\left[\int_{-\infty}^0 [(t-u)^{H(t)-1/2} - (-u)^{H(t)-1/2}] dB(u) \right. \\ &\left. + \int_0^t (t-u)^{H(t)-1/2} dB(u) \right]. \end{aligned} \quad (22)$$

Here we shall adopt the above definition but remark that the results are applicable for the harmonizable form [16] as well. The variance of MBM is given by

$$\langle (B_{H(t)}(t))^2 \rangle = \sigma_{H(t)}^2 |t|^{2H(t)}, \quad (23)$$

where

$$\sigma_{H(t)}^2 = \frac{\Gamma(1-2H(t))\cos(\pi H(t))}{\pi H(t)}. \quad (24)$$

Due to the fact that σ is now time dependent, it will be desirable to normalize the process such that $\langle (B_{H(t)})^2 \rangle = |t|^{2H(t)}$. This requires the normalized standard MBM to be given by $\hat{B}_{H(t)} = B_{H(t)} / \sqrt{\sigma_{H(t)}}$.

For notational convenience, we shall denote $B_{H(t)}$ as the normalized MBM. The covariance of MBM is then given by [17]

$$\begin{aligned} \langle B_{H(t_1)}(t_1) B_{H(t_2)}(t_2) \rangle &= \frac{C(H(t_1) + H(t_2)/2)}{2\sigma_{H(t_1)}\sigma_{H(t_2)}} \\ &\times (|t_1|^{H(t_1)+H(t_2)} + |t_2|^{H(t_1)+H(t_2)} - |t_1 - t_2|^{H(t_1)+H(t_2)}), \end{aligned} \quad (25)$$

where

$$C\left(\frac{H(t_1) + H(t_2)}{2}\right) = \frac{\Gamma(1 - H(t_1) - H(t_2))\cos((H(t_1) + H(t_2))\pi/2)}{\pi \left(\frac{H(t_1) + H(t_2)}{2}\right)}. \quad (26)$$

Locally, $B_{H(t)}$ behaves very much like FBM, B_H . Due to the time dependency of the Hurst exponent, MBM does not satisfy the self-similarity property globally, and its increment process is no more stationary. However, if an additional assumption is imposed on $H(t)$ such that $H(t) \in C^r(R, (0,1))$, $t \in R$ for some positive r with $r > \sup H(t)$, then it can be shown that $H(t_0)$ is almost surely the local Hurst exponent and the box dimensions of the graphs of $B_{H(t)}$ at t_0 is $2 - H(t_0)$.

Since the scaling exponent is time dependent, MBM fails to satisfy the global self-similarity property. However, one can show that MBM is locally asymptotically self-similar in the following sense:

$$\lim_{\rho \rightarrow 0^+} \left[\frac{B_{H(t_0+\rho u)}(t_0 + \rho u) - B_{H(t_0)}(t_0)}{\rho^{H(t_0)}} \right]_{u \in R} = (B_{H(t_0)}(u))_{u \in R}. \quad (27)$$

Here we remarked that RL-FBM can be generalized in a similar way to give the RL-MBM:

$$X_{H(t)}(t) = \frac{1}{\Gamma(H(t) + 1/2)} \int_0^t (t-u)^{H(t)-1/2} dB(u). \quad (28)$$

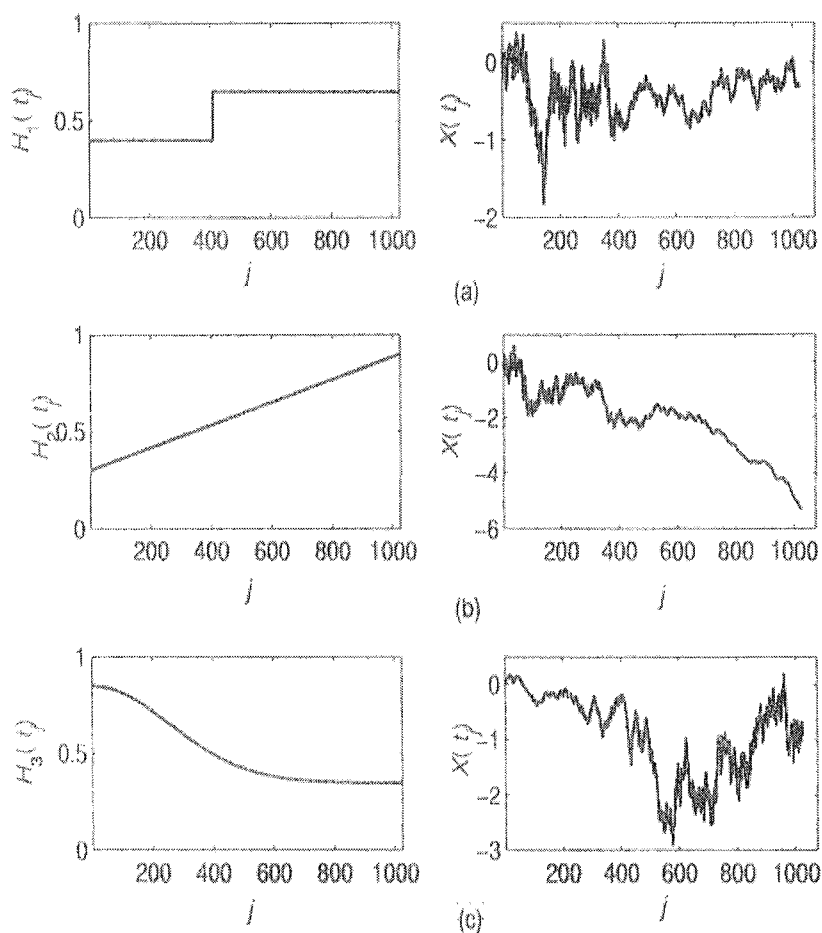


Figure 1. The sample paths of RL-MBM for three different time-varying Hurst exponents: (a) piecewise constant, (b) $H(t)=0.6t + 0.3$, and (c) $H(t) = 0.8\exp(-t)$.

It has been shown that the RL-MBM is also locally asymptotically self-similar [17]. Since the stationarity of the increment is no longer possible in the standard MBM, one may use RL-MBM for the study of locally self-similar processes, in particular those begin at time origin, $t = 0$. Figure 1 shows three examples of the time-varying Hurst exponents and the corresponding sample paths of the RL-MBM. Next, we discuss the application of FBM and MBM in the statistical analysis of DNA walk.

4 FBM Model of DNA Walk

In this section, we apply the theory of FBM to model DNA sequence, which is to be regarded as a discrete time series in the sense explained below. As far as information content is concerned, a DNA sequence can be represented as symbolic sequence of four alphabets A, C, G and T, which represent the four nucleotides. These nucleotides form two base pairs of two purines and two pyrimidines. The two purines are adenine (A) and guanine (G) and the two pyrimidines are cytosine (C) and thymine(T).

In order to study the stochastic properties such as correlation and moments of a DNA sequence one usually introduces a graphical representation of the base-pair sequence as DNA walk or fractal landscape [18]. By converting the (A, C, G, T) symbol text into a binary sequence based on purine (A, G) and pyrimidine (C, T), a DNA walk can be constructed by assigning the following binary mapping rule: the displacement of a random walker $u(i)$ at step i increases or decreases by 1 if the DNA site is occupied by a purine or pyrimidine respectively. Thus the DNA walk allows one to visualize directly the fluctuations of the purine-pyrimidine content in the DNA sequence. Figure 2 shows a DNA walk constructed from the sequence of 10,000 nucleotides from the noncoding sequences of retinoblastoma DNA.

Denoting $Y(n)$ as the “net displacement” of the DNA walker after n steps, then

$$Y(n) = \sum_{i=1}^n u(i), \quad (29)$$

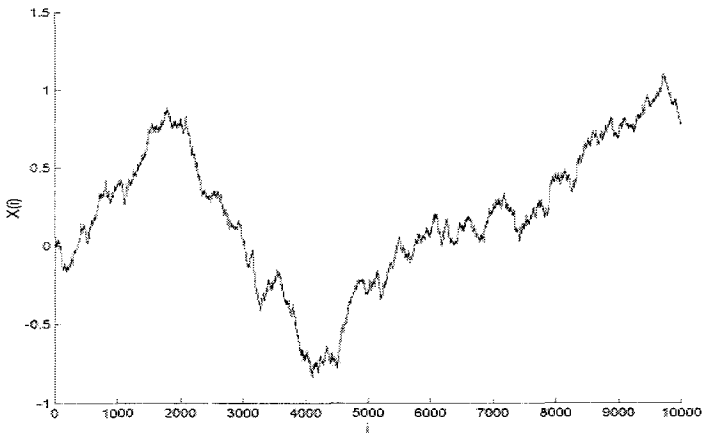


Figure 2. DNA walk constructed from the human retinoblastoma noncoding DNA sequence.

which is the sum of the step for each step i . The mean square fluctuation about the average of the displacement is defined by

$$\begin{aligned} F(n)^2 &= \left\langle (\Delta Y(n) - \langle \Delta Y(n) \rangle)^2 \right\rangle \\ &= \left\langle (\Delta Y(n))^2 \right\rangle - \langle \Delta Y(n) \rangle^2, \end{aligned} \quad (30)$$

where $\Delta Y(n)$ is the increment

$$\Delta Y(n) = Y(n_0 + n) - Y(n_0). \quad (31)$$

Various analyses [19] have shown that

$$F(n)^2 \sim n^{2\alpha} \quad (32)$$

with $\alpha > 1/2$ for noncoding DNA or introns, with the DNA walker performing anomalous diffusion. On the other hand, $\alpha \approx 1/2$ for coding DNA or exons, which corresponds to normal diffusion (i.e., standard random walk). More detailed studies based on detrended fluctuation analysis [20] and wavelet analysis [21] show that the DNA walk is multifractal, with values of α depending on n such that there exist patches of sequence with different scaling exponent $\alpha(n)$.

In order to model the DNA walk using FBM, it is necessary to consider the discrete version of FBM. Let η_i be the random variable representing a step taken at the discrete time i . The total displacement in time $t = n\tau$ after n steps is

$$Y(n) = \sum_{i=1}^n u(i). \quad (33)$$

We can express (33) as a right difference equation

$$X(t) - X(t - \tau) = (1 - L)X(t) = \eta_n(t), \quad (34)$$

where L is the backward shift or lag operator. By letting $\tau = 1$ (for simplicity), the right difference equation can be rewritten using discrete indices for arbitrary step i :

$$(1 - L)X_i = \eta_i. \quad (35)$$

Following [22, 23], (35) can be generalized to become fractional right difference process:

$$(1 - L)^\alpha X_i = \eta_i, \quad (36)$$

where $\alpha > 0$. If η_i is the white noise process, then (36) can be regarded as the discrete time analog of FBM. Inverting the fractional right difference operator in (36) gives

$$\begin{aligned}
 X_i &= (1-L)^{-\alpha} \eta_i \\
 &= \sum_{k=0}^{\infty} \frac{(k+\alpha-1)!}{k!(\alpha-1)!} L^{-k} \eta_i \\
 &= \sum_{k=0}^{\infty} \frac{(k+\alpha-1)!}{k!(\alpha-1)!} \eta_{i-k} \\
 &\approx \sum_{k=0}^{\infty} \frac{k^{\alpha-1}}{(\alpha-1)!} \eta_{i-k}
 \end{aligned} \tag{37}$$

for $k \rightarrow \infty$ and since $k \gg \alpha$. We can show that α lies in the interval $-1/2 < \alpha < 1/2$. The spectrum of the process $S(\omega)$ can be computed as follows:

$$S(\omega) = \langle |\tilde{X}(\omega)|^2 \rangle, \tag{38}$$

where $\tilde{X}$ denotes the Fourier transform of X . We have

$$\begin{aligned}
 S(\omega) &= \left| \sum_{k=0}^{\infty} \frac{(k+\alpha-1)!}{k!(\alpha-1)!} e^{ik\omega} \right|^2 \\
 &= \left| (1 - e^{-i\omega})^{-\alpha} \right|^2 \\
 &= \frac{1}{[2 \sin \frac{\omega}{2}]^{2\alpha}}.
 \end{aligned} \tag{39}$$

Therefore,

$$S(\omega) \approx \frac{1}{\omega^{2\alpha}} \quad \text{as } \omega \rightarrow 0. \tag{40}$$

Expressed in term of Hurst exponent, $\alpha = H - 1/2$ and $S(\omega) \approx \omega^{1-2H}$ as $\omega \rightarrow 0$. This corresponds to the spectrum of fractional Gaussian noise associated with FBM. It can then be shown that the generalized spectrum for FBM is given by $S(\omega) \approx \omega^{-(2H+1)}$ as $\omega \rightarrow 0$.

If one consider the site number as “discrete time”, the DNA walk can then be regarded as a fractal time series. By carrying out statistical analysis of this time series, it is possible to determine the correlation and other stochastic properties of

the sequence. Based on various studies carried out, it is widely held that there exist long-range power law correlations for noncoding DNA sequences, whereas such long-range correlations are absent in coding sequences. Since mathematically, power law is equivalent to scale invariance or self-similarity, and the evidence that DNA walk exhibits anomalous diffusion properties, it is quite natural for one to apply FBM in the modeling of DNA sequence where the Hurst exponent can be directly link to the long-range dependence parameter.

5 Scaling Analysis and Numerical Parameter Estimation

As we have stated earlier, the two properties self-similarity and stationary increments ensure the power-law type spectrum for FBM. Mainly for this reason, FBM has become a popular model especially in the study of processes with long-range dependence. One can associate the values $0 < H < 1/2$ to the anti-persistent behavior in the time series, where any positive increment in the past will be followed by negative increment in the future and vice versa. Meanwhile the range $1/2 < H < 1$ corresponds to persistent characteristics in the time series, i.e., any positive increment in the past will persist in the future. Such a description has been found to be useful in the predictability and study of trends in financial and meteorological time series [24]. On the other hand, the Hurst exponent H indicates the degree of irregularity of the landscape or fractal time series, and the global fractal dimension, D_H of the graph is given by $D_H = 2 - H$.

A practical method widely used in the estimation of Hurst exponent is the rescaled-range (R/S) method. The rescaled range is calculated by first rescaling the data by subtracting the sample mean:

$$Z_r = (X_r - X_{mean}), \quad r = 1, \dots, n \quad (41)$$

and a cumulative time series, Y is calculated from (41). The adjusted range is then the maximum minus the minimum value of the Y_r namely

$$R_n = \max(Y_1, \dots, Y_n) - \min(Y_1, \dots, Y_n). \quad (42)$$

Equation (42) is then divided by the standard deviation $S_n = n^{-1/2} \sqrt{(X_r - X_{mean})^2}$ to give the rescaled range relationship:

$$(R/S)_n = Cn^H, \quad (43)$$

where C is a constant. The slope of the log-log plot of (43) gives the H value of the time series $\{X(t)\}$. The result of the estimation of Hurst exponents for the DNA walk is illustrated in Fig. 3. The R/S method produces a single value for the Hurst exponent, $H = 0.60 \pm 0.05$ or the global fractal dimension, $D = 1.40 \pm 0.05$. We

remark that R/S analysis method is only suitable for fractal analysis of time series exhibiting monofractal behavior where the local irregularity of the graph is uniform everywhere. In other words, the single value of the Hurst exponent obtained from the R/S analysis describes the global long-range dependence of a fixed trend in the time series.

Detailed analyses of local irregularity and the short-term dependence in the DNA walks have shown that the global description based on single Hurst exponent is inadequate [20,21]. The local Hurst exponent estimation has revealed the presence of different H values that is location dependent. This characteristic often make the long-range correlation analysis based on a single Hurst exponent (monofractal character) difficult and unreliable due to the mosaic structure of DNA sequences which are also known as "patches" or "strand bias" of different underlying composition [20,21]. Such inhomogenities can be observed as manifestation of the breaking of global scale invariance. In this situation, FBM model is inadequate as the sample path of FBM is a monofractal function with global fractal dimension, D , and its local regularity is uniform throughout the graph and is characterized by a single Hurst exponent.

On the other hand, a function or time series with singularity exponent that vary from point to point is said to be multifractal. The local regularity of multifractal function is described by the spectrum of singularity measure. The capacity dimension, $f(\alpha)$, of a multifractal function $X(t)$ is defined such that if one makes a

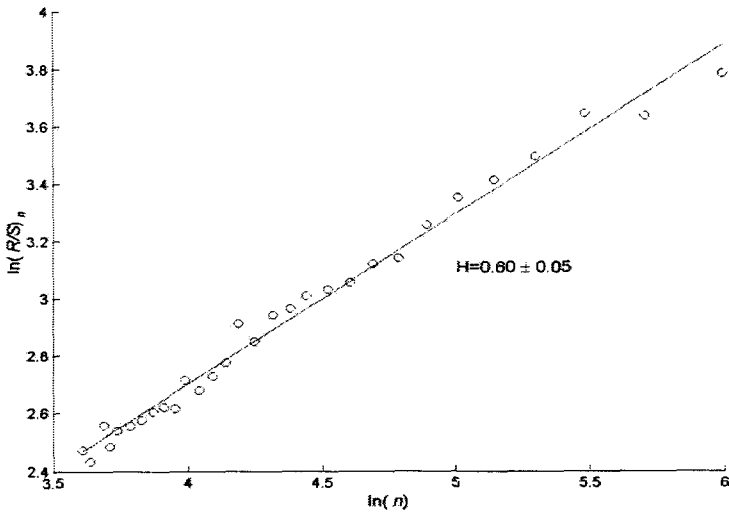


Figure 3. Monofractal R/S analysis of DNA walk.

disjoint cover of the support of f with intervals of size s then the number of intervals that intersect S_α satisfies the powerlaw, $N(s) \sim s^{-f(\alpha)}$ where S_α is a set of all points $t \in R$ where the pointwise Hölder-Lipschitz regularity of f is equal to α . The singularity spectrum describes the proportion of Hölder-Lipschitz α -singularities that appear at any scale s [25].

A multifractal function is said to be homogeneous if all singularities have the same Lipschitz exponent α_0 which means the support of $f(\alpha)$ is restricted to $\{\alpha_0\}$. A numerical scheme to determine the singularity spectrum using the wavelet transform method is briefly described [26]:

- i. compute the wavelet transform, $W_f(u, s)$ and the modulus maxima at each scale s are estimated. These quantities are chained to produce continuous ridges;
- ii. calculate the partition function $Z(q, s) = \sum_n |W_f(u, s)|^q$;
- iii. compute the scaling exponent measure,

$$\tau(q) = \liminf_{s \rightarrow 0} \left[\frac{\log Z(q, s)}{\log s} \right];$$

- iv. compute the singularity spectrum,

$$f(\alpha) = \min_{q \in R} (q(\alpha + 1/2) - \tau(q)).$$

Applying the multifractal analysis on the DNA walk given in Fig. 2, we are able to show that the DNA walk of retinoblastoma noncoding DNA sequence is multifractal based on the wide support of the singularity spectrum $f(\alpha)$ (see Fig. 4). The empirical range of the singularity exponents (equivalent to Hurst exponent, H) is roughly 0.62 – 0.72. The occurrence of wide-range of Hurst exponent in the time series can be modeled by allowing the Hurst exponent to vary with the location of the nucleotides, denoted by $H(t)$ and therefore the MBM serves as a suitable model for multifractal DNA walk.

We consider the time-scale approach to the estimation of the local scaling exponent using the continuous wavelet transform defined as

$$T_{X_{H(t)}}(t, a) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} X_{H(t)}(s) \Psi^* \left(\frac{t-s}{a} \right) ds, \quad (44)$$

where Ψ is the mother wavelet and a is the scaling parameter. Here we use the Morlet wavelet. Since the two MBMs have been shown to exhibit similar locally asymptotic behaviors, the following discussion applies for both models. Consider

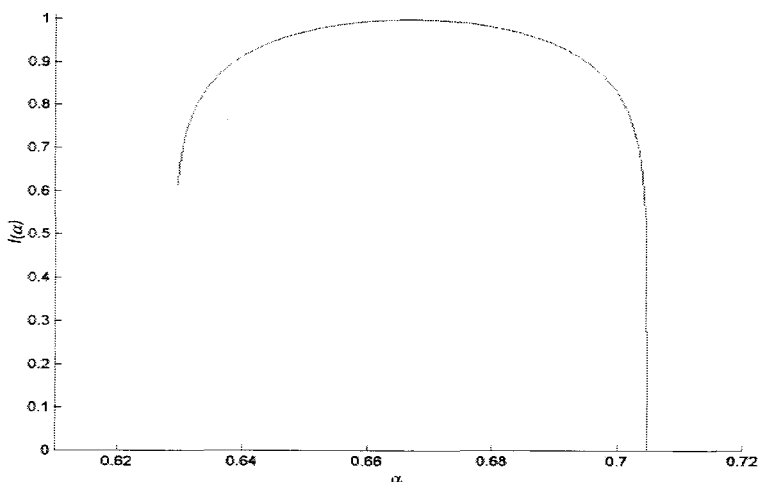


Figure 4. Large deviation Legendre multifractal singularity spectrum of DNA walk.

the local scaling of the increment processes of MBM at t' , where $t' \in [t-\varepsilon/2, t+\varepsilon/2]$ such that the increments are locally stationary with constant local H_t kept frozen in the neighborhood of t ,

$$\left\langle |X_{H_t}(t'+\tau) - X_{H_t}(t')|^2 \right\rangle \sim |\tau|^{H_t}, \quad 0 < \tau < \varepsilon < 1. \quad (45)$$

Therefore, the wavelet coefficient can be written as

$$T_X(t, a) = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} [X_{H_t}(t'+\tau) - X_{H_t}(t')] \psi^*\left(\frac{\tau - (t-t')}{a}\right) d\tau, \quad (46)$$

thus giving the scalogram with power-law type scaling:

$$\Omega_{X_{H(t)}}(t, a) = \left\langle |T_{X_{H(t)}}(t, a)|^2 \right\rangle = a^{2H(t)+1} C_\psi(t), \quad a \rightarrow 0. \quad (47)$$

where $C_\psi(t)$ is a function that depends on the correlation of the two wavelets with overlapping supports. The Hurst exponent is then given by

$$H(t) = \frac{\frac{k}{2} \int a \Omega_{X_{H(t)}}(t, a) e^{-ka} da}{\int \Omega_{X_{H(t)}}(t, a) e^{-ka} da} - 1 \quad (48)$$

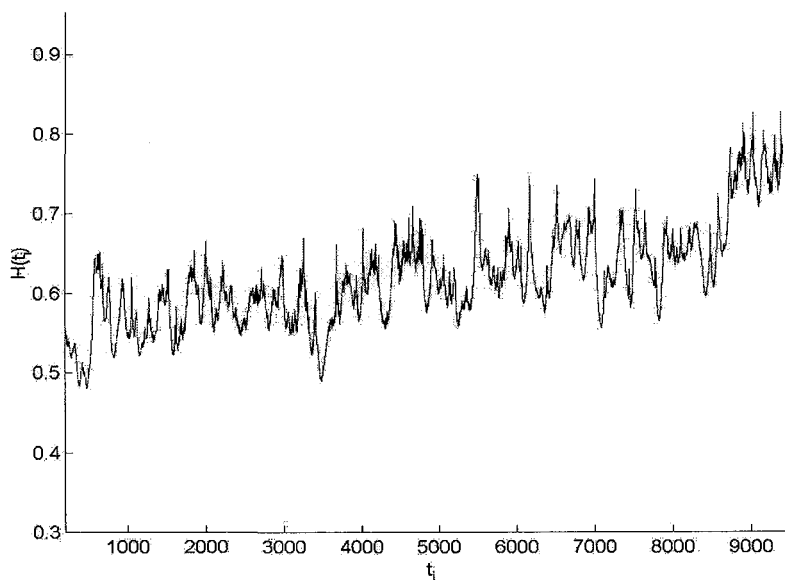


Figure 5. Location dependent Hurst exponents of the DNA walk.

with the smoothing function $g(a)=e^{-ka}$, $k > 0$. The value for k is chosen such that the local behavior (47) is feasible within the domain of integration. Using (48), we can estimate the local Hurst exponent of the DNA walk, $X(i)$, $i = 1, \dots, N$. For illustration purpose, consider $N=10,000$ nucleotides and the result of the estimation is shown in Fig. 5. It is found that the local Hurst exponent, $H(i)$, $i = 1, \dots, N$ is location dependent with an approximately linear modulation and for variation falls within the range estimated in the multifractal analysis.

One may also consider an alternative approach to explain the existence of non-compact set of Hurst exponents by consider multiscale fractional Brownian model,

$$Y(t) = \sum_{k=1}^M c_k B_{H_k}(t), \quad (49)$$

where $\{B_{H_k}\}$ is a finite set of independent FBMs with different global Hurst exponent H_k and c_k is the weighting coefficient.

In order to estimate the different values of H present in the time series, we use the technique of regularization dimension [27] explained below. Consider a function f with support K and let $\chi(t)$ be a kernel function of Schwartz class S such that

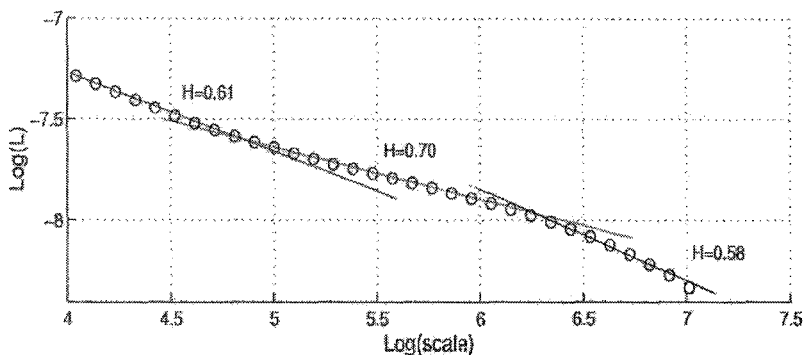


Figure 6. The log-log plot of L_a versus scale showing the coexistence of three different values of Hurst exponent at different scale regimes.

$$\int \chi(t) dt = 1. \quad (50)$$

Let $\chi_a(t) = \frac{1}{a} \chi(\frac{t}{a})$ be the dilated version of χ at scale a . The convolution of f with χ_a is $f_a = f * \chi_a$. Since $f_a \in S$, the length of its graph on K is finite and given by

$$L_a = \int_K \sqrt{1 + \left(\frac{df_a}{dt}\right)^2} dt. \quad (51)$$

The regularization dimension is then defined as [25]:

$$D_R = 1 + \lim_{a \rightarrow 0} \left[-\frac{\log(L_a)}{\log(a)} \right]. \quad (52)$$

with $D_R \leq D_B$ where D_B is box-counting dimension. It is interesting to note that the regularization dimension is related to the large deviation multifractal spectrum $f(\alpha)$ in the following form:

$$D_R = \max[1, 2 - (\inf(\alpha - f(\alpha)))]. \quad (53)$$

The multiscale analysis on the DNA walk using the method of regularization dimension shows the coexistence of different Hurst exponents in the location axis as

illustrated in Fig. 6 by the presence of three different slopes of $H = 0.58, 0.61$ and 0.7 at different scale regimes. The results are in close agreement with that based on the multifractal approach.

6 Conclusion

FBM has been proved to be well suited for modeling phenomena that exhibit scale invariance and inverse power-law type spectra. In this paper, we have applied FBM to model noncoding DNA sequence. Our analysis also showed that the DNA walk is more appropriately modeled by a multifractal model such the MBM due to the presence of patches with different long-range dependence parameters at different segments of the nucleotide chain. There also exist other processes that can be used to model fractal phenomena; for example, non-Gaussian Levy stable process [28] and other colored noises [29]. In order to see which particular process correctly describes the phenomenon under consideration, properties in addition to self-similarity and power-law correlation need to be taken into consideration.

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References

1. Peng C. K., Havlin S. Stanley H. H. and Goldberger A. L., Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos* **5** (1995) 82 - 87.
2. Hausdorf J. M., Purdon P. L., Peng C. K., Ladin Z., Wei J. Y. and Goldberger A. L., Fractal dynamics of human gait: stability of long range correlations in stride interval fluctuations. *J. Appl. Physiol.* **80** (1996) pp. 1448 -1457.
3. Baumann G., Dolinger J., Losa G. A. and Nonnenmacher T. F., Fractal analysis in medicine. In *Fractals in Biology and Medicine, Vol II*, ed. by G. A. Losa, D. Merlini, T. F. Nonnenmacher and E. R. Weibel (Birkhauser, Boston, 1998) pp. 97-113.
4. Li H., Liu R. and Lo S., Fractal modeling and segmentation for the enhancement of microcalcifications in digital mammograms. *IEEE Medical Imaging* **16** (1997) pp. 785-798.

5. Melendez R. Melendez H. E. and Canela E. I., The fractal structure of glycogen: a clever solution to optimize cell metabolism. *Biophys. J.* **77** (1999) pp. 1327-1332.
6. Voss R., Evolution of long-range fractal correlations and $1/f$ noise in DNA base sequences. *Phys. Rev. Lett.* **68** (1992) pp. 3805-3808.
7. Koepf M., Metzler K., Haferkamp O. and Nonnenmacher T. F., NMR studies of anomalous diffusion in biological tissues: experimental observation of Levy stable processes. In *Fractals in Biology and Medicine, Vol II*, ed. by G. A. Losa, D. Merlini, T. F. Nonnenmacher and E. R. Weibel (Birkhauser, Boston, 1998) pp. 354 - 364.
8. Stanley H. E. Amaral L. A. N., Goldberger A. L., Havlin S., Ivanov P. Ch. and Peng C. K., Statistical physics and physiology: monofractal and multifractal approaches. *Physica* **A270** (1999) pp. 309-324.
9. Amblard F., Maggs A. C., Yurke B., Pargellis A. N. and Leibler S., Subdiffusion and anomalous local viscoelasticity in actin network. *Phys. Rev Lett.* **77** (1996) pp. 4470 - 4473.
10. Mandelbrot B. B. *Fractal Geometry of Nature* (Freeman, San Francisco, 1983).
11. Mandelbrot B. B. and Van Ness, J. W. Fractional Brownian motion, fractional noises and applications. *SIAM Rev.* **10** (1968) pp. 422 - 437.
12. Levy P., Random function: general theory with special references to Laplacian random function. *University of California Publ. Statist.* **1** (1953) 331 -390.
13. Barnes J. A. and Allan D. W., A statistical model of flicker noise. *Proc. IEEE.* **54** (1966) pp. 176 - 178.
14. Lim S. C. and Sithi V. M., Asymptotic properties of the fractional Brownian motion of Riemann-Liouville type. *Phys. Lett.* **A206** (1995) pp. 311-317.
15. Peltier R. F. and Levy Vehel J., Multifractal Brownian motion: definition and preliminary results. *INRIA Report* **2645** (1995) pp.1-40.
16. Benassi A., Jaffard S. and Roux D., Elliptic gaussian random processes. *Revista Matematica Iberoamerica* **13** (1997) pp. 19-90.
17. Lim S. C. and Muniandy S. V., On some possible generalization of fractional Brownian motion. *Phys. Lett.* **A266** (2000) pp.140-145.
18. Peng C. K., Buldyrev S. V., Hausdorff J. M. Havlin, S. Mietus J. E. M., Simons, M. Stanley, H. E. and Goldberger A. L., Fractal lanscape in physiology & medicine: long range correlations in DNA sequences and heart rate intervals. In *Fractals in Biology and Medicine*. ed. by T. F. Nonnenmacher, G. A. Losa, E. R. Weibel (Birkhaser Verlag, Basel, 1994) pp. 55 - 66.
19. Stanley H. E., Buldyrev S. V., Goldberger A. L., Havlin, S., Peng, C. K. and Simons M., Scaling features of noncoding DNA. *Physica* **A273** (1999) pp.1-18.

20. Viswanathan G. M. Buldyrev S. V., Havlin, S. and Stanley, H. E., Long-range correlation measures for quantifying patchiness: deviations from uniform power-law scaling in genomic DNA. *Physica A***249** (1998) pp. 591-586.
21. Arneodo A. Aubenton-Carafa Y. D. Audit B., Bacry E., Muzy, J. F. and Thermes, C., What can we learn with wavelets about DNA sequences? *Physica A***249** (1998) pp.439 - 448.
22. Hosking J. T. M. Fractional Differencing. *Biometrika* **68** (1981) pp. 165 - 176.
23. West B. J. and Grigolini P., Fractional difference, derivative and fractional time series. In *Application of Fractional Calculus in Physics*. ed. by R. Hilfer (World Scientific, Singapore, 2000) pp. 171-201.
24. Peters E. E., *Fractal Market Analysis: Applying Chaos Theory to Investment and Economics* (John Wiley, New York, 1994).
25. Muzy J. F., Bacry E. and Arneodo A. The multifractal formalism revisited with wavelets. *Int. J. Bifurcation and Chaos* **4** (1994) pp. 245-302.
26. Mallat S., *A Wavelet Tour of Signal Processing*. (Academic Press, San Diego, 1998).
27. Roueff F. and Levy Vehel J., A regularization approach to fractional dimension estimation. *INRIA Preprint* (2000) pp. 1-14.
28. Allegrini P., Buiatti M., Grigolini P. and West B. J., Non-Gaussian statistics of anomalous diffusion: the DNA sequences of prokaryotes. *Phys. Rev.* **E58** (1998) pp. 3640-3648.
29. Wang K. G. and Tokuyama M., Nonequilibrium statistical description of anomalous diffusion. *Physica A***265** (1999) 341 - 351.

MYOGLOBIN—THE SMALLEST CHEMICAL REACTOR

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Myoglobin confines small hydrophobic molecules such as CO, NO, and O₂ in internal cavities, causing them to rapidly react with one another at the catalytic heme-iron atom. These small molecules play a variety of roles in muscle and vascular function. Furthermore, myoglobin exists in distinct taxonomic substates which catalyze different reactions, suggesting a mechanism for myoglobin to control reactions in an environment-dependent manner.

For many years textbooks [1] have considered myoglobin (Mb) the prototype of a simple protein, with just one function, storage of dioxygen: "Myoglobin is a single-chain, oxygen binding protein found in muscle. The oxygen-binding curve of myoglobin has the characteristic of a simple equilibrium reaction: $E + O_2 \leftrightarrow EO_2$ " [2]. The sophisticated construction of Mb is explained as being needed to exclude CO which is toxic and can bind more tightly than O₂. Mb may not be so simple, however; it may have several functions. Many mammalian Mb have exactly the same number of residues, 153, while oxygen storage can be accomplished with a smaller number [3]. For many Mb with rather different dioxygen affinities, the ratio of the CO to the O₂ affinities is approximately constant. If a small CO affinity were the ultimate goal, Mb could do better, for example by replacing Leu-29 with a phenylalanine [4]. Finally, a cross section through Mb, as shown in Fig. 1, suggests that Mb may indeed have more functions than O₂ storage.

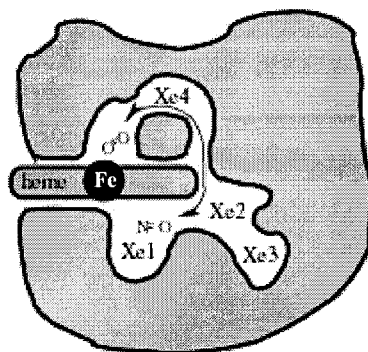


Figure 1. A schematic cross section through myoglobin as NO is reacting with a bound O₂. The distal cavity is on the upper (distal) side of the heme group, the Xe1 cavity is on the lower (proximal) side.

The cross section exhibits some prominent features. Embedded in the protein is a heme group with a central iron atom. Small molecules such as O_2 , CO, and NO can bind covalently to the heme iron, occupying the heme cavity. Four other cavities, denoted by Xe1 to Xe4, are also evident. They are called xenon cavities because, under pressure, xenon atoms occupy these cavities and they can then be seen in X-ray diffraction [5]. A very long look at the cross section in Fig. 1 suggests that Mb is built like a nanoscale chemical reactor [6]. The heme iron is the reaction center where one reaction partner can be bound. The second reaction partner can be enriched in the xenon cavities, particularly Xe1. The structure of the passage from Xe1 to the heme cavity may be designed so that the reaction of the two molecules, one bound at the heme iron and one coming from Xe1, is optimized. The reaction can be controlled by the heme group and by the residues that form the passage from Xe1 to the bound O_2 .

For a long time, a vast number of experiments have given little or no evidence that Mb is a physiologically important chemical reaction chamber, as shown in Fig. 1. A hint that Mb may be more complex in its structure and function came, however, from the exploration of the energy or conformation landscape of proteins [7], discussed in the overview. These studies show conclusively that proteins do not exist in a unique structure, but can assume a very large number of different conformations. The energy landscape is actually organized in a hierarchy, with valleys within valleys within valleys. At the top of the hierarchy are taxonomic substates. Mb can assume a small number of conformations or substates that are distinct enough to be studied individually. Three such taxonomic substates are clearly recognizable. They are denoted by A_0 , A_1 , and A_3 . In Mb with CO bound at the heme iron (MbCO), they are characterized unambiguously by their stretch frequency [8]. They also have different reactive properties and, for example, bind CO with different rates [8,9]. Of particular importance for the discussion here are two features of the taxonomic substates that have emerged from the study of Mb.

The first feature is that the structure of Mb in the taxonomic substates A_0 and A_1 is different [10]. The most dramatic difference is exhibited by the distal histidine, His-64, a residue that in A_1 extends into the heme pocket. In A_0 , His-64 swings out of the heme pocket. Cross-sections of the structures of A_1 and A_0 around the heme group are shown in Fig. 2. The differences are clearly recognizable. The structure of A_3 has not yet been determined.

The second feature is that the ratio of the populations of the substates A_0 and A_1 depends on external factors such as temperature, pressure, and hydration. Particularly strong is the dependence on pH. At a pH above about 7, the population of A_0 is negligible while at a pH below about 5, A_0 dominates. Less is known about the parameters that influence A_3 .

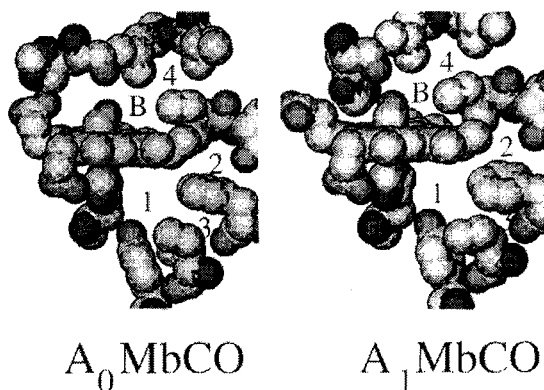
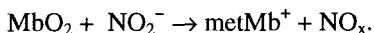


Figure 2. Cross section through the central part of Mb at pH~5 (left) and pH~7 (right). At pH~5, A_0 dominates, at pH~7, A_1 dominates (from reference 8). In A_0 , the distal histidine has moved out of the heme pocket. B denotes the site occupied by a CO molecule immediately after a photon breaks its bond to the heme iron. The numbers 1-4 denote the xenon cavities.

The question that now emerges concerns the importance of the taxonomic substates. Are they simply an accident of evolution, or do they fulfill an important role? Could the substates A_0 and A_1 have different functions? A possible role involves NO. NO is toxic at high concentrations. At low concentrations it is a neurotransmitter and is involved in the relaxation of muscles [11]. Mb is abundant in muscles and the question consequently arose if it is involved in the control of NO. To test this speculation, and to learn at the same time if the substates A_0 and A_1 could possibly have different functions, we measured the reaction rate for the reaction



This reaction is a one-electron oxidation of Mb by the nitrite ion, resulting in another oxide of nitrogen and consuming two protons. The rate of this reaction is given as a function of pH in Fig. 3. The result is unambiguous. In the taxonomic substate A_1 the reaction does not occur, in A_0 it is remarkably fast.

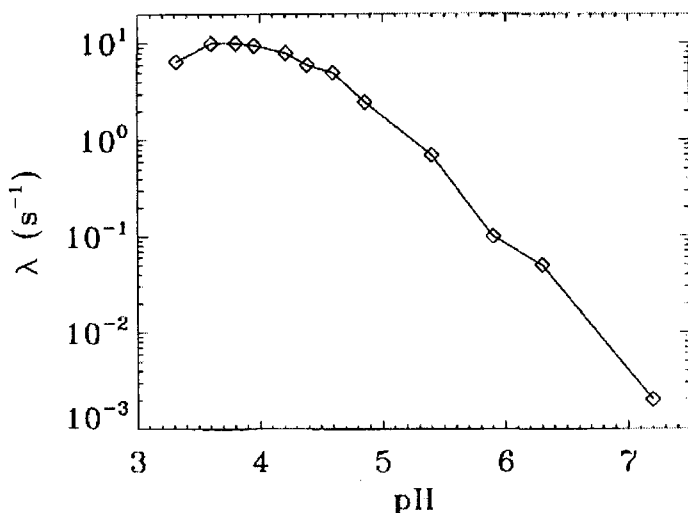


Figure 3. pH dependence of the rate of MbO₂ oxidation by nitrite.

This preliminary result suggests that Mb has (at least) two roles. In the substate A₁ it may well perform the role ascribed to it for more than a century, storage of O₂. In the substate A₀, it may be involved in the control of NO. Mb thus can be considered to be an allosteric enzyme.

This study suggests a number of avenues for more experiments. The details of the reaction of Mb and MbO₂ with NO remain to be explored. NO reactions are notoriously complex and the result shown in Fig. 3 is only a teaser. Can the interaction of other molecules or of other proteins change the populations of A₁ and A₀? Does A₃ play a role? More generally, many other proteins possess taxonomic substates. Do they all perform more than one function and what controls the functions? We are only at the beginning of understanding how Mb works.

Acknowledgments

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References

1. Stryer, L., *Biochemistry* (W. H. Freeman and Company, New York, 1995).
2. Lodish, H. et al., *Molecular Cell Biology* (Scientific American Books, New York, 1995).
3. Grandori, R., Schwarzsinger, S., and Muller, N., Cloning, overexpression and characterization of micro-myoglobin, a minimal heme-binding fragment, *Europ. J. Biochem.* **267** (2000) pp. 1168-1172.
4. Springer, B. A., Sligar, S. G., Olson, J. S., and Phillips, G. N., Mechanisms of ligand recognition in myoglobin, *Chem. Rev.* **94** (1994) pp. 699-714.
5. Tilton, Jr., R. F., Kuntz, Jr., I. D., and Petsko, G. A., Cavities in proteins: Structure of a metmyoglobin – xenon complex solved to 1.9 Å, *Biochem.* **23** (1984) pp. 2849-2857.
6. Frauenfelder, H., McMahon, B. H., Austin, R. H., Chu, K., and Groves, J. T., the role of structure, energy landscape, dynamics, and allostery in the enzymatic function of myoglobin, *Proc. Natl. Acad. Sci. USA* **98** (2001) pp. 2370-2374.
7. Austin, R. H., Beeson, K. W., Eisenstein, L., Frauenfelder, H., Gunsalus, I. C., Dynamics of ligand-binding to myoglobin, *Biochem.* **14** (1975) pp. 5355-5373.
8. Ansari, A. et al., Rebinding and relaxation in the myoglobin pocket, *Biophys. Chem.* **26** (1987) pp. 337-355.
9. Johnson, J. B., et al., Ligand binding to heme proteins .6. Interconversion of taxonomic substates in carbonmonoxymyoglobin, *Biophys. J.* **71** (1996) pp. 1563-1573.
10. Yang, F. and Phillips, G. N., Crystal structures of CO-, deoxy- and met-myoglobins at various pH values, *J. Mol. Biol.* **256** (1996) pp. 762-774.
11. *LES PRIX NOBEL: The Nobel Prizes 1998* (Almqvist and Wiksell International. Stockholm, Sweden, 1999).

OBSERVING CONFORMATIONAL CHANGES OF INDIVIDUAL RNA MOLECULES USING CONFOCAL MICROSCOPY

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Recent years have seen enormous advances in single molecule detection and spectroscopy by laser-induced fluorescence. Without a doubt, the most exciting applications are in the area of Biological Physics, as the technique can readily be applied to investigations of individual biological molecules under physiological conditions. Biological macromolecules are complex physical systems that are characterized by a huge number of conformational states, and transitions among these states are intimately linked to their function. Using single molecule spectroscopy, time trajectories of physical observables can be obtained from single molecules, and conformational heterogeneity and dynamics can be investigated in a direct fashion. As an example, we discuss the measurement of conformational changes of individual 3-helix junction RNA molecules induced by the binding of Mg^{2+} ions. The transition from an open to a folded configuration was monitored by the change of fluorescence resonance energy transfer in a pair of dye molecules attached to different ends of two helices in the RNA junction. The two conformational states of the RNA can be clearly distinguished at the single molecule level, and transitions between the states can be monitored on the millisecond time scale.

1 Introduction

In 1952, Erwin Schrödinger made the remark that it would never be possible to perform experiments on individual electrons, atoms, or molecules¹. Seven years later, however, Richard Feynman, in a classic talk at the Annual Meeting of the American Physical Society at the California Institute of Technology entitled "There is plenty of room at the

bottom” pointed out that “the principles of physics, as far as I can see, do not speak against the possibility of maneuvering things atom by atom².” Indeed, during the previous two decades, a variety of approaches have been developed for the exploration of single atoms and molecules. Scanning probe techniques, such as scanning tunneling microscopy (STM) or atomic force microscopy (AFM) allow one to observe and manipulate individual molecules on surfaces by bringing the object under study in contact with a sharp tip. In this article, we will focus on the optical probing of single molecules. Owing to the weak perturbation involved, this approach is quite powerful and has therefore become very popular in the study of biological macromolecules³.

1.1 Why Study Individual Biomolecules?

Living systems function through the subtle interplay of large numbers of biological macromolecules, which interact via complex signaling pathways and enzymatic reactions. Our knowledge of biomolecules at a descriptive level is becoming larger every day, thanks to sophisticated physical techniques to solve molecular structures with atomic resolution. Our understanding of the mechanisms by which functional processes are carried out in these nanomachines remains, however, elusive. This problem is deeply rooted in the inherent complexity of biomolecules⁴. They are highly flexible entities and can assume a huge number of conformational states which can be viewed as local minima in a rugged energy landscape, separated by energy barriers⁵. To perform a particular function, such as an enzymatic reaction, a protein molecule fluctuates thermally among the many states, and a rare fluctuation may take the protein to a particular conformation that is crucially involved in the function. Since such transient structures are populated to a negligible extent in equilibrium, they can, in general, not be inferred from the average structures. Unfortunately, computer simulations of biomolecular dynamics only extend to the nanosecond scale, whereas most functional processes are much slower. Therefore, to obtain a better understanding of biomolecular function, one has to resort to experimental observations of conformational motions that can reveal the sequence of transient intermediate structures involved in function. Traditionally, these issues have been addressed with time-resolved spectroscopic studies on large ensembles of biomolecules and, more recently, using crystallography of cryogenically trapped intermediate states^{6,7}. Experimental studies have favored systems for which the molecular ensemble can be perturbed and thus synchronized by a short laser pulse; popular examples are heme proteins and proteins involved in photosynthesis. Time-resolved studies on bulk samples, however, yield distributions only if the conformational changes causing heterogeneous physical properties are slow compared with the time scale of the experiment. This condition can be met experimentally by

measuring dynamics with high time resolution and/or slowing these processes by cooling the sample to low temperature.

By contrast, single molecule experiments provide information about distributions of parameters and enable one to observe time trajectories of observables in real time. Individual members of a heterogeneous population can be examined, sorted and classified. Moreover, fluctuations of individual molecules under equilibrium conditions can be observed as well as relaxations of nonequilibrated molecules towards equilibrium. Single molecule studies offer new possibilities especially for systems that cannot be synchronized by external perturbations in ensemble studies. New insights have already been obtained into a variety of proteins, including motor proteins^{8,9,10}, enzymes¹¹, and structural proteins^{12,13}, as well as DNA molecules^{14,15}. Since the rich set of conformational pathways involved in biomolecular folding and function can in principle be made accessible with single molecule experiments, it is of utmost importance to further develop these techniques.

2 Experimental Approach

2.1 Confocal Microscopy and Fluorescence Resonance Energy Transfer

For the optical detection of fluorescence emission from individual molecules it is necessary to rigorously minimize unwanted background. Most importantly, the volume from which light is collected has to be made as small as possible because background from solvent Rayleigh and Raman scattering as well as fluorescent impurities cannot be completely suppressed. Using a confocal microscope with a high numerical aperture objective lens, sample volumes below 1 fL can be achieved, which enables one to observe fluorescence emission from individual chromophores well above background when using low noise detectors such as avalanche photodiodes.

To measure conformational changes on the scale of a few nanometers, the technique of fluorescence resonance energy transfer (FRET) has recently been applied to single biomolecules^{3,16,17,18}. Two dye molecules are attached to the biomolecule at specific locations. In our experiments, one of the dye molecules, the so-called donor, absorbs in the green and is efficiently excited by Ar ion laser light (514.5 nm). Without a second dye in close proximity, the donor re-emits the light, red-shifted by typically a few ten nanometers (Stokes shift). However, if there is another dye molecule absorbing further in the red, termed the acceptor, located within a few nanometers from the donor, and there is sufficient spectral overlap of the donor emission and acceptor absorption

spectra, the excitation energy can be transferred in a radiationless process. The FRET efficiency is given by

$$\varepsilon = \frac{1}{1 + (R/R_0)^6}, \quad (1)$$

where R denotes the donor-acceptor distance, and the characteristic separation R_0 depends on the spectral overlap, orientation and donor quantum yield of the dye pair chosen. Thus, a measurement of the ratio of the fluorescence emission from the two dyes is a sensitive reporter of their relative orientation and spatial separation in the host molecule. If conformational motions are present, they change the FRET efficiency and thus reveal themselves in intensity fluctuations that can be analyzed by fluorescence correlation spectroscopy (FCS)^{19,20,21}.

To monitor the intensity fluctuations in the confocal microscope, the fluorescence emission is separated into a green and a red channel using dichroic filters. The photons are counted with two separate photodiode detectors, and for each photon the arrival time is stored in the computer. For the experiments presented here, photon events were collected with 100 ns time resolution and later regrouped in wider bins as appropriate for the dynamics investigated.

2.2 Ribosomal RNA 3-Helix Junction as A Model System

We have studied the dynamics of an RNA 3-helix junction from the 30S ribosomal subunit of *Thermus thermophilus*. This subunit consists of a 1540 nucleotide 16S rRNA and 21 ribosomal proteins. It is assembled in a cascade of RNA conformational changes induced by a sequence of protein binding events²². The structure of the 16S RNA complexed with proteins S15, S6, and S18 has recently been determined by x-ray crystallography²³. Our model system is a fragment of 16S that contains the binding site for ribosomal protein S15. Binding of S15 is accompanied by a large conformational change in the junction region (Fig. 1). The free RNA junction has been shown to be nearly planar with $\sim 120^\circ$ angles between each pair of helices using gel mobility shift and transient electric birefringence experiments^{24,25,26}. Binding of S15 protein or metal ions such as Mg^{2+} , Ca^{2+} and Co^{3+} causes one of the helices to rotate by 60° , becoming collinear with another one, as shown in Fig. 1.

To observe conformational changes in the RNA junction, a Cy3 dye was attached to the end of one helix as the donor (D) and a Cy5 dye to the end of another helix as the acceptor (A). In the open conformation, donor and acceptor are 8.5 nm apart, whereas

the separation is 5 nm in the folded conformation. For the single molecule experiments, the RNA junctions were biotinylated at the end of the remaining helix and immobilized on a glass surface using a biotin-streptavidin linkage to adsorbed biotinylated bovine serum albumin (BSA). All measurements were done in 10 mM TRIS buffer, pH 8, 50 mM NaCl and at varying concentrations of MgCl_2 .

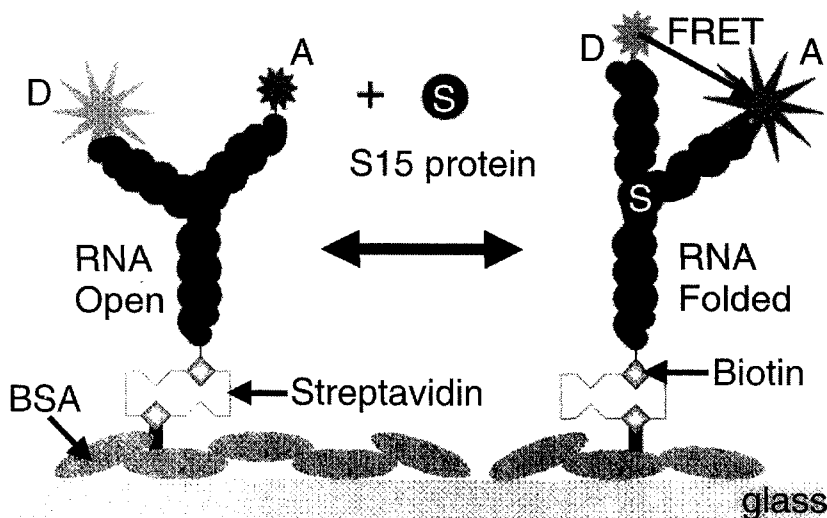


Figure 1. Sketch of the 3-helix junction bound to the surface in the open (left hand side) and folded (right hand side) conformation. Upon folding, the donor (D) and acceptor (A) dye labels at the end of the helices approach each other so that efficient fluorescence resonance energy transfer (FRET) occurs.

3 Results and Discussion

3.1 Titrating Individual RNA Molecules with Magnesium

Figure 2 shows an image created by scanning an RNA sample over an area of $8\ \mu\text{m} \times 8\ \mu\text{m}$ and collecting the intensity in the red detection channel. A number of diffraction-limited spots are visible; they originate from individual fluorophores as evidenced by the fact that the intensity disappears suddenly and completely within a few seconds of

excitation under the usual illumination conditions. This effect is called “digital photobleaching.” By proper adjustment of the excitation power it is possible to scan an image many times before photobleaching occurs. To measure the equilibrium between



Figure 2. Image of an $8\ \mu\text{m} \times 8\ \mu\text{m}$ area from a sample of immobilized 3-helix RNA junctions obtained by scanning the sample across the sensitive volume of the confocal microscope.

the open and folded conformation of the 3-helix junction as a function of the magnesium concentration, we have taken consecutive scans and varied the magnesium concentration *in situ* between scans. Successive images were spaced 10 minutes apart in time to ensure that the sample was indeed in equilibrium. For the individual spots, we have integrated the number of counts in the donor and the acceptor channel and subtracted the background as determined from the counts near the spots. From the background-corrected photon counts in the donor and acceptor channels, I_D and I_A , we calculated the proximity factor, $P = I_A/(I_A + I_D)$, which is an experimentally directly accessible quantity related to the FRET efficiency. Because it takes a few seconds during an image scan to collect the photons from a single molecule spot, the data

represent an average over this time interval and reflect equilibrium properties only if the fluctuations between the open and closed conformation are fast compared to the acquisition time. The data discussed in the following subsection confirm that this assumption is indeed valid. Figure 3 shows the dependence of the proximity factors on the magnesium ion concentration for two representative RNA molecules. Each of the RNA junctions studied had a different apparent transition midpoint in the range of a few hundred μM , suggesting that each molecule had a different microenvironment. We note that, within the precision of the data, the $[\text{Mg}^{2+}]$ dependence can be described well in the framework of a bimolecular reaction between the open conformation, $\text{RNA}(\text{O})$, and Mg^{2+} to form the folded molecule, $\text{RNA}(\text{F})\text{-Mg}^{2+}$, which would imply that the folding rate speeds up with increasing $[\text{Mg}^{2+}]$ owing to the higher collision probability of the reactants. The kinetic data discussed in the following subsection show, however, that the transition from the open to the folded conformation depends only weakly on $[\text{Mg}^{2+}]$, whereas the transition in the opposite direction has a strong dependence on $[\text{Mg}^{2+}]$.

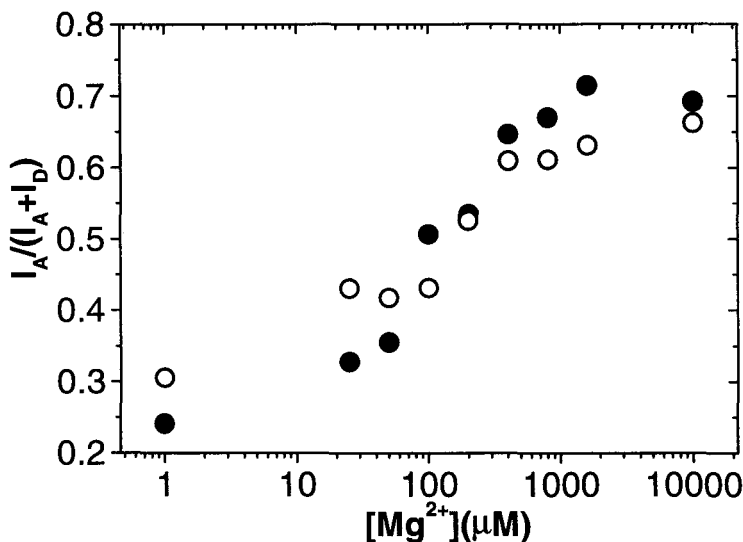


Figure 3. Plot of the proximity factors, P , for two representative RNA junctions as a function of the magnesium ion concentration.

3.2 Dynamics of RNA Interconversion between Open and Folded States

To examine the kinetics of interconversion between the open and folded conformation, we have taken time traces of the photon emission from individual RNA junctions. Figure 4 shows a time trace of a single RNA molecule in the green (solid line) and red (dotted line) channel after binning the record of photon arrival times in intervals of 5 ms. The excitation power was kept low ($\sim 1 \mu\text{W}/\mu\text{m}^2$) to minimize additional photophysical fluctuations (“blinking”). Usually, the acceptor dye (Cy5) photobleached faster than the donor (Cy3), typically after detection of 50,000 photons. Despite the noise arising from the low photon counts in each bin, an anticorrelation between the donor and acceptor channel is clearly visible, suggesting a conformational change that modulates the FRET efficiency. Binning in shorter time intervals only increases the noise and obscures the anticorrelation whereas binning on longer time scales leads to a smoothing of the fluctuations. Thus, the binning time of 5 ms in Fig. 4 is expected to be the approximate time scale of the dynamics.

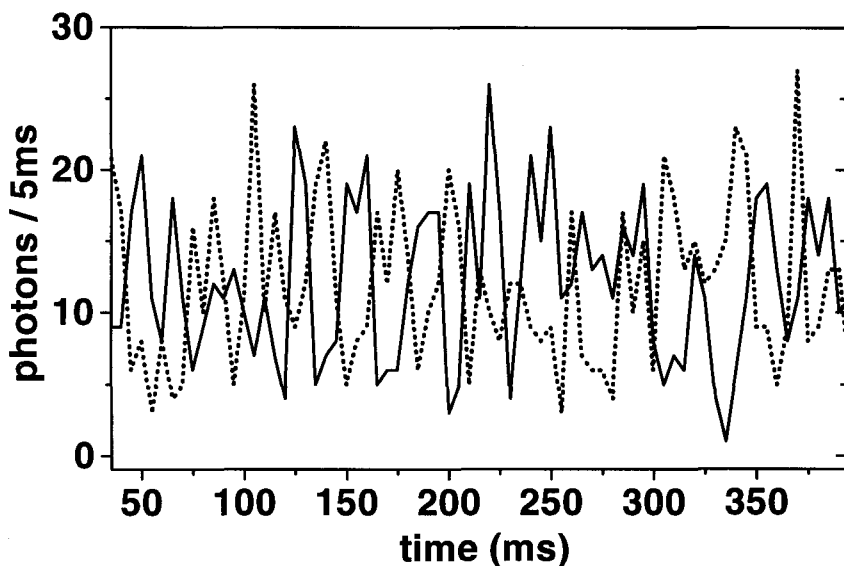


Figure 4. Time trace of the fluorescence fluctuations of a single RNA molecule at $[\text{Mg}^{2+}] = 200 \mu\text{M}$, solid line: donor, dotted line: acceptor. With photon counts binned in 5 ms intervals, an anticorrelation between the donor and acceptor fluctuations is clearly visible.

To quantitatively analyze the stochastic processes underlying the observed intensity fluctuations, it is useful to calculate autocorrelation functions of the donor intensity, $AC_D(\tau)$, or the acceptor intensity, $AC_A(\tau)$,

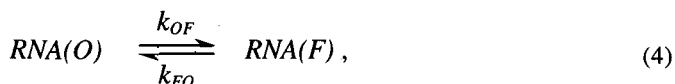
$$AC_{D(A)}(\tau) = \frac{\langle \delta I_{D(A)}(t) \delta I_{D(A)}(t+\tau) \rangle}{\langle I_{D(A)}(t) \rangle \langle I_{D(A)}(t) \rangle} = \frac{\langle I_{D(A)}(t) I_{D(A)}(t+\tau) \rangle}{\langle I_{D(A)}(t) \rangle \langle I_{D(A)}(t) \rangle} - 1 \quad (2)$$

or the crosscorrelation function $CC(\tau)$

$$CC(\tau) = \frac{\langle \delta I_D(t) \delta I_A(t+\tau) \rangle}{\langle I_D(t) \rangle \langle I_A(t) \rangle} = \frac{\langle I_D(t) I_A(t+\tau) \rangle}{\langle I_D(t) \rangle \langle I_A(t) \rangle} - 1 \quad (3)$$

from the experimental data. Figure 5 shows autocorrelations of donor and acceptor intensities as well as the crosscorrelation function calculated from the time trace of an individual RNA molecule at $[Mg^{2+}] = 200 \mu M$. To obtain better statistics, fluorescence time traces from typically 50 single molecules were collected at each of the magnesium concentrations chosen until photobleaching occurred, and correlation functions were calculated for the ensemble by time-weighted averaging of the individual correlation functions.

To compare the correlation functions with theoretical expressions, we assume that the dynamics of the 3-helix junction is properly described by a two-state interconversion between an open (O) and a folded (F) conformation,



characterized by the two rate coefficients, k_{OF} and k_{FO} , as depicted in Fig. 6. With the intensity levels of the donor in the open and folded conformation, I_D^O and I_D^F , and the corresponding ones, I_A^O and I_A^F , of the acceptor, the correlation functions can be calculated in a straightforward manner as

$$AC_A(\tau) = \frac{(I_A^F - I_A^O)^2}{(k_{FO} I_A^F + k_{OF} I_A^O)^2} k_{OF} k_{FO} \exp[-\lambda \tau], \quad (5)$$

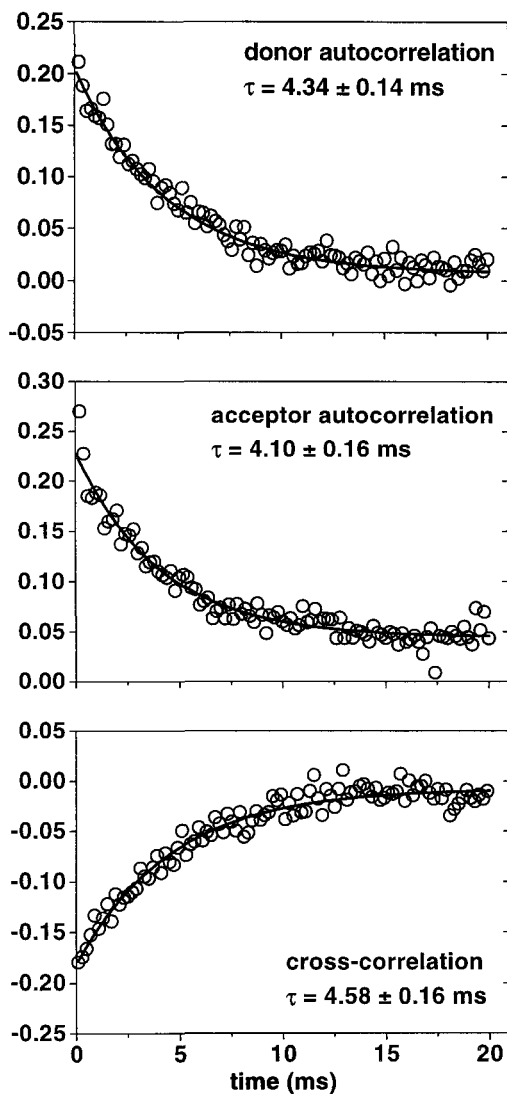


Figure 5. From top to bottom: autocorrelation functions of the donor and acceptor intensity and the donor-acceptor crosscorrelation function, calculated from the time trace of an individual RNA molecule. The decay times, τ , given in the figure panels were obtained by fitting exponentials to the data.

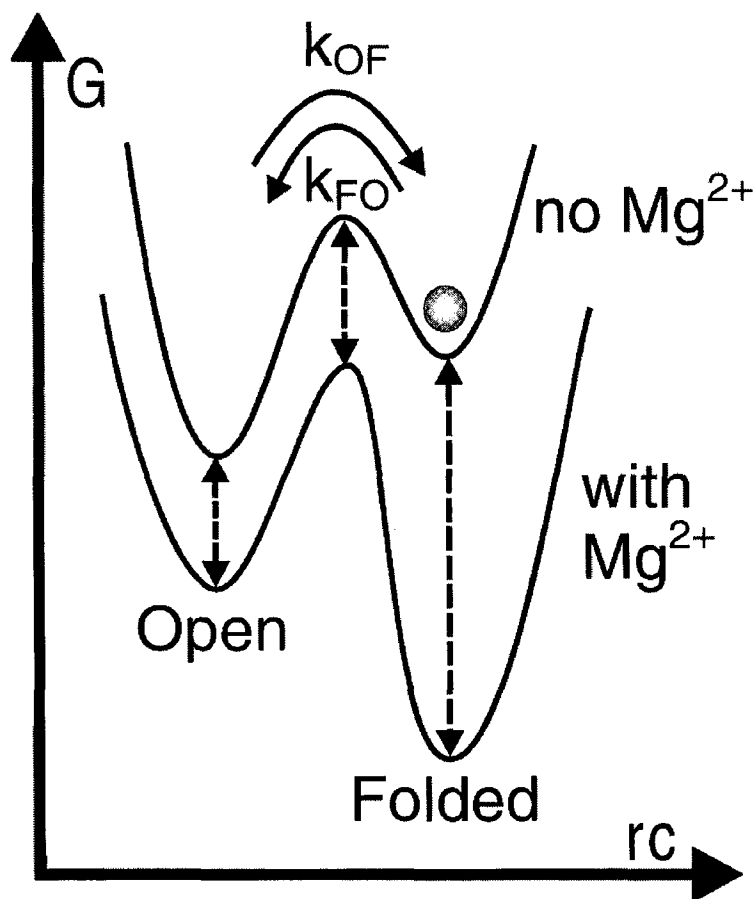


Figure 6. Free energy surfaces governing the transitions between the two RNA conformations in the absence (upper curve) and in the presence of Mg^{2+} ions (lower curve). The folded state is more efficiently stabilized by Mg^{2+} ions than the open state.

$$AC_D(\tau) = \frac{(I_D^F - I_D^O)^2}{(k_{FO}I_D^F + k_{OF}I_D^O)^2} k_{OF}k_{FO} \exp[-\lambda\tau], \quad \text{and} \quad (6)$$

$$CC(\tau) = \frac{(I_D^F - I_D^O)(I_A^F - I_A^O)}{(k_{FO}I_D^F + k_{OF}I_D^O)(k_{FO}I_A^F + k_{OF}I_A^O)} k_{OF}k_{FO} \exp[-\lambda\tau]. \quad (7)$$

Here, the apparent rate coefficient, λ , is given by

$$\lambda = k_{FO} + k_{OF}. \quad (8)$$

Within the two-state model, the correlations decay exponentially. Figure 5 shows an exponential fit to the data, from which both amplitudes and rate coefficients λ can be determined. Note that the decay times given in Fig. 5 are the inverse of the apparent rate coefficients, $\tau = \lambda^{-1}$. Due to the noise, there are slight differences in the decay times determined from time traces of individual molecules. These differences are negligible for the correlation functions obtained by averaging over the ensemble.

The intensities of the folded conformation, I_D^F and I_A^F , were measured with S15 bound to the RNA, whereas those of the open conformation, I_D^O and I_A^O , were measured in buffer without magnesium. Therefore, k_{OF} and k_{FO} are the only unknowns in Eqs. 5-8 and thus can be determined from a fit of these expressions to the data. The microscopic rate coefficients for magnesium ion concentrations between 50 and 300 μM are compiled in Fig. 7. Note that the rate coefficient for folding, k_{OF} , is only weakly dependent on $[\text{Mg}^{2+}]$, in contrast to what one would expect if this were a bimolecular (or higher order) reaction. The pronounced dependence on $[\text{Mg}^{2+}]$ enters via the opening rate coefficient, k_{FO} , which increases sharply towards low $[\text{Mg}^{2+}]$. The midpoint of the transition is where the two rate coefficients are equal, that is, at $[\text{Mg}^{2+}] \approx 140 \mu\text{M}$.

Conformational changes at physiological temperatures occur by thermal activation over a barrier, and the rate coefficients governing the transitions are related to the free energy difference between a particular state and the transition state at the top of the barrier, ΔG ,

$$k = \nu \exp\left[\frac{-\Delta G}{RT}\right].$$

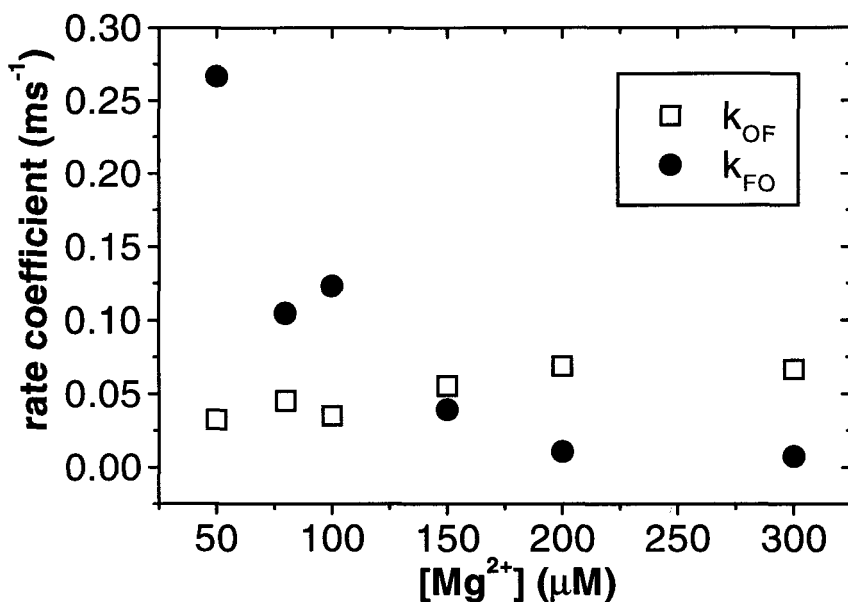


Figure 7. Microscopic rate coefficients, k_{OF} and k_{FO} , governing the conformational fluctuations of the 3-helix junction, plotted as a function of the magnesium ion concentration.

Here, ν is a frequency factor, R is the gas constant and T the absolute temperature in Kelvin. The observed dependencies in the rate coefficients k_{OF} and k_{FO} imply a change of the free energy surface with $[Mg^{2+}]$, as depicted in Fig. 6. The open and closed states as well as the transition state will decrease in free energy due to charge stabilization by $[Mg^{2+}]$ ions. The weak dependence of k_{OF} on $[Mg^{2+}]$ indicates, however, that there is only little change in the *free energy difference* between the open state and the transition state. The strong $[Mg^{2+}]$ dependence of k_{FO} reflects that the *free energy difference* between the folded state and the transition state decreases markedly as $[Mg^{2+}]$ decreases, thus reducing the barrier to the open state. Stabilization of the negative charges on the nucleic acid backbone by magnesium ions is thus energetically most significant in the folded state, as expected from basic electrostatic considerations.

4 Conclusions

Conformational interconversions between the open and the folded state of a 3-helix RNA junction have been investigated at the single molecule level through measurements of the fluorescence resonance energy transfer efficiency between two dye molecules attached to the ends of the junction. By measurement of the FRET efficiency as a function of $[\text{Mg}^{2+}]$, we have explored the equilibrium between the open and closed conformation. From the analysis of the correlation functions calculated from time traces of fluorescence emission from individual molecules, we determined the dependence of the kinetic rate coefficients on the concentration of magnesium. The folding rate was only weakly dependent on $[\text{Mg}^{2+}]$, whereas a pronounced increase of the opening transition was observed with decreasing $[\text{Mg}^{2+}]$. This behavior indicates that the magnesium ions function so as to stabilize the negative charges on the RNA molecule preferentially in the folded conformation.

The single-molecule techniques developed with this model system will be useful in the future for the study of a wide variety of problems in Biological Physics at the molecular level.

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References

1. E. Schrödinger, *Br. J. Philos.* (1952), 233.
2. R. P. Feynman, in *Miniaturization*, H. D. Gilbert, Ed. (Reinhold, New York, 1961).
3. S. Weiss, *Science* **283**, 1676 (1999).
4. H. Frauenfelder, J. Deisenhofer, P. Wolynes, Eds., *Simplicity and complexity in proteins and nucleic acids* (Dahlem University Press, Berlin, 1999).
5. G. U. Nienhaus, R. D. Young, in *Encyclopedia of Applied Physics*, Vol. 15, G. L. Trigg, Ed., (VCH Publishers, New York, 1996), 163.
6. H. J. Sass, G. Büldt, R. Gessenich, D. Hehn, D. Neff, R. Schlesinger, J. Berendzen, P. Ormos, *Nature* **406**, 649 (2000).
7. A. Ostermann, R. Waschipky, F. G. Parak, G. U. Nienhaus, *Nature* **404**, 205 (2000).
8. J. T. Finer, R. M. Simmons, J. A. Spudich, *Nature* **368**, 113 (1994).
9. H. Noji, R. Yasuda, M. Yoshida and K. Kinosita, *Nature* **386**, 299 (1997).
10. K. Kitamura, M. Tokunaga, A. H. Iwane, T. Yanagida, *Nature* **397**, 129 (1999).
11. H. P. Lu, L. Y. Xun, X. S. Xie, *Science* **282**, 1877 (1998).
12. M. S. Z. Kellermayer, S. B. Smith, H. L. Granzier, C. Bustamante, *Science* **276**, 1112 (1997).
13. M. Rief, M. Gautel, F. Oesterhelt, J. M. Fernandez, H. E. Gaub, *Science* **276**, 1109 (1997).
14. T. T. Perkins, D. E. Smith, S. Chu, *Science* **276**, 2016 (1997).
15. D. E. Smith, S. Chu, *Science* **281**, 1335 (1998).
16. T. Ha, Th. Enderle, D. F. Ogletree, D. S. Chemla, P. R. Selvin, S. Weiss, *Proc. Natl. Acad. Sci. USA* **93**, 6264 (1996).
17. T. Ha, X. Zhuang, H. D. Kim, J. W. Orr, J. R. Williamson, S. Chu, *Proc. Natl. Acad. Sci. USA* **96**, 9077 (1999).
18. H. D. Kim, G. U. Nienhaus, T. Ha, J. W. Orr, J. R. Williamson, S. Chu, S., submitted.
19. R. Rigler, U. Mets, J. Widengren, P. Kask, *Eur. Biophys. J.* **22**, 169 (1993).
20. G. Bonnet, O. Krichevsky, A. Libchaber, *Proc. Natl. Acad. Sci. USA* **95**, 8602 (1998).
21. D. C. Lamb, A. Schenk, C. Röcker, C. Scalfi-Happ, G. U. Nienhaus, *Biophys. J.* **79**, 1129 (2000).
22. W. A. Held, B. Ballou, S. Mizushima and M. Nomura, *J. Biol. Chem.* **249**, 3103, (1974).

23. A. C. Agalarov, G. S. Prasad, P. M. Funke, C. D. Stout, J. R. Williamson, *Science* **288**, 107 (2000).
24. R. T. Batey, J. R. Williamson, *J. Mol. Biol.* **261**, 536 (1996).
25. R. T. Batey, J. R. Williamson, *J. Mol. Biol.* **261**, 550 (1996).
26. J. W. Orr, P. J. Hagerman, J. R. Williamson, *J. Mol. Biol.* **275**, 453 (1998).

PATH INTEGRAL APPROACH TO A SINGLE POLYMER CHAIN WITH EXCLUDED VOLUME EFFECT

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The effect of a polymer chain with excluded volume representing the long-range interaction between segments along the chain is studied using the Feynman path integral method. The main problem is to calculate the mean square distance, $\langle R^2 \rangle$, that is

$$\langle R^2 \rangle = AN^\nu$$

where N is the degree of polymerization, ν is a scaling exponent varying from 1-2 for free chain and stiff chain, respectively; and A is a coefficient that depends on the details of the polymer. In the case that the excluded volume interactions are present, $\nu = 6/5$.

The model proposed by Edwards and Singh [1] and Muthukumar and Nickel [2] are employed. Instead of using the idea of an effective step-length technique and the perturbation technique, the idea of Feynman [3] in relation to the polaron problem is used and developed by handling a disordered system. The idea is to model the polymer action as a model of quadratic trial action and consider the differences between the polymer action and the trial action as the first cumulant approximation in one parameter. The variational principle is used to find the optimal values of the variational parameters and the mean square distance is obtained. A comparison between these approaches and effective step-length and perturbation approach will be discussed.

1 Introduction

The theory of the excluded volume effect in a polymer chain is one of the central problems in the field of polymer solution theory representing the effect of the interaction between segments which are far apart along the chain. This interaction is often called the **long-range interaction** in contrast to the **short-range interaction** representing the interaction among a few neighbouring segments.

The polymer excluded volume problem is of the same form and difficulty as the general many body problem, first discussed by Kuhn [4]. The modern development was initiated by Flory [5]. It is recognized that the long-range interaction changes the statistical property of the chain entirely.

The main problem is to calculate the mean square end-to-end distance $\langle R^2 \rangle$, that is

$$\langle R^2 \rangle = AN^\nu \quad (1.1)$$

where the **exponent** $1 < \nu < 2$, is to be determined and appears to be solely a function of the dimensionality of space.

Over past decades many attempts have endeavoured to understand the excluded volume effect. One of the theories of the polymer excluded volume problem is the Edwards and Singh's method. A simple method capable of adoption in more complicated problems is developed using the idea of an effective step-length. The mean square value of R^2 is developed as a series which for large L is

$$R^2 = \omega^{\frac{2}{5}} L^{\frac{6}{5}} l^{\frac{2}{5}} (1.12 + 1.05 + 1.03 + \dots) \quad (1.2)$$

where length $L = Nl$ and ω is the self repulsion.

Another theory is the perturbation which represents a simple derivation of the mean square end-to-end distance $\langle R^2 \rangle$ of a linear-flexible chain as a perturbation series in the dimensionless excluded volume parameter z_d . In an essential way the method used Laplace transforms with respect to the contour length L . The results, to orders six and four in space dimension $d = 3$ and 2, respectively, are

$$\begin{aligned} \langle R^2 \rangle &= Ll \left[1 + \frac{4}{3} z_3 - 2.075385396 z_3^2 + 6.296879676 z_3^3 - 25.05725072 z_3^4 \right. \\ &\quad \left. + 116.134785 z_3^5 - 594.71663 z_3^6 + \dots \right], \quad d = 3, \\ \langle R^2 \rangle &= Ll \left[1 + \frac{1}{2} z_2 - 0.12154525 z_2^2 + 0.02663136 z_2^3 - 0.13223603 z_2^4 + \dots \right], \\ &\quad d = 2. \end{aligned}$$

The purpose of this paper is to calculate the mean square end-to-end distance, employing the model proposed by Edwards [6] and using the idea of Feynman in the polaron problem and one developed by us for handling disordered systems [7]. The idea is to model the system with a trial action containing adjustable, to be determined, parameters. Once the trial action is introduced, the average distribution G can be calculated by expanding in first cumulant approximation about the corresponding trial average distribution G_0 . G , approximated by the first cumulant, G_1 can be obtained.

As mentioned above, the parameters should be determined by minimizing the exponent of G_1 . However, this procedure leads to a complication because the parameters will depend parametrically on the initial and final position of the polymer chain. To avoid such a complication, a simpler approximation in which only the diagonal contribution of the exponent of G_1 is used in the minimization. This approximation is equivalent to minimizing the free energy. Once the

parameters are obtained they will be used to calculate the mean square end-to-end distance.

The outline of the paper is as follows: section 2 is a brief review of an effective step-length technique and the perturbation technique. In section 3, Edwards' model is reviewed. The path integral approach with one parameter model is introduced in section 4. In section 5, the results are discussed and compared with other presentations. Finally, in order to be able to carry out the calculations arising in section 4, an appendix gives a detailed derivation of a characteristic functional corresponding to the trial action $S_0(\omega)$.

2 The Mean Square Distance

2.1 The Effective Step Length Technique

The excluded volume problem is a central part of polymer solution theory, the mean-square end-to-end distance agreed on $\langle R^2 \rangle \propto L^\alpha$ where $\alpha = 6/5$ is shown to within one to two percent. Analytic theories based on self consistent fields give α to be exactly 6/5. The chain is considered a locus in space $\mathbf{r}(s)$, s the arc length, and the random walk constraint is represented by the Wiener measure

$$\exp\left[-\frac{3}{2l} \int_0^L \dot{\mathbf{r}}^2(s) ds\right] \quad (2.1.1)$$

and the interaction

$$\exp\left\{-\omega \int_0^L \int_0^L \delta[\mathbf{r}(s) - \mathbf{r}(s')] ds ds'\right\}. \quad (2.1.2)$$

This allows the consideration of $\omega > 0$. The step-length is l , and ω has the dimensions of volume. If the symbol $(D(r))$ denotes integration over all paths, then

$$\langle R^2 \rangle = \frac{\int_{r(0)=0}^{r(L)=R} D(\mathbf{r}) [\mathbf{r}(L) - \mathbf{r}(0)]^2 \exp(A)}{\int d^3R \int_{r(0)=0}^{r(L)=R} \exp(A) D(\mathbf{r})} \quad (2.1.3)$$

Instead it can be argued that an effective step-length l_l be introduced, so that, by definition

$$\langle R^2 \rangle = Ll_1. \quad (2.1.4)$$

Therefore one can write

$$\begin{aligned} \frac{3}{2l} \int \dot{r}^2 ds + \omega \int [D[r(s) - r(s')] ds ds' \\ = \frac{3}{2l_1} \int \dot{r}^2 ds + \left\{ \frac{3}{2} \left(\frac{1}{l} - \frac{1}{l_1} \right) \int \dot{r}^2 ds + \omega \int [D[r(s) - r(s')] ds ds' \right\} \end{aligned} \quad (2.1.5)$$

$$= \frac{3}{2l_1} \int \dot{r}^2 ds + B, \text{ say} \quad (2.1.6)$$

$$= C + B. \quad (2.1.7)$$

Then

$$\langle R^2 \rangle = Ll_1 + O(B) + O(B^2) + O(B^3) + \dots \quad (2.1.8)$$

At this point, choose l_1 so that

$$\langle R^2 \rangle = Ll_1$$

so that to first order in B , it gives the equation

$$Ll_1^2 \left(\frac{1}{l} - \frac{1}{l_1} \right) = 2 \sqrt{\frac{6}{\pi^3}} \omega \frac{L^{\frac{3}{2}}}{l_1^{\frac{1}{2}}}. \quad (2.1.9)$$

The solution to this equation clearly subsumes perturbation theory, for if ω is small, $l \cong l_1$,

$$\langle R^2 \rangle = Ll + 2 \sqrt{\frac{6}{\pi^3}} \omega L^{\frac{3}{2}} l^{-\frac{1}{2}}. \quad (2.1.10)$$

But for $L^{\frac{1}{2}} > \omega$ there is a Flory type equation with solution

$$l_1 = (2)^{\frac{2}{5}} \left(\frac{6}{\pi^3} \right)^{\frac{1}{5}} \omega^{\frac{2}{5}} L^{\frac{2}{5}} L^{\frac{6}{5}} \quad (2.1.11)$$

so that

$$\langle R^2 \rangle = (2)^{\frac{2}{5}} \left(\frac{6}{\pi^3} \right)^{\frac{1}{5}} \omega^{\frac{2}{5}} l^{\frac{2}{5}} L^{\frac{6}{5}}. \quad (2.1.12)$$

To establish the stability of the index α against higher approximations, $\langle R^2 \rangle$ can be written as follows

$$\langle R^2 \rangle = Ll + \frac{A\omega L^{\frac{3}{2}}}{l^{\frac{1}{2}}} + \frac{B\omega^2 L^2}{l^2} + \frac{C\omega^3 L^{\frac{5}{2}}}{l^{\frac{7}{2}}} \quad (2.1.13)$$

where A , B , and C are numbers. Now effective step length l can be introduced to give

$$l = l_1 [1 - l_1 (\frac{1}{l} - \frac{1}{l_1}) + l_1^2 (\frac{1}{l} - \frac{1}{l_1})^2 \dots] \quad (2.1.14)$$

Using equation (2.1.14) in (2.1.13) it get to the third in ω

$$\begin{aligned} \langle R^2 \rangle = & Ll_1 - Ll_1^2 (\frac{1}{l} - \frac{1}{l_1}) + Ll_1^3 (\frac{1}{l} - \frac{1}{l_1})^2 + \frac{A\omega L^{\frac{3}{2}}}{l^{\frac{1}{2}}} + \frac{A\omega L^{\frac{3}{2}}}{2} l_1^{\frac{1}{2}} (\frac{1}{l} - \frac{1}{l_1}) \\ & + \frac{B\omega^2 L^2}{l^2} + 2B\omega^2 \frac{l^2}{l_1} (\frac{1}{l} - \frac{1}{l_1}) + \frac{C\omega^3 L^{\frac{5}{2}}}{l^{\frac{7}{2}}} - \dots \end{aligned} \quad (2.1.15)$$

Thus the first order approximation gives

$$\alpha = 1.059 \omega^{\frac{1}{5}} L^{\frac{1}{10}} l^{\frac{-3}{10}} \quad (2.1.16)$$

$$\langle R^2 \rangle \sim \alpha^2 \sim Ll_1 = 1.12 \omega^{\frac{2}{5}} L^{\frac{6}{5}} l^{\frac{2}{5}}.$$

Now to the second order contained is

$$\alpha = 1.025 \omega^{\frac{1}{5}} L^{\frac{1}{10}} l^{\frac{-3}{10}} \quad (2.1.17)$$

$\langle R^2 \rangle$ still retains the form $\sim \omega^{\frac{2}{5}} L^{\frac{6}{5}} l^{\frac{2}{5}}$. Now to the third order where

$$\begin{aligned} \alpha &= 1.015 \omega^{\frac{1}{5}} L^{\frac{1}{10}} l^{\frac{-3}{10}}, \\ \langle R^2 \rangle &\sim \alpha^2 = 1.03 \omega^{\frac{2}{5}} L^{\frac{6}{5}} l^{\frac{2}{5}}. \end{aligned} \quad (2.1.18)$$

Finally this can be expressed in a series for $\langle R^2 \rangle$, the additions coming from the order of expansion

$$\langle R^2 \rangle = \omega^{\frac{2}{3}} L^{\frac{6}{5}} l^{\frac{2}{5}} (1.12 + 1.05 + 1.03 + \dots). \quad (2.1.19)$$

2.2 The Perturbation Technique

The net effect of the excluded volume interaction between segments of the polymer chain is usually repulsive and leads to an expansion of the chain size. When the excluded volume interaction is very weak, a perturbation theory for the ratio of the mean square end-to-end distance is $\langle R^2 \rangle$ of the chain. Its unperturbed value $\langle R^2 \rangle_0$ can be developed and reduced to a varies in a single dimensionless interaction parameter z_d [8] as

$$\frac{\langle R^2 \rangle}{\langle R^2 \rangle_0} = 1 + C_1 z_d + C_2 z_d^2 + C_3 z_d^3 + \dots \quad (2.2.1)$$

In describing the approach to equation (2.2.1) the equation starts directly with the continuum model and works entirely with the Laplace transform functions $\tilde{G}(E) = \int_0^\infty G(L) e^{-EL} dL$, where $G(L)$ are probability functions for a chain's contour length L .

For the standard discrete Gaussian chain model of $N+1$ segments, the probability distribution function $G_0(R, n)$ can be evaluated and a term of the contour length $L = Nl$ is given by

$$G_0(R, L) = \left(\frac{d}{2\pi Ll} \right)^{d/2} \exp\left(-\frac{dR^2}{2Ll} \right). \quad (2.2.2)$$

The mean square end-to-end distance of the Gaussian chain is

$$\begin{aligned} \langle R^2 \rangle_0 &= \int d^d R R^2 G_0(R, L) / \int d^d R G_0(R, L) \\ &= Ll. \end{aligned} \quad (2.2.3)$$

When interactions are introduced, the bare distribution will be modified to a non-Gaussian $G(R, L)$ with corresponding characteristic function $\hat{G}(k, L)$ and propagator $\tilde{G}(k, E_0)$. The mean square end-to-end chain distance is given by

$$\langle R^2 \rangle = \frac{\int d^d R R^2 G(R, L)}{\int d^d R G(R, L)} = l \frac{\int \frac{dE_0}{2\pi} e^{E_0 l} \left[-\frac{2d}{l} \frac{\partial}{\partial k^2} \tilde{G}(k, E_0) \right]_{k=0}}{\int \frac{dE_0}{2\pi} e^{E_0 l} \tilde{G}(0, E_0)} . \quad (2.2.4)$$

This can define a new variable, The so-call “renormalized” energy E by

$$E \equiv E_0 + \Sigma(0, E_0) . \quad (2.2.5)$$

This definition can be reverted order-by-order in perturbation theory to yield $E_0(E)$. If this is defined as

$$\tilde{\Sigma}(k, E) \equiv \tilde{\Sigma}[k, E_0(E)] , \quad (2.2.6)$$

then the exact propagator becomes

$$\tilde{G}[k, E_0(E)] = \frac{1}{E + \frac{k^2 l}{2d} + \tilde{\Sigma}(k, E) - \tilde{\Sigma}(0, E)} . \quad (2.2.7)$$

These results can be expressed as functions of E and the equation (2.2.4) rewritten as

$$\langle R^2 \rangle = l \frac{\int \frac{dE}{2\pi} e^{E l} F_{2(E)}}{\int \frac{dE}{2\pi} e^{E l} F_1(E)} , \quad (2.2.8)$$

where

$$\begin{aligned} F_1(E) &= E^{-1} J \exp[(E_0 - E)L], \\ F_2(E) &= E^{-1} K F_1(E)L, \end{aligned} \quad (2.2.9)$$

where

$$\begin{aligned} J &= \frac{dE_0}{dE} = 1 - \frac{d}{dE} \tilde{\Sigma}(0, E), \\ K &= 1 + \frac{2d}{l} \frac{\partial}{\partial k^2} \tilde{\Sigma}(k, E) |_{k=0} . \end{aligned} \quad (2.2.10)$$

Solving these simultaneous equations in (2.2.9) for F_1 and F_2 , $d = 3$ is obtained and

$$\langle R^2 \rangle = Ll[1 + \sum_{m=1} C_m z_3^m] , \quad (2.2.11)$$

where the coefficients through order $m = 6$ are

$$z_3 \equiv \left(\frac{3}{2\pi}\right)^{3/2} \frac{\omega L^{1/2}}{l^{3/2}}, C_1 = 4/3, C_2 = \frac{28\pi}{27} - \frac{16}{3}, C_3 \approx 6.296879676 , \\ C_4 \approx -25.05725072 , C_5 \approx 116.134785 , C_6 \approx -594.71663 . \quad (2.2.12)$$

For $d = 2$, the calculation of $\langle R^2 \rangle$ runs in an exactly analogous manner to the above derivation in $d = 3$. Then

$$\langle R^2 \rangle = Ll[1 + \sum_{m=1} C_m z_2^m] \quad (2.2.13)$$

where

$$z_2 \equiv \frac{\omega L}{\pi l}, C_1 = 1/2, C_2 = -0.1215452 , \\ C_3 = 0.0266313 , C_4 = -0.13223603 . \quad (2.2.14)$$

3 Edwards' Model

In an ideal polymer, there is only a short-range interaction and the action can be written as

$$S = \frac{3}{2l^2} \int_0^N d\tau \dot{\mathbf{R}}^2(\tau) \quad (3.1)$$

where l is the effective bond length representing the short-range interaction, and $\mathbf{R}(\tau)$ is the position of segment τ of the polymer.

Since there are many effects in real polymers, the long-range interaction is quite complicated: steric effects, Van der Waals attraction, and solvent molecules effect. However, for the large-length scale concerned, the details of the interaction can be omitted. Thus the interaction between the polymer segments τ and σ can be expressed as

$$k_B T v(\mathbf{R}_\tau - \mathbf{R}_\sigma) .$$

This can be approximated even further by a delta function

$$v k_B T \delta(\mathbf{R}_\tau - \mathbf{R}_\sigma) ,$$

where v is the excluded volume which represents the long-range interaction, and has the dimension of volume.

The total interaction energy is thus written as

$$U_1 = \frac{v}{2} k_B T \int_0^N \int_0^N d\tau d\sigma \delta(\mathbf{R}(\tau) - \mathbf{R}(\sigma)) \quad (3.2)$$

using the local concentration of the segments

$$c(\mathbf{r}) = \int_0^N d\tau \delta(\mathbf{r} - \mathbf{R}(\tau)) . \quad (3.3)$$

Thus, equation (3.2) may be rewritten

$$U_1 = \int d\mathbf{r} \frac{1}{2} v k_B T c(\mathbf{r})^2 . \quad (3.4)$$

This statement indicates that equation (3.2) is the first term in the virial expansion of the free energy with respect to the local concentration $c(\mathbf{r})$. Now if the interaction equation (3.2) is taken into account, the action becomes

$$S = \frac{3}{2l^2} \int_0^N d\tau \dot{\mathbf{R}}^2(\tau) + \frac{v}{2} \int_0^N \int_0^N d\tau d\sigma \delta(\mathbf{R}(\tau) - \mathbf{R}(\sigma)) . \quad (3.5)$$

The second term accounts for excluded volume interactions between segments of the polymer. The probability distribution of the end-to-end distance is given by

$$G(\mathbf{R}_2, \mathbf{R}_1; N) = \int_{\mathbf{R}_1}^{\mathbf{R}_2} D[\mathbf{R}(\tau)] e^{-S} . \quad (3.6)$$

4 Path Integral Approach

In this section, we apply the idea of Feynman, developed for the polaron problem, and Sa-yakanit[9], applied to disorder system, to polymer problem. This idea is to model the system with a model trial action which can be solved exactly for one parameter model.

The Edwards action is exactly solvable only in the limit, $v = 0$, where no excluded volume interactions are presented. In this case, the polymer exhibits Gaussian statistics. If v is not zero, the model is no longer exactly solvable;

however, this problem is similar to a polaron problem, which can be solved by path integral and variation method, by introducing the trial action $S_0(\omega)$,

$$S_0(\omega) = \frac{m}{2} \int_0^N d\tau \dot{\mathbf{R}}^2(\tau) + \frac{m\omega^2}{4N} \int_0^N \int_0^N d\tau d\sigma (\mathbf{R}(\tau) - \mathbf{R}(\sigma))^2 \quad (4.1)$$

where $m = 3/l^2$ and ω is a parameter.

Once the trial action $S_0(\omega)$ has been introduced, it is possible to find the average distribution which, from equation (3.6) can be rewritten as

$$G(\mathbf{R}_2, \mathbf{R}_1; N) = G_0(\mathbf{R}_2, \mathbf{R}_1; N, \omega) \langle \exp[S_0(\omega) - S] \rangle_{S_0(\omega)}, \quad (4.2)$$

where the trial distribution $G_0(\mathbf{R}_2, \mathbf{R}_1; N, \omega)$ is defined by

$$G_0(\mathbf{R}_2, \mathbf{R}_1; N) = \int_{\mathbf{R}_1}^{\mathbf{R}_2} D[\mathbf{R}(\tau)] e^{-S_0(\omega)}, \quad (4.3)$$

and the average $\langle x \rangle_{S_0(\omega)}$ is defined as

$$\langle X \rangle_{S_0(\omega)} = \frac{\int_0^N D[\mathbf{R}(\tau)] X e^{-S_0(\omega)}}{\int_0^N D[\mathbf{R}(\tau)] e^{-S_0(\omega)}}. \quad (4.4)$$

Approximating equation (4.2) by the first cumulant, we get

$$G_1(\mathbf{R}_2, \mathbf{R}_1; N) = G_0(\mathbf{R}_2, \mathbf{R}_1; N, \omega) \langle \exp[S_0(\omega) - S] \rangle_{S_0(\omega)}. \quad (4.5)$$

To obtain $G_1(\mathbf{R}_2, \mathbf{R}_1; N)$, find $G_0(\mathbf{R}_2, \mathbf{R}_1; N)$ and the average. Firstly, consider the average $\langle S_0(\omega) - S \rangle_{S_0(\omega)}$. Since the first term in S and $S_0(\omega)$ always cancel each other, this denotes $\langle S \rangle_{S_0(\omega)}$ and $\langle S_0(\omega) \rangle_{S_0(\omega)}$ for convenience as the averages of the second term respectively. The average of $\langle S \rangle_{S_0(\omega)}$ can be evaluated by making a Fourier decomposition of $\delta(\mathbf{R}(\tau) - \mathbf{R}(\sigma))$. Thus,

$$\langle S \rangle_{S_0(\omega)} = \frac{v}{2} \int_0^N \int_0^N d\tau d\sigma \left(\frac{1}{2\pi} \right)^3 \int d\mathbf{k} \langle \exp[i\mathbf{k} \cdot (\mathbf{R}(\tau) - \mathbf{R}(\sigma))] \rangle_{S_0(\omega)}. \quad (4.6)$$

The average on the right-hand side of equation (4.6) can be expanded in cumulants, and because $S_0(\omega)$ is quadratic, only the first two cumulants are non-zero [10]. Equation (4.6) becomes

$$\langle S \rangle_{S_0(\omega)} = \frac{V}{2} \int_0^N \int_0^N d\tau d\sigma \left(\frac{1}{2\pi} \right)^3 \int d\mathbf{k} \exp(x_1 + x_2) , \quad (4.7)$$

where

$$x_1 = i\mathbf{k} \cdot \langle \mathbf{R}(\tau) - \mathbf{R}(\sigma) \rangle_{S_0(\omega)} , \quad (4.8)$$

and

$$x_2 = \frac{-k^2}{2} \left[\frac{1}{3} \langle (\mathbf{R}(\tau) - \mathbf{R}(\sigma))^2 \rangle_{S_0(\omega)} - \langle \mathbf{R}(\tau) - \mathbf{R}(\sigma) \rangle_{S_0(\omega)} \right] . \quad (4.9)$$

Note that the second term inside the square brackets of equation (4.9) represents only one component of the coordinates. Performing the $\mathbf{k}$ -integration results in

$$\langle S \rangle_{S_0(\omega)} = \frac{V}{2} \int_0^N \int_0^N d\tau d\sigma \left(\frac{1}{2\pi} \right)^3 A^{-3/2} \exp\left(\frac{-\mathbf{B}^2}{4A} \right) \quad (4.10)$$

where

$$A = \frac{1}{2} \left[\frac{1}{3} \langle (\mathbf{R}(\tau) - \mathbf{R}(\sigma))^2 \rangle_{S_0(\omega)} - \langle \mathbf{R}(\tau) - \mathbf{R}(\sigma) \rangle_{S_0(\omega)} \right] \quad (4.11)$$

and

$$\mathbf{B} = i \langle \mathbf{R}(\tau) - \mathbf{R}(\sigma) \rangle_{S_0(\omega)} . \quad (4.12)$$

Next we consider the average of $\langle S_0(\omega) \rangle_{S_0(\omega)}$ which is easily written as

$$\langle S_0(\omega) \rangle_{S_0(\omega)} = \frac{m\omega^2}{4N} \int_0^N \int_0^N d\tau d\sigma \langle (\mathbf{R}(\tau) - \mathbf{R}(\sigma))^2 \rangle_{S_0(\omega)} . \quad (4.13)$$

4.1 The Characteristic Functional

From equations (4.10) and (4.13) it can be seen that the average $\langle S_0(\omega) - S \rangle_{S_0(\omega)}$ can be expressed solely in terms of the following averages: $\langle \mathbf{R}(\tau) \rangle_{S_0(\omega)}$ and $\langle \mathbf{R}(\tau)\mathbf{R}(\sigma) \rangle_{S_0(\omega)}$. Such averages can be obtained from a characteristic functional of

$\langle \exp(\int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}(\tau)) \rangle_{S_0(\omega)}$. From Feynman and Hibbs, the characteristic functional can be expressed as

$$\langle \exp(\int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}(\tau)) \rangle_{S_0(\omega)} = \exp(-[S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega) - S_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)]) \quad (4.1.1)$$

where $S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ and $S_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ are two classical actions which we have derived from the calculation in the Appendix. Once the classical action $S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ is obtained, we can differentiate expression (4.1.1) with respect to $\mathbf{f}(\tau)$ to obtain

$$\begin{aligned} \langle \mathbf{R}(\tau) \rangle_{S_0(\omega)} &= - \frac{\delta S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)}{\delta \mathbf{f}(\tau)} \Big|_{\mathbf{f}(\tau)=0} \\ &= \frac{m\omega}{2 \sinh \omega N} \left[\frac{2\mathbf{R}_2}{m\omega} \left(\sinh \omega \tau + 2 \sinh \frac{\omega N}{2} \sinh \frac{\omega(N-\tau)}{2} \sinh \frac{\omega \tau}{2} \right) \right. \\ &\quad \left. + \frac{2\mathbf{R}_1}{m\omega} \left(\sinh \omega(N-\tau) + 2 \sinh \frac{\omega N}{2} \sinh \frac{\omega(N-\tau)}{2} \sinh \frac{\omega \tau}{2} \right) \right] \quad (4.1.2) \end{aligned}$$

where the symbol $|_{\mathbf{f}(\tau)=0}$ implies that after the differentiation, $\mathbf{f}(\tau)=0$ must be set. Continuing the differentiation,

$$\begin{aligned} \langle \mathbf{R}(\tau) \mathbf{R}(\sigma) \rangle_{S_0(\omega)} &= \left[- \frac{\delta^2 S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)}{\delta \mathbf{f}(\tau) \delta \mathbf{f}(\sigma)} \right. \\ &\quad \left. + \frac{\delta S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)}{\delta \mathbf{f}(\tau)} \cdot \frac{\delta S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)}{\delta \mathbf{f}(\sigma)} \right] \Big|_{\mathbf{f}(\tau)=0} \quad (4.1.3) \end{aligned}$$

Set $\sigma = \tau$ in equation(4.1.3) to obtain

$$\begin{aligned}
\langle R^2(\tau) \rangle_{s_o(\omega)} = & \frac{3}{m\omega} \left\{ \frac{\sinh \omega(N-\tau) \sinh \omega\tau}{\sinh \omega N} + \frac{4 \sinh^2 \frac{\omega}{2} (N-\tau) \sinh^2 \frac{\omega}{2} \tau}{\sinh \omega N} \right\} \\
& + [R_2 \left(\frac{\sinh \omega\tau}{\sinh \omega N} + \frac{2 \sinh \frac{\omega N}{2} \sinh \frac{\omega(N-\tau)}{2} \sinh \frac{\omega\tau}{2}}{\sinh \omega N} \right) \\
& + R_1 \left(\frac{\sinh \omega(N-\tau)}{\sinh \omega N} + \frac{2 \sinh \frac{\omega N}{2} \sinh \frac{\omega(N-\tau)}{2} \sinh \frac{\omega\tau}{2}}{\sinh \omega N} \right)]^2 .
\end{aligned} \tag{4.1.4}$$

Equation (4.1.4) is the mean square end-to-end distance of the polymer. This method is more general than another methods because the mean square can be found at any point along the polymer chain.

Using equations (4.1.2) and (4.1.3) and performing the integration in equation (4.13) the following is obtained:

$$\begin{aligned}
\langle S_o(\omega) \rangle_{s_o(\omega)} = & \frac{3}{2} \left(\frac{\omega N}{2} \coth \frac{\omega N}{2} - 1 \right) \\
& + \frac{m}{2} \left[\frac{\omega N}{2} \coth \frac{\omega N}{2} - \left(\frac{\omega N}{2} \operatorname{sech} \frac{\omega N}{2} \right)^2 \right] \frac{(\mathbf{R}_2 - \mathbf{R}_1)^2}{2N} .
\end{aligned} \tag{4.1.5}$$

Collecting the above results, the following is obtained:

$$\begin{aligned}
G_1(\mathbf{R}_2 - \mathbf{R}_1; N, \omega) = & \left(\frac{3}{2\pi N l^2} \right)^{3/2} \left(\frac{\omega N}{2 \sinh \frac{\omega N}{2}} \right) \\
& \exp \left[\left(\frac{-3\omega}{4l^2} \right) (\mathbf{R}_2 - \mathbf{R}_1)^2 \left(\frac{1}{2} \coth \frac{\omega N}{2} + \frac{\omega N}{4} \operatorname{sech}^2 \frac{\omega N}{2} \right) \right. \\
& - \frac{3}{2} + \frac{3\omega N}{4} \coth \frac{\omega N}{2} \\
& \left. - \frac{v}{2} \int_0^N \int_0^N d\tau d\sigma \left(\frac{1}{4\pi} \right)^{3/2} A^{-3/2} \exp \left(\frac{-\mathbf{B}^2}{4A} \right) \right] ,
\end{aligned} \tag{4.1.6}$$

where we find for $\tau > \sigma$

$$A = \frac{\sinh \frac{\omega(\tau - \sigma)}{2} \sinh \frac{\omega(N - (\tau - \sigma))}{2}}{m\omega \sinh \frac{\omega N}{2}} \quad (4.1.7)$$

and

$$B = \frac{i \sinh \frac{\omega(\tau - \sigma)}{2} \cosh \frac{\omega(N - (\tau + \sigma))}{2}}{\sinh \frac{\omega N}{2}} \cdot (\mathbf{R}_2 - \mathbf{R}_1). \quad (4.1.8)$$

$G_1(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ is the average distribution in the first cumulant. To determine $\langle R^2(\tau) \rangle$, ω must be found first. Three cases are considered:

Case I ($\nu = 0$ and $\omega = 0$)

This case is the free polymer chain or the chain without excluded volume effect.

$\frac{\partial \ln G_1(\mathbf{R}_2, \mathbf{R}_1; N, \omega)}{\partial \mathbf{R}_2} = 0$ was calculated, This approximation is equivalent to

minimizing the free energy, then $\mathbf{R}_2 = \mathbf{R}_1$ was obtained. If one end of the polymer chain at the origin ($\mathbf{R}_1 = 0$) is fixed and taken to the limit $\omega \rightarrow 0$ in equation (4.1.4), then

$$\langle R^2(\tau) \rangle \approx \frac{l^2(N - \tau)\tau}{N}. \quad (4.1.9)$$

Equation (4.1.9) represents the mean square distance at any points along the chain without volume effect—free polymer chain. This result corresponds to the experiment and another methods, but is more general as can be seen for $N \rightarrow \infty$:

$$\langle R^2(\tau) \rangle \approx l^2 \tau.$$

Cases II and III

In cases II and III, the variational method was used by minimizing the diagonal contribution of the exponent of $G_1(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$. This approximation is equivalent to minimizing the free energy:

$$\frac{\partial \ln \text{Tr} G_1(\mathbf{R}_2, \mathbf{R}_1; N, \omega)}{\partial \omega} = 0. \quad (4.1.10)$$

Thus

$$\begin{aligned}
& \left(1 - \frac{\omega N}{2} \coth \frac{\omega N}{2} \right) \\
& + \frac{1}{2} \left(\left(\frac{\omega N}{2} \coth \frac{\omega N}{2} \right) - \left(\frac{\omega N}{2} \operatorname{cosech} \frac{\omega N}{2} \right)^2 \right) = \frac{vN}{4m} \left(\frac{1}{4\pi} \right)^{3/2} \int_0^N dx A^{-5/2} \\
& \left[\frac{\sinh \frac{\omega x}{2} \sinh \frac{\omega(N-x)}{2}}{\omega \sinh \frac{\omega N}{2}} - \frac{N \sinh^2 \frac{\omega x}{2}}{2 \sinh^2 \frac{\omega N}{2}} \right. \\
& \quad \left. - \frac{x \sinh \frac{\omega(N-2x)}{2}}{2 \sinh \frac{\omega N}{2}} \right], \quad (4.1.11)
\end{aligned}$$

where $x = \tau - \sigma$ and $\tau > \sigma$. Equations (4.1.6) and (4.1.11) represent a complete determination of $G_1(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$; however, they can not be solved exactly.

In Case II (ω is small and v is not zero):

Equation (4.1.11) was approximated as

$$1 - \frac{\omega N}{4} - \frac{(\omega N)^2 e^{-\omega N}}{2} = \frac{1}{4} \left(\frac{m}{2\pi} \right)^{3/2} v N \omega^{1/2} (1 + \omega N) (1 - e^{-\omega N}). \quad (4.1.12)$$

By substituting equation (4.1.12) in equation (4.1.4) the following was obtained:

$$\langle R^2 \rangle = N l^2 \left[\frac{1}{4} + \frac{7 v m^{3/2} N^{1/2}}{20 \cdot 2\pi^{3/2}} - \frac{273 m^3 v^2 N}{2000 \pi^3} + \dots \right]. \quad (4.1.13)$$

In Case III (ω is large and v is not zero):

Equation (4.1.11) can be expressed in asymptotic form as

$$\begin{aligned}
\left(1 - \frac{\omega N}{2} \right) + \frac{\omega N}{4} &= \frac{vN}{4m} \left(\frac{1}{4\pi} \right)^{3/2} \int_0^N dx \left(\frac{1}{2m\omega} \right)^{-5/2} \left(\frac{1}{2\omega} \right) \\
1 - \frac{\omega N}{4} &= \left(\frac{vN^2}{4} \right) \left(\frac{m}{2\pi} \right)^{3/2} \omega^{3/2} \\
\omega &= \left(\frac{2\pi}{m} \right) \left(\frac{4}{v} \right)^{2/3} N^{-4/3} \left(1 - \frac{\omega N}{4} \right)^{2/3}. \quad (4.1.14)
\end{aligned}$$

Asymptotically substituting equation (4.1.14) into (4.1.4) to obtain

$$\langle R^2 \rangle \approx \left(\frac{3}{2\pi} \right) \left(\frac{v}{4} \right)^{2/3} N^{4/3}. \quad (4.1.15)$$

5 Discussion and Conclusions

The paper studied the polymer-excluded volume employing the Feynman path integral method with the model proposed by Edwards. The calculation that follows is developed by us for handling the disorder system. The average mean square displacement at any length in the polymer is obtained. Therefore the result is more general than other methods where only the end-to-end point is calculated. In order to be able to appreciate the result of these calculations see Table 1.

Table 1.

Model Case	Perturbation	Edwards	Our method
Free chain	Nl^2	Nl^2	$\frac{l^2(N - \tau)\tau}{N}$
Weak interaction	$Nl^2 \left(1 + \frac{4}{3} \left(\frac{3}{2\pi l} \right)^{\frac{3}{2}} \omega L^{\frac{1}{2}} \right)$	-----	$\frac{Nl^2}{4}$
Strong interaction	-----	$1.12 \omega^{\frac{2}{5}} N^{\frac{6}{5}} l^{\frac{8}{5}}$	$\left(\frac{3}{2\pi} \right) \left(\frac{v}{4} \right)^{\frac{2}{3}} N^{\frac{4}{3}}$

Table 1: Present results are compared with the perturbation method and the method developed by Edwards. From the table for free chain all approaches lead to Nl^2 . Note that since the present results give detailed information along the chain. Therefore $N \rightarrow \infty$, $\frac{(N - \tau)\tau}{N} \rightarrow N$ and the present results will coincide with the free chain. For weak interaction the present results differ from the perturbation by a factor of 1/4. This is due to our approximation by using harmonic approximation. It is well known that a harmonic approximation always leads to unphysical results for weak ω . Finally, for strong interaction the present result is $N^{4/3}$ instead of $N^{6/5}$ as

obtained by Edwards. It is noticeable that the harmonic approximation is also not very good for strong interaction. The reason is that a harmonic trial action cannot model the delta function in this excluded volume problem because the delta function has a bounded state at minus infinity. If our excluded volume has a finite range then the harmonic trial action will be able to model the long-range problem. Future research will consider more of this problem. Although the use of a harmonic trial action does not correctly produce the weak and strong coupling in the exponent, it does give the prefactor A correctly which is important for calculating the magnitude of the mean square displacement. This result can be recognized by noticing that

$$\langle R^2 \rangle = AN^\nu.$$

Then the exponent can be obtained by plotting ν against $\ln \left[\frac{\langle R^2 \rangle}{l^2} \right] / \ln[N]$ for

large N. The $\langle R^2 \rangle$ can be taken from equation (4.1.4) and the result is given in Fig. 1.

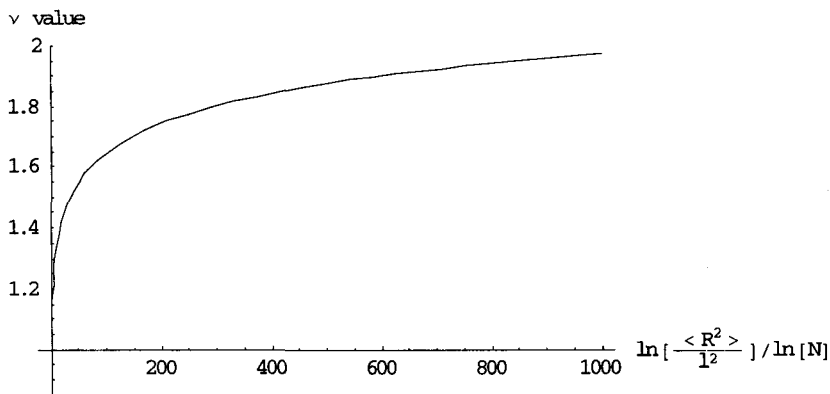


Figure 1.

The advantage of the variational method is that the mean square displacement of the polymer chain with excluded volume for any coupling strength and fluctuation along the length of the polymer can be obtained. The intermediate strength is shown graphically in Fig. 1.

Finally, this method can be generalized to two parameters as in the case of polaron.

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7 Appendix

In order to evaluate the averages of $\langle \mathbf{R}(\tau) \rangle_{S_0(\omega)}$ and $\langle \mathbf{R}(\tau)\mathbf{R}(\sigma) \rangle_{S_0(\omega)}$, it is necessary to establish a characteristic functional, as

$$\langle \exp \left(\int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}(\tau) \right) \rangle_{S_0(\omega)} = \frac{\int_0^N D[\mathbf{R}(\tau)] \exp(-S_0(\omega) + \int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}(\tau))}{\int_0^N D[\mathbf{R}(\tau)] \exp(-S_0(\omega))} \quad (\text{A1})$$

where $\mathbf{f}(\tau)$ is any arbitrary function of time, equation (A1) suggests that if the trial action $S_0(\omega)$ is quadratic, then the action of $S'_0(\omega) = S_0(\omega) - \int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}(\tau)$.

Using Feynman and Hipps, the path integral of equation (A1) can be carried out exactly as

$$\langle \exp \left(\int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}(\tau) \right) \rangle_{S_0(\omega)} = \exp \left(- \left[S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega) - S_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega) \right] \right) \quad (\text{A2})$$

where $S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ and $S_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ are the corresponding classical actions of $S'_0(\omega)$ and $S_0(\omega)$. When the classical action $S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ is obtained, the classical action $S_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ can be obtained from it by setting $\mathbf{f}(\tau) = 0$.

To obtain the classical action $S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ a variation is required for $S'_{0,cl}(\omega)$ to obtain the equation

$$\frac{d^2 \mathbf{R}_c(\tau)}{d\tau^2} - \frac{\omega^2}{N} \int_0^N d\sigma (\mathbf{R}_c(\tau) - \mathbf{R}_c(\sigma)) = -\frac{\mathbf{f}(\tau)}{m}. \quad (\text{A3})$$

This equation may be rewritten in the form

$$\frac{d^2 \mathbf{R}_c(\tau)}{d\tau^2} - \omega^2 \mathbf{R}_c(\tau) = -\frac{\omega^2}{N} \int_0^N d\sigma \mathbf{R}_c(\sigma) - \frac{\mathbf{f}(\tau)}{m}. \quad (\text{A4})$$

By introducing a Green Function

$$\left(\frac{d^2}{d\tau^2} - \omega^2 \right) g(\tau, \sigma) = \delta(\tau - \sigma) \quad (\text{A5})$$

where

$$g(\tau, \sigma) = - \left[\frac{\sinh \omega(N - \tau) \sinh \omega \sigma H(\tau - \sigma) + \sinh \omega(N - \sigma) \sinh \omega \tau H(\sigma - \tau)}{\omega \sinh \omega N} \right] \quad (\text{A6})$$

with H denoting the Heviside step function, then the general solution of equation (A5) with the boundary condition $\mathbf{R}(0) = \mathbf{R}_1$ and $\mathbf{R}(N) = \mathbf{R}_2$ can be written as

$$\begin{aligned} \mathbf{R}_c(\tau) = & \frac{[\mathbf{R}_2 \sinh \omega \tau + \mathbf{R}_1 \sinh \omega(N - \tau)]}{\sinh \omega N} \\ & - \int_0^N \left[\frac{\mathbf{f}(\sigma)}{m} + \frac{\omega^2}{N} \int_0^N \mathbf{R}(\sigma) d\sigma \right] g(\tau, \sigma) d\sigma \end{aligned} \quad (\text{A7})$$

This equation (A7) is an integral equation which can be solved. The solution is

$$\begin{aligned} \mathbf{R}_c(\tau) = & \frac{[\mathbf{R}_2 \sinh \omega \tau + \mathbf{R}_1 \sinh \omega(N - \tau)]}{\sinh \omega N} - \int_0^N \frac{\mathbf{f}(\sigma)}{m} g(\tau, \sigma) d\sigma \\ & + \frac{(\mathbf{R}_1 + \mathbf{R}_2) \sinh \frac{\omega \tau}{2} \sinh \frac{\omega(N - \tau)}{2}}{\cosh \frac{\omega N}{2}} \end{aligned}$$

$$+ \frac{4 \sinh \frac{\omega \tau}{2} \sinh \frac{\omega(N-\tau)}{2} \int_0^N \mathbf{f}(\sigma) \sinh \frac{\omega \sigma}{2} \sinh \frac{\omega(N-\sigma)}{2} d\sigma}{m\omega \sinh \omega N} \quad (\text{A8})$$

The classical action of $S'_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ is simply obtained by substituting $\mathbf{R}_c$ into the expression

$$\begin{aligned} S'_0(\mathbf{R}_2 - \mathbf{R}_1; \omega, N) &= \frac{m}{2} \int_0^N d\tau \dot{\mathbf{R}}_c^2(\tau) + \frac{m\omega^2}{4N} \int_0^N \int_0^N d\tau d\sigma (\mathbf{R}_c(\tau) - \mathbf{R}_c(\sigma))^2 \\ &\quad - \int_0^N d\tau \mathbf{f}(\tau) \mathbf{R}_c(\tau) \\ &= \frac{m}{2} \left[\dot{\mathbf{R}}_c(N) \mathbf{R}_c(N) - \dot{\mathbf{R}}_c(0) \mathbf{R}_c(0) \right] \end{aligned} \quad (\text{A9})$$

to give

$$\begin{aligned} S'_0(\mathbf{R}_2 - \mathbf{R}_1; \omega) &= \frac{m\omega}{4} \coth \frac{\omega N}{2} |\mathbf{R}_2 - \mathbf{R}_1|^2 \\ &\quad - \frac{m\omega}{2 \sinh \omega N} \left[\frac{2\mathbf{R}_2}{m\omega} \int_0^N d\tau \mathbf{f}(\tau) \chi \sinh \frac{\omega \tau}{2} \right. \\ &\quad + 2 \sinh \frac{\omega N}{2} \sinh \frac{\omega \tau}{2} \sinh \frac{\omega(N-\tau)}{2} \\ &\quad + \frac{2\mathbf{R}_1}{m\omega} \int_0^N d\tau \mathbf{f}(\tau) \chi \sinh \frac{\omega(N-\tau)}{2} \\ &\quad + 2 \sinh \frac{\omega N}{2} \sinh \frac{\omega \tau}{2} \sinh \frac{\omega(N-\tau)}{2} \\ &\quad + \frac{2}{m^2 \omega^2} \int_0^N \int_0^\tau d\tau d\sigma \mathbf{f}(\tau) \mathbf{f}(\sigma) \chi \sinh \omega(N-\tau) \sinh \omega \sigma \\ &\quad \left. + 4 \sinh \frac{\omega \tau}{2} \sinh \frac{\omega(N-\tau)}{2} \sinh \frac{\omega \sigma}{2} \sinh \frac{\omega(N-\sigma)}{2} \right] \end{aligned} \quad (\text{A10})$$

The classical action $S_{0,cl}(\mathbf{R}_2 - \mathbf{R}_1; N, \omega)$ is then obtained by setting $\mathbf{f}(\tau) = 0$ in equation (A10):

$$S_0(\mathbf{R}_2 - \mathbf{R}_1; \omega) = \frac{m\omega}{4} \coth \frac{\omega N}{2} |\mathbf{R}_2 - \mathbf{R}_1|^2. \quad (\text{A11})$$

Next, to evaluate the trial propagator $G_0(\mathbf{R}_1, \mathbf{R}_2; N, \omega)$, the trial action is rewritten as $S_0(\omega)$ in the form

$$S_0(\omega) = S_0(HP) - \frac{m\omega}{2N} \left(\int_0^N d\tau \mathbf{R}(\tau) \right)^2 \quad (\text{A12})$$

where $S_0(HP)$ is the simple harmonic potential action as shown in

$$S_0(HP) = \int_0^N d\tau \frac{m}{2} \left(\dot{\mathbf{R}}^2(\tau) + \omega^2 \mathbf{R}^2(\tau) \right). \quad (\text{A13})$$

The second term of equation (A12) can be converted to an integral form by the identity

$$\exp \left[\frac{m\omega^2}{2N} \left(\int_0^N d\tau \mathbf{R}(\tau) \right)^2 \right] = \left(\frac{N}{2\pi m\omega^2} \right)^{3/2} \int d\mathbf{f} \exp \left(\frac{-N\mathbf{f}^2}{2m\omega^2} - \int_0^N d\tau \mathbf{R}(\tau) \mathbf{f} \right). \quad (\text{A14})$$

Form equations (A12) and (A14), it can be found that the propagator G_0 can be expressed as

$$G_0(\mathbf{R}_2, \mathbf{R}_1; N, \omega) = \left(\frac{N}{2\pi m\omega^2} \right)^{3/2} \int_{\mathbf{R}_1}^{\mathbf{R}_2} D[\mathbf{R}(\tau)] \int d\mathbf{f} \exp \left[- (S_0(HP) + \frac{N\mathbf{f}^2}{2m\omega^2} + \int_0^N d\tau \mathbf{R}(\tau) \mathbf{f}) \right]. \quad (\text{A15})$$

Changing the order of integration, (A15) becomes

$$G_0(\mathbf{R}_2, \mathbf{R}_1; N, \omega) = \left(\frac{N}{2\pi m \omega^2} \right)^{3/2} \int d\mathbf{f} \exp \left[-\frac{N\mathbf{f}^2}{2m\omega^2} \right] G_{\mathbf{f}}(\mathbf{R}_2, \mathbf{R}_1; N, \mathbf{f}) \quad (\text{A16})$$

where

$$G_{\mathbf{f}}(\mathbf{R}_2, \mathbf{R}_1; N, \mathbf{f}) = \int_{\mathbf{R}_1}^{\mathbf{R}_2} D[\mathbf{R}(\tau)] \exp \left[-\left(S_0(HP) + \int_0^N d\tau \mathbf{R}(\tau) \mathbf{f} \right) \right]. \quad (\text{A17})$$

The propagator (A17) is the force hamonic oscillator propagator with a constant external force $\mathbf{f}$, which is

$$\begin{aligned} G_{\mathbf{f}}(\mathbf{R}_2, \mathbf{R}_1; N, \mathbf{f}) = & \left(\frac{m\omega}{2\pi \sinh \omega N} \right)^{3/2} \exp \left[-\left(\frac{m\omega}{4} \coth \frac{\omega N}{2} |\mathbf{R}_2 - \mathbf{R}_1|^2 \right. \right. \\ & + \tanh \frac{\omega N}{2} (\mathbf{R}_2 + \mathbf{R}_1)^2 + \frac{1}{\omega} \tanh \frac{\omega N}{2} (\mathbf{R}_2 + \mathbf{R}_1) \mathbf{f} \\ & \left. \left. + \left(\frac{1}{m\omega^3} \tanh \frac{\omega N}{2} - \frac{N\mathbf{f}^2}{2m\omega^2} \right) \right] \right]. \quad (\text{A18}) \end{aligned}$$

Substituting (A18) into (A16), and performing the $\mathbf{f}$ -integration, equation (A19) is obtained:

$$G_0(\mathbf{R}_2, \mathbf{R}_1; N, \omega) = \left(\frac{m}{2\pi N} \right)^{3/2} \left(\frac{\omega N}{2 \sinh \frac{\omega N}{2}} \right)^3 \exp \left[-\frac{m\omega}{4} \coth \frac{\omega N}{2} |\mathbf{R}_2 - \mathbf{R}_1|^2 \right] \quad (\text{A19})$$

References

1. F. Edwards and Pooran Singh, Size of Polymer Molecule in Solution (1978).
2. M. Muthukumar and Bernie G. Nickel, Perturbation theory for a polymer chain with excluded volume interaction, J. Chem. Phys. 80(11), 1 June (1984).
3. R. P. Feynman and A. R. Hibbs, Quantum Mechanics and Path Integrals (Taiwan: McGraw-Hill, 1995).
4. Kuhn, W., *Kolloid Z.* 68, 2 (1934)
5. P. J. Flory, J. Chem. Phys. 17, 303(1949).

6. M. Doi and S. F. Edwards, *The Theory of Polymer Dynamics*. (1986).
7. V. Samathiyakanit, *Phys. C: Solid state Phys*, 2849 (1974).
8. H. Yamakawa, *Modern Theory of Polymer Solutions* (Harper and Row, New York, 1971).
9. V. Sa-yakanit, The Feynman effective mass of the polaron, *Physical Review B*. 2377-2380, v19 (1979).
10. R. Kubo, *J. Phys. Soc. Japan*, 1100-20 (1962).

DNA AND MICROTUBULES AS VORTEX-STRINGS IN SUPERCONDUCTOR-LIKE DYNAMICS

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The non-stationary one-dimensional model of DNA replication and cell cycle control derived in my previous lecture, also included in this book, is generalized here to three spatial dimensions with a complex scalar matter field and an electromagnetic field. The leading order interaction obtained is superconductor-like and the DNA duplex and MTs appear as stringlike macromolecular configurations in the form of vortex solutions. Contrary to thermotropic and *ad hoc* type models, these vortex solutions emerge as a result of dependence on the initial reactant concentrations. A further, spherically symmetric generalization of that model with three scalar field components (three order parameters) yields a particular hedgehog solution which could be taken as a pre-replication conformation of the cytoskeleton.

1 Long range interaction between proteins in DNA controls the cell cycle

In the previous lecture a nonstationary model that controls DNA replication and cell cycle progression was derived in terms of manybody physics [1]. That model, in which the DNA duplex and MTs behave like elastically braced strings, predicts a long range force between the origin recognition complexes (ORCs), bound to DNA origins [2]. The molecular complexes of these string like lattices are squeezed together by a long range force F , which is attractive in G_1 , such that mobile electrons in the same complexes can transfer, a prerequisite for oxidation-reduction processes encompassing replication to take place. Initiation of replication thus depends critically on the classical assembly of the pre-replication complexes (pre-RC), as well as their phosphorylation by cyclin-dependent kinases which could not function without electron transfer. All one-dimensional equations employed are given in my previous lecture presented in this book [1].

The long range force $F(\varphi)$, which acts as a driving force for DNA replication and the cell cycle progression, is attractive (+), hence condensating, in the (G_1) assembly state ($0 < \varphi < N$) as expected, φ being the number of ORCs, and N the threshold number for initiation. DNA replication is initiated by a switch of sign of interaction at $\varphi = N$, from attraction (-) to repulsion (+). During the DNA replication ($N < \varphi < 2N$) F is repulsive (-), thus explaining the disassembly with release of licensing factors (LFs) and hence also providing a mechanism for the prevention of re-replication during the S phase. This is one of the most essential prerequisites for the genome to be duplicated just one time. The termination of DNA replication at

the S - G_2 interface is due to a vanishing of the driving force at $\varphi = 2N$, when all primed replicons are duplicated once. Thus the model makes sure that the DNA content of G_2 cells is exactly twice that of G_1 cells. With zero DNA origins engaged, $\varphi = 0$, the cell stays at rest in the absence of a driving force, $F(0) = 0$. The model also yields a nonlinear expression for the threshold N in the terms of initial concentrations of protein complexes and DNA origins.

The same form of interaction between protein complexes is obtained for the interacting receptors at the surface membrane [3], and for the tubulin dimers complexed with guanosine-tri-phosphate (GTP) in cytoskeletal microtubules (MTs) [4]. The switch of sign of interaction at $\varphi = N$ then similarly explains the dynamic instability [5] in growing MTs and the switch of interaction of interleukin-2 (IL2) with its receptor in late G_1 , at the R -point, after which the cell proceeds to replication without further exposure to IL2. Shape, slope and scale of the response curves derived agree well with experimental data from dividing T cells and polymerizing MTs, the variable length of which is due to a nonlinear dependence of the growth amplitude on the initial concentrations of guanosine-triphosphate (GTP) and tubulin dimers.

The agreement with data [1] is good and the elastically braced string model could hence be taken as a reasonable one-dimensional approximation of the activated cell, and each of its main subsystems: the surface membrane, the DNA duplex, and the cytoskeletal MTs:

$$\rho^* \frac{\partial^2 \varphi}{\partial t^2} - \varepsilon \frac{\partial^2 \varphi}{\partial x^2} = -\varepsilon \frac{\partial V_A}{\partial \varphi^*} = \quad (1.1)$$

$$F(\varphi) = \frac{\kappa}{2} \varphi \left(1 - \frac{\varphi}{N}\right) \left(2 - \frac{\varphi}{N}\right)$$

The parameter ρ^* is the mass density in the actual subsystem, ε the elastic modulus, κ the compressibility modulus, $V_A(\varphi) = 1/2 g^2 \varphi^2 (\varphi - 2\mu/g)^2$ the chemical driving potential obtained after the spontaneous symmetry breakdown, μ and g coupling coefficients [1]. The Hamiltonian energy density [6] contributed by the N complexes is

$$H(\varphi) = \varepsilon \left\{ \frac{1}{2s^2} \left(\frac{\partial \varphi}{\partial t} \right)^2 + \frac{1}{2} \left(\frac{\partial \varphi}{\partial x} \right)^2 + V(\varphi) \right\} \quad (1.2)$$

However, the generalization to three spatial dimensions proposed here starts out from the Lagrangian density [6]

$$\begin{aligned} L(\varphi) &= \varepsilon \frac{1}{s^2} \left(\frac{\partial \varphi}{\partial t} \right)^2 - H(\varphi) = \\ &\varepsilon \left\{ \frac{1}{2s^2} \left(\frac{\partial \varphi}{\partial t} \right)^2 - \frac{1}{2} \left(\frac{\partial \varphi}{\partial x} \right)^2 - V(\varphi) \right\} \end{aligned} \quad (1.3)$$

Apart from the dissipative effects induced by binding of the molecular complexes, which makes the ordered system supercritical [1], the prereplication conductor system should be essentially free from dissipations. The preceding stabilization of the conductor system - the assembled stringlike lattices of molecular complexes in the cytoskeletal MTs, the DNA duplex, and the surface membrane - is the main requirement for the rapid electron transfer that actually ignites DNA replication. This does not automatically imply vanishing of resistance for the transferring electrons.

2 Extension to three spatial dimensions

In an electric conductor of inanimate matter the assembly of its constituent molecular complexes is usually given from the start. On the contrary, in the actual subsystems of a living cell, the assembly of molecular complexes is the essential part of the event. When the definite threshold number of receptors and origins are engaged, DNA replication is initiated [3]. In other words, given the stabilized string like conductors in the actual subsystems of an activated cell - the membrane, the DNA duplex, and the MTs - the rapid transfer of electrons automatically takes place. Inclusion of electromagnetic effects, however, requires a generalization to three spatial dimensions. In an isotropic system the second order equation of harmonic-displacive motion in any of the actual conductor lattices, before the spontaneous symmetry breakdown, could then be written on the form

$$\rho * \frac{\partial^2 \varphi}{\partial t^2} - \varepsilon \nabla \cdot \nabla \varphi = -\varepsilon \frac{\partial V}{\partial \varphi *} = F_0(\varphi) \quad (2.1)$$

where $\nabla = (\partial/\partial x_1, \partial/\partial x_2, \partial/\partial x_3)$ is the gradient operator. In (2.1) the order parameter φ from the one-dimensional model is replaced by a two-component, complex scalar field in a first attempt to include charged and polarized particles. In order to keep the equations short, a four vector notation $x_\mu = (x_0, \underline{x})$ with $x_0 = st$, a metric (1, -1, -1, -1) for scalar products, a four-gradient $\partial_\mu = \partial/\partial x_\mu$, and the convention to sum over repeated indices, for instance $\partial_\mu \partial^\mu = \partial x_0^2 - \partial x_1^2 - \partial x_2^2 - \partial x_3^2 = s^2 \partial_t^2 - \nabla \cdot \nabla$, are introduced. The Lagrangian corresponding to (2.1) then takes the form

$$L(\varphi) = \varepsilon \left\{ \frac{1}{2} \partial_\mu \varphi * \partial^\mu \varphi - V(\varphi^2) \right\} = \varepsilon \frac{1}{2} \left\{ \frac{1}{s^2} \left(\frac{\partial \varphi}{\partial t} \right)^2 - \left(\frac{\partial \varphi}{\partial x} \right)^2 - \left(\frac{\partial \varphi}{\partial y} \right)^2 - \left(\frac{\partial \varphi}{\partial z} \right)^2 - 2V(\varphi^2) \right\} \quad (2.2)$$

Instead of a discrete symmetry at reflections $\varphi \rightarrow -\varphi$, which was the case in the one-dimensional model, the potential $V(\varphi^2) = 1/2 g^2 (\mu^2/g^2 - \varphi^2)^2$ and Lagrangian in (2.2) are invariant under continuous $SO(2)$ rotations in the complex φ -plane. This

corresponds to global (x -independent) $U(1)$ gauge transformations, $\varphi(x) \rightarrow e^{-i\Lambda} \varphi(x)$. The potential V in (2.2) has the same form as that given in one spatial dimension with a nonlinear coupling $g = (\rho_0 - r_0)/(\rho_0 + r_0)$ in the terms of the initial reactant concentrations, ρ_0 and r_0 . But, with a complex scalar field $V(\varphi)$ has now an infinite set of minima located on a circle around origo in the complex φ -plane (Fig. 1). This is the so called Goldstone model [7], however, which lacks finite energy solutions [6], a prerequisite to reobtain the elastically braced strings in three spatial dimensions. It also has the flaw of global charge conservation, such that the matter field φ carries an electric charge of same sign in the entire system.

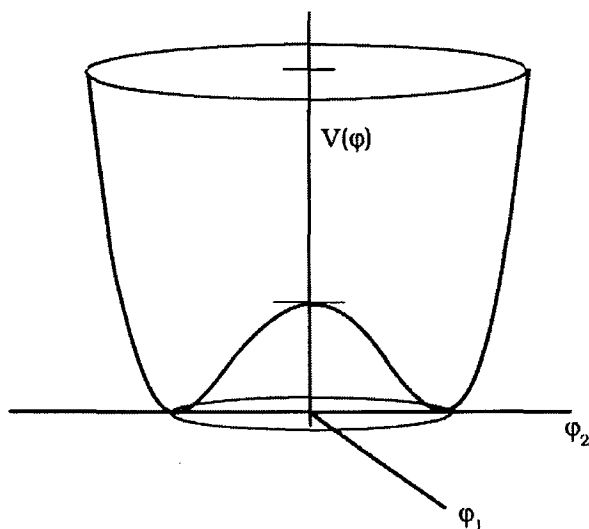


Figure 1. The symmetric lyotropic potential in the complex order parameter space.

In order to obtain finite energy solutions and conservation of electric charge, independent of sign at any point in the system, the theory must be invariant under local gauge transformations, implying an x -dependent gauge parameter $\Lambda(x)$. This in turn requires the introduction of a four-vector gauge field $A_\mu = (A_0, \underline{A})$ [6]. A_0 and $\underline{A}$ may as usual be interpreted as the electric and magnetic potentials, in which the mobile electrons can transfer along the stringlike assemblies of molecular complexes at the initiation of replication.

Before that, the mobile electrons are confined in their neutral molecular complexes in a harmonic bound state motion which induces a polarization such as in the GTP-tubulin dimers [8]. However, the transferring of electrons, which actually ignites the DNA replication, and for which a stabilized nondissipative

conductor system of molecular complexes is an indispensable prerequisite, is not a classical harmonic type phenomenon, it obeys the laws of quantum mechanics.

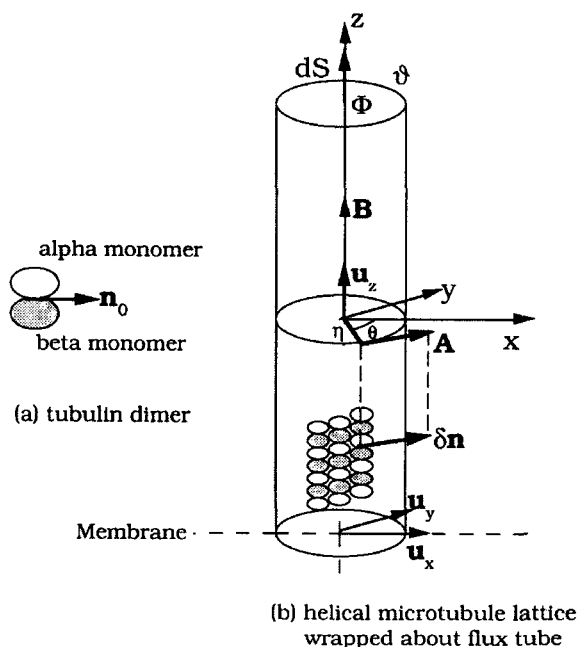


Figure 2. (a) Tubulin dimer with director $\mathbf{n}_0$. (b) Helical polymerization of hexagonal microtubule lattice wrapped about a quantized magnetic flux tube.

Like in a thermotropic superconductor and other inanimate condensed matter, the lyotropically controlled displacement effects induce masses and screening of the interacting fields. In analogy with a superconductor, the complex matter field could also be interpreted as a displacement and layer fluctuation in a smectic liquid crystal [9]. The gauge field may then represent the fluctuation $\delta \mathbf{n} = \mathbf{n} - \mathbf{n}_0$ of the director $\mathbf{n}_0$, for instance, of the tubulin dimers in a cytoskeletal MT (Fig. 2). In order to analyze this situation, and to recover the stringlike finite energy solutions, obtained already in one spatial dimension, the four-vector gauge field must then first be implemented in the harmonic type dynamics (2.2).

Quite irrespective of physical interpretation, the basic problem is then to find the covariant dependence of the classical harmonic-displacive type dynamics (2.2) on the gauge vector potential A_μ . In quantum electrodynamics of a pointlike charge Q , the rule of minimal substitution prescribes a replacement of the four-momentum

p_μ by $p_\mu - Q/cA_\mu$, with p_μ corresponding to $p_\mu = \hbar/i\partial_\mu = -i\hbar\partial/\partial x_\mu$ and $\hbar$ being Planck's constant divided by 2π . In this case we are only interested in the relation between the electromagnetic and massive field couplings, and except for the electric charge, all constants are therefore chosen equal to one. The four-gradient ∂_μ is then replaced by the covariant derivative

$$D_\mu(x) = \partial_\mu - iQA_\mu(x) \quad (2.3)$$

and the Lagrangian (2.2) becomes

$$L(\varphi) = \frac{1}{2} (D_\mu \varphi)^* (D^\mu \varphi) - V(\varphi^2) + \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \quad (2.4)$$

which is invariant under local gauge transformations

$$\varphi(x) \rightarrow e^{-i\Lambda(x)} \varphi(x) \quad (2.5)$$

$$A_\mu(x) \rightarrow A_\mu(x) - \frac{1}{Q} \partial_\mu \Lambda(x) \quad (2.6)$$

Interaction (2.4) thus conserves electric charge independent on sign at any point in space, a property required also for living condensed matter. The last term in (2.4), with a field strength tensor

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (2.7)$$

is needed here in order to have a closed canonical system. The gauge field A_μ may then be interpreted as the intrinsic electromagnetic potential, induced by the mobile electric charges in the molecular complexes of the activated cell, and hence indirectly generated by the ligand-receptor interaction at the surface membrane.

The screening of interacting fields is developed like in a superconductor, through a spontaneous symmetry breakdown much the same as in one spatial dimension. Like in that case the symmetry breakdown is also required here to obtain a positive definite quantum of energy, $\kappa = 4e\mu^2$, stored per origin, per receptor engagement [10], or per GTP-tubulin dimer in an MT. The MTs are so called electrets, of neutral but polarized tubulin dimers, in which the mobile electrons [8] can take up energy as bound state oscillations, and finally transfer rapidly when the energy attains the scattering threshold such as at the initiation of DNA replication. A more realistic cell model, however, should allow a cytoskeleton with MTs pointing in all spatial directions about the centrosome. Moreover, the two-dimensional surface membrane receptor system should be a spherically symmetric structure embedded in the ordinary three-dimensional space. However, first we analyze the model (2.4) with a 'flat' simplified surface membrane.

2 Elastically braced strings in three spatial dimensions

Using polar field coordinates to represent the complex scalar field, $\varphi = f \exp(i\theta)$, expanding about one of the minima of the potential

$$f = \frac{\mu}{g} + \sigma \quad (3.1)$$

and treating σ and θ as small variables, the leading order Lagrangian (2.4) becomes

$$L(\varphi) = \frac{1}{2} (\partial_\mu \sigma \partial^\mu \sigma - 4\mu^2 \sigma^2) + \frac{1}{2} Q^2 \frac{\mu^2}{g^2} A_\mu A^\mu - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \quad (3.2)$$

Like in the one-dimensional case, in (3.2) σ has acquired a "screening" mass, $m = 2\mu$, which defines a natural scale $\xi = 1/(2\mu)$ for spatial variations

$$\xi = \frac{1}{2\mu} = \frac{1}{m} \quad (3.3)$$

The parameter ξ is called the Ginzburg-Landau coherence length. According to (3.2), through (3.1), the spontaneous symmetry breakdown, has now also generated a positive photon mass

$$m_V = Q \frac{\mu}{g} = QN \quad (3.4)$$

which induces a finite penetration depth for the magnetic field, or for the fluctuation of the liquid crystal director [9], in the pre-conductor system.

$$\lambda = \frac{1}{m_V} = \frac{1}{QN} \quad (3.5)$$

N is the number of molecular complexes, each of which contains a mobile electron. Like in a superconductor, the actual oxidation-reduction reactions involve the transfer of pairs of electrons that are correlated over long distances [11]. However, contrary to thermotropic superconductors and thermotropic liquid crystals, in this lyotropic model the Ginzburg-Landau parameter $\lambda\xi$ is controlled by the initial reactant concentrations

$$\frac{\lambda}{\xi} = \frac{2g}{Q} = \frac{1}{Q} \frac{\rho_0 - r_0}{\rho_0 + r_0} \quad (3.6)$$

This is illustrated in figure 3.

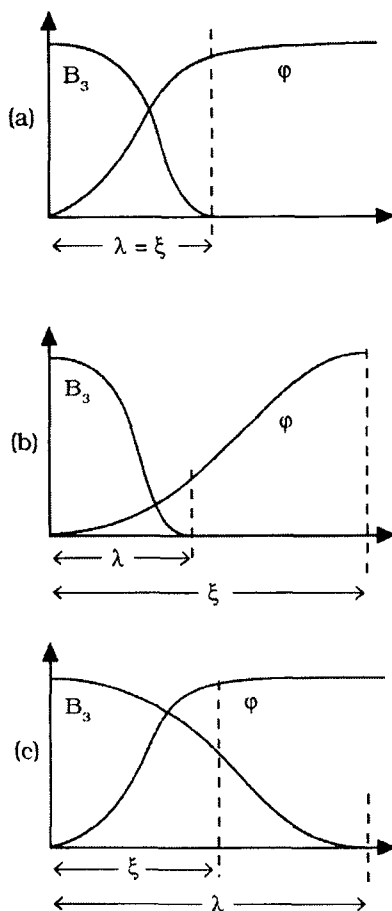


Figure 3. Interpolation between electromagnetic and materialized modes according to the vortex solutions. The magnetic field is related to the magnetic potential through $\underline{B} = \nabla \times \underline{A}$.

In certain directions from the centrosome, at favourable initial concentrations of tubulin dimers and GTP, the nonlinear coupling g could thus become of the same order of magnitude as $Q/2$, such that $\xi \approx \lambda$ (Fig. 3a), which is the criterion for stringlike vortex solutions [13] that could here be interpreted as cytoskeletal MTs. A similar discussion should lead to a stringlike vortex approximation for the DNA duplex. The elastically braced string picture could hence be reobtained also in three spatial dimensions, in terms of stringlike vortex solutions to (3.2). However, contrary to the *ad hoc* case [13] and the thermotropic liquid crystal analogy [9], in the actual nonstationary system the values of coupling coefficients, required to

obtain a sufficiently thin vortex-string with radius λ , are not chosen *ad hoc* by an external observer. In the activated cell these parameters are regulated lyotropically, through the coupling g , by the preceding synthesis of adequate reactant concentration.

The dynamics of low energy excitations in a closepacked stringlike lattice is harmonic, hence Lorentz invariant, both in one (1.1) and three spatial dimensions (2.4). The analysis of displacement and anharmonic corrections must therefore be treated in a Lorentz invariant formalism consistent with local charge conservation, however, in this case generalized to a lyotropically regulated cell system. Development of screening masses and materialization of stringlike vortices, i. e., the assembly of pre-conductors for rapid electron transfer in three spatial dimensions, are hence also in this case collective phenomena due to changes of the initial reactant concentrations.

In the $\partial_\mu A^\mu = 0$ gauge the equations of motion of (3.2) are

$$D_\mu D^\mu \varphi = -\frac{\partial V}{\partial \varphi^*} \quad (3.7)$$

$$\begin{aligned} \left(\frac{1}{s^2} \frac{\partial^2}{\partial t^2} - \nabla \cdot \nabla\right) A_\nu &= j_\nu \\ &= \frac{1}{2} iQ(\varphi^* \partial_\nu \varphi - \varphi \partial_\nu \varphi^*) - Q^2 \varphi^* \varphi A_\nu \end{aligned} \quad (3.8)$$

Inserting $\varphi = f \exp(i\theta)$ (3.8) becomes

$$A_\nu = -\frac{1}{Q^2} \frac{j_\nu}{f^2} - \frac{1}{Q} \partial_\nu \theta(x) \quad (3.9)$$

With $j_\nu = 0$ along the closed curve ϑ about the z -axis (Fig 2b), the magnetic flux in the z -direction is given by

$$\Phi = \int \underline{B} \cdot d\underline{S} = \oint_{\vartheta} \underline{A} \cdot d\underline{x} = \frac{1}{Q} \oint_{\vartheta} \nabla \theta(\underline{x}) \cdot d\underline{x} = \pm \nu \frac{2\pi}{Q} \quad (3.10)$$

The requirement of a single valued scalar field $\varphi = f \exp(i\theta)$ enforces ν to be an integer and the flux tube in the z -direction to be quantized [13,14].

The solutions to (3.7) and (3.8) corresponding to ν flux units are

$$\varphi(\underline{x}) = e^{-i\nu\theta} f(\eta) \quad (3.11)$$

$$\underline{A}(\underline{x}) = -\frac{(\underline{\eta} \times \underline{u}_z)}{\eta} A(\eta) \quad (3.12)$$

where $\underline{u}_z$ is a unit vector in the z -direction and (η, θ) are cylindrical coordinates about the same axis.

The flux quantization (3.10), however, is not restricted to superconductors only. It is a property common to most materials [15]. Also in the analogy with liquid crystals, where the gauge field $\underline{A}$ is replaced by the fluctuation of the crystal director, $\delta \underline{n} = \underline{n} - \underline{n}_0$, the integral (3.10) should become an integer [9]. The dislocation-free case, which could be assumed to be a necessary requirement for rapid transfer of electrons, for instance in a stabilized MT, should then correspond to $\nu = 0$. To become more realistic, however, the cell model with its membrane and cytoskeleton should be made spherically symmetric with a nucleus/centrosome singularity located at origo. As shown subsequently, in a spherically symmetric generalization of (3.2), a hedgehog cytoskeleton conformation about a pointlike nucleus/centrosome singularity emerges directly from the model.

4 Hedgehog cytoskeleton solution from non-Abelian model

A realistic cell model should be spherically symmetric and permit polarization, vibrations and displacements in all three spatial dimensions. Instead of the $U(1) \approx SO(2)$ invariant formulation (3.2), with a two-component scalar field, the manifold of internal states therefore requires an $SO(3)$ invariant formulation with a triplet of scalar fields $\underline{\varphi} = (\varphi_1, \varphi_2, \varphi_3)$.

One such generalization of the string like interactions (1.1) and (2.4) could be obtained through the $SO(3)$ invariant model

$$L(\underline{\varphi}) = \frac{1}{2} (D_\mu \underline{\varphi}) \cdot (D^\mu \underline{\varphi}) - V(\underline{\varphi}) + \frac{1}{4} F_{a\mu\nu} F_a^{\mu\nu} \quad (4.1)$$

where

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a - Q\varepsilon^{abc} A_\mu^b A_\nu^c \quad (4.2)$$

$$D_\mu \varphi^a = \partial_\mu \varphi^a - Q\varepsilon^{abc} A_\mu^b \varphi^c \quad (4.3)$$

$$V(\underline{\varphi}) = \frac{1}{g^2} (\mu^2 - g^2 \underline{\varphi} \cdot \underline{\varphi})^2 \quad (4.4)$$

The totally antisymmetric elements ε_{abc} , with $\varepsilon_{123} = 1$, are the three $SU(2)$ generators $(\tau_a)_{bc} = i\varepsilon_{abc}$ in a 3×3 matrix representation, and the isospin generators τ^i are related directly to the Pauli spin matrices $\sigma^i = 2\tau^i$.

The model (4.1) could be regarded as a lyotropic generalization of the Georgi-Glashow model without fermions [16], with the coefficients being functions of the initial reactant concentrations. The equations of motion are given by

$$D_\mu F_a^{\mu\nu} = -Q\varepsilon_{abc} \varphi_b D^\nu \varphi_c \quad (4.5)$$

$$D_\mu D^\mu \varphi_\alpha = -2g^2 \varphi_\alpha (\underline{\varphi} \cdot \underline{\varphi} - \frac{\mu^2}{g^2}) \quad (4.6)$$

In this model the points that minimize the potential (4.4) span a sphere in the three-dimensional internal order parameter space. As usual the system spontaneously chooses one of these lowest energy states in the order parameter space and for convenience the point $(0, 0, \mu/g)$ on the 3-axis is selected here. A new triplet of displaced scalar fields is then defined by

$$\underline{\varphi} = (\varphi_1, \varphi_2, \frac{\mu}{g} + \varphi_3) \quad (4.7)$$

The correspondingly displaced part of (4.1), invariant under $U(1) \approx SO(2)$ rotations about the 3-axis in φ -space, could still be interpreted as the electromagnetic interaction, and the corresponding massless gauge bosons as photons $A_\mu^3 = A_\mu$. The electric and magnetic fields are then defined by

$$\begin{aligned} F_3^{0i} &= E^i \\ F_3^{ij} &= \varepsilon^{ijk} B^k \end{aligned} \quad (4.8)$$

Static solutions could be found on the form [17,18]

$$\varphi_i = -\delta_{ia} \frac{x_a}{Qr^2} H\left(\frac{\mu}{g} Qr\right) \quad (4.9)$$

$$A^b_i = -\varepsilon_{bij} \frac{x_j}{Qr^2} \{1 - K\left(\frac{\mu}{g} Qr\right)\} \quad (4.10)$$

where r is the length of the vector $\underline{x}$. To have finite energies requiring that $K \rightarrow 0$ and $H \rightarrow Qr\mu/g$ for large r -values, these solutions corresponding to a magnetic monopole with magnetic charge $-4\pi/Q$ become

$$\varphi_i = \frac{\mu}{g} \frac{x_i}{r} \quad (4.11)$$

$$A^b_i = -\varepsilon_{bij} \frac{x_j}{Qr^2} \quad (4.12)$$

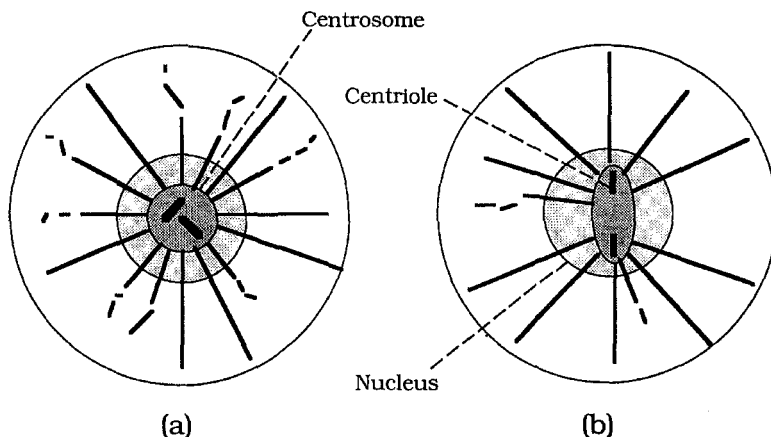


Figure 4. (a) Cytoskeletal hedgehog conformation of microtubules prior to DNA replication according to the magnetic monopole solution of the lyotropic nonabelian model proposed. (b) Same hedgehog solution at the transition to the *S* phase with separated centrioles.

The lyotropic, non-Abelian, spherically symmetric model (4.1), with a triplet of scalar fields, thus yields a particular cytoskeletal hedgehog solution, with a centrosome singularity located at origo (Fig. 4). The post-replication part of the cell cycle is characterized by a bipolar mitotic spindle with two separate centrosomes (Fig. 5c). Such multiple-monopole solutions have been derived [19,20], and a computer graphic representation of a two-monopole solution [21] shows that this type of model (4.1) could also describe the morphological development of a dividing cell (Fig. 5).

5 Summary and outlook

The perplexing discovery that an activated cell actually counts the number of receptor engagements, irrespective to their location on the surface membrane, considerably restricts the choice of dynamics that controls DNA replication and the cell cycle progression. The exact timing of the initiation of DNA replication, indicates the presence of nonlocal correlations, hence, a long range interaction between the molecular complexes, in the DNA double helix and at the surface

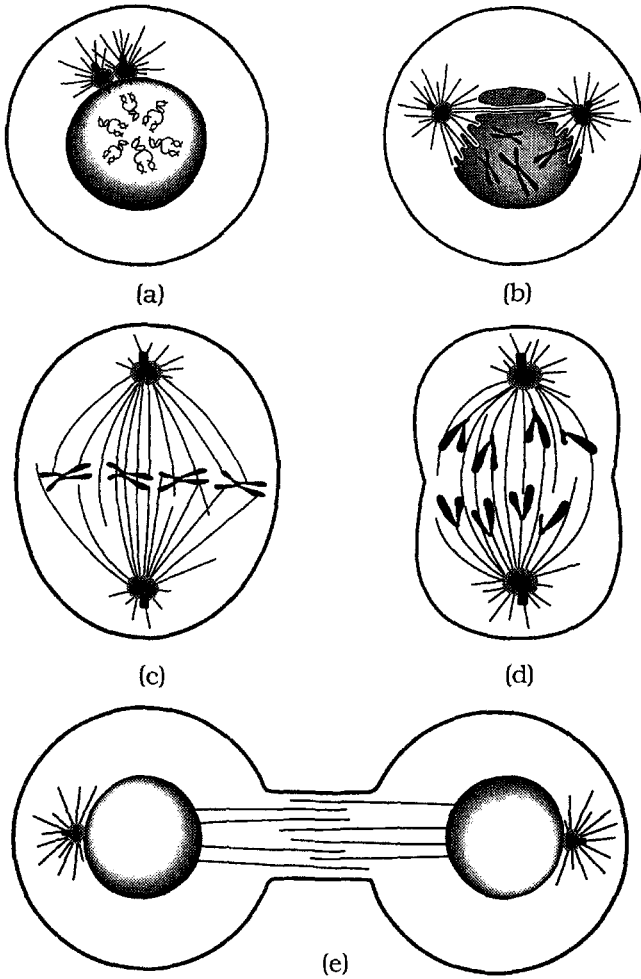


Figure 5. Mitotic cell division: (a) interphase, (b) prophase, (c) metaphase, (d) anaphase, (e) telophase.

membrane, the dynamics of which enslaves the entire cell at the transition to *S* phase. The enslavement accounts for the major dissipative forces and the remaining friction in the system should hence be small compared to the leading order interaction.

However, neither the surface membrane alone nor the DNA double helix would be sufficiently rigid to explain the presence of nonlocal correlations. A reactant molecule at one end of the system would know very little about a reactant molecule at the other end. The solid character of the system, indicated at the transition to *S* phase, is provided by the cytoskeleton, the most rigid structure in the cell. What happens is that the MTs, and the MT-associated proteins (MAPs), behave like rigid struts in the cell cytoplasm, thus making the DNA double helix, the individual MTs, and the membrane receptor system itself, behave like rigid lattices. Also the histones may contribute to the stabilization of DNA. The cytoskeleton dynamics, which thus works together with the surface membrane system until a critical point in late G_1 , the *R*-point, after which the cell proceeds to replication without any further exposure to ligands, and then together with the DNA double helix at the transition to *S* phase, must hence also be enslaved by the same long range forces because only one long range interaction could possibly control the cell at these critical points of the cell cycle. In the one dimensional approximation all three subsystems behave like elastically braced strings in the continuum approximation, however, only after a spontaneous symmetry breakdown which provides the correct sign of the compressibility modulus, and of the energy carried by each molecular complex [1].

To my knowledge this lyotropic, nonlinear elastically braced string dynamics is the only selfconsistent model that correctly explains shape, slope and scale of the response data of a dividing cell as a function of the two reactant concentrations [1], yields the quantum of excitation energy per receptor engagement [10], per GTP-tubulin dimer, and per ATP-ORC-origin, defines the quantal threshold for transition into the *S* phase, and explains the dynamic instability and the variable length of growing MTs [1]. Moreover, as discussed in the introduction, already in one spatial dimension the long range force $F(\varphi)$ between protein complexes in DNA (1.1) explains the gross behaviour of DNA replication and the cell cycle progression in terms of the molecular interaction. The model also makes sure that the DNA content of G_2 cells is exactly twice that of G_1 cells.

The superconduction like generalization of the harmonic string like interaction (1.1) into three spatial dimension, with a complex order parameter φ , is a direct consequence of local charge conservation, a requirement that should apply also to living condensed matter systems. This does not automatically imply that the resistance at the rapid electron transfer along the stringlike conductors vanishes. However, if the pair correlation between the transferring electrons in the actual oxidation-reduction processes is due to the same type of phonon coupling rearrangements as in a superconductor, with a corresponding lowering of the resistance, then a superconduction like current of electron pairs could be the thing that actually ignites DNA replication albeit at physiological temperature. In any

case, the possible lowering of resistance would then be an effect of variation of the reactant concentrations and not due to change of temperature. Irrespective of if the electric resistance is high or low, the current of correlated electrons, which follows as a result of assembly/ordering of the required threshold number N of pre-RCs in the string like DNA lattice [1], is the crucial event in hydrolysis and phosphorylation reactions encompassing DNA replication at critical points such as the initiation.

As discussed here before, this three-dimensional generalization of the leading order dynamics also yields an analogy with liquid crystals [9]. It could therefore be asked which one of these two pictures should apply to the living cell, the electromagnetic interaction of moving charges, or the liquid crystal analogy, for instance with MT filaments. As demonstrated in the $SO(3)$ invariant model (4.1) both these pictures can coexist after the symmetry breakdown and, moreover, this model yields a particular hedgehog solution that could be interpreted as a prereplication conformation of the cytoskeleton, with a centrosome singularity at origo.

It may look unusual to use a covariant field theory formulation in this context, however, already in one spatial dimension the stringlike interaction has the harmonic hence Lorentz invariant character. Moreover, there seems to be no alternative book-keeping method for the analysis of displacement and unharmonic corrections to the leading order harmonic dynamics in three dimensions, and in order to understand the role of symmetry breakdown, a natural and indispensable ingredient also in living condensed matter.

References

1. Matsson L., Long range Interaction between Protein Complexes in DNA Controls Replication and Cell Cycle Progression: the Double Helix and Microtubules Behave Like Elastically Braced Strings; in this volume: Biological Physics 2000. (Ed:s H. Frauenfelder, L. Matsson, and V. Sayakanit, World Scientific, Singapore, 2001).
2. Dutta A. and Bell S.P., Initiation of DNA replication in eukaryote cells. *Annu. Rev. Cell Dev. Biol.* **13** (1997), pp 293-332.
3. Cantrell D.A. and Smith K.A. The interleukin-2 T-cell system: A new cell growth model. *Science* **224** (1984) pp 1312-1316.
4. Voter W. A. and Erickson H. P., The kinetics of microtubule assembly, *J. Biol. Chem.* **259** (1984) pp 10430-10438.
5. Mitchison T. and Kirschner M., Dynamic instability of microtubule growth. *Nature (London)*, **312** (1984) pp 237-242.

6. Rajaraman R., Solitons and Instantons. (North Holland, Amsterdam 1982) pp 1-83.
7. Goldstone J., Field Theories with "Superconductor" Solutions. *Nuovo Cimento* **19** (1961) pp 154-164.
8. Melki R., Carlier M.-F, Pantaloni D, and Timasheff S.N., Cold Depolymerization of Microtubules to Double Rings: Geometric Stabilization of Assemblies. *Biochemistry* **28** (1989) pp 9143-9152.
9. de Gennes P.G., An Analogy between Superconductors and Smectics A. *Solid State Commun.* **10**, (1972) pp 753-756.
10. Paton W.D.M., A theory of drug action based on the rate of drug- receptor combination. *Proc. R. Soc. London Ser. B* **154** (1961) pp. 21-69.
11. Bohinski R.C., Modern Concepts in Biochemistry. (Allyn and Bacon Inc., Boston, 1983) pp 335-365.
12. Fetter A.L. and Walecka J.D., Quantum Theory of Many-Particle Systems. (McGraw-Hill Publishing Company, New York, 1971) pp 430-439.
13. Nielsen H. B. and Olesen P., Vortex-Line Models for Dual Strings. *Nuclear Physics B* **61** (1973) pp 45-61.
14. Abrikosov A.A., On the Magnetic Properties of Superconductors of the Second Group. *Soviet Physics JETP* **5** (1957) pp 1174-1182.
15. Bloch F., Simple Interpretation of the Josephson Effect, *Phys. Rev. Lett.* **21**, (1968) pp 1241-1243.
16. Georgi H. and Glashow S.L., Unified Weak and Electromagnetic Interaction Without Neutral Currents, *Phys. Rev. Lett.* **28**, (1972) pp 1494-1497.
17. 'tHooft G., Magnetic Monopoles in Unified Gauge Theories, *Nucl. Phys. B* **79**, (1974) pp 276-284.
18. Polyakov A.M., Particle Spectrum in Quantum Field Theory, *JETP Lett.* **20**, (1974) pp 194-195.
19. Forgács P., Horváth Z. and Palla L., Physicist's techniques for multimonopole solutions, in; Monopoles in Quantum Field Theory. (Ed:s N. S. Craigie, P. Goddard, and W. Nahm, World Scientific, Singapore, 1982) pp 21-50.
20. Völkel A.R., Mertens F.G., Wysin G.M., Bishop A.R., and Schnitzer H.J., Collective variable approach for a magnetic N -vortex system, in: Nonlinear Coherent Structures in Physics and Biology, (Eds. K. H. Spatschek and F. G. Mertens, NATO ASI Series B: Physics Vol 329, Plenum Press, New York 1994) pp 199-206.
21. Hey A.J.G., Merlin J.H., Ricketts M.W., Vaughn M.T., and Williams D.C., Topological Solutions in Gauge Theory and Their Computer Graphic Representation. *Science* **240**, (1988) pp 1163-1168.

THEORY OF STRETCHING INDIVIDUAL POLYNUCLEOTIDE MOLECULE

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We review the recent results of theoretical investigations on the elasticity of both single-stranded DNA (ssDNA) and double-stranded DNA (dsDNA) molecules. The path integral method and Monte Carlo technique are used to calculate the thermodynamics of pulling single biopolymer. The theoretical calculations are in good agreement with experimental measurements of stretching the torsionally relaxed and supercoiled dsDNA, and of pulling ssDNA with random and designed poly(dA-dT)/(dG-dC) nucleotide sequences.

1 Introduction

The elastic properties of polynucleotide molecules of double-stranded DNA and single-stranded RNA/DNA are of vital importance in many basic life processes. They affect, e.g., how DNA packs into chromosomes or serves as a template during the processes of transcription and replication, and how RNA folds into stable native patterns [1]. These mechanical properties have been studied experimentally for a long time through bulk methods, such as light scattering, sedimentation velocity, and ligase-catalyzed cyclization [2]. The macroscopic measurements represent ensemble averages over all accessible molecular configurations, thereby providing little information on the intermolecular and intramolecular forces that develop in the biopolymers during the course of their biological reactions.

In the past decade, the revolutionary progresses have been brought forth with the development of new micromanipulation techniques combining high force sensitivity (piconewtons) with accurate positioning (angstroms). These techniques allow researchers to stretch single biological macromolecule from both ends, and directly monitor its time-dependent elastic response under the external force. Until now, both double-stranded DNA (dsDNA) and single-stranded DNA (ssDNA) molecules in different circumstances have been pulled in various laboratories (see,

e.g., Refs. [3-9]). Depending on the concerned range of force and elasticity of the molecules, the applied force fields have been imposed through different mediums, for example, magnetic beads (0.01~10 pN), optical tweezers (0.1~100 pN), and atomic force microscopy (10~10,000 pN). Complex behavior has been revealed by these mechanical studies of the polynucleotide molecules. An excellent review of the experimental measurements can be found in Ref. [10].

In this talk, we will focus on the theoretical understanding of the above elasticity measurements of both dsDNA and ssDNA molecules. In our theoretical studies, the dsDNA molecule is modeled as two interwound worm-like chains bound tightly by permanent hydrogen bonds [11]. The structural deformation is characterized by the folding angle between side chains and the central axis. The stacking interaction of base pairs is taken into account by using Lennard-Jones potential with the result of quantum chemical calculations. For ssDNA, we model the molecule as a freely jointed chain (FJC) of elastic bonds with electrostatic, secondary structure (i.e. hairpins formed by the pairing of complementary base of A-T and G-C), and base-stacking interactions all taken into account. The stretching thermodynamics of our DNA models under applied force are calculated through path integral method of partition function and Metropolis Monte Carlo techniques. The computed force-elongation characteristics are in comprehensive agreement with above experimental measurements by different laboratory on both dsDNA and ssDNA in different circumstances.

2 Elastic Properties of Double-Stranded DNA

2.1 Model of dsDNA

2.1.1 Bending and Folding Deformations

Natural DNA molecule in a living cell is a double-stranded biopolymer, in which two complementary sugar-phosphate chains twist around each other to form a right-handed double helix. Each chain is a linear polynucleotide consisting of the following four bases: two purines (A, G) and two pyrimidines (C, T) [1,12]. The two chains are joined together by hydrogen bonds between pairs of nucleotides A-T and G-C.

In our mathematical model [11,13], the embeddings of two backbones are defined by $\mathbf{r}_1(s)$ and $\mathbf{r}_2(s')$. The ribbon structure of DNA is enforced by having $\mathbf{r}_2(s')$ separated from $\mathbf{r}_1(s)$ by a distance $2R$, i.e., $\mathbf{r}_2(s') = \mathbf{r}_1(s) + 2R\mathbf{b}(s)$ where the hydrogen-bond-director unit vector $\mathbf{b}(s)$ points from $\mathbf{r}_1(s)$ to $\mathbf{r}_2(s')$. As the result of the wormlike backbones, the bending energy of two backbones can be written as

$$E_B = \frac{\kappa}{2} \int_0^L \left[\left(\frac{d\mathbf{r}_1}{ds} \right)^2 ds + \left(\frac{d\mathbf{r}_2}{ds} \right)^2 ds \right]. \quad (1)$$

The formation of base pairs leads to rigid constraints between the two backbones and at the same time they hinder considerably the bending freedom of DNA central axis because of the strong steric effect. In the assumption of permanent hydrogen bonds [14,15], $|\mathbf{s}' \cdot \mathbf{s}| = 0$. The relative sliding of backbones is prohibited and the base pair orientation lies perpendicular to the tangent vectors $\mathbf{t}_1 = d\mathbf{r}_1/ds$ and $\mathbf{t}_2 = d\mathbf{r}_2/ds$ of the two backbones and that of the central axis, $\mathbf{t}$: $\mathbf{b} \cdot \mathbf{t}_1 = \mathbf{b} \cdot \mathbf{t}_2 = \mathbf{b} \cdot \mathbf{t} = 0$. By defining the folding angle θ as half of the rotation angle from $\mathbf{t}_2(s)$ to $\mathbf{t}_1(s)$, i.e., the intersection angle between tangent vector of backbones $\mathbf{t}_{1(2)}$ and DNA central axis $\mathbf{t}$, we have

$$\begin{cases} \mathbf{t}_1 = \cos\theta \mathbf{t} + \sin\theta \mathbf{b} \times \mathbf{t} \\ \mathbf{t}_2 = \cos\theta \mathbf{t} - \sin\theta \mathbf{b} \times \mathbf{t} \end{cases} \quad (2)$$

Therefore, the bending energy of the two backbones can be rewritten as [11]

$$E_B = \int_0^L \left[\kappa \left(\frac{d\mathbf{t}}{ds} \right)^2 + \kappa \left(\frac{d\theta}{ds} \right)^2 + \kappa \frac{\sin^4 \theta}{R^2} \right] d\theta, \quad (3)$$

where ds denotes arc length element of the backbones, L the total contour length of each backbone, and κ the persistence length of one DNA backbone.

Bearing in mind that the pairing and stacking enthalpy of the bases significantly increase bending stiffness of polymer axis, the experimental value of persistent length of dsDNA polymer is considerably larger than that of a DNA single strand [5]. By incorporating the steric effect and also considering the typical experiment value of persistent length of dsDNA $p = 53$ nm [16], we phenomenologically replace the bending rigidity κ in the first term of Eq. (3) with a new parameter κ^* . It is required that $\kappa^* > \kappa$, and the precise value of κ^* will then be determined self-consistently by the best fitting with experimental data (see below).

2.1.2 Base-Stacking Interactions

The vertical stacking interactions between base pairs originate from the weak Van der Waals attraction between the polar groups in adjacent nucleotide base pairs. Such interactions are short-ranged and their total effect is usually described by a potential energy of the Lennard-Jones form (6-12 potential) [12]). In a continuum theory of elasticity, the summed total base-stacking potential energy is converted into the form of the following integration:

$$E_{\text{L}} = \sum_{i=1}^{N_{\text{bp}}-1} U_{i,i+1} = \int_0^L \rho(\theta) ds, \quad (4)$$

where $U_{i,i+1}$ is the base-stacking potential between the i -th and the $(i+1)$ -th base pair, N_{bp} is the total number of base pairs, and the base-stacking energy density ρ is expressed as

$$\rho(\theta) = \begin{cases} \frac{\varepsilon}{r_0} \left[\left(\frac{\cos \theta_0}{\cos \theta} \right)^{12} - 2 \left(\frac{\cos \theta_0}{\cos \theta} \right)^6 \right] & (\text{for } \theta \geq 0), \\ \frac{\varepsilon}{r_0} [\cos^{12} \theta_0 - 2 \cos^6 \theta_0] & (\text{for } \theta < 0), \end{cases} \quad (5)$$

In Eq. (5), the parameter r_0 is the *backbone* arclength between adjacent bases ($r_0 = L/N$); θ_0 a parameter related to the equilibrium distance between a DNA dimer ($r_0 \cos \theta_0 \sim 0.34$ nm); and ε the base-stacking intensity which is generally base-sequence specific. In this work we focus on macroscopic properties of long DNA chains composed of relatively random sequences, therefore we just consider ε in the average sense and take it as a constant, with $\varepsilon \sim 14.0 k_{\text{B}}T$ as averaged over quantum-mechanically calculated results on all the different DNA dimers [12].

The asymmetric base-stacking potential in Eq. (5) ensures a relaxed DNA to take on a right-handed double-helix configuration (i.e., the B-form) with its folding angle $\theta \sim \theta_0$. To deviate the local configuration of DNA considerably from its B-form generally requires a free energy of the order of ε per base pair. Thus, DNA molecule will be very stable under normal physiological conditions and thermal energy can only make it fluctuate very slightly around its equilibrium configuration, since $\varepsilon \gg k_{\text{B}}T$. Nevertheless, although the stacking intensity ε in dsDNA is very strong compared to thermal energy, the base-stacking interaction by its nature is short-ranged and hence sensitive to the distance between the adjacent base pairs. If dsDNA chain is stretched by large external forces, which cause the average inter-base pair distance to exceed some threshold value determined intrinsically by the molecule, the restoring force provided by the base-stacking interactions will no longer be able to offset the external forces. Consequently, it will be possible that the B-form configuration of dsDNA will collapse and the chain will turn to be highly extensible. Thus, on one hand, the strong base-stacking interaction ensures the standard B-form configuration to be very stable upon thermal fluctuations and small external forces (this is required for the biological functions of DNA molecule to be properly fulfilled [1]); but on the other hand, its short-rangedness gives it considerable latitude to change its configuration to adapt to possible severe environments (otherwise, the chain may be pulled break by external forces, for example, during DNA segregation [1]). This property of base-stacking interactions is very important in the determination of elasticity of dsDNA and the conformation

of secondary structure in single-stranded DNA/RNA molecules, as will shown in following sections.

2.1.3 External Forces Field

In the previous two subsections, we have described the intrinsic energy of DNA double helix. Experimentally, to probe the elastic response of linear DNA molecule, the polymer chain is often pulled by external force fields. Here we constrain ourselves to the simplest situation where one terminal of DNA molecule is fixed and the other terminal is pulled with a force $\mathbf{F} = f\mathbf{z}_0$ along the direction of unit vector $\mathbf{z}_0$ [3-9]. (In fact, hydrodynamic fields or electric fields are also frequently used to stretch semiflexible polymers [17], but we will not discuss such cases here.) The end-to-end vector of a DNA chain is expressed as $\int_0^L \mathbf{t}(s) \cos \theta(s) ds$. Then the total “potential” energy of the chain in the external force field is

$$E_f = -\int_0^L \mathbf{t} \cos \theta ds \cdot \mathbf{F} = -\int_0^L f \mathbf{t} \cdot \mathbf{z}_0 \cos \theta ds. \quad (6)$$

To conclude this section, the total energy of a dsDNA molecule under the action of an external force is expressed in our model as

$$E = E_b + E_L + E_f$$

$$= \int_0^L \left[\kappa \left(\frac{d\mathbf{t}}{ds} \right)^2 + \kappa \left(\frac{d\theta}{ds} \right)^2 + \frac{\kappa}{R^2} \sin^4 \theta + \rho(\theta) - f \mathbf{t} \cdot \mathbf{z}_0 \cos \theta \right] ds. \quad (7)$$

2.2 Extensibility and Entropic Elasticity

According to the path integral method of polymer chain (see Appendix A of Ref. [13]), the Green equation of Eq. (7) is

$$\frac{\partial \Psi(\mathbf{t}, \theta; s)}{\partial s} = \left[\frac{\partial^2}{4l_p^* \partial \mathbf{t}^2} + \frac{\partial^2}{4l_p^* \partial \theta^2} + \frac{f \cos \theta}{k_B T} \mathbf{t} \cdot \mathbf{z}_0 - \frac{\rho(\theta)}{k_B T} - \frac{l_p}{R^2} \sin^4 \theta \right] \Psi(\mathbf{t}, \theta; s), \quad (8)$$

where $l_p^* = \kappa^*/k_B T$, $l = \kappa/k_B T$, and $\Psi(\mathbf{t}, \theta; s)$ is an auxiliary function for the configuration of dsDNA system. The spectrum of the above Green equation is discrete and for a long dsDNA molecule its average extension can be obtained either by differentiation of the ground-state eigenvalue, g_0 , of Eq. (7) with respect to f :

$$\langle Z \rangle = \int_0^L \langle \mathbf{t} \cdot \mathbf{z}_0 \cos \theta \rangle ds = L k_B T \frac{\partial g_0}{\partial f}, \quad (9)$$

or by a direct integration with the normalized ground-state eigenfunction, $\Phi_0(\mathbf{t}, \theta)$, of Eq. (8):

$$\langle Z \rangle = L \int |\Phi_0|^2 \mathbf{t} \cdot \mathbf{z}_0 \cos \theta d\mathbf{t} d\theta. \quad (10)$$

Both g_0 and $\Phi_0(\mathbf{t}, \theta)$ can be obtained numerically through standard diagonalization methods and identical results are obtained by Eqs. (9) and (10).

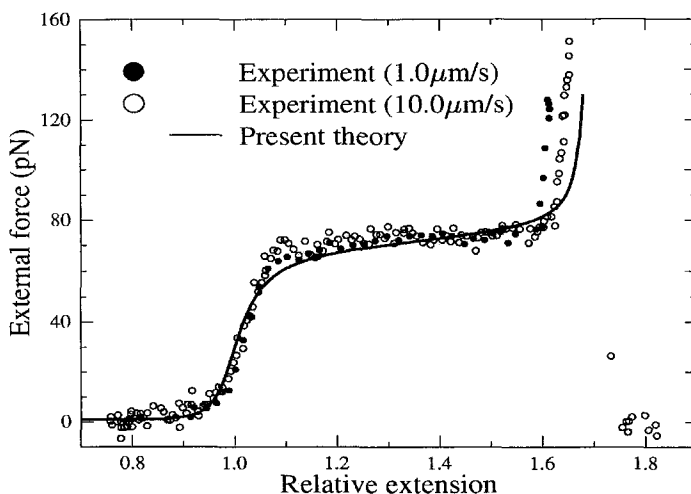


Figure 1. Force-extension relation of torsionally relaxed DNA molecule. Experimental data taken from Ref. [4] (symbols). Theoretical curve obtained by the following considerations: (a) $l_p = 1.5$ nm and $\epsilon = 14.0 k_B T$; (b) $l_p^* = 53.0 / 2 \langle \cos \theta \rangle_{f=0}$ nm, $\eta_0 = 0.34 / \langle \cos \theta \rangle_{f=0}$ nm and $R = (0.34 \times 10.5 / 2\pi) \langle \tan \theta \rangle_{f=0}$ nm; (c) adjust the value of θ_0 to fit the data. For each θ_0 , the value of $\langle \cos \theta \rangle_{f=0}$ is obtained self-consistently. The present curve is drawn with $\theta_0 = 62.0^\circ$ (in close consistence with the structural property of DNA and $\langle \cos \theta \rangle_{f=0}$ is determined to be 0.573840. DNA extension is scaled with its B-form contour length $L \langle \cos \theta \rangle_{f=0}$.

Figure 1 shows the calculated force versus extension relation in the whole relevant force range in comparison with the experimental observations [4,5]. The theoretical curve in this figure is obtained with just one adjustable parameter (see

caption of Fig. 1); the agreement with experiment is excellent. Figure 1 demonstrates that the highly extensibility of DNA molecule under large external forces can be quantitatively explained by the present model.

To further understand the force-induced extensibility of DNA, in Fig. 2 we show the folding angle distribution of dsDNA molecule at different external force, i.e.,

$$P(\theta) = \int |\Phi_0(\mathbf{t}, \theta)|^2 dt. \quad (11)$$

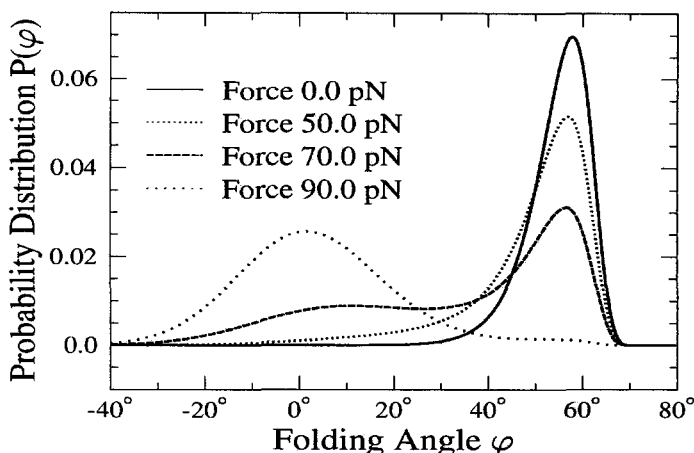


Figure 2. Folding angle distributions for negatively supercoiled DNA molecule pulled with a force of 1.3 pN.

Figures 1 and 2, taking together, demonstrate that the elastic behaviors of dsDNA molecule are radically different under the condition of low and large applied forces:

The low-force region. When external force is low (≤ 10 pN), the folding angle is distributed narrowly around the angle of $\theta \sim +57^\circ$, and there is no probability for the folding angle to take on values less than 0° (Fig. 3), indicating that DNA chain is completely in the right-handed B-form configuration with small axial fluctuations. This should be attributed to the strong base-stacking intensity, as pointed out above. Consequently, the elasticity of DNA is solely caused by thermal fluctuations in the axial tangent $\mathbf{t}$ (Fig. 1), and DNA molecule can be regarded as an inextensible chain. This is the physical reason why, in this force region, the elastic behavior of DNA can be well described by the wormlike chain model [16,18]. Indeed, as shown in

Fig. 3, for forces ≤ 10 pN, the wormlike chain model and the present model give identical results. Thus, we can conclude with confidence that, when external fields are not strong, the wormlike chain model is a good approximation of the present model to describe the elastic property of dsDNA molecules.

The large-force region. With the continuous increase of external pulling forces, the axial fluctuations become more and more significant. For example, for forces ~ 50 pN, although the folding angle distribution is still peaked at $\theta \sim 57^\circ$, there is also considerable probability for the folding angle to be distributed in the region $\theta \sim 0^\circ$ (Fig. 2). Therefore, at this force region, DNA polymer can no longer be regarded as inextensible. For $f \sim 65$ pN, another peak in the folding angle distribution begins to emerge at $\theta \sim 0^\circ$, marking the onset of cooperative transition from B-form DNA to overstretched S-form DNA [4,5]. This is closely related to the short-ranged nature of the base-stacking interactions [12]. At even higher forces ($f \geq 80$ pN), the DNA molecule becomes completely into the overstretched form with its folding angle peaked at $\theta \sim 0^\circ$. This threshold f_t of over-stretch force is also consistent with a plain evaluation from base-stacking potential of $\epsilon \sim f_t r_0$, i.e., $f_t \sim 90$ pN.

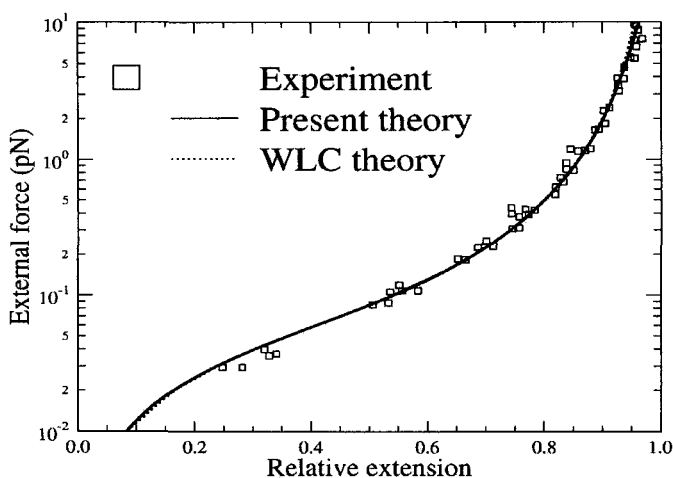


Figure 3. Low-force elastic behavior of DNA. Here experimental data is from Fig. 5B of [3], the dotted curve is obtained for a wormlike chain with bending persistence length of 53.0 nm and the parameters for the slid curve are the same as those in Fig. 1.

It should be mentioned that in the experiments [4,5] the transition to S-DNA occurs even more cooperatively and abruptly than predicted by the present theory (see Fig. 1). This may be related to the existence of single-stranded breaks (nicks) in

the dsDNA molecules used in the experiments. Nicks in DNA backbones can lead to strand separation or relative sliding of backbones [4,5], and they can make the transition process more cooperative. However, the comprehensive agreement achieved in Figs. 1 and 3 indicates that such effects are only of limited significance. The elasticity of DNA is mainly determined by the competition between folding angle fluctuation and tangential fluctuation, which are governed, respectively, by the base-stacking interactions (ϵ) and the axial bending rigidity (κ^*) in Eq. (7).

2.3 Elongation of Supercoiled DNA

The number of times the two strands of DNA double helix are interwound, i.e., the linking number Lk , is a topologic invariant quantity for closed DNA molecule. It is also topologically invariant for a linear DNA polymer in case that the orientations of two extremities of the linear polymer are fixed and any part the of polymer is forbidden to go around the extremities of the polymer (as performed in Strick *et al.*'s supercoiling experiments [6]). An unstressed B-DNA molecule has one right-handed twist per 3.4 nm along its length, i.e., $Lk_0 = L_B/3.4$. Under some twist stress, the linking number of DNA polymer may deviate from its torsionally relaxed value [20]. In all cases, when $\Delta Lk = Lk - Lk_0 \neq 0$, the DNA polymer is called "supercoiled." The relative difference in the linking number, $\sigma = (Lk - Lk_0)/Lk_0$, signifies the degree of supercoiling which is independent upon the length of DNA polymer. In this section we investigate the elasticity of the supercoiled DNA double helix through Monte Carlo simulation, based on the same model but in a discrete form [21].

2.3.1 Discrete dsDNA Model

In the simulation, the double-stranded DNA molecule is modeled as a chain of discrete cylinders, or two discrete wormlike chains constrained by base pairs of fixed length $2R$. The conformation of the chain is specified by the space positions of vertices of its central axis, $\mathbf{r}_i = (x_i, y_i, z_i)$ in 3-D Cartesian coordinate system, and the folding angle of the sugar-phosphate backbones around the central axis, θ_i , $i = 1, 2, \dots, N$. Each segment is assigned the same number of base pairs, n_{bp} , so that the length of the i th segment satisfies $\Delta s_i = |\mathbf{r}_i - \mathbf{r}_{i-1}| = 0.34n_{bp} \frac{\cos \theta_i}{\langle \cos \theta \rangle_0}$, where $\langle \dots \rangle_0$ means the thermal average for a relaxed DNA molecule.

According to Eq. (7), the total energy of dsDNA molecule with N segments in our discrete computery model is expressed as

$$E = \alpha \sum_{i=1}^{N-1} \gamma_i^2 + \alpha' \sum_{i=1}^{N-1} (\theta_{i+1} - \theta_i)^2 + \frac{\kappa}{R^2} \sum_{i=1}^N \Delta s_i \sin^3 \theta_i \tan \theta_i + \sum_{j=1}^{N_{bp}} U(\theta_j) - f z_N, \quad (12)$$

where γ_i is the bending angle between the $(i-1)$ th and the i th segments, and z_N the total extension of the DNA central axis along the direction of the external force f (assumed in the z -direction). Here, the bending rigidity constant α corresponds to the persistent length $p = 53$ nm of dsDNA according to the direct discretization of Eq. (1) or (3), i.e.,

$$\alpha = \frac{p}{2b} k_B T, \quad (13)$$

where b is the average length of segments.

In earlier approaches [21,22], the bending rigidity constant α of discrete chain was determined according to the Kuhn statistic length of wormlike chain, which is twice the persistent length p . In fact, the Kuhn length l_{kuhn} of discrete wormlike chain with rigidity α is written as (see, e.g., [21])

$$l_{\text{kuhn}} = b \frac{1 + \langle \cos \gamma \rangle}{1 - \langle \cos \gamma \rangle}, \quad (14)$$

where

$$\langle \cos \gamma \rangle = \frac{\int_0^\pi \cos \gamma \exp(-\alpha \gamma^2 / k_B T) \sin \gamma d\gamma}{\int_0^\pi \exp(-\alpha \gamma^2 / k_B T) \sin \gamma d\gamma}. \quad (15)$$

The rigidity constant α defined in this way is only the function of $m = l_{\text{kuhn}}/b$, the number of links within one Kuhn length. The dependence of α versus m obtained from Eqs. (14) and (15) is shown numerically in Fig. 4. The rigidity constant α follows very well the linear dependence upon m , especially in the reasonable region of $m > 5$. As a comparison, we also show the line of Eq. (13), i.e., $\alpha = (m/4)k_B T$. It is obvious that the rigidity constants of discrete chain are numerically equivalent in two algorithms. However, the processes of discretization of Eq. (3) is more convenient to be used, especially to the models with complicated elasticity.

The constant α' in the second term of Eq. (12) should be associated with stiffness of the DNA single strand. As a crude approximation, we have taken here $\alpha' = \alpha$. Our unpublished data show that, the amount of second term of Eq. (12) is quite small compared with the other four terms. And the result of simulation is not sensitive to α' .

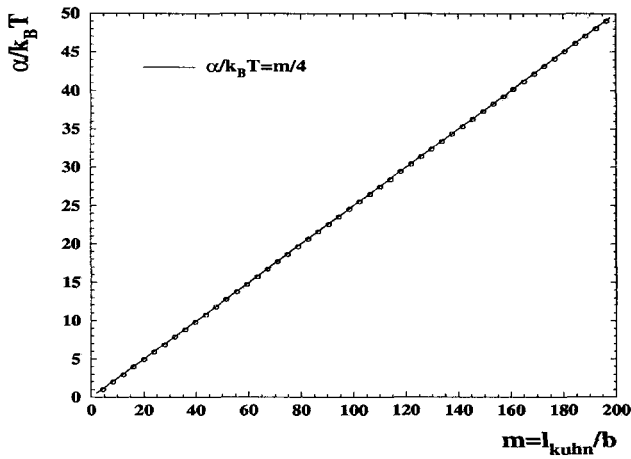


Figure 4. The bending rigidity constant α of discrete wormlike chain as function of number of segments within one Kuhn length $m = l_{\text{kuhn}}/b$. The circle points denote the result calculated from Eqs. (14) and (15); the solid line from direct discretization of Eq. (3).

2.3.2 Simulation Procedure

The equilibrium sets of conformations of dsDNA chain are constructed using the Metropolis MC procedure [23]. The conformational space is sampled through a Markov chain process. Three kinds of movements are considered: (1) the length of a randomly chosen segment is modified; (2) a portion of the chain is rotated around the axis connecting the two ends of rotated chain; (3) The segments from a randomly chosen vertex to the free end are rotated around an arbitrary orientation axis that passes the chosen vertex [21]. A trial move from a conformation (or state) i to a conformation (or state) j is accepted on the basis of the probability $p_{i \rightarrow j} = \min(1, \rho_j / \rho_i)$, where ρ_i is the probability density of conformation i . Energetic importance sampling is realized in the Metropolis MC method by choosing the probability density ρ_i as the Boltzmann probability: $\rho_i = \exp(-E_i/k_B T)$, where E_i is the energy of conformation i calculated according to Eq. (12).

The starting conformation of chain is unknotted. To avoid knotted configuration in the Markov process, we calculate the Alexander polynomial of each trial conformation [24,25]. In the case where the trial movement knots the chain, the trial movement will be rejected. To incorporate the excluded-volume effect for each trial conformation, we calculate the distance between any point on the axis of a segment

and any point on the axis of another nonadjacent segment. If the minimum distance for any to chosen segment is less than the DNA diameter $2R$, the energy of trial conformation is set infinite and the movement is then rejected.

During the evolution of DNA chain, the supercoiling degree σ may distribute around all the possible values. In order to avoid the waste of computation events and also for the comparison of Strick *et al.*'s supercoiling experiments [6,19], we bound the supercoiling σ of DNA chain inside the region of $-0.12 \leq \sigma \leq 0.12$. When the torsional degree of trial conformation is beyond the chosen range, we simply neglect the movement and reproduce a new trial movement again.

The linking number Lk of each conformation is calculated according to White's formula [26], $Lk = Tw + Wr$, where twisting number Tw can be directly calculated from

$$Tw = \frac{1}{2\pi R} \sum_{i=1}^N \Delta s_i \tan \theta_i. \quad (16)$$

To enclose the linear DNA molecule without changing its linking number, we add three long flat ribbons to the two ends for each conformation and keep the ribbons in the same planar. The writhing number Wr can therefore be calculated through the Gaussian integral

$$Wr = \frac{1}{4\pi} \iint ds ds' \frac{\partial_s \mathbf{r}(s) \times \partial_{s'} \mathbf{r}(s') \cdot [\mathbf{r}(s) - \mathbf{r}(s')]}{|\mathbf{r}(s) - \mathbf{r}(s')|^3}. \quad (17)$$

2.3.3 Elasticity of supercoiled dsDNA

To obtain equilibrium ensemble of DNA evolution, 10^7 elementary displacements are produced for each chosen applied force f . The relative extension x and supercoiling degree σ of each accepted conformation of DNA chain are calculated. When the trial movement is rejected, the current conformation is counted twice (see Ref. [23]).

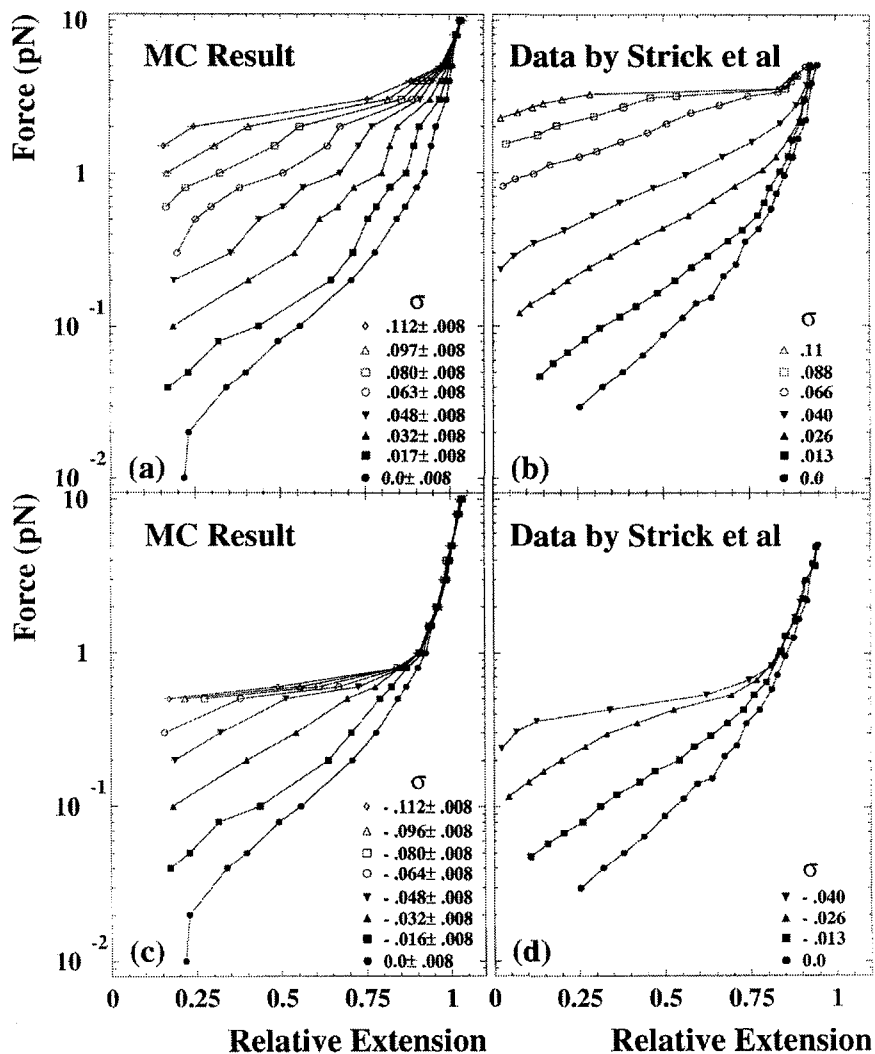


Figure 5. Force versus relative extension curves for negatively (a,b) and positively (c,d) supercoiling DNA molecule. The left two plots (a) and (c) are the results of our Monte Carlo simulation, and the horizontal bars of points denote the statistic error of relative extension in our simulations. The right two plots (b) and (d) are the experimental data from Ref. [19]. The solid curves serve as eye guides.

In order to see the dependence of mechanical property of dsDNA upon supercoiling degree, the whole sample is partitioned into 15 subsamples according to the value of the supercoiling degree σ . For each subsample, we calculate the averaged extension

$$\langle z_N \rangle_j = \frac{1}{N_j} \sum_{i=1}^{N_j} \frac{z_N(i)}{L_B}, \quad j = 1, \dots, 15 \quad (18)$$

and the averaged torsion

$$\langle \sigma \rangle_j = \frac{1}{N_j} \sum_{i=1}^{N_j} \sigma_i, \quad j = 1, \dots, 15, \quad (19)$$

where N_j is the number of supercoiling movements of which belong to j th subsample.

We display the force versus relation extension for all positive and negative supercoiling in Fig. 5a and c respectively. As a comparison, the experimental data [6,19] are shown in Fig. 5b and d. Fig. 6 shows the averaged extension as a function of supercoiling degree for 3 typical applied forces. In spite of quantitative difference between Monte Carlo results and experimental data, the qualitative coincidence is striking. Especially, three evident regimes exist in both experimental data and our Monte Carlo simulations:

- 1) At a low force, the elastic behavior of DNA is symmetrical under positive or negative supercoiling. This is understandable, since the DNA torsion is the cooperative result of hydrogen-bond constrained bending of DNA backbones and the base-stacking interaction in our model. At very low force, the contribution from applied force and the thermodynamic fluctuation perturbs the folding angle θ of base pair to deviate just a little from the equilibrium position, θ_0 . Therefore, the DNA elasticity is achiral at this region. For a fixed applied force, the increasing torsion stress tends to produce plectonemic state which shortens the distance of two ends causing the relative extension of linear DNA polymer. These features can be also understood by the traditional approaches with harmonic twist and bending elasticity [22,27].
- 2) At intermediate force, the folding angle of base pairs are pulled slightly further away from equilibrium value θ_0 where Van der Waals potential is not symmetric around θ_0 . So the chiral nature of elasticity of the DNA molecule appears. In negative supercoiling region, i.e., $\theta < \theta_0$, the contribution of the applied force dominates that of the potential because of the low plateaus of $U(\theta)$ (see Eq. (5)). So the extension is insensitive to negative supercoiling degree. On the other hand, the positive supercoiling still tends to contract the molecule.
- 3) At higher force, the contribution of the applied force to the energy dominates that of Van der Waals potential in both over- and underwound

DNA. The extension of DNA accesses to its B-form length. Therefore, the plectonemic DNA is fully converted into extended DNA, the writhe is essentially entirely converted into twist and the force-extension behavior reverts to that of untwisted $\sigma = 0$ DNA as expected from a torsionless worm-like chain model [3,11,18].

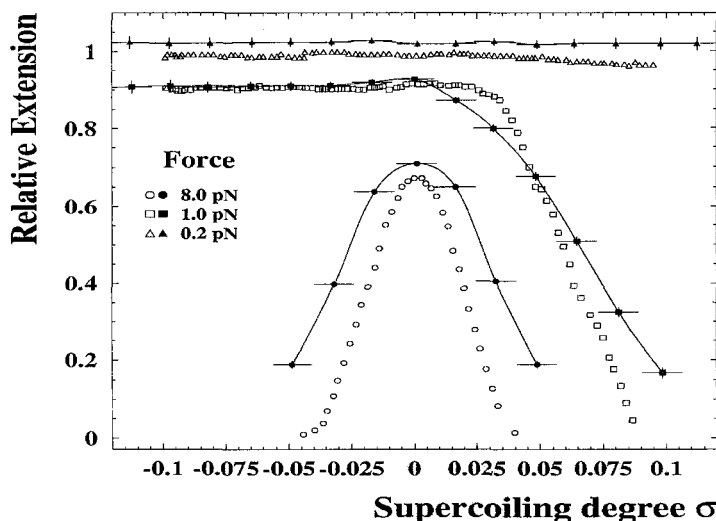


Figure 6. Relative extension versus supercoiling degree of DNA polymer for three typical stretch forces. Open points denote the experimental data [19], and solid points the results of our Monte Carlo simulation. The vertical bars of the solid points signify the statistic error of the simulations, and the horizontal ones denote the bin-width that we partition the phase space of supercoiling degree. The solid lines connect the solid points to guide the eye.

It should be mentioned that there is an upper limit of supercoiling degree for extended DNA in current approach, i.e., $\sigma_{\max} \sim 0.14$, which corresponds to $\theta = 90^\circ$ of the folding angle. In recent experiments, Allemand *et al.* [28] twisted the plasmid up to the range of $-5 < \sigma < 3$. They found that at this “unrealistically high” supercoiling, the curves of force versus extension for different σ split again at higher stretch force (> 3 pN). As argued by Allemand *et al.*, in the extremely under- and overwound torsion stress, two new DNA forms, denatured DNA and P-DNA with exposed bases, will appear. In fact, if the deviation of the angle which specifies DNA twist from its equilibrium value exceeds some threshold, the corresponding torsional stress causes local distraction of the regular double helical structure. So the emergence of these two striking forms is essentially associated with the broken processes of some base pairs under super-highly torsional stress. In this case, the

permanent hydrogen constrain will be violated and the configuration of base stacking interactions be varied considerably.

3 Mechanics of Pulling Single-Stranded DNA

By attaching dsDNA between beads and melting off the unlabeled strand with distilled water or formaldehyde, a single stranded can be obtained [10]. Because of its thin diameter and high flexibility, ssDNA is more contractile than dsDNA in low force. However, it can be stretched to a greater length at high force since it no longer forms a helix. In 150 mM NaCl solution, the force/extension curve of ssDNA, melted from a λ phage DNA molecule, was found to be able to fit with a simple freely jointed chain (FJC) of Kuhn length of 1.5 nm including a stretch modulus [5]. However, more detailed measurements show that the elongation characteristics of ssDNA is very sensitive to the ionic concentration of solution, and the FJC is not valid in both high ionic (e.g., 5 mM $MgCl_2$) and low ionic (e.g., 2 mM NaCl) solutions [10,9]. On the other hand, the measurements by Rief *et al.* [7] shows that the force/extension characteristics of ssDNA are strongly sequence-dependent. When a single designed poly(dA-dT) or poly(dG-dC) strand is pulled with an atomic force microscope, they found that, at some stretched force [9 pN for poly(dA-dT) and 20 pN for poly(dG-dC)], the distance of the two ends of the designed molecules suddenly elongates from nearly zero to a value comparable to its total contour length in a very cooperative manner, which is drastically different from the gradual elongation of the ssDNA in nature (within a relative random sequence).

Here, we present our recent Monte Carlo calculations of a modified freely jointed chain with elastic bonds. In order to attain an unified understanding of reported force/extension data of ssDNA molecule in different ionic atmospheres and for different nucleotide sequences, we have incorporated three possible interactions of base pairing [29], base stacking [30] and electrostatic interactions in our calculations. In the next section, we will at first determine the electrostatic potential between DNA strands through numerically solving the nonlinear Poisson-Boltzmann equation.

3.1 Electrostatic Interaction between ssDNA

Under the assumptions of (1) the solute in a solution of strong electrolyte is completely dissociated into ions; (2) all deviations from the properties of an ideal solution (ions are uniformed distributed) are due to the electrostatic forces which exist between the ions, the electrostatic potential $\psi(\mathbf{r})$ at a space point $\mathbf{r}$ can be submitted to the Poisson-Boltzmann equation [32]:

$$\nabla^2 \psi(\mathbf{r}) = -\frac{4\pi}{D} \sum_{i=1}^n v_i e c_i \exp(-v_i e \psi(\mathbf{r}) / k_B T). \quad (20)$$

Here the solution is assumed to contain $N_1, \dots, N_n$ different ions with valences $v_1, \dots, v_n$, and $c_i (=N_i/V)$ is the bulk concentration of the ionic species i , where V and D are the volume and dielectric constant of the solution, and e is the protonic charge.

Equation (20) cannot be solved in closed form. Here, we calculate the electrostatic potential of ssDNA cylinder immersed in the solutions of NaCl and MgCl_2 , through numerically solving Eq. (20) according to the series expansion method used earliest by Pierce [31,32]. As illustrative examples, we show in Fig. 7 the electrostatic potential of ssDNA cylinder in 2 mM NaCl and 5 mM MgCl_2 solutions, where the potential function is expanded up to 17th order for symmetrical electrolyte (NaCl) and 14th order for unsymmetrical electrolyte (MgCl_2) in our calculations.

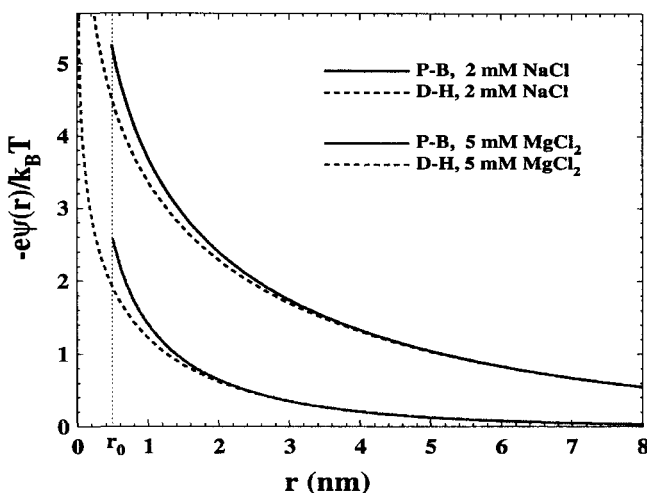


Figure 7. Electrostatic potential of ssDNA cylinder versus the radial distance from the cylinder axis in the solutions of 2 mM NaCl (black) and 5 mM MgCl_2 (grey). The solid curves are the numerical solutions of Poisson-Boltzmann (P-B) up to the expansion of 17th order for NaCl and 14th order for MgCl_2 ; the dashed curves denote the Debye-Hückel approximation (D-H) with effective linear charge density v along the axis listed in Table 1. The dotted line corresponds to the surface of ssDNA cylinder of $r_0 = 0.5$ nm.

However, the numerical solution of straight charged cylinder of Poisson-Boltzmann equation can not be directly used in the calculations of ssDNA molecule, since the real molecule actually takes a variety of irregular configurations. To approach the problem, we consider the first-order approximation of Eq. (20), i.e., linear Poisson-Boltzmann equation, the solution of which can be implicitly

expressed. Around a point charge q , the electrostatic potential in the linear equation can be written in Debye-Hückel form as

$$\psi_{DH}(\mathbf{r}) = \frac{q}{D|\mathbf{r}|} \exp(-\kappa|\mathbf{r}|), \quad (21)$$

where $\mathbf{r}$ is the position vector from q , and the inverse Debye length $\kappa = \left(\frac{8\pi\epsilon_0 e^2}{Dk_B T}\right)^{1/2}$

for NaCl solution, and $\kappa = \left(\frac{24\pi\epsilon_0 e^2}{Dk_B T}\right)^{1/2}$ for MgCl_2 .

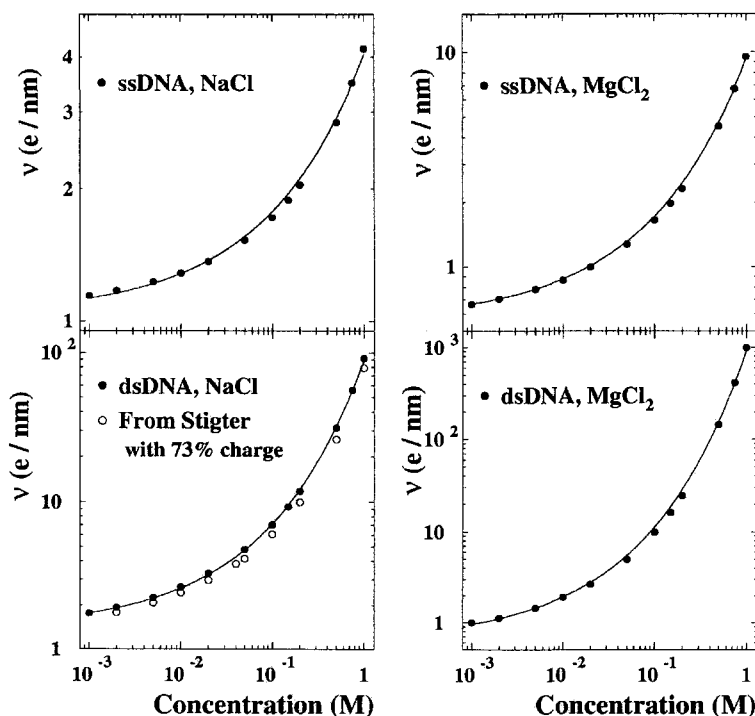


Figure 8. The effective linear charge density v of both ssDNA and dsDNA as function of ionic concentrations of NaCl and MgCl_2 solutions. The solid circles are the results in present calculations; The opened circles are Stigter's results [34], where electrophoretic charge of $-0.73e$, which is required to fit Stigter's electrophoresis theory to experimental data, were used. In present calculations, the full charge per phosphate group is assumed. The curve is a fit of Eq. (23) with fit parameters listed in Table 1.

In order to count the influence of higher expansion terms of Poisson-Boltzmann equation, one can phenomenologically change the amplitude of the Debye-Hückel potential of Eq. (21) to match the numerical solution of Poisson-Boltzmann equation according to Brenner and Parsegian [33] and Stigter [34]. According to Eq. (21), the electrostatic potential of a straight charged cylinder of infinite length can be written as

$$\psi_{\text{DH}}(r) = \int_{-\infty}^{\infty} \frac{v d\lambda}{D} \frac{\exp(-\kappa \sqrt{\lambda^2 + r^2})}{\sqrt{\lambda^2 + r^2}} = \frac{2v}{D} K_0(\kappa r), \quad (22)$$

where the integral of λ is along the cylinder axis, r is the radial distance from cylinder axis, v the linear charged density, and K_0 the first-order modified Bessel function. By comparing the Eq. (22) with Poisson-Boltzmann solution in the overlap region far from the cylinder surface, we can determine the effective linear charge density v in different bulk ionic concentrations c of both NaCl and MgCl₂ (see Table 1). In Table 1 we also show the effective charge density of dsDNA.

As shown in Fig. 8, all the data of v can be very well fitted by the formula

$$v = \exp(\alpha + \beta c^{2/5}), \quad (23)$$

with the fit parameters α and β listed in Table 1. As a comparison, Stigter's calculation for dsDNA in NaCl solution, where 73% of electrophoretic charge was assumed [34], is also shown in Fig. 8.

Table 1. The effective linear charge density v (in unit of e/nm) of DNA molecules, calculated from the comparison of Poisson-Boltzmann solution and the modified Bessel function (see text). α and β are the parameters of Eq. (23) fitted to the data of v (see Fig. 8).

Ionic Concentration c_0 (M)	SsDNA		DsDNA	
	NaCl	MgCl ₂	NaCl	MgCl ₂
1.	4.18	9.50	91.85	993.16
0.75	3.50	6.74	56.15	410.67
0.5	2.84	4.51	31.22	144.10
0.2	2.04	2.31	11.73	24.52
0.15	1.89	1.97	9.29	16.22
0.1	1.73	1.64	7.02	9.82
0.05	1.53	1.27	4.78	4.98
0.02	1.37	0.99	3.29	2.66
0.01	1.29	0.86	2.66	1.91
0.005	1.23	0.78	2.26	1.45
0.002	1.17	0.71	1.93	1.13
0.001	1.14	0.67	1.76	1.00
α	0.0338	-0.577	0.300	-0.505
β	1.36	2.80	4.18	7.33

3.2 Model and Method of Calculations

3.2.1 Model of Single-Stranded DNA

In the simulation, the single-stranded DNA molecule is modeled as a freely jointed chain with N elastic bonds. The conformation of the chain is specified by the space position of its vertices, $\mathbf{r}_i = (x_i, y_i, z_i)$, $i = 0, 1, \dots, N$, in three-dimensional Cartesian coordinate system with $\mathbf{r}_0$ fixed at the original point. The equilibrium features of stretched ssDNA in salt solution are determined by the interplay of following five energies within the frame of canonical Boltzmann statistical mechanics.

The first energy, called E_{ele} , is the electrostatic interaction energy between strands. As discussed in the last section, the electrostatic energy of ssDNA molecule can be calculated according to Debye-Hückel approximation

$$\frac{E_{\text{ele}}}{k_{\text{B}}T} = \frac{v^2}{k_{\text{B}}TD} \int d\lambda_i \int d\lambda_j \frac{\exp(-\kappa |\mathbf{r}_i - \mathbf{r}_j|)}{|\mathbf{r}_i - \mathbf{r}_j|}, \quad (24)$$

where the effective charged density v is taken from Table 1. The integration is performed along the strand, and $|\mathbf{r}_i - \mathbf{r}_j|$ is the distance between the current positions at the strand to which the integration parameters λ_i and λ_j correspond.

The second energy describes pairing interactions of complementarity bases in ssDNA. Two elementary pairings in ssDNA are the G-C and A-T base pairs of Watson and Crick. As a G-C pair is formed through three hydrogen bonds while an A-T one involves only two hydrogen bonds, the base pairing potential should be sequence-dependent. Since the Kuhn length of ssDNA is rather longer than its sugar-phosphate backbone between two bases, each node in our model actually includes several bases. We approximate the base pairing interaction by the node-pairing

energy, $E_{\text{pair}} = \sum_{i=1}^{N_p} V_p$, where N_p is the number of node-pairs and V_p is a

sequence-dependent parameter. The pairing rule in our simulation is similar to the standard one which keeps only nested structure [35,36,29,37,30]: (a) Two nodes (i, j) can be paired only when their distance $|\mathbf{r}_i - \mathbf{r}_j|$ is less than 2 nm, which corresponds approximately to the interaction range of Watson-Crick hydrogen bond in dsDNA [12]; (b) each node can be pairing to at most one other node; (c) two-node pairs (i, j) and (k, l) can coexist only when they are either nested (i.e., $i < k < l < j$) or independent (i.e., $i < j < k < l$); this condition neglects the formation of pseudoknots which are known to be very rare in real RNA structure. (d) $|j - i| \geq 4$; this restriction permits flexibility of the chain and it is also necessary to rule out entirely the influence of phase space on the number of pairings, as confirmed by our following MC simulations.

The third energy describes the vertical stacking interactions of neighboring base pairs. We approximate the base pairing energy by $E_{sta} = \sum_{i=1}^{N_S} V_S$, where N_S is the number of stacked node-pairs and V_S is the stacking potential between two neighboring node-pairs. Two node-pairs are considered as 'stacked' only when they are nearest neighbors to each other, e.g., pairs of (i,j) and $(i+1,j+1)$.

The fourth energy in our model is the deformation energy of the chain when the length of its individual linker deviates from its equilibrium length b . According to Hooke's law, for small rod deformation the elastic energy of the ssDNA can be written as

$$E_{ela} = \frac{1}{2} \sum_{i=1}^N Y (|\mathbf{r}_i - \mathbf{r}_{i-1}| - b_0)^2, \quad (25)$$

where b_0 is the Kuhn statistic length of ssDNA and Y characterizes the stretch stiffness of the ssDNA backbone.

The fifth energy, i.e., work done by the external force F , is written as $E_{for} = -Fz_N$, where z_N is the coordinate of the last vertex, i.e., the distance of the two termini of ssDNA molecule in the direction of external force F . Here we have chosen the orientation of F along the z-axis of the Cartesian system.

3.2.2 MC Procedure of ssDNA

The Monte Carlo procedure of ssDNA is similar to that of dsDNA described above. Beside those three updates used in dsDNA, however, an additional movement is used here, which involves a permutation between a subchain of 2 segments at position i and another subchain of 3 segments at position j . This move was first adopted by Vologodskii *et al.* [38] in the MC calculation of supercoiled dsDNA. Before the permutation, the conformations of both subchains of doublet and triplet should be deformed so that the length of the subchains could be incorporated to their new positions. The net result of this permutation is a translation of random chosen subchain of (i, j) by 1 segment along the chain axis. Even though the acceptance probability of this movement can be quite low, it can substantially increase the probability of extrusion and resorption of hairpin structures and help the simulation to go out of some local traps.

As mentioned above, the trial movements in the canonical Metropolis algorithm is accepted or rejected according to the Boltzmann weight of $\rho_i = \exp(E_i/k_B T)$, where the total energy $E_i = E_{ele} + E_{pair} + E_{sta} + E_{ela} + E_{for}$ in the case of ssDNA. However, the energy landscape of ssDNA with hairpin structures is characterized by numerous local minima separated by energy barriers, and the probability of canonical Metropolis procedure to cross the energy barrier of height ΔE is proportional to $\exp$

$(\Delta E/k_B T)$. When the pairing and stacking energy are rather large, e.g., in the case of poly(dG-dC), the energy barriers around some special conformations can be very high so that the simulation tends to get trapped in these conformations although they are by no means of the lowest energy. During the finite CPU time, only small parts of the canonical ensemble of DNA conformations can therefore be explored, rendering the calculation of physical quantities unreliable.

In order to overcome this problem of “ergodicity breaking” of poly(dG-dC) ssDNA, we produce an artificial ensemble according to a modified weight factor [39],

$$\rho(E) = \exp[(E + \frac{\sqrt{2}}{\sigma} |E - \langle E \rangle|) / k_B T], \quad (26)$$

where $\sigma^2 = n_F/2$ is the mean squared derivation of energy of the canonical thermodynamic system, and n_F the number of degree of freedom of the chain, $\langle E \rangle$ is the averaged energy of the system which can be calculated in a simple iteration procedure [39,40]. In Eq. (26), the probability of both high- and low-energy are exponentially reinforced and sharp peak of the canonical ensemble around $\langle E \rangle$ is damped, which can efficiently help the simulation to jump out from local energy basins. Since one configuration in the artificial ensemble of Eq. (26), in fact, represents $n(E) = \exp[-\sqrt{2} |E - \langle E \rangle| / (k_B T \sigma)]$ configurations in canonical system, we should reweight the artificial sample to obtain the expectation value of considered quantity. For example, the averaged extension z_N should be calculated by [41]

$$\langle z_N \rangle = \frac{\sum_{i=1}^{N_{MC}} z_N(E_i) n(E_i)}{\sum_{i=1}^{N_{MC}} n(E_i)}, \quad (27)$$

where N_{MC} is the number of sweeps of the artificial sample.

Table 2. The parameters used in our modeling calculation of ssDNA to fit the experimental data presented in Fig. 9.

Sequence	$V_P(k_B T)$	$V_S(k_B T)$	b	$Y(k_B T/nm^2)$
Random	4.6	0	1.6	123.5
Poly(dA-dT)	4.1	4.0	1.6	123.5
Poly(dG-dC)	10.4	6.0	1.6	123.5

3.3 MC Results of ssDNA

3.3.1 Force/Extension of ssDNA

Until now, there are three groups who have pulled ssDNA of both random and designed sequence and presented their force/extension data in different salt environments [5,7,9,10]. These data offer us good opportunity to check the theoretical model and meanwhile determine the 4 main parameters in our model, i.e., the Kuhn length b , stiffness of ssDNA backbone Y , pairing potential V_P , and stacking potential V_S (see Table 2). In the following calculations, we make 20,000,000 Monte Carlo runs with $N = 100$ nodes at each given external force for each case, according to our CPU capacity. We have also confirmed that the larger value of N , e.g., $N = 200$, with more MC runs will not lead to different results.

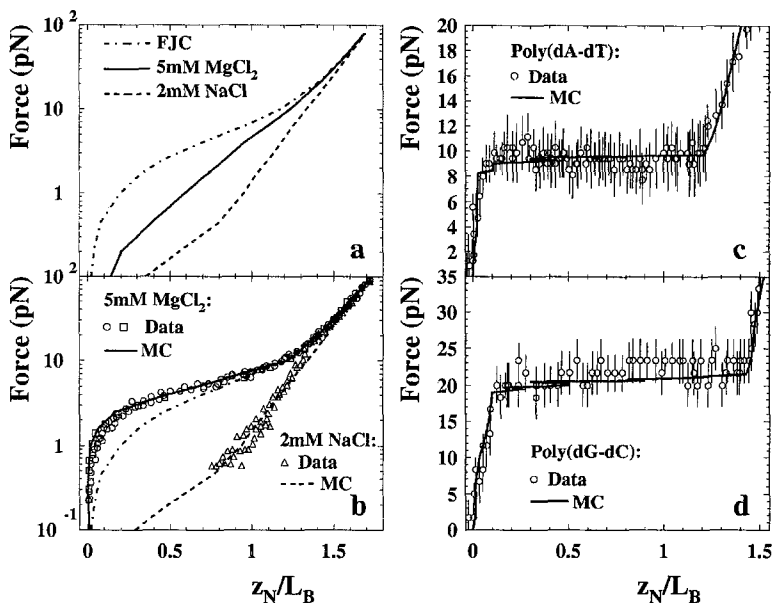


Figure 9. External force as the function of distance of two ends of ssDNA, scaled by its B-form length L_B . $L_B = Nb/1.58769$ in the modeling calculation, and other model parameters used in our Monte Carlo calculation are listed in Table 2. (a) Monte Carlo results of pure freely jointed chain (FJC) without electrostatic, pairing and stacking potential (dash-dotted line), and that with electrostatic interaction considered in 2 mM NaCl (dashed line) and 5 mM $MgCl_2$ (solid line), but $V_P = V_S = 0$. (b) λ -phage DNA in 2 mM NaCl and 5 mM $MgCl_2$ solutions, but $V_P = V_S = 0$. Data are taken from Ref. [10] (triangles and squares), and from Ref. [9] (circles). (c) and (d), for designed poly(dA-dT) and poly(dG-dC) sequences respectively in 150 mM NaCl solution. Data are from Ref. [7].

We at first calculate force/extension characteristics of FJC with elastic bonds in different solutions, but without including pairing and stacking potentials. As shown in Fig. 9a, the electrostatic interactions tend to swell the volume of chain and make the segments more easily aligned along the force direction. This is formally equivalent to enlarging the Kuhn length of ssDNA. The lower the ionic concentration becomes, the larger the effective Kuhn length is, and the more rigid the molecule looks like. Since ssDNA can be entirely pulled back in very high force and all data become the same in this force range, we can determine accordingly the stiffness of sugar-phosphate backbone of ssDNA as $Y = 123.5 \text{ } k_B T / \text{nm}^2$ in our calculations.

Figure 9b shows the force/extension data of a plasmid ssDNA fragment of 10.4 kilobases under 5 mM MgCl_2 and 2 mM NaCl solutions. Bearing in mind that the sequence is relatively random and that the formed base pairs in ssDNA are usually separated spatially from each other along the molecule, the stacking interaction is negligible in this case. We therefore take the stacking potential $V_S = 0$ in our calculations. The best fit to the data is $V_P = 4.6 \text{ } k_B T$ and $b = 1.6 \text{ nm}$. In the lower salt case (2 mM NaCl), the force-elongation curve is not influenced much by the pairing potential compared with that in Fig. 9a, since the dominant electrostatic repulsive potential excludes the bases from getting close to pairing interaction range and the node-pairing probability is very rare. However, in high salt solution of 5 mM MgCl_2 , the force/extension property is the result of completion of two opposite interactions of pairing and electrostatic repulsive interactions. The experimental data suggest that the pairing effect is slightly larger in this case.

In the case of designed poly(dA-dT)/poly(dG-dC) ssDNA, two types of unitary base pairs of A-T and G-C can be formed in respective sequences. Any two bases are possible to be paired and strong stacking interaction exists between two base pairs when they are nearest neighbors (i.e., at the Van der Waals distance range [12]). The stacking potential can dramatically change the conformation of secondary structure of ssDNA in low force. When the stacking potential is absent, the base pairs are formed quite randomly with the formation of many interior loops and branched structures. So the dispersed hairpin structure can be easily pulled open in a medium force [42]. When the stacking interaction exists, on the other hand, the base pairs are encouraged energetically to be neighboring, and therefore lead to the formation of bulk hairpin structure. Both experimental data and modeling calculation suggest that it needs a threshold force to pull back the bulk hairpin structure with the existence of stacking interaction. The value of threshold force, i.e., the height of plateau in force/extension curve, is dependent on the pairing and stacking potentials of the nodes. The best fit to the data, as in Fig. 9c and d, are $V_P = 4.1 \text{ } k_B T$ and $V_S = 4 \text{ } k_B T$ in poly(dA-dT) sequence, $V_P = 10.4 \text{ } k_B T$ and $V_S = 6 \text{ } k_B T$ in poly(dA-dT) sequence.

3.3.2 Criticalness of Force-Induced Phase Transition

It has been shown from force/extension data that the elongation of ssDNA in nature with random sequence under external force is gradual, but that of designed sequence is in a cooperative and discontinuous manner when stacking interaction is involved. This first-order-like phase transition of the designed ssDNA system can also manifest itself in other aspects.

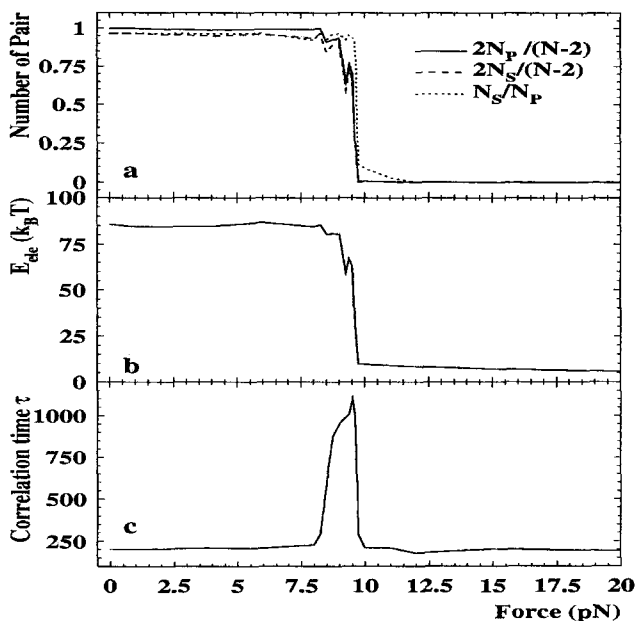


Figure 10. The order parameters and autocorrelation time as the function of external force for poly(dA-dT) sequence in 150 mM NaCl solution. (a) Number of node pairs (N_p), stackings (N_s), and the ratio of pairings and stackings, all of which are scaled to be the maximum value of the pairings of $(N-2)/2$. (b) Electrostatic Debye-Hückel potential. (c) The autocorrelation time of extension of ssDNA polymer calculated by Eqs. (28) and (29). The MC time is scaled by the number of nodes N .

In Fig. 10a we present the average number of pairings scaled by the maximum pairing number of $(N-2)/2$ for poly(dA-dT) sequence in 150 mM NaCl. When the external force is smaller than the critical force ($F_c \sim 9.5$ pN), the number of pairing $2N_p/(N-2) \sim 1$, denoting that almost all the bases are paired. The number of stacked pairs N_s has also its maximum value, signifying that all the pairs aggregate into a compact pattern. Around the external force of F_c (~ 9.5 pN), the bulk hairpin is suddenly pulled back and the numbers of pairings and stackings sharply decrease from their maxima to zero. This behavior is a reminiscence of temperature induced

first-order transition in, e.g., a two-dimensional spin system as described by Ising model. But there the order parameter is magnetization or number of spins with specified orientation. And here the order parameter is number of paired nodes, and the transition is force induced and takes place in the one-dimensional system.

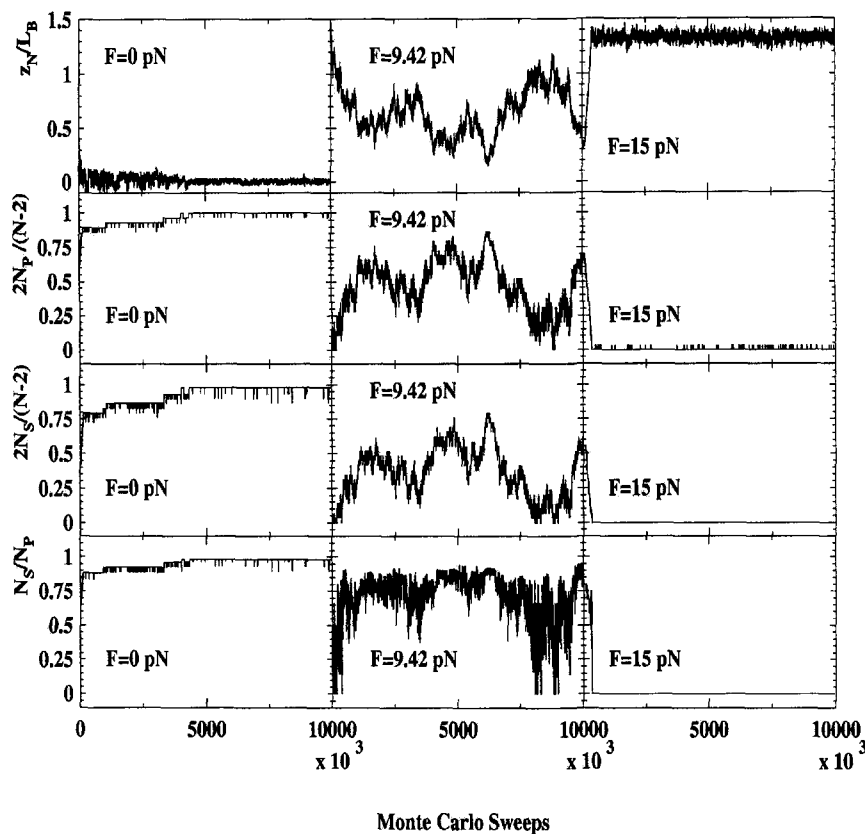


Figure 11. Monte Carlo time series of scaled extension z_N/L_B , scaled pairing numbers $2N_p/(N-2)$, scaled stacking pairs $2N_s/(N-2)$, and the ratio of pairs and stacking numbers N_s/N_p for poly(dA-dT) sequence in 150 mM NaCl solution. All the order parameters fluctuate around their thermal equilibrium values when the force is far away from critical force $F_c = 9.42$ pN; however, the fluctuations diverge around the critical point.

The electrostatic energy of ssDNA changes with external force also in a cooperative manner (Fig. 10b), because of discontinuous jump of the averaged distance between the ssDNA backbone at critical point. One can notice that there are

irregular doglegs for the values of averaged distance of two ends, number of pairings and the electrostatic energy, when the external force approaches the critical point. This is because of the so-called critical fluctuation in our simulations of ssDNA chain at the critical point. Figure 11 shows the time series of some order quantities such as the extension z_N and number of pairings. The order quantities keep staying around their thermodynamic equilibrium values when the external force is away from the critical point. When the external force approaches F_c , however, the fluctuation of order quantities sharply enlarge, since the correlation length may diverge at this point.

As a confirmation, we calculate integrated autocorrelation time τ of the distance z_N of two ends of the ssDNA, i.e.,

$$\tau = \int_0^\infty \frac{\chi(t)}{\chi(0)} dt, \quad (28)$$

where time t is the Monte Carlo steps, which has been scaled by the magnitude N of the chain, and the time-displaced autocorrelation function $\chi(t)$ is calculated as

$$\chi(t) = \int_0^{t_{\max}} dt' [z_N(t') - \langle z_N \rangle][z_N(t'+t) - \langle z_N \rangle]. \quad (29)$$

Here $t_{\max}$ is the total time sweeps of Monte Carlo simulation, and $\langle \cdots \rangle$ denotes the time average along the Monte Carlo series.

In Fig. 10c, the time correlation length as the function of external force. At the critical point, the correlation time indeed diverges. So the number of independent measurements, $n = t_{\max}/2\tau$, is very small in the simulation, which renders the MC calculation at the critical point unreliable. This effect, known as critical slowing down, is an inherent property of Monte Carlo algorithm used to perform the simulation for phase transition system. Some techniques, such as cluster-flipping algorithm [43,44], have been proposed to alleviate the problem of critical slowing down for a number of spin systems. However, an efficient algorithm for that of biopolymer system still lacks. Having in mind the pronounced double-peak structure of the sample action density near the critical point, which is the main reason of critical slowing down in our simulation, it is possible to construct a new weight factor and enhance the tunneling between these two metastable states at the critical point. The detail into this problem is being discussed somewhere else [45].

4 Summary and Conclusions

We present detailed theoretical studies of the elasticities of DNA molecules, and comparisons of modeling calculations and experimental measurements of stretching

single- and double-stranded DNA molecules. In various laboratories, the dsDNA molecule has been stretched through different media in high- and low-force ranges with torsionally relaxed and supercoiled status [3-6,10,46]), while the ssDNA was pulled in different ionic atmosphere with relatively random and designed poly(dA-dT)/(dG-dC) sequences [7,8,9,10,46]).

In order to attain a comprehensive understanding of the experimental data, we model the dsDNA molecule as two interwound wormlike chain, bound by dominant hydrogen bonds. The structural deformation of dsDNA is characterized by the folding angle of sugar-phosphate strand and the molecule central axis. The stacking interaction is calculated through Lennard-Jones potential, determined by the distance of neighboring base pairs or the folding angle of backbones. The ssDNA molecule in our calculation is modeled as a freely jointed chain with elastic bonds. Base-pairing (i.e., hairpins formed by the pairing of complementary bases of ssDNA) and vertical base-stacking potentials are incorporated in the FJC. In order to calculate the electrostatic interaction, Poisson-Boltzmann equation is numerically solved in both mono- and bi-valent ionic atmosphere. The amplitude, or effective charge density, of Debye-Hückel electrostatic potential is determined through matching the first-order modified Bessel function with the Poisson-Boltzmann potential in the overlap region.

The thermodynamics of DNA molecules in external fields of force and torque is calculated by both path integral method and Monte Carlo simulations. We conclude by summarizing the main results of the calculations.

Double-Stranded DNA:

1. In low force range (0 ~ 10 pN), the elasticity of the molecule is entropy-dominated. A simple inextensible wormlike chain model with persistent length of 53 nm can give excellent description of the experimental results in this range [16,18,11,13].
2. In high force region (starting as several tens of pN), the back-stacking potential can be overcome and the helically stacked base pairs are pulled apart at this stretched force, and therefore a structural transition from canonical B-form to a new overstretched conformation called S-DNA can be triggered [11,13]. This structural transition has been manifested by the cooperative elongation of dsDNA at about 70 pN, as observed by Cluzel *et al.* [4] and Smith *et al.* [5].
3. The elasticity of supercoiled DNA is decided by the interplay of external force/torque field and the stacking potential of base pairs. In low force and torque, the inherent helix structure is not perturbed and the elastic behavior of DNA is symmetrical under positive or negative supercoilings [21,22,27]; in intermediate force, the chirality of the elasticity appears since the base pairs are pulled back a little and the Van der Waals stacking potential is

asymmetric in this region [21]; in high force field, the contribution of the external field dominates that of Van der Waals potential in both over- and underwound DNA, and the plectonemic DNA is fully converted to extended DNA. All these results are in qualitative agreements with the experimental measurements of Strick *et al.* [6,19].

Single-Stranded DNA:

1. In high salt solution, secondary structure abounds and hairpins can be formed when the ssDNA bends back onto itself and complementary bases A-T and G-C are paired, gaining energy of several $k_B T$ per pair. This interaction makes it needed a slightly larger external force to pull back a λ -ssDNA than that expected in a pure FJC [9,10,29,30].
2. In low salt solution, the electrostatic repulsive interaction dominates. There are little base pairs formed since the strong electrostatic repulsive potential excludes the bases from getting close to the Watson-Crick base pairing range. This interaction makes the chain more easily to be aligned and more subjected to stretch; this is formally equivalent to enlarging the stiffness of the chain.
3. For designed poly(dA-dT)/poly(dG-dC) chain, the stacking interactions between base pairs encourage base pairs to aggregate into a compact pattern, and it needs a threshold force to pull back the bulk hairpin structure, characterized by a plateau in the force/extension curve. The height of plateau is determined by the pairing potential V_P and stacking potential V_S in our model. The best fit to the experimental data shows that $V_P = 4.1 k_B T$ and $V_S = 4 k_B T$ for poly(dA-dT) sequence, $V_P = 10.4 k_B T$ and $V_S = 6 k_B T$ for poly(dG-dC) sequence. Bearing in mind that each Kuhn length (~ 1.6 nm) contains about three nucleotide bases, we can infer that the pairing energy of each A-T base pair is about $1.37 k_B T$ and that for each G-C base pair is about $3.47 k_B T$. These values are comparable with the measurements of Bockelmann *et al.* [47] when they pulled apart the two strands of a double-stranded DNA helix.
4. Opposite to the gradual elongating of ssDNA of random sequence, the hairpin-coil transition of designed ssDNA is discontinuous. The calculated thermodynamics of stretching designed ssDNA sequence shows typical critical characteristics, as happened in well-known two-dimensional Ising spin model. However, the transition in our model is force-induced and takes place in one-dimensional system. All the order parameters, such as distance of two ends of the ssDNA chain, number of pairings, and electrostatic potential, discontinuously jump when the external force pass through the critical force. At the critical point, the fluctuation of the order parameters and the autocorrelation time diverge. This effect, known as

critical slowing down, renders the canonical Metropolis Monte Carlo calculations unreliable. All these features make the designed ssDNA an excellent laboratory for the study of first-order phase transition in one-dimensional system.

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References

1. J. D. Watson, N. H. Hopkins, J. W. Roberts, J. A. Steitz, and A. M. Weiner, *Molecular Biology of the Gene*. Benjamin/Cummings Pub., California, (1987), 4th edition.
2. P. J. Hagerman, *Ann. Rev. Biophys. Biophys. Chem.* **17**, 265 (1988).
3. S. B. Smith, L. Finzi, and C. Bustamante, *Science* **258**, 1122 (1992).
4. P. Cluzel, A. Lebrun, C. Heller, R. Lavery, J. Viovy, D. Chatenay, and F. Caron, *Science* **271**, 792 (1996).
5. S. B. Smith, Y. Cui, and C. Bustamante, *Science* **271**, 795 (1996).
6. T. R. Strick, J. F. Allemand, D. Bensimon, A. Bensimon, V. Croquette, *Science* **271**, 1835 (1996).
7. M. Rief, H. Clausen-Schaumann, and H. E. Gaub, *Nat. Str. Bio.* **6**, 346 (1999).
8. G. J. Wuite, S. B. Smith, M. Young, D. Keller, and C. Bustamante, *Nature* **404**, 103 (2000).
9. B. Maier, D. Bensimon, and V. Croquette, *Proc. Natl. Acad. Sci. USA* **97**, 12002 (2000).
10. C. Bustamante, S. B. Smith, J. Liphardt, and D. Smith, *Curr. Opi. Struc. Biol.* **10**, 279 (2000).
11. H. J. Zhou, Y. Zhang, and Z. C. Ou-Yang, *Phys. Rev. Lett.* **82**, 4560 (1999).
12. W. Saenger, *Principles of Nucleic Acid Structure*. Springer-Verlag, New York, (1984).
13. H. J. Zhou, Y. Zhang, and Z. C. Ou-Yang, *Phys. Rev. E* **62**, 1045 (2000).
14. R. Everaers, R. Bundschuh and K. Kremer, *Europhys. Lett.* **29**, 263 (1995).
15. T. B. Liverpool, R. Golestanian and K. Kremer, *Phys. Rev. Lett.* **80**, 405 (1998).
16. C. Bustamante, J. F. Marko, E. D. Siggia, and S. B. Smith, *Science* **265**, 1599 (1994).

17. T. T. Perkins, D. E. Smith, S. Chu, *Science* **276**, 2016 (1997), and references cited therein.
18. J. F. Marko, and E. D. Siggia, *Macromolecules* **28**, 8759 (1995).
19. T. R. Strick, J. F. Allemand, D. Bensimon, and V. Croquette, *Biophys. J.* **74**, 2016 (1998).
20. H. J. Zhou, Y. Zhang, Z. C. Ou-Yang, X. Z. Feng, S. M. Lindsay, P. Balagurumoorthy, and R. E. Harrington, *J. Mol. Biol.*, **306**, 227 (2001).
21. Y. Zhang, H. J. Zhou, Z. C. Ou-Yang, *Biophys. J.* **78**, 1979 (2000).
22. A. V. Vologodskii, and J. F. Marko, *Biophys. J.* **73**, 123 (1997).
23. N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, and A. H. Teller, *J. Chem. Phys.* **21**, 1087 (1953).
24. A. V. Vologodskii, A. V. Lukashin, M. D. Frank-Kamenetskii and V. V. Anshelevich, *Sov. Phys. JETP* **39**, 1059 (1974).
25. B. A. Harris, and S. C. Harvey, *J. Comput. Chem.* **20**, 813 (1999).
26. J. H. White, *Am. J. Math.* **91**, 693 (1969).
27. C. Bouchiat, and M. Mèzard, *Phys. Rev. Lett.* **80**, 1556 (1998).
28. J. F. Allemand, D. Bensimon, R. Lavery, and V. Croquette, *Proc. Natl. Acad. Sci. USA* **95**, 14152 (1998).
29. A. Montanari and M. Mèzard, *Cond-mat/0006166* (preprint).
30. H. J. Zhou, and Y. Zhang, *J. Chem. Phys.* (Submitted).
31. P. Pierce, *Ph. D. Dissertation*, Department of Chemistry, Yale University (1958).
32. S. A. Rice, and M. Nagasawa, *Polyelectrolyte solutions*, Academic Press, New York and London (1961).
33. S. L. Brenner, and V. A. Parsegian, *Biophys. J.* **14**, 327 (1974).
34. D. Stigter, *Biopolymers* **16**, 1435 (1977).
35. P. G. Higgs, *Phys. Rev. Lett.* **76**, 704 (1996).
36. R. Bundschuh, and T. Hwa, *Phys. Rev. Lett.* **83**, 1479 (1999).
37. H. J. Zhou, Y. Zhang, and Z. C. Ou-Yang, *Phys. Rev. Lett.* **86**, 356 (2001).
38. A. V. Vologodskii, S. D. Levene, K. V. Klenin, M. Frank-Kamenetskii, and N. R. Cozzarelli. *J. Mol. Biol.* **227**, 1224 (1992).
39. Y. Zhang, *Phys. Rev. E* **62**, R5923 (2000).
40. U. H. E. Hansmann, and Y. Okamoto. *Phys. Rev. E* **56**, 2228 (1997).
41. A. M. Ferrenberg, and Swendsen, R. H. *Phys. Rev. Lett.* **61**, 2635 (1988).
42. Y. Zhang, H. J. Zhou, and Z. C. Ou-Yang, 'Stretching single-stranded DNA: Interplay of electrostatic and base-pairing and stacking interactions', (to be published).
43. R. H. Swendsen, and J. S. Wang. *Phys. Rev. Lett.* **58**, 86 (1987).
44. U. Wolff, *Phys. Rev. Lett.* **62**, 361 (1989).
45. Y. Zhang, and H. J. Zhou, (in preparation).

46. H. Clausen-Schaumann, M. Seitz, R. Krautbauer, and H. E. Gaub. *Curr. Opi. Chem. Biol.* **4**, 524 (2000).
47. U. Bockelmann, B. Essevaz-Roulet, and F. Heslot. *Phys. Rev. Lett.* **79**, 4489 (1997).

THE PROPAGATION OF ELECTRONIC EXCITATION IN MOLECULAR AGGREGATES

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The influence of intra-molecular vibrations on the propagation of Frenkel excitons along polymer chains of identical monomers is considered. Two regimes can be distinguished according to the strength of the monomer interaction compared to the vibrational bandwidth. The shape of the polymer spectrum and the extent of the exciton propagation are very different in the two cases.

1 Introduction

The propagation of electronic excitation in molecular aggregates is of interest in biological physics for problems as diverse as the formation of dimers on DNA¹ or the function of the photosynthetic unit². Unfortunately the detailed application of theory to such complex systems is bedevilled by problems such as the lack of precise knowledge on conformational structures, solvent interactions, finite temperature effects and so on. Fortunately, nature has provided certain 'clean' systems, which can be studied to test theories of electronic propagation on molecular aggregates. Paramount amongst these are the class of photosensitizing cyanine dyes, which aggregate readily into simple well-defined structures. In such polymeric systems electronic excitation is described by de-localised Frenkel excitons due to resonant coupling between identical monomer units and such a theoretical model has been applied to study excitation transfer in a bacteriochlorophyll system².

In the absence of coupling to vibrational degrees of freedom, electronic excitation can propagate unhindered along the chain. Coupling to vibrations hinders this propagation. The coupling can be either to external vibrations (e.g., of the surrounding medium), to lattice vibrations of the polymeric aggregate or to internal vibrations of the monomers. Many investigations of the first two types of vibrational-electronic (vibronic) coupling have been undertaken over the years, see for example^{3,4,5}. Studies of the explicit effect of intramolecular vibrations are less common, simply because, at least where only one vibrational mode is involved, these do not lead to energy dissipation. Here, a Green function method developed earlier⁶ will be used to describe excitation formation and propagation where both types of vibronic coupling are present. The intramolecular vibrations will be included explicitly so that electronic excitation is accompanied always by vibrational excitation in the exciton band. This leads to very different absorption

band profiles in the case of weak and strong coupling between monomers. A very simple model can explain these structural changes. The coupling to external vibrations is then included also in a phenomenological way by broadening each vibronic line to yield a continuous monomer spectrum. Then, using a time-dependent Green function, it is shown that the distance of propagation of the exciton (degree of de-localisation) depends crucially on the ratio of the width of the vibronic monomer spectrum to the strength of the monomer-monomer interaction (the exciton bandwidth).

2 Exciton Formation; the CES Approximation

In general when monomers aggregate there is a change in the polymer spectrum compared to that of the constituent monomers. That the extent of this change depends upon the ratio of monomer bandwidth to exciton bandwidth was recognised long ago by Simpson and Peterson⁷. One speaks of weak or strong coupling as this ratio is greater than or less than unity. In the case of weak coupling, there is only a small change in the absorption bandshape when the monomers stack up in a regular way. Such behaviour is typified by the spectrum⁸ of the polypeptide poly-L-lysine hydrochloride shown in Fig. 1.

By contrast, in strong coupling there is a drastic shift and narrowing of the spectrum⁹ as in the case of the cyanine dye TDBC shown in Fig. 2. In ref.⁶ a simple approximation was suggested which can reproduce both these limiting cases. This approximation is called the coherent exciton scattering (CES) approximation, whose derivation is outlined below. In particular, the CES approximation reproduces well the spectral changes occurring when the dye pseudo-isocyanine aggregates in aqueous solution¹⁰. In this class of cyanine dyes the excitonic state with small vibrational width is called the J-band (for historical reasons).

In¹¹ the light absorption cross-section is related to the polarisability tensor (either for monomer or polymer) in the following way;

$$\sigma(\omega) = \frac{4\pi\omega}{c} \text{Im}(\underline{\varepsilon}^* \cdot \underline{\alpha}(\omega) \cdot \underline{\varepsilon}), \quad (1)$$

where $\underline{\varepsilon}$ is the photon's polarisation vector, ω its frequency and $\underline{\alpha}$ the dipole polarisability tensor of the absorbing aggregate. For the polymer the polarisability tensor is defined as

$$\underline{\alpha}^p = -\sum_{nm} \underline{\mu}_n \langle G_{nm} \rangle \underline{\mu}_m, \quad (2)$$

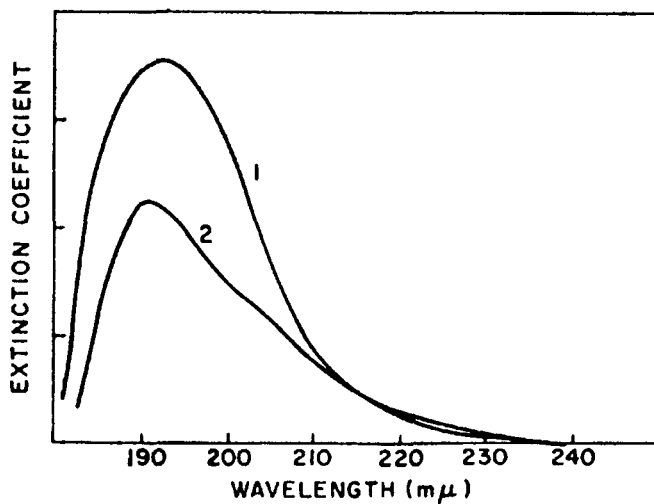


Figure 1. The absorption spectrum of poly-L-lysine hydrochloride in (1) the random coil form, (2) the helical form (from ⁸).

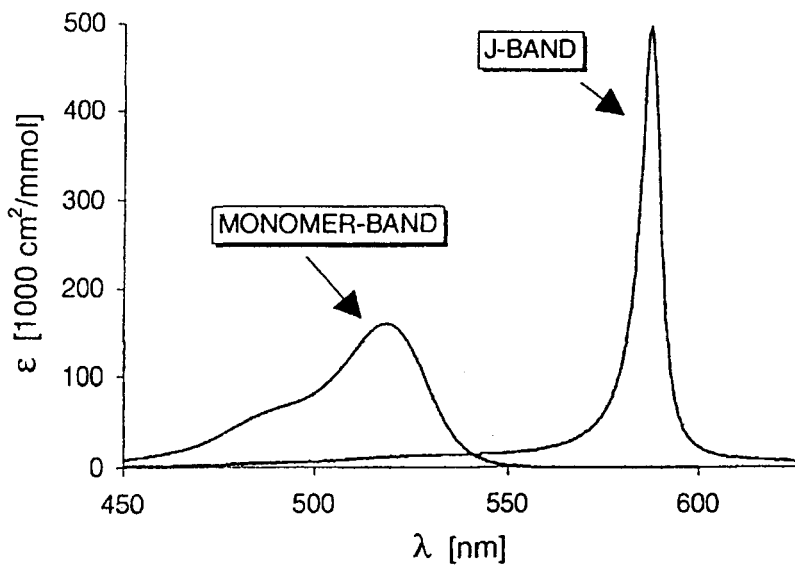


Figure 2. The absorption spectrum of the dye TDBC (from ⁹).

where $\underline{\mu}_n$ is the vector electronic transition dipole from the electronic ground state and the brackets $\langle \dots \rangle$ denote an average over the initial vibrational state of the nuclei. The matrix element is defined by

$$G_{nm} = (\pi_n | G(E) | \pi_m), \quad (3)$$

where brackets (...) denote integration over electronic degrees of freedom only. The state $|\pi_m\rangle$ is one in which the m 'th monomer is in the excited electronic state and all others in their electronic ground state. Hence G_{nm} is still an operator in the space of nuclear co-ordinates. The full Green operator $G(E)$ is

$$G(E) = (E - H + i\delta)^{-1}, \quad (4)$$

where H is the complete polymer Hamiltonian operator, i.e.,

$$\begin{aligned} H &= \sum_n (H_n^{el} + T_n) + \sum_{nm} V_{nm} \\ &\equiv H_0 + V. \end{aligned} \quad (5)$$

Here H_n^{el} and T_n are the electronic and nuclear Hamiltonians for monomer n and V_{nm} is the interaction potential between monomers n and m . In deriving Eq. (2) the following approximations have been made:

1. The Born-Oppenheimer approximation for the polymer ground state.
2. The assumption that $\underline{\mu}_n$ is independent of nuclear vibrations.
3. The rotating-wave approximation for the interaction of light with the polymer, i.e., in Eq. (3), $E = E_0 + \hbar\omega$ where E_0 is the polymer ground-state energy.

When the interaction between the monomers is switched off (or, for a random polymer, is assumed to average to zero) one can put $V = 0$ in Eq.(5) to give the polarisability tensor of an aggregate of non-interacting monomers

$$\underline{\alpha}^M = - \sum_n \underline{\mu}_n \langle g_{nm} \rangle \underline{\mu}_n, \quad (6)$$

where now

$$g_n(E) = (\pi_n | g(E) | \pi_n) \quad (7)$$

and

$$g(E) = (E - H_0 + i\delta)^{-1}. \quad (8)$$

Then we note that for random orientation of the monomers the frequency dependence of the absorption spectrum is simply that of a single monomer and given by $\text{Im}\langle g_n(E_0 + \hbar\omega) \rangle$. Using the identity

$$g(E) = P \frac{1}{(E - H_0)} - i\pi\delta(E - H_0), \quad (9)$$

one sees that absorption occurs at the poles in the monomer Green operator or, from (6), in the monomer polarisability. Correspondingly polymer absorption occurs at the poles in $\underline{\alpha}^P$ or, equivalently, in the Green operator $G(E)$. The problem then is to calculate the position of these poles. To this end one begins with the identity

$$G = g + gVG \quad (10)$$

whose matrix element $\langle \pi_n | \dots | \pi_m \rangle$ is given by

$$\langle G_{nm} \rangle = \langle g_n \rangle \delta_{nm} + \left\langle g_n \sum_{n'} V_{nn'} G_{n'm} \right\rangle \quad (11)$$

or, as an operator in the space of electronic states,

$$\langle G \rangle = \langle g \rangle + \langle gVG \rangle, \quad (12)$$

where $\langle g \rangle$ is proportional to the unit operator.

The key assumption to solve Eq. (12) in a simple way are:

- a) Assume that $V_{nn'}$ is independent of nuclear co-ordinates. This approximation is equivalent to ii) above since, if the monomers do not overlap strongly, $V_{nn'}$ can be considered to arise from a dipole-dipole interaction between $\underline{\mu}_n$ and $\underline{\mu}_{n'}$.
- b) The replacement of g by its vibrational ground-state expectation value $\langle g \rangle$.

This approximation b) is the key approximation, the CES approximation and corresponds to the assumption that the monomers do not depart significantly from their ground-state nuclear configuration during electronic excitation or de-excitation.

With approximations a) and b), Eq. (12) becomes

$$\langle G \rangle = \langle g \rangle + \langle g \rangle V \langle G \rangle \quad (13)$$

or

$$\langle G \rangle = \frac{1}{1 - \langle g \rangle V} \langle g \rangle. \quad (14)$$

In ref. ¹¹, it is shown that, when the absorption cross-section is evaluated according to Eqs. (1), (2) and (6) either for a linear or a helical array of monomers, the polymer cross-section is proportional to $\text{Im}\langle G(E) \rangle$ and the monomer cross-section is proportional to $\text{Im}\langle g(E) \rangle$. Hence the simple equation (14) can be used to discuss the changes in absorption spectra in going from monomer to polymer.

The physical interpretation of Eq. (14) is clear. The monomer absorbs where $\langle g \rangle$ has poles. The polymer absorption occurs where the r.h.s. of (14) has poles. Clearly this is not where $\langle g \rangle$ has poles (here $\langle G \rangle$ is smooth and proportional to V^{-1}). Hence there is a shift in the absorption to new resonance positions where $(1 - \langle g \rangle V)^{-1}$ has poles, i.e., where

$$\langle g \rangle = V^{-1} \quad (15)$$

or, since V is real

$$\text{Re}\langle g \rangle = V^{-1}. \quad (16)$$

Furthermore, from the structure of (14) one can interpret the factor $(1 - \langle g \rangle V)^{-1}$ as providing a new "dielectric constant" for the absorption characteristics of the polymer. As will emerge presently the formation of an excitation of the polymer as a whole is formally similar to the formation of other collective oscillations known in Physics, e.g., the plasmon or the giant dipole resonance in nuclei.

3 The Effect of Intra-Monomer Vibrations

First the very simplest case, the electronic excitation from a ground vibrational state into a single vibrational mode in the excited electronic state, will be discussed. This implies neglect of all dissipative effects due to coupling to other vibrational modes

of the monomer, to vibrations of the polymer lattice and to possible solvent vibrational modes. The principal feature to emerge will be to show how the narrow excitonic J-band is produced in the case of strong coupling.

In the one-mode approximation, the monomer absorption spectrum is a set of infinitely-resolved spectral lines, i.e., we take

$$\text{Im}\langle g \rangle = -\pi \sum_j f_j \delta(E - \epsilon_j), \quad (17)$$

where f_j is the Franck-Condon factor for exciting the j 'th vibrational level, of energy ϵ_j , of the upper electronic state. Since the real and imaginary parts of $\langle g \rangle$ are connected by a dispersion relation, one can derive from (17) that

$$\text{Re}\langle g \rangle = P \sum_j \frac{f_j}{(E - \epsilon_j)} \quad (18)$$

which can then be used in Eq. (16) to calculate the polymer absorption spectrum. The emergence of shifted absorption lines for the polymer is best illustrated graphically. The $\text{Re}\langle g \rangle$ for a typical spectrum is shown in Fig. 3. The function is singular at $j = 0, 1, 2$, and so on where the monomer has absorption lines (the energy

in Fig. 3 is measured in units of the vibrational quantum in the upper state). In these units the monomer spectrum has width $\sqrt{2}$. The polymer absorbs where the different branches of the curve intersect the horizontal line V^{-1} . Clearly for weak coupling (V^{-1} large) there is only a very small shift from the monomer positions. However for large coupling (V^{-1} small) something rather dramatic occurs. The new poles, corresponding to polymer absorption are shown in Fig. 3. There is polymer absorption still in the region where the monomer absorbs, however on the high-energy side of the monomer band, between $j = 8$ and 9 in Fig. 3, a completely new pole appears. Not only that, one can show that the fractional strength associated with each polymer level is given by

$$F_j = -\pi \left[V^2 \frac{\partial \langle g \rangle}{\partial E} \bigg|_{E_j} \right]^{-1}, \quad (19)$$

where E_j is the position of the pole. That is, the strength of the polymer absorption at each pole is inversely proportional to the slope of the $\text{Re}\langle g \rangle$ curve at the pole position.

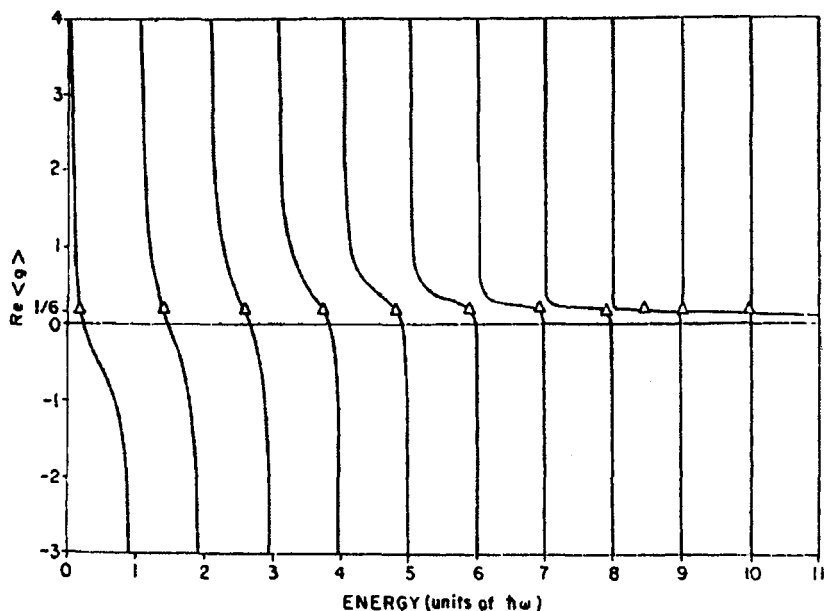


Figure 3. The function $\text{Re}\langle g \rangle$. The poles of $\langle G \rangle$ are indicated by open triangles for the case $V = 6\hbar\omega$.

From Fig. 3 one sees that for the pole between $j = 8$ and 9 , the curve is relatively flat, giving rise to large absorption strength into that level. This is confirmed by a direct calculation of the corresponding spectra (Fig. 4). As in the case of the plasmon and the nuclear giant dipole resonance, the state which splits off from the monomer band is interpreted as a collective excitation of the whole polymer and carries almost all the oscillator strength of the transition. This is the exciton. In the limit of very strong coupling the spectrum is a single absorption line, as would be the case were one to ignore the vibrational structure of the electronic transition altogether.

4 Inclusion of Vibrational Broadening

Due to the coupling with other modes and with the surroundings, the monomer vibronic absorption lines are broadened, in the extreme case into a continuum as shown in Fig. 5, for example. The exciton absorption (J-band) is much narrower.

This behaviour is reproduced in the CES approximation and, as in the previous section, finds its explanation in the formation of a de-localised co-operative state.

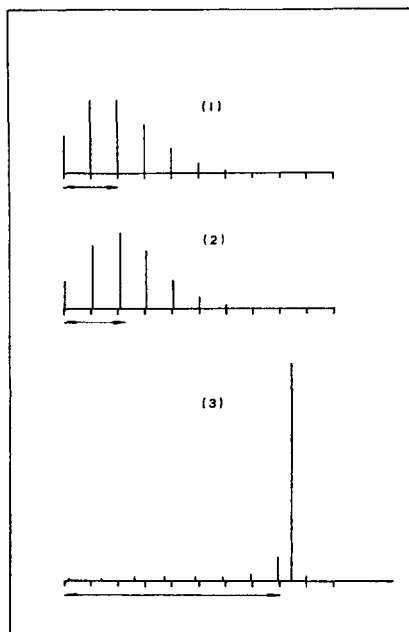


Figure 4. The calculated vibrational spectrum of 1) the monomer, 2) the polymer with $V = \hbar\omega/4$, 3) the polymer with $V = 6\hbar\omega$.

For simplicity, a continuous monomer vibronic band is approximated by a Gaussian form

$$\text{Im}\langle g(E) \rangle = -\frac{\sqrt{\pi}}{\Delta} \exp(-\varepsilon^2), \quad (20)$$

where $\varepsilon = (E_0 + \hbar\omega - E_1)/\Delta$ is the energy of the electronic state and Δ the Franck-Condon width of the monomer band. With the help of the dispersion relation the full function $\langle g(E) \rangle$ can be calculated and then the polymer spectrum from (14). The result is shown in Fig. 5. In this case the coupling strength is measured by the parameter V/Δ . As the coupling strength increases, the polymer spectrum shifts to higher energies and narrows considerably. Clearly there is a close qualitative

correspondence between the behaviour in Fig. 5 and that shown in Fig. 4. In the case of pseudo-isocyanine it has been shown¹⁰ that the CES approximation reproduces quantitatively the measured spectrum.

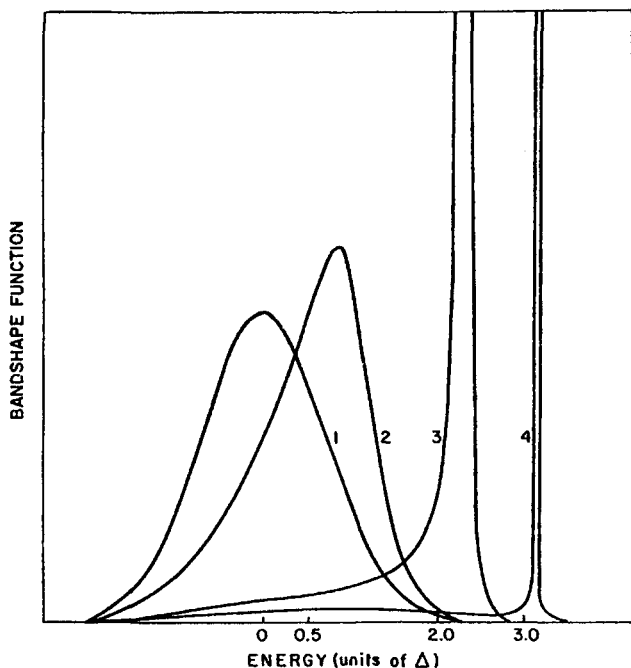


Figure 5. The absorption spectrum of the monomer is the Gaussian curve 1. The polymer spectrum is shown for 2) $V = \Delta/2$, 3) $V = 2\Delta$, 4) $V = 3\Delta$.

The narrowing of the polymer spectrum has been explained mathematically with the help of Fig. 3. The question remains as to a physical explanation. The answer lies in a consideration of the times involved. The time for vibrational relaxation is given roughly by $\hbar/\Delta$. The time for electronic excitation to pass from one monomer to the next is given roughly by $\hbar/V$. Hence the coupling strength parameter V/Δ is the ratio of these two times. For weak coupling, dissipation occurs before the exciton has passed on and the polymer spectrum remains broad. For strong coupling the electronic excitation transfer time is much less than the vibrational relaxation time. Transfer takes place before the nuclei have time to move out of their initial configuration and the polymer spectrum shows no vibrational structure. Similar considerations explain the absence of vibrational splitting when only intra-monomer vibrations are present. That this explanation is correct is

supported by the TDBC fluorescence spectra ⁹ shown in Fig. 6. The monomer spectrum shows a large Stokes shift, indicating vibrational relaxation before fluorescent emission. The polymer spectra show almost no Stokes shift corresponding to fluorescence from a nuclear configuration identical to that of the initial state.

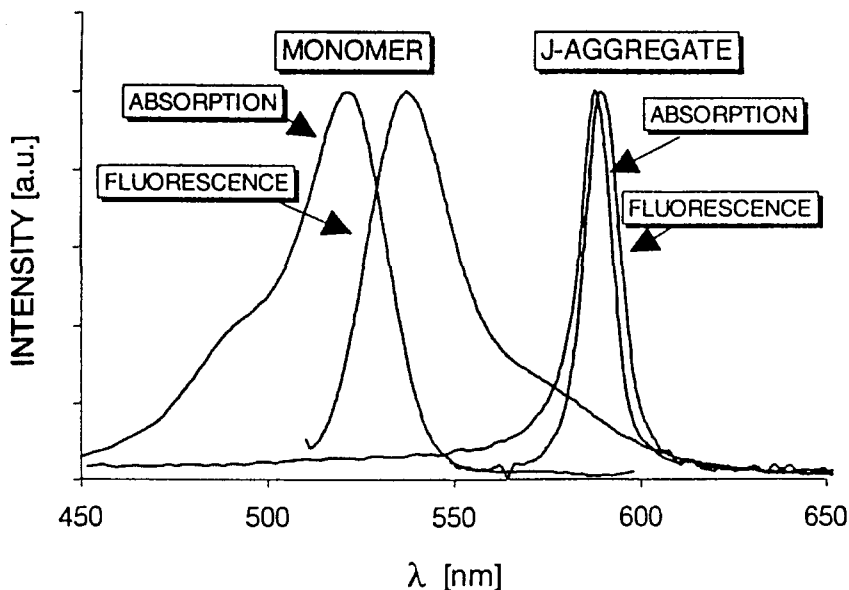


Figure 6. The absorption and fluorescence spectra of monomeric and aggregated TDBC dye (from ⁹).

5 Propagation of Excitation

Since the spectra of monomer and polymer give information on the propagation characteristics of electronic excitation, it is of interest to calculate the propagation length along a linear chain of monomers. Clearly this will depend also on the vibrational dissipation time compared with the transfer time, i.e., on the ratio V/Δ . For weak coupling, excitation will be localised on one or two monomers; in the limit of infinitely strong coupling it will be completely de-localised. A simple estimate of the propagation length is obtained in the following way. The amplitude that a monomer n becomes electronically excited after initial excitation of monomer m is clearly proportional to $\langle G_{nm} \rangle$. The various exciton modes along the linear

chain are characterised by an exciton wave vector k , corresponding to a travelling wave of excitation. Hence, restricting to a single mode k , translational invariance requires that

$$\langle G_{nm} \rangle = A_k \exp[ik(n - m)] \quad (21)$$

where A_k is a constant.

In the CES approximation and restricting to nearest-neighbour interaction only, one can show that

$$k = \cos^{-1} \left(1 / (2V \langle g \rangle) \right). \quad (22)$$

Note that due to vibrational damping, k is complex $k = k' + ik''$ and the probability of excitation travelling from monomer m to monomer n is proportional to

$$|\langle G_{nm} \rangle|^2 = |A_k|^2 \exp[-2k''(n - m)]. \quad (23)$$

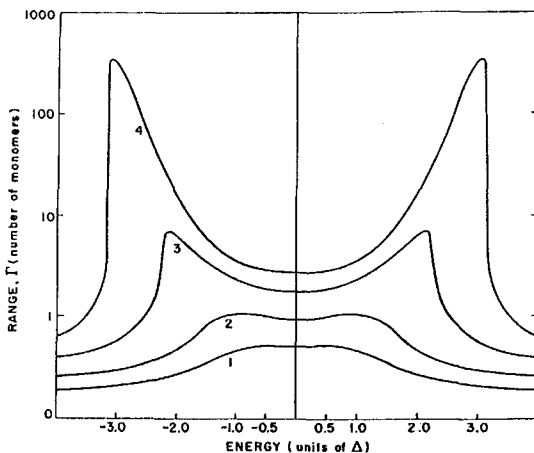


Figure 7. The range of exciton propagation for 1) $V = \Delta/2$, 2) $V = \Delta$, 3) $V = 2\Delta$, 4) $V = 3\Delta$.

Hence we define a range of energy transfer $\Gamma = (2k'')^{-1}$. This range is plotted in Fig. 7 as a function of the energy (or k value) across the exciton band of width $2V$ for various ratios V/Δ . As expected one sees that for weak coupling the exciton is localised, whilst for strong coupling excitation can propagate over hundreds of

monomers before dissipating into vibrations. Propagation is maximum at the band edges and in particular at the upper-energy band edge which, in the geometry considered here, is the exciton state which absorbs light (since for this k -vector all dipoles are in phase).

6 Conclusions

The effect of both intra- and intermolecular vibrations on the absorption spectrum and propagation characteristics of an exciton band formed when identical monomers aggregate has been studied. In the case where broadening of the vibrational levels can be neglected it has been shown that a narrow exciton line, carrying almost all the oscillator strength, appears when the excitonic coupling energy exceeds the monomer vibrational bandwidth. The main characteristics of the polymer spectrum are preserved when coupling to external vibrations are also taken into account. In particular it has been shown explicitly that in strong coupling the exciton can propagate over several hundred monomers before its energy is dissipated into vibrations.

References

1. A. A. Voityuk, M. Michel-Beyerle and N. Rösch, *J. A. C. S.* **118**, 9750 (1996).
2. S. Tretiak, C. Middleton, V. Chernyak and S. Mukamel, *J. Phys. Chem.* **B104**, 4519 (2000).
3. H. Haken and G. Strobl, *Z. Phys.* **262**, 135 (1973).
4. H. Sumi, *J. Chem. Phys.* **67**, 2943 (1977).
5. E. W. Knapp, *Chem. Phys.* **85**, 73 (1984).
6. J. S. Briggs and A. Herzenberg, *J. Phys.* **B3**, 1663 (1970).
7. W. T. Simpson and D. L. Peterson, *J. Chem. Phys.* **26**, 588 (1995).
8. W. B. Gratzner, G. M. Holzwarth and P. Doty, *Proc. N. A. S.* **47**, 1775 (1961).
9. J. Moll, *Forschungsbericht* 214, B. A. M. (Berlin) (1995).
10. J. S. Briggs, *Z. Phys.* **75**, 214 (1971).
11. J. S. Briggs and A. Herzenberg, *Mol. Phys.* **21**, 865 (1971).

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